

# Compare Results

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USB\_PD\_R3\_2 V1.1 2024-10\_Chapter 6 Protocol Layer.pdf

198 pages (5.10 MB)

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Total Changes

4088

Text only comparison

Content

2966 Replacements

703 Insertions

419 Deletions

Styling and Annotations

0 Styling

0 Annotations

Go to First Change (page 1)

## 6 Protocol Layer

### 6.1 Overview

This chapter describes the requirements of the USB Power Delivery Specification's *Protocol Layer* including:

- Details of how *Messages* are constructed and used.
- Use of timers and timeout values.
- Use of *Message* and retry counters.
- Reset operation.
- Error handling.
- *State* behavior.

Refer to [Section 2.6, "Architectural Overview"](#) for an overview of the theory of operation of USB Power Delivery.

## 6.2 Messages

This specification defines three types of *Messages*:

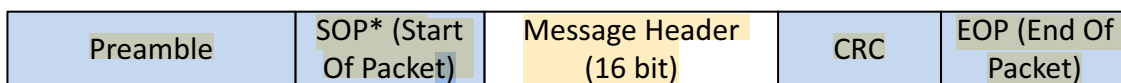
- *Control Messages* that are short and used to manage the *Message* flow between *Port Partners* or to exchange *Messages* that require no additional data. *Control Messages* are 16 bits in length.
- *Data Messages* that are used to exchange information between a pair of *Port Partners*. *Data Messages* range from 48 to 240 bits in length.
  - Some examples of *Data Messages* are:
    - Those used to expose *Capabilities* and *Negotiate* power.
    - Those used for the *BIST*.
    - Those that are *Vendor Defined Messages*.
- *Extended Messages* that are used to exchange information between a pair of *Port Partners*. *Extended Messages* are up to **MaxExtendedMsgLen** bytes.
  - Some examples of *Extended Messages* are:
    - Those used for *Source* and *Battery* information.
    - Those used for Security.
    - Those used for Firmware Update.
    - Those that are *Vendor Defined Extended Messages*.

### 6.2.1 Message Construction

All *Messages* **shall** be composed of a *Message Header* and a variable length (including zero) data portion. A *Message* either originates in the *Protocol Layer* and is passed to the *PHY Layer*, or it is received by the *PHY Layer* and is passed to the *Protocol Layer*.

[Figure 6.1, "USB Power Delivery Packet Format for a Control Message"](#) illustrates a *Control Message* as part of a *Packet* showing the parts are provided by the *Protocol Layer* and *PHY Layer*.

**Figure 6.1 USB Power Delivery Packet Format for a Control Message**



Legend:



[Figure 6.2, "USB Power Delivery Packet Format including Data Message Payload"](#) illustrates a *Data Message* as part of a *Packet* showing the parts are provided by the *Protocol Layer* and *PHY Layer*.

**Figure 6.2 USB Power Delivery Packet Format including Data Message Payload**

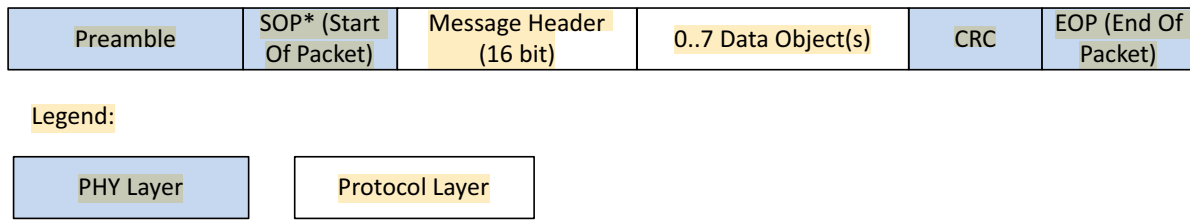
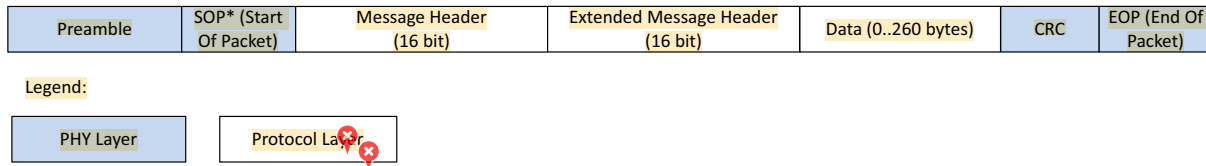


Figure 6.3, "USB Power Delivery Packet Format including an Extended Message Header and Payload" illustrates an Extended Message as part of a Packet showing the parts are provided by the Protocol Layer and PHY Layer.

**Figure 6.3 USB Power Delivery Packet Format including an Extended Message Header and Payload**



### 6.2.1.1 Message Header

Every Message **Shall** start with a Message Header as shown in:

- [Figure 6.1, "USB Power Delivery Packet Format for a Control Message"](#)
- [Figure 6.2, "USB Power Delivery Packet Format including Data Message Payload"](#)
- [Figure 6.3, "USB Power Delivery Packet Format including an Extended Message Header and Payload"](#)

and as defined in [Table 6.1, "Message Header"](#). The Message Header contains basic information about the Message and the PD Port Capabilities.

The Message Header **May** be used standalone as a Control Message when the **Number of Data Objects** field is zero or as the first part of a Data Message when the **Number of Data Objects** field is non-zero.

**Table 6.1 Message Header**

| Bit(s)  | Start of Packet | Field Name             | Reference                         |
|---------|-----------------|------------------------|-----------------------------------|
| 15      | SOP*            | Extended               | <a href="#">Section 6.2.1.1.1</a> |
| 14...12 | SOP*            | Number of Data Objects | <a href="#">Section 6.2.1.1.2</a> |
| 11...9  | SOP*            | MessageID              | <a href="#">Section 6.2.1.1.3</a> |
| 8       | SOP only        | Port Power Role        | <a href="#">Section 6.2.1.1.4</a> |
|         | SOP'/SOP''      | Cable Plug             | <a href="#">Section 6.2.1.1.7</a> |
| 7...6   | SOP*            | Specification Revision | <a href="#">Section 6.2.1.1.5</a> |
| 5       | SOP only        | Port Data Role         | <a href="#">Section 6.2.1.1.6</a> |
|         | SOP'/SOP''      | Reserved               | <a href="#">Section 1.4.2</a>     |
| 4...0   | SOP*            | Message Type           | <a href="#">Section 6.2.1.1.8</a> |

#### 6.2.1.1.1 Extended

The 1-bit **Extended** field **Shall** be set to zero to indicate a Control Message or Data Message and set to one to indicate an Extended Message.



The **Extended** field **Shall** apply to all *SOP\* Packet* types.

#### 6.2.1.1.2 Number of Data Objects

When the **Extended** field is set to zero the 3-bit **Number of Data Objects** field **Shall** indicate the number of 32-bit *Data Objects* that follow the *Message Header*. When this field is zero the *Message* is a *Control Message* and when it is non-zero, the *Message* is a *Data Message*.

The **Number of Data Objects** field **Shall** apply to all *SOP\* Packet* types.

When both the **Extended** bit and **Chunked** bit are set to one, the **Number of Data Objects** field **Shall** indicate the number of *Data Objects* in the *Message* padded to the 4-byte boundary including the *Extended Message Header* as part of the first *Data Object*.

When the **Extended** bit is set to one and **Chunked** bit is set to zero, the **Number of Data Objects** field **Shall** be **Reserved**.

**Note:** In this case, the *Message* length is determined solely by the **Data Size** field in the *Extended Message Header*.

#### 6.2.1.1.3 MessageID

The 3-bit **MessageID** field is the value generated by a rolling counter maintained by the originator of the *Message*. The **MessageIDCounter** **Shall** be initialized to zero at power-on as a result of a *Soft Reset*, or a *Hard Reset*. The **MessageIDCounter** **Shall** be incremented when a *Message* is successfully received as indicated by receipt of a **GoodCRC** *Message*.

**Note:** The usage of **MessageID** during testing with *BIST Messages* is defined in [\[USBPDCompliance\]](#).

The **MessageID** field **Shall** apply to all *SOP\* Packet* types.

#### 6.2.1.1.4 Port Power Role

The 1-bit **Port Power Role** field **Shall** indicate the *Port's* present *Power Role*:

- 0b Sink
- 1b Source

*Messages*, such as **Get\_Sink\_Cap\_Extended**, that are only ever sent by a *Source*, **Shall** always have the **Port Power Role** field set to *Source*. Similarly, *Messages* such as the **Request** *Message* that are only ever sent by a *Sink* **Shall** always have the **Port Power Role** field set to *Sink*.

During the *Power Role Swap AMS*, for the *Initial Source Port*, the **Port Power Role** field **Shall** be set to *Sink* in the **PS\_RDY** *Message* indicating that the *Initial Source's* power supply is turned off (see [Table 8.60, "Steps for a Successful Source Initiated Power Role Swap Sequence"](#) and [Table 8.63, "Steps for a Successful Sink Initiated Power Role Swap Sequence"](#)).

During the *Power Role Swap AMS*, for the *Initial Sink*, the **Port Power Role** field **Shall** be set to *Source* for *Messages* initiated by the *Policy Engine* after receiving the **PS\_RDY** *Message* from the *Initial Source* (see [Table 8.60, "Steps for a Successful Source Initiated Power Role Swap Sequence"](#) and [Table 8.63, "Steps for a Successful Sink Initiated Power Role Swap Sequence"](#)).

During the *Fast Role Swap AMS*, for the *Initial Source Port*, the **Port Power Role** field **Shall** be set to *Sink* in the **PS\_RDY** *Message* indicating that *V<sub>BUS</sub>* is not being driven by the *Initial Source* and is within **vSafe5V** (see [Figure 8.39, "Successful Fast Role Swap Sequence"](#)).

During the *Fast Role Swap AMS*, for the *Initial Sink Port*, the **Port Power Role** field **Shall** be set to *Source* for *Messages* initiated by the *Policy Engine* after receiving the **PS\_RDY** *Message* from the *Initial Source* (see [Figure 8.39, "Successful Fast Role Swap Sequence"](#)).

**Note:** The **GoodCRC** *Message* sent by the *Initial Sink* in response to the **PS\_RDY** *Message* from the *Initial Source* will have its **Port Power Role** field set to *Sink* since this is initiated by the *Protocol Layer*. Subsequent

Messages initiated by the Policy Engine, such as the **PS\_RDY** Message sent to indicate that **VBUS** is ready, will have the **Port Power Role** field set to *Source*.

The **Port Power Role** field of a received Message **Shall Not** be verified by the receiver and **Shall Not** lead to *Soft Reset*, *Hard Reset* or *Error Recovery* if it is incorrect.

The **Port Power Role** field **Shall** only be defined for *SOP* Packets.

### 6.2.1.1.5 Specification Revision

The **Specification Revision** field **Shall** be one of the following values (except 11b):

- 00b - *Revision 1.0 (Deprecated)*
- 01b - *Revision 2.0*
- 10b - *Revision 3.x*
- 11b - **Reserved, Shall Not** be used.

To ensure interoperability with existing **PDUSB** products, **PDUSB** products **Shall** support every PD *Specification Revision* starting from **[USB 2.0]** for *SOP\**; the only exception to this is a *VPD* which **Shall Ignore** Messages sent with PD *Specification Revision* 2.0 and earlier.

After a physical or logical (*USB Type-C® Error Recovery*) *Attach*, a *Port* discovers the common *Specification Revision* level between itself and its *Port Partner* and/or the *Cable Plug(s)*, and uses this *Specification Revision* level until a *Detach*, *Hard Reset* or *Error Recovery* happens.

After detection of the *Specification Revision* to be used, all PD communications **Shall** comply completely with the relevant *Revision* of the PD specification.

The 2-bit **Specification Revision** field of a **GoodCRC** Message does not carry any meaning and **Shall** be considered as *don't care* by the recipient of the Message. The sender of a **GoodCRC** Message **Shall** set the **Specification Revision** field to 01b (*Revision 2.0*) when responding to a Message that contains 01b in the **Specification Revision** field of the Message Header. The sender of a **GoodCRC** Message **May** set the **Specification Revision** field to 01b or 10b when responding to a Message that contains 10b (*Revision 3.x*) in the **Specification Revision** field of the Message Header.

The **Specification Revision** field **Shall** apply to all *SOP\** Packet types.

An *Attach* event or a *Hard Reset* **Shall** cause the detection of the applicable *Specification Revision* to be performed for both *Ports* and *Cable Plugs* according to the rules stated below:

When the *Source Port* first communicates with the *Sink Port* the **Specification Revision** field **Shall** be used as described by the following steps:

- 1) The *Source Port* sends a **Source Capabilities** Message to the *Sink Port* setting the **Specification Revision** field to the highest *Revision* of the Power Delivery Specification the *Source Port* supports.
- 2) The *Sink Port* responds with a **Request** Message setting the **Specification Revision** field to the highest *Revision* of the Power Delivery Specification the *Sink Port* supports that is equal to or lower than the *Specification Revision* received from the *Source Port*.
- 3) The *Source* and *Sink Ports* **Shall** use the *Specification Revision* in the **Request** Message from the *Sink* in step 2 in all subsequent communications until a *Detach*, *Hard Reset*, or *Error Recovery* happens.

Prior to entering the *First Explicit Contract*, the *VCONN Source* **Shall** use the following steps to establish a *Specification Revision* level:

- 1) The *VCONN Source* sends a **Discover Identity REQ** to the *Cable Plug (SOP)* setting the **Specification Revision** field in the Message to the highest *Revision* of the Power Delivery Specification the *VCONN Source* supports. After a *VCONN Swap* the required **Soft\_Reset / Accept** Message exchange is used for the same purpose (see [Section 6.3.13, "Soft Reset Message"](#)).

- 2) The *Cable Plug* responds with a **Discover Identity ACK** setting the **Specification Revision** field in the *Message* to the highest *Revision* of the Power Delivery Specification the *VCONN Source* supports that is equal to or lower than the *Specification Revision* it received from the *Source Port*.
- 3) The *Cable Plug* and *VCONN Source* **Shall** communicate using the lower of the two revisions until an *Explicit Contract* has been established.
- 4) [Table 6.2, "Revision Interoperability during an Explicit Contract"](#) shows the *Specification Revision* that **Shall** be used between the *Port Partners* and the *Cable Plugs* when the *Specification Revision* has been discovered and an *Explicit Contract* is in place.

**Notes:**

- A *VCONN Source* that does not communicate with the *Cable Plug(s)* **May** skip the above procedure.
- When a *Cable Plug* does not respond to a *Revision 3.x Discover Identity REQ* with a **Discover Identity ACK** or **BUSY** the *VCONN Source* **May** repeat steps 1-4 using a *Revision 2.0 Discover Identity REQ* in step 1 before establishing that there is no *Cable Plug* to communicate with.

A *VCONN Source* that supports *Revision 3.x* of the Power Delivery Specification **May** communicate with a *Cable Plug* also supporting *Revision 3.x* using *Revision 3.x* Compliant Communications regardless of the *Specification Revision* of its *Port Partner* while no *Explicit Contract* exists. After an *Explicit Contract* has been established the *Port Partners* and *Cable Plug(s)* **Shall** use [Table 6.2, "Revision Interoperability during an Explicit Contract"](#) to determine the *Revision* to be used.

All data in all *Messages* **Shall** be consistent with the **Specification Revision** field in the *Message Header* for that particular *Message*.

A *Cable Plug* **Shall Not** save the state of the agreed *Specification Revision*. A *Cable Plug* **Shall** respond with the highest *Specification Revision* it supports that is equal to or lower than the *Specification Revision* contained in the *Message* received from the *VCONN Source*.

*Cable Plugs* **Shall** operate using the same *Specification Revision* for both *SOP'* and *SOP''*. Cable assemblies with two *Cable Plugs* **Shall** operate using the same *Specification Revision* for both *Cable Plugs*.

See [Table 6.2, "Revision Interoperability during an Explicit Contract"](#) for details of how various *Revisions* **Shall** inter-operate.

**Table 6.2 Revision Interoperability during an Explicit Contract**

| Port 1 Revision | Cable Plug Revision | Port 2 Revision | Port to Port Operating Revision | Port to Cable Plug Operating Revision |
|-----------------|---------------------|-----------------|---------------------------------|---------------------------------------|
| 2               | 2                   | 2               | 2                               | 2                                     |
| 2               | 2                   | 3               | 2                               | 2                                     |
| 2               | 3                   | 2               | 2                               | 2                                     |
| 2               | 3                   | 3               | 2                               | 2                                     |
| 3               | 2                   | 2               | 2                               | 2                                     |
| 3               | 2                   | 3               | 3                               | 2                                     |
| 3               | 3                   | 2               | 2                               | 2                                     |
| 3               | 3                   | 3               | 3                               | 3                                     |

### 6.2.1.1.6 Port Data Role

The 1-bit **Port Data Role** field **Shall** indicate the *Port's* present *Data Role*:

- 0b *UFP*
- 1b *DFP*

The **Port Data Role** field **Shall** only be defined for *SOP Packets*. For all other *SOP\* Packets* the **Port Data Role** field is **Reserved** and **Shall** be set to zero.

If a *USB Type-C Port* receives a *Message* with the **Port Data Role** field set to the same *Data Role* as its current *Data Role*, except for the *GoodCRC Message*, *USB Type-C Error Recovery* actions as defined in [\[USB Type-C 2.4\]](#) **Shall** be performed.

For a *USB Type-C Port* the **Port Data Role** field **Shall** be set to the default value at *Attachment* after a *Hard Reset*: 0b for a *Port* with  $R_d$  asserted and 1b for a *Port* with  $R_p$  asserted.

In the case that a *Port* is not *USB Communications capable*, at *Attachment* a *Source Port* **Shall** default to *DFP* and a *Sink Port* **Shall** default to *UFP*.

#### 6.2.1.1.7 Cable Plug

The 1-bit **Cable Plug** field **Shall** indicate whether this *Message* originated from a *Cable Plug* or *VPD*:

- 0b *Message* originated from a *DFP* or *UFP*.
- 1b *Message* originated from a *Cable Plug* or *VPD*

The **Cable Plug** field **Shall** only apply to *SOP' Packet* and *SOP'' Packet* types.

#### 6.2.1.1.8 Message Type

The 5-bit **Message Type** field **Shall** indicate the type of *Message* being sent. To fully decode the **Message Type**, the **Number of Data Objects** field is first examined to determine whether the *Message* is a *Control Message* or a *Data Message*. Then the specific **Message Type** can be found in [Table 6.5, "Control Message Types"](#) or [Table 6.6, "Data Message Types"](#).

The **Message Type** field **Shall** apply to all *SOP\* Packet* types.

#### 6.2.1.2 Extended Message Header

*Extended Messages* (indicated by the **Extended** field being set in the *Message Header*) **Shall** contain an *Extended Message Header* following the *Message Header* as shown in [Figure 6.3, "USB Power Delivery Packet Format including an Extended Message Header and Payload"](#) and defined in ["Table 6.3, "Extended Message Header"](#).

*Extended Messages* contain *Data Blocks* of **Data Size**, defined in the *Extended Message*, that are either sent in a single *Message* or as a series of *Chunks*. When the *Data Block* is sent as a series of *Chunks*, each *Chunk* in the series, except for the last *Chunk*, **Shall** contain **MaxExtendedMsgChunkLen** bytes. The last *Chunk* in the series **Shall** contain the remainder of the *Data Block* and so could be less than **MaxExtendedMsgChunkLen** bytes and **Shall** be padded to the next 4-byte *Data Object* boundary.

##### 6.2.1.2.1 Chunked

Table 6.3 Extended Message Header

| Bit(s)  | Start of Packet | Field Name           | Reference                         |
|---------|-----------------|----------------------|-----------------------------------|
| 15      | <i>SOP*</i>     | <b>Chunked</b>       | <a href="#">Section 6.2.1.2.1</a> |
| 14...11 | <i>SOP*</i>     | <b>Chunk Number</b>  | <a href="#">Section 6.2.1.2.2</a> |
| 10      | <i>SOP*</i>     | <b>Request Chunk</b> | <a href="#">Section 6.2.1.2.3</a> |
| 9       | <i>SOP*</i>     | <b>Reserved</b>      |                                   |
| 8...0   | <i>SOP*</i>     | <b>Data Size</b>     | <a href="#">Section 6.2.1.2.4</a> |

The *Port Partners* **Shall** use the **Unchunked Extended Messages Supported** field in the *Source\_Capabilities Message* and **Unchunked Extended Messages Supported** field in the *Request Message* to determine whether to send *Messages* of **Data Size** > **MaxExtendedMsgLegacyLen** bytes in a single *Unchunked Extended Message* (see [Section 6.4.1.2.1.6, "Unchunked Extended Messages Supported"](#) and [Section 6.4.2.6, "Unchunked Extended Messages Supported"](#)).

When either *Port Partner* only supports *Chunked Extended Messages*:

- The **Chunked** bit in every *Extended Message* **Shall** be set to one.
- Every *Extended Message* of **Data Size** > **MaxExtendedMsgLegacyLen** **Shall** be transmitted between the *Port Partners* in *Chunks*
- The **Number of Data Objects** in the *Message Header* **Shall** indicate the number of *Data Objects* in the *Message* padded to the 4-byte boundary including the *Extended Message Header* as part of the first *Data Object*.

The conditions listed above **Shall** apply until the *Port Pair* is *Detached*, there is a *Hard Reset*, there is *Error Recovery* or the *Source* removes power (except during a *Power Role Swap* or *Fast Role Swap* when the *Initial Source* removes power in order to for the *New Source* to apply power).

When both *Port Partners* support *Unchunked Extended Messages*:

- The **Chunked** bit in every *Extended Message* **Shall** be set to zero.
- Every *Extended Message* **Shall** be transmitted between the *Port Partners* *Unchunked*.
- The **Number of Data Objects** in the *Message Header* is **Reserved**.

The conditions listed above **Shall** apply until the *Port Pair* is *Detached*, there is a *Hard Reset*, there is *Error Recovery* or the *Source* removes power (except during a *Power Role Swap* or *Fast Role Swap* when the *Initial Source* removes power in order to for the *New Source* to apply power).

When sending *Extended Messages* to the *Cable Plug* the *VCONN Source* **Shall** only send *Chunked Extended Messages*. *Cable Plugs* **Shall** always send *Extended Messages* of **Data Size** > **MaxExtendedMsgLegacyLen** *Chunked* and **Shall** set the **Chunked** bit in every *Extended Message* to one.

When *Extended Messages* are supported *Chunking* **Shall** be supported.

#### 6.2.1.2.2 Chunk Number

The **Chunk Number** field **Shall** only be **Valid** in a *Message* if the **Chunked** flag is set to one. If the **Chunked** flag is set to zero the **Chunk Number** field **Shall** also be set to zero.

The **Chunk Number** field is used differently depending on whether the *Message* is a request for *Data*, or a requested *Data Block* being returned:

- In a request for *data* the **Chunk Number** field indicates the number of the *Chunk* being requested. The requester **Shall** only set this field to the number of the next *Chunk* in the series (the next *Chunk* after the last received *Chunk*).
- In the requested *Data Block* the **Chunk Number** field indicates the number of the *Chunk* being returned. The **Chunk Number** for each *Chunk* in the series **Shall** start at zero and **Shall** increment for each *Chunk* by one up to a maximum of 9 corresponding to 10 *Chunks* in total.

#### 6.2.1.2.3 Request Chunk

The **Request Chunk** bit **Shall** only be used for the *Chunked* transfer of an *Extended Message* when the **Chunked** bit is set to 1 (see [Figure 6.7, "Example Security Request sequence Chunked \(Chunked bit = 1\)"](#)). For *Unchunked Extended Message* transfers, *Messages* **Shall** be sent and received without the request/response mechanism (see [Figure 6.4, "Example Security Request sequence Unchunked \(Chunked bit = 0\)"](#)).

The **Request Chunk** bit **Shall** be set to one to indicate that this is a request for a *Chunk* of a *Data Block* and **Shall** be set to zero to indicate that this is a *Chunk* response containing a *Chunk*. Except for *Chunk zero*, a requested *Chunk* of a *Data Block* **Shall** only be returned as a *Chunk* response to a corresponding request for that *Chunk*. Both the *Chunk* request and the *Chunk* response **Shall** contain the same value in the **Message Type** field. When the **Request Chunk** bit is set to one the **Data Size** field **Shall** be zero.



#### 6.2.1.2.4 Data Size

The **Data Size** field **Shall** indicate how many bytes of data in total are in *Data Block* being returned. The total number of data bytes in the *Message* **Shall Not** exceed **MaxExtendedMsgLen**.

If the **Data Size** field is less than **MaxExtendedMsgLegacyLen** and the **Chunked** bit is set then the *Packet Payload* **Shall** be padded to the next 4-byte *Data Object* boundary with zeros (0x00).

If the **Data Size** field is greater than expected for a given *Extended Message* but less than or equal to **MaxExtendedMsgLen** then the expected fields in the *Message* **Shall** be processed appropriately and the additional fields **Shall** be **Ignored**.

#### 6.2.1.2.5 Extended Message Examples

The following examples illustrate the transmission of *Extended Messages* both *Chunked* (**Chunked** bit is one) and *Unchunked* (**Chunked** bit is zero). The examples use a **Security\_Request** *Message* of **Data Size** 7 bytes which is responded to by a **Security\_Response** *Message* of **Data Size** 30 bytes. The sizes of these *Messages* are arbitrary and are used to illustrate *Message* transmission; they are not intended to correspond to genuine security related *Messages*.

During *Negotiation* of the *Explicit Contract* after connection, the *Port Partners* use the **Unchunked Extended Messages Supported** field in the **Source\_Capabilities** *Message* and **Unchunked Extended Messages Supported** field in the **Request** *Message* to determine the value of the **Chunked** bit (see [Table 6.4, "Use of Unchunked Message Supported bit"](#)). When both *Port Partners* support *Unchunked Extended Messages* then the **Chunked** bit is zero otherwise the **Chunked** bit is one.

The **Chunked** bit is used to determine whether:

- ✖ The *Chunk* request/response mechanism is used.
- *Extended Messages* are *Chunked*.
- Padding is applied.
- The **Number of Data Objects** field is used.

The following examples illustrate the expected usage in each case.

Table 6.4 Use of Unchunked Message Supported bit

|                                |                                       | Source: <b>Source_Capabilities</b> Message |                                     |
|--------------------------------|---------------------------------------|--------------------------------------------|-------------------------------------|
|                                |                                       | ✖ Unchunked Message Supported bit = 0      | Unchunked Message Supported bit = 1 |
| ✖ Sink: <b>Request</b> Message | Unchunked Message Supported bit = 0   | <b>Chunked</b> bit = 1                     | <b>Chunked</b> bit = 1              |
|                                | ✖ Unchunked Message Supported bit = 1 | <b>Chunked</b> bit = 1                     | <b>Chunked</b> bit = 0              |

#### 6.2.1.2.5.1 Security\_Request/Security\_Response Unchunked Example

[Figure 6.4, "Example Security\\_Request sequence Unchunked \(Chunked bit = 0\)"](#) illustrates a typical sequence for a **Security\_Request** *Message* responded to by a **Security\_Response** *Message* using *Unchunked Extended Messages* (**Chunked** bit is zero) between a *USB Host* and a *Charger*. The entire *Data Block* is returned in one *Message*. The *Chunk* request/response mechanism is not used.

Figure 6.4 Example Security\_Request sequence Unchunked (Chunked bit = 0)

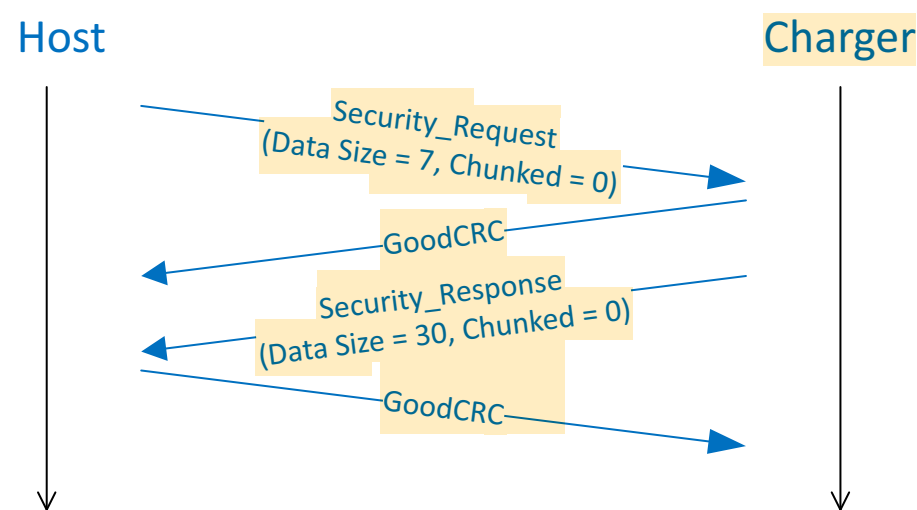


Figure 6.5, "Example byte transmission for Security\_Request Message of Data Size 7 (Chunked bit is set to zero)" details the **Security\_Request** Message shown in Figure 6.4, "Example Security\_Request sequence Unchunked (Chunked bit = 0)". The figure shows the byte ordering on the bus as well as the fact that there is no padding in this case. The **Number of Data Objects** field has a value of 0 since it is **Reserved** when the **Chunked** bit is zero. The **Data Size** field indicates the length of the **Extended Message** when the **Chunked** bit is set to 0, which in this case is 7 bytes.

Figure 6.5 Example byte transmission for Security\_Request Message of Data Size 7 (Chunked bit is set to zero)

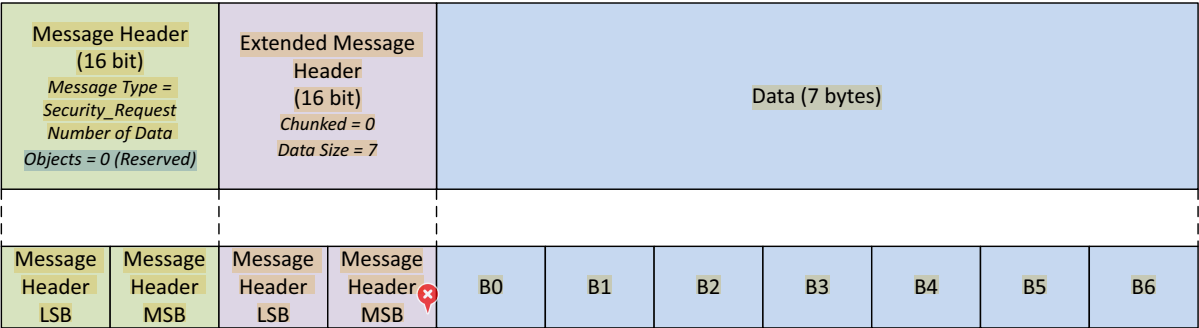
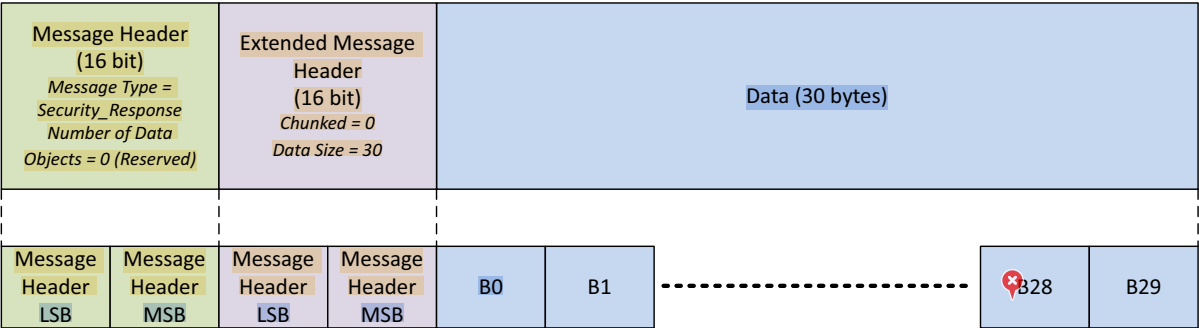


Figure 6.6, "Example byte transmission for Security\_Response Message of Data Size 30 (Chunked bit is set to zero)" details the **Security\_Response** Message shown in Figure 6.4, "Example Security\_Request sequence Unchunked (Chunked bit = 0)". The figure shows the byte ordering on the bus as well as the fact that there is no padding in this case. The **Number of Data Objects** field has a value of 0 since it is **Reserved** when the **Chunked** bit is zero. The **Data Size** field indicates the length of the **Extended Message** when the **Chunked** bit is set to zero, which in this case is 30 bytes.

Figure 6.6 Example byte transmission for Security\_Response Message of Data Size 30 (Chunked bit is set to zero)



6.2.1.2.5.2 Security\_Request/Security\_Response Chunked Example

Figure 6.7, "Example Security\_Request sequence Chunked (Chunked bit = 1)" illustrates a typical sequence for a Security\_Request Message responded to by a Security\_Response Message using Chunked Extended Messages (Chunked bit is one) between a USB Host and a Charger.

**Note:** Chunk Number zero in every Extended Message is sent without the need for a Chunk Request, but Chunk Number one and following need to be requested with a Chunk request.



Figure 6.7 Example Security\_Request sequence Chunked (Chunked bit = 1)

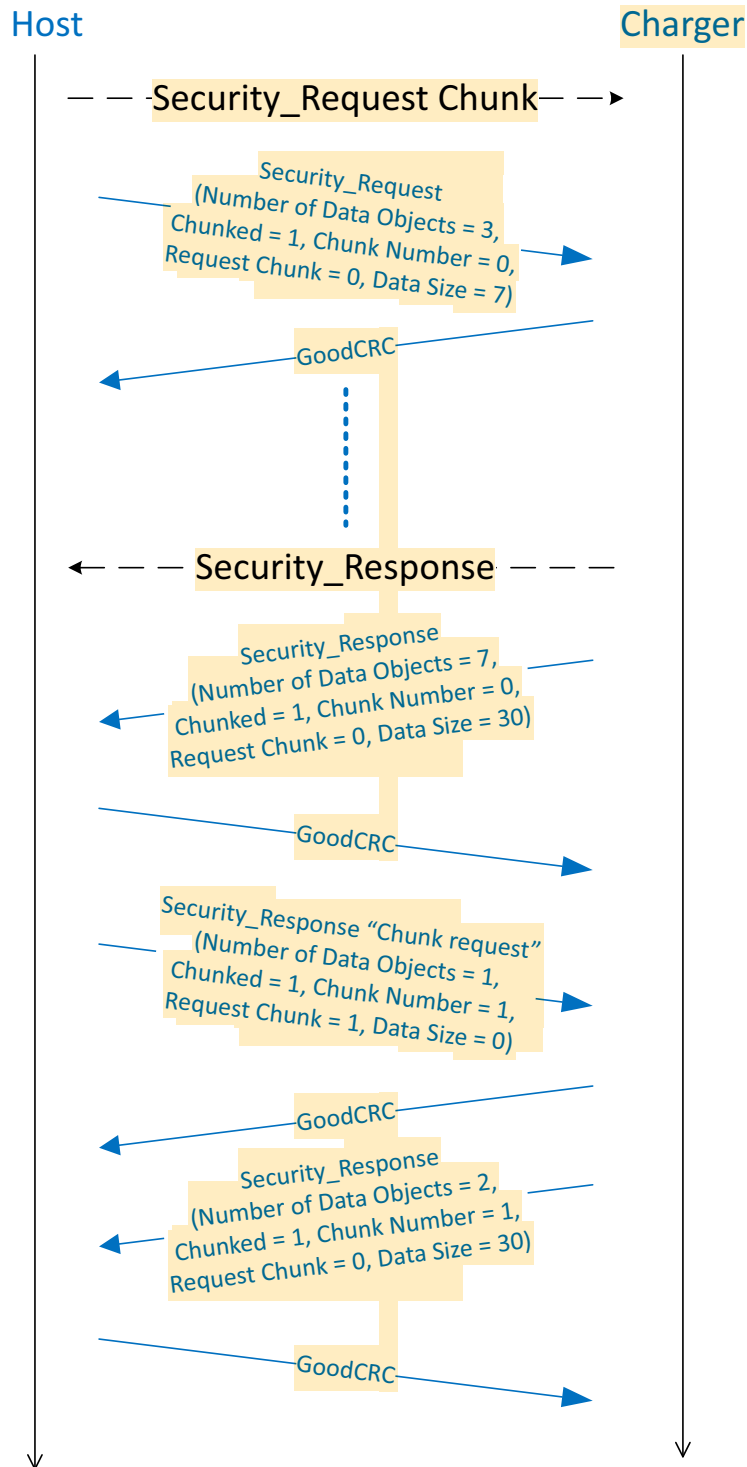


Figure 6.8, "Example Security Request Message of Data Size 7 (Chunked bit set to 1)" shows the **Security\_Request** Message shown in Figure 6.7, "Example Security Request sequence Chunked (Chunked bit = 1)" in more detail including the byte ordering on the bus and padding. Three bytes of padding have been added to the Message so that the total number of bytes is a multiple of 32-bits, corresponding to 3 Data Objects. The **Number of Data Objects** field is set to 3 to indicate the length of this **Chunk**. The **Chunk Number** is set to zero and the **Data Size** field is set to 7 to indicate the length of the whole **Extended Message**.

Figure 6.8 Example Security\_Request Message of Data Size 7 (Chunked bit set to 1)

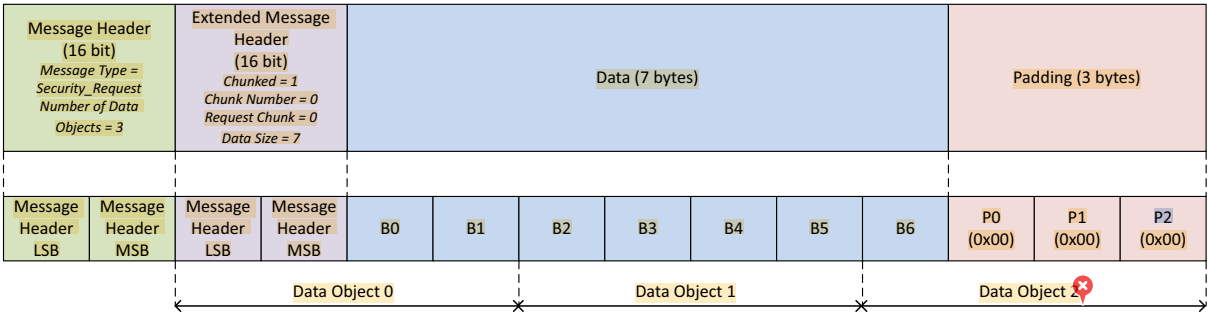


Figure 6.9, "Example Chunk 0 of Security\_Response Message of Data Size 30 (Chunked bit set to 1)" shows **Chunk Number** zero of the **Security\_Response** Message shown in Figure 6.7, "Example Security\_Request sequence Chunked (Chunked bit = 1)" in more detail including the byte ordering on the bus and padding. No padding is need for this **Chunk** since the full 26-byte **Payload** plus 2-byte **Extended Message Header** is a multiple of 32-bits, corresponding to 7 **Data Objects**. The **Number of Data Objects** field is set to 7 to indicate the length of this **Chunk** and the **Data Size** field is set to 30 to indicate the length of the whole **Extended Message**.

Figure 6.9 Example Chunk 0 of Security\_Response Message of Data Size 30 (Chunked bit set to 1)

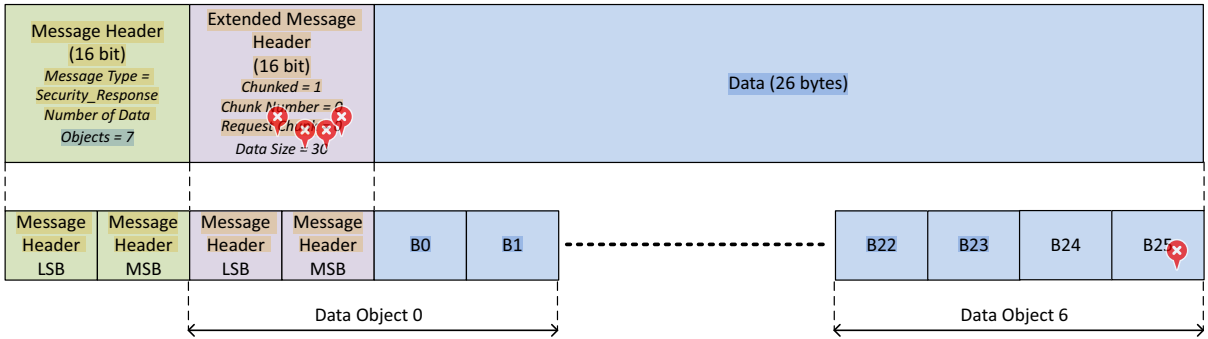


Figure 6.7, "Example Security\_Request sequence Chunked (Chunked bit = 1)" shows an example of the **Message** format, byte ordering and padding for the **Security\_Response** Message **Chunk** request for **Chunk Number** one shown in Figure 6.7, "Example Security\_Request sequence Chunked (Chunked bit = 1)". In the **Chunk** request the **Number of Data Objects** field in the **Message** is set to 1 to indicate that the **Payload** is 32 bits equivalent to 1 data object (see Figure 6.10, "Example byte transmission for a Security\_Response Message **Chunk** request (Chunked bit is set to 1)"). Since the **Chunked** bit is set to 1 the **Chunk** request/**Chunk** response mechanism is used. The **Message** is a **Chunk** request so the **Request Chunk** bit is set to one, and in this case **Chunk** one is being requested so **Chunk Number** is set to one. **Data Size** is set to zero indicating the length of the **Data Block** being transferred. Two bytes of padding are added to ensure that the **Payload** is a multiple of 32 bits.

Figure 6.10 Example byte transmission for a Security\_Response Message Chunk request (Chunked bit is set to 1)

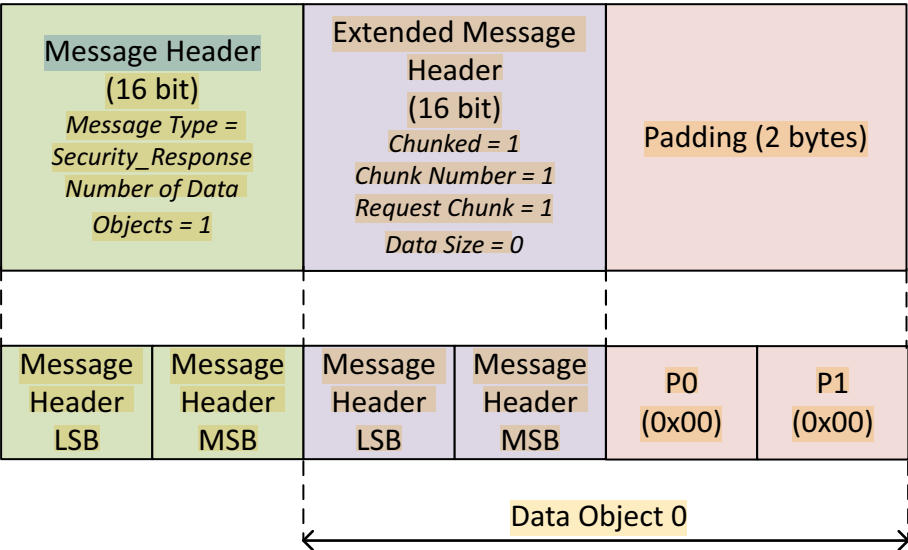
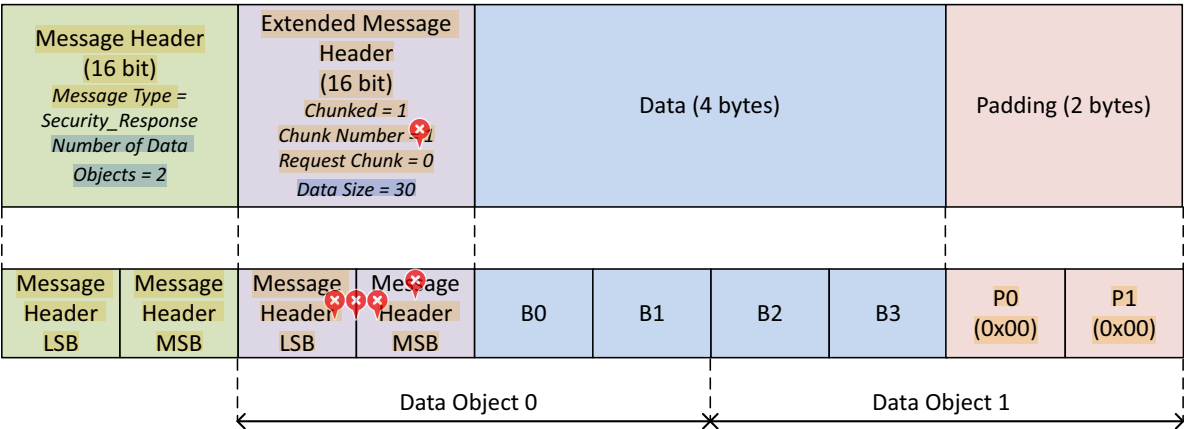


Figure 6.11, "Example Chunk 1 of Security\_Response Message of Data Size 30 (Chunked bit set to 1)" shows **Chunk Number** one of the **Security\_Response** Message shown in Figure 6.7, "Example Security\_Request sequence Chunked (Chunked bit = 1)" in more detail including the byte ordering on the bus and padding. Two bytes of padding are added to ensure that the **Payload** is a multiple of 32 bits, corresponding to 2 **Data Objects**. The **Number of Data Objects** field is set to 2 to indicate the length of this **Chunk** and the **Data Size** field is set to 30 to indicate the length of the whole **Extended Message**.

Figure 6.11 Example Chunk 1 of Security\_Response Message of Data Size 30 (Chunked bit set to 1)



## 6.3 Control Message

A Message is defined as a Control Message when the **Number of Data Objects** field in the Message Header is set to zero. The Control Message consists only of a Message Header and a CRC. The Protocol Layer originates the Control Messages (i.e., **Accept** Message, **Reject** Message etc.).

The Control Message types are specified in the Message Header's **Message Type** field (bits 4...0) and are summarized in [Table 6.5, "Control Message Types"](#). The Sent by column indicates entities which **May** send the given Message (Source, Sink or Cable Plug); entities not listed **Shall Not** issue the corresponding Message. The "Valid Start of Packet" column indicates the Messages which **Shall** only be issued in SOP Packets and the Messages which **May** be issued in SOP\* Packets.

**Table 6.5 Control Message Types**

| Bits 4...0          | Message Type                   | Sent by                                 | Description                                                                         | Valid Start of Packet |
|---------------------|--------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------|-----------------------|
| 0_0000              | <b>Reserved</b>                | N/A                                     | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                       |
| 0_0001              | <b>GoodCRC</b>                 | Source, Sink <sup>1</sup> or Cable Plug | See <a href="#">Section 6.3.1</a> .                                                 | SOP*                  |
| 0_0010              | <b>GotoMin (Deprecated)</b>    | <b>Deprecated</b>                       | See <a href="#">Section 6.3.2</a> .                                                 | N/A                   |
| 0_0011              | <b>Accept</b>                  | Source, Sink or Cable Plug              | See <a href="#">Section 6.3.3</a> .                                                 | SOP*                  |
| 0_0100              | <b>Reject</b>                  | Source, Sink or Cable Plug              | See <a href="#">Section 6.3.4</a> .                                                 | SOP*                  |
| 0_0101              | <b>Ping (Deprecated)</b>       | <b>Deprecated</b>                       | See <a href="#">Section 6.3.5</a> .                                                 | SOP only              |
| 0_0110              | <b>PS_RDY</b>                  | Source or Sink                          | See <a href="#">Section 6.3.6</a> .                                                 | SOP only              |
| 0_0111              | <b>Get_Source_Cap</b>          | Sink or DRP                             | See <a href="#">Section 6.3.7</a> .                                                 | SOP only              |
| 0_1000              | <b>Get_Sink_Cap</b>            | Source or DRP                           | See <a href="#">Section 6.3.8</a> .                                                 | SOP only              |
| 0_1001              | <b>DR_Swap</b>                 | Source or Sink                          | See <a href="#">Section 6.3.9</a> .                                                 | SOP only              |
| 0_1010              | <b>PR_Swap</b>                 | Source or Sink                          | See <a href="#">Section 6.3.10</a> .                                                | SOP only              |
| 0_1011              | <b>VCONN_Swap</b>              | Source or Sink                          | See <a href="#">Section 6.3.11</a> .                                                | SOP only              |
| 0_1100              | <b>Wait</b>                    | Source or Sink                          | See <a href="#">Section 6.3.12</a> .                                                | SOP only              |
| 0_1101              | <b>Soft_Reset</b>              | Source or Sink                          | See <a href="#">Section 6.3.13</a> .                                                | SOP*                  |
| 0_1110              | <b>Data_Reset</b>              | Source or Sink                          | See <a href="#">Section 6.3.14</a> .                                                | SOP only              |
| 0_1111              | <b>Data_Reset_Complete</b>     | Source or Sink                          | See <a href="#">Section 6.3.15</a> .                                                | SOP only              |
| 1_0000              | <b>Not_Supported</b>           | Source, Sink or Cable Plug              | See <a href="#">Section 6.3.16</a> .                                                | SOP*                  |
| 1_0001              | <b>Get_Source_Cap_Extended</b> | Sink or DRP                             | See <a href="#">Section 6.3.17</a> .                                                | SOP only              |
| 1_0010              | <b>Get_Status</b>              | Source or Sink                          | See <a href="#">Section 6.3.18</a> .                                                | SOP*                  |
| 1_0011              | <b>FR_Swap</b>                 | Sink <sup>1</sup>                       | See <a href="#">Section 6.3.19</a> .                                                | SOP only              |
| 1_0100              | <b>Get_PPS_Status</b>          | Sink                                    | See <a href="#">Section 6.3.20</a> .                                                | SOP only              |
| 1_0101              | <b>Get_Country_Codes</b>       | Source or Sink                          | See <a href="#">Section 6.3.21</a> .                                                | SOP only              |
| 1_0110              | <b>Get_Sink_Cap_Extended</b>   | Source or DRP                           | See <a href="#">Section 6.3.22</a> .                                                | SOP only              |
| 1_0111              | <b>Get_Source_Info</b>         | Sink or DRP                             | See <a href="#">Section 6.3.23</a> .                                                | SOP Only              |
| 1_1000              | <b>Get_Revision</b>            | Source or Sink <sup>1</sup>             | See <a href="#">Section 6.3.24</a> .                                                | SOP*                  |
| 1_1001...<br>1_1111 | <b>Reserved</b>                | N/A                                     | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                       |

- 1) In this case the Port is providing **vSafe5V** however it will have  $R_d$  asserted rather than  $R_p$  and sets the **Port Power Role** field to Sink, until the Fast Role Swap AMS has completed.

### 6.3.1 GoodCRC Message

✖ The **GoodCRC** Message **Shall** be sent by the receiver to acknowledge that the previous *Message* was correctly received (i.e., had a **GoodCRC** Message). The **GoodCRC** Message **Shall** return the *Message*'s **MessageID** so the sender can determine that the correct *Message* is being acknowledged. The first bit of the **GoodCRC** Message **Shall** be returned within **tTransmit** after receipt of the last bit of the previous *Message*.

BIST does not send the **GoodCRC** Message while in a *Continuous BIST Mode* (see [Section 6.4.3, "BIST Message"](#)).

The retry mechanism is triggered when the *Message* sender fails to receive a **GoodCRC** Message before the **CRCReceiveTimer** expires. It is used by the *Message* sender to detect that the *Message* was not correctly received by the *Message* recipient due to noise or other disturbance on the *Configuration Channel (CC)*. The retry mechanism **Shall Not** be used for any other purpose such as a means of gaining time for processing the required response to the received *Message*.

### 6.3.2 GotoMin Message (Deprecated)

The **GotoMin (Deprecated)** Message has been **Deprecated**. The 0\_0010 **Message Type** is no longer **Valid** and **Shall** be responded to by a **Not\_Supported** Message. ✖

### 6.3.3 Accept Message

The **Accept** Message is a **Valid** response in the following cases:

- It **Shall** be sent by the *Source*, in *SPR Mode*, to signal the *Sink* that the *Source* is willing to meet the **Request** Message.
- It **Shall** be sent by the *Source*, in *EPR Mode*, to signal the *Sink* that the *Source* is willing to meet the **EPR\_Request** Message.
- It **Shall** be sent by the recipient of the **PR\_Swap** Message to signal that it is willing to do a *Power Role Swap* and has begun the *Power Role Swap AMS*.
- It **Shall** be sent by the recipient of the **DR\_Swap** Message to signal that it is willing to do a *Data Role Swap* and has begun the *Data Role Swap AMS*.
- It **Shall** be sent by the recipient of the **VCONN\_Swap** Message to signal that it is willing to do a *VCONN Swap* and has begun the *VCONN Swap AMS*.
- It **Shall** be sent by the recipient of the **FR\_Swap** Message to indicate that it has begun the *Fast Role Swap AMS*.
- It **Shall** be sent by the recipient of the **Soft\_Reset** Message to indicate that it has completed its *Soft Reset*.
- It **Shall** be sent by the recipient of the **Enter\_USB** Message to indicate that it has begun the *Enter USB AMS*.
- It **Shall** be sent by the recipient of the **Data\_Reset** Message to indicate that it has begun the *Data Reset AMS*.

The **Accept** Message **Shall** be sent within **tReceiverResponse** of the receipt of the last bit of the *Message* (see [Section 6.6.2, "SenderResponseTimer"](#)).

### 6.3.4 Reject Message

The **Reject** Message is a **Valid** response in the following cases:

- It **Shall** be sent to signal the *Sink*, in *SPR Mode*, that the *Source* is unable to meet the **Request** Message. This **May** be due an **Invalid** request or because the *Source* can no longer provide what it previously *Advertised*.

- It **Shall** be sent to signal the *Sink*, in *EPR Mode*, that the *Source* is unable to meet the *EPR\_Request Message*. This **May** be due an *Invalid* request or because the *Source* can no longer provide what it previously *Advertised*.
- It **Shall** be sent by the recipient of a *PR\_Swap Message* to indicate it is unable to do a *Power Role Swap*.
- It **Shall** be sent by the recipient of a *PR\_Swap Message* while in *EPR Mode*.
- It **Shall** be sent by the recipient of a *DR\_Swap Message* to indicate it is unable to do a *Data Role Swap*.
- It **Shall** be sent by the recipient of a *VCONN\_Swap Message* that is not presently the *VCONN Source*, to indicate it is unable to do a *VCONN Swap*.
- It **Shall** be sent by *UFP* on receiving an *Enter\_USB Message* to indicate it is unable to enter the requested *USB Mode*.

The sender of a *Request*, *EPR\_Request*, *PR\_Swap*, *DR\_Swap*, *VCONN\_Swap*, or *Enter\_USB Message*, on receiving a *Reject Message* response, **Shall Not** send this same *Message* to the recipient until one of the following has occurred:

- A New *Explicit Contract Negotiation* as a result of the *Source* sending a *Source\_Capabilities Message* or *EPR\_Source\_Capabilities Message*. This can be triggered by:
  - The *Source's Device Policy Manager*.
  - A *Get\_Source\_Cap Message* sent from the *Sink* to the *Source* in *SPR Mode*.
  - An *EPR\_Get\_Source\_Cap Message* sent from the *Sink* to the *Source* in *EPR Mode*.
  - A *Power Role Swap*.
  - A *Soft Reset*.
  - A *Hard Reset*.
  - A *Disconnect/Re-connect*.
- A *Data Role Swap*.
- A *Data Reset*.

The *Sink* **May** send a different *Request Message* to the one which was rejected but **Shall Not** repeat the same *Request Message*, using the same *RDO*, unless there has been a New *Explicit Contract Negotiation*, *Data Role Swap* or *Data Reset* as described above.

The *Reject Message* **Shall** be sent within *tReceiverResponse* of the receipt of the last bit of *Message* (see [Section 6.6.2, "SenderResponseTimer"](#)).

**Note:** The *Reject Message* is not a *Valid* response when a *Message* is not supported. In this case the *Not\_Supported Message* is returned (see [Section 6.3.16, "Not\\_Supported Message"](#)).

### 6.3.5 Ping Message

The *Ping (Deprecated) Message* has been deprecated. The 0\_0101 *Message Type* is no longer *Valid*.

A *Port* that receives a *Ping (Deprecated) Message* **May** respond with a *Not\_Supported Message* or *Ignore* the *Ping (Deprecated) Message*. A *Cable Plug* that receives a *Ping (Deprecated) Message* **Shall Ignore** the *Ping (Deprecated) Message*.

### 6.3.6 PS\_RDY Message

The *PS\_RDY Message* **Shall** be sent by the *Source* (or by both the *New Sink* and *New Source* during the *Power Role Swap AMS* or *Fast Role Swap AMS*) to indicate its power supply has reached the desired operating condition (see [Section 8.3.2.2, "Power Negotiation"](#)).



### 6.3.7 Get\_Source\_Cap Message

The **Get\_Source\_Cap** (Get Source Capabilities) Message **May** be sent by a Port to request the Source Capabilities and Dual-Role Power capability of its Port Partner (e.g., Dual-Role Power capable). The Port **Shall** respond by returning a **Source\_Capabilities** Message (see [Section 6.4.1.5, "SPR Source Capabilities Message"](#)).

### 6.3.8 Get\_Sink\_Cap Message

The **Get\_Sink\_Cap** (Get Sink Capabilities) Message **May** be sent by a Port to request the Sink Capabilities and Dual-Role Power capability of its Port Partner (e.g., Dual-Role Power capable). The Port **Shall** respond by returning a **Sink\_Capabilities** Message (see [Section 6.4.1.6, "SPR Sink Capabilities Message"](#)).

### 6.3.9 DR\_Swap Message

The **DR\_Swap** Message is used to exchange DFP and UFP operation between Port Partners while maintaining the direction of power flow over **VBUS**. The **Data Role Swap** process can be used by Port Partners whether or not they support USB Communications capability. A DFP that supports USB Communication capability starts as the USB Host on Attachment. A UFP that supports USB Communication capability starts as the USB Device on Attachment.

**[USB Type-C 2.4]** Dual-Role Data (DRD) Ports **Shall** have the capability to perform a Data Role Swap from the **PE\_SRC\_Ready** or **PE\_SNK\_Ready** states. DFPs and UFPs **May** have the capability to perform a Data Role Swap from the **PE\_SRC\_Ready** or **PE\_SNK\_Ready** states. A Data Role Swap **Shall** be regarded in the same way as a cable **Detach/Re-attach** in relation to any USB Communication which is ongoing between the Port Partners. If there are any Active Modes between the Port Partners when a **DR\_Swap** Message is received, then a Hard Reset **Shall** be performed (see [Section 6.4.4.3.4, "Enter Mode Command"](#)). If the Cable Plug has any Active Modes then the DFP **Shall Not** issue a **DR\_Swap** Message and **Shall** cause all Active Modes in the Cable Plug to be exited before accepting a **Data Role Swap** request.

The source of **VBUS** and **VCONN Source** **Shall** remain unchanged as well as the  $R_p/R_d$  resistors on the CC wire during the Data Role Swap process.

The **DR\_Swap** Message **May** be sent by either Port Partner. The recipient of the **DR\_Swap** Message **Shall** respond by sending an **Accept** Message, a **Wait** Message or a **Reject** Message (see [Section 6.9, "Accept, Reject and Wait"](#)).

- If an **Accept** Message is sent, the Source and Sink **Shall** exchange Data Roles.
- If a **Reject** Message is sent, the requester is informed that the recipient is unable, or unwilling, to do a Data Role Swap and no action **Shall** be taken.
- If a **Wait** Message is sent, the requester is informed that a Data Role Swap might be possible in the future but that no immediate action **Shall** be taken.

Before a Data Role Swap the initial DFP **Shall** have its **Port Data Role** bit set to DFP, and the initial UFP **Shall** have its **Port Data Role** bit set to UFP.

After a successful Data Role Swap the DFP/Host **Shall** become the UFP/Device and vice-versa; the new DFP **Shall** have its **Port Data Role** bit set to DFP, and the new UFP **Shall** have its **Port Data Role** bit set to UFP. Where USB Communication is supported by both Port Partners a USB data connection **Should** be established according to the new Data Roles.

If the Data Role Swap, after having been accepted by the Port Partner, is subsequently not successful, in order to attempt a re-establishment of the connection, USB Type-C Error Recovery actions, such as disconnect, as defined in **[USB Type-C 2.4]** will be necessary.

See [Section 8.3.2.9, "Data Role Swap"](#).

### 6.3.10 PR\_Swap Message

The **PR\_Swap** Message **May** be sent by either Port Partner to request an exchange of Power Roles. The recipient of the Message **Shall** respond by sending an **Accept** Message, a **Wait** Message or a **Reject** Message (see [Section 6.9, "Accept, Reject and Wait"](#)).



- If an **Accept** Message is sent, the Source and Sink **Shall** do a Power Role Swap.
- If a **Reject** Message is sent, the requester is informed that the recipient is unable, or unwilling, to do a Power Role Swap and no action **Shall** be taken.
- If a **Wait** Message is sent, the requester is informed that a Power Role Swap might be possible in the future but that no immediate action **Shall** be taken.

The **PR\_Swap** Message **Shall Not** be sent while in EPR Mode. While in EPR Mode if a Power Role Swap is required, an EPR Mode exit **Shall** be done first.

After a successful Power Role Swap the Port Partners **Shall** reset their respective Protocol Layers (equivalent to a Soft Reset): resetting their **MessageIDCounter**, **RetryCounter** and Protocol Layer state machines before attempting to establish the First Explicit Contract. At this point the New Source **Shall** also reset its **CapsCounter**.

The New Source **Shall** have  $R_p$  asserted on the CC wire and the New Sink **Shall** have  $R_d$  asserted on the CC wire as defined in [USB Type-C 2.4]. When performing a Power Role Swap from Source to Sink, the Port **Shall** change its CC wire resistor from  $R_p$  to  $R_d$ . When performing a Power Role Swap from Sink to Source, the Port **Shall** change its CC wire resistor from  $R_d$  to  $R_p$ . The DFP (Host), UFP (Device) Data Roles and VCONN Source **Shall** remain unchanged by the Power Role Swap process.

**Note:** During the Power Role Swap process the Initial Sink does not disconnect even though  $V_{BUS}$  drops below  $v_{Safe5V}$ .

For more information regarding the Power Role Swap, refer to:

- [Section 7.3.2, "Transitions Caused by Power Role Swap"](#)
- [Section 8.3.2.5, "Data Reset"](#).
- [Section 8.3.3.19.3, "Policy Engine in Source to Sink Power Role Swap State Diagram"](#).
- [Section 8.3.3.19.4, "Policy Engine in Sink to Source Power Role Swap State Diagram"](#).
- [Section 9.1.2, "Mapping to USB Device States"](#).

## 6.3.11 VCONN\_Swap Message

The **VCONN\_Swap** Message **Shall** be supported by any Port that can operate as a VCONN Source.

The **VCONN\_Swap** Message **May** be sent by either Port Partner to request an exchange of VCONN Source. The recipient of the Message **Shall** respond by sending an **Accept** Message, **Reject** Message, **Wait** Message (see [Section 6.9, "Accept, Reject and Wait"](#)) or **Not\_Supported** Message.

- If an **Accept** Message is sent, the Port Partners **Shall** perform a VCONN Swap. The new VCONN Source **Shall** send a **PS\_RDY** Message within  $tVCONNSourceOn$  to indicate that it is now sourcing VCONN. The initial VCONN Source **Shall** cease sourcing VCONN within  $tVCONNSourceOff$  of receipt of the last bit of the **EOP** of the **PS\_RDY** Message.
- If a **Reject** Message is sent, the requester is informed that the recipient is unable, or unwilling, to do a VCONN Swap and no action **Shall** be taken. A **Reject** Message **Shall** only be sent by the Port that is not presently the VCONN Source in response to a **VCONN\_Swap** Message. The Port that is presently the VCONN Source **Shall Not** send a **Reject** Message in response to **VCONN\_Swap** Message.
- If a **Wait** Message is sent, the requester is informed that a VCONN Swap might be possible in the future but that no immediate action **Shall** be taken. A Port after losing the VCONN Source role due to incoming VCONN Swap request **Shall Not** initiate a VCONN Swap until at least  $tVCONNSwapDelayDFP$ / $tVCONNSwapDelayUFP$  after completing the previous VCONN Swap AMS.
- If a **Not\_Supported** Message is sent, the requester is informed that VCONN Swap is not supported. The Port that is not presently the VCONN Source **May** turn on VCONN when a **Not\_Supported** Message is received in response to a **VCONN\_Swap** Message.

The *DFP (Host)*, *UFP (Device)* *Data Roles* and *Source* of *VBUS* **shall** remain unchanged as well as the  $R_p/R_d$  resistors on the *CC* wire during the *VCONN Swap* process.

*VCONN* **shall** be continually sourced during the *VCONN Swap* process to maintain power to the *Cable Plug(s)* i.e., make before break.

Before communicating with a *Cable Plug* a *Port* **shall** ensure that it is the *VCONN Source* and that the *Cable Plugs* are powered, by performing a *VCONN Swap* if necessary. Since it cannot be guaranteed that the present *VCONN Source* is supplying *VCONN*, the only means to ensure that the *Cable Plugs* are powered is for a *Port* wishing to communicate with a *Cable Plug* to become the *VCONN Source*. If a *Not\_Supported* Message is returned in response to the *VCONN\_Swap* Message, then the *Port* is allowed to become the *VCONN Source* until a *Hard Reset* or *Detach*.

A *VCONN Source* that is also a *Source* can attempt to send a *Discover Identity* Command using *SOP'* to a *Cable Plug* prior to the establishment of the *First Explicit Contract*.

**Note:** Even when it is presently the *VCONN Source*, the *Sink* is not permitted to initiate an *AMS* with a *Cable Plug* unless  $R_p$  is set to *SinkTxOK* (see [Section 6.9, "Accept, Reject and Wait"](#)).

## 6.3.12 Wait Message

The *Wait* Message is a *Valid* response to one of the following Messages:

- It **shall** be sent to signal the *Sink*, in response to a *Request* Message in *SPR Mode* during *Negotiation*, to indicate that the *Source* is currently unable to meet the request.
- It **shall** be sent to signal the *Sink*, in response to a *EPR\_Request* Message in *EPR Mode* during *Negotiation*, to indicate that the *Source* is currently unable to meet the request.
- It **shall** be sent by the recipient of a *PR\_Swap* Message to indicate it is currently unable to do a *Power Role Swap*.
- It **shall** be sent by the recipient of a *DR\_Swap* Message to indicate it is currently unable to do a *Data Role Swap*.
- It **shall** be sent by the recipient of a *VCONN\_Swap* Message that is not presently the *VCONN Source* to indicate it is currently unable to do a *VCONN Swap*.
- It **shall** be sent by the recipient of an *Enter\_USB* Message to indicate it is currently unable to enter the requested *USB Mode*.

The *Wait* Message **shall** be sent within *tReceiverResponse* of the receipt of the last bit of the Message (see [Section 6.9, "Accept, Reject and Wait"](#)).

### 6.3.12.1 Wait in response to a Request Message

The *Wait* Message allows the *Source* time to recover the power it requires to meet the request, e.g., through *Re-negotiation* with other *Sinks* or an upstream *Source*. A *Source* **should** only send a *Wait* Message in response to a *Request* Message when an *Explicit Contract* exists between the *Port Partners*.

The *Sink* is allowed to repeat the *Request* Message using the *SinkRequestTimer* and **shall** ensure that there is *tSinkRequest* after receiving the *Wait* Message before sending another *Request* Message.

### 6.3.12.2 Wait in response to a PR\_Swap Message

The *Wait* Message is used when responding to a *PR\_Swap* Message to indicate that a *Power Role Swap* might be possible in the future. This can occur in any case where the device receiving the *PR\_Swap* Message needs to evaluate the request further e.g., by requesting *Sink Capabilities* from the originator of the *PR\_Swap* Message. Once it has completed this evaluation one of the *Port Partners* **should** initiate the *Power Role Swap* process again by sending a *PR\_Swap* Message.

The *Wait* Message is also used where a *Hub* is operating in hybrid mode when a request cannot be satisfied (see [\[UCSI\]](#)).

A Port that receives a **Wait Message** in response to a **PR\_Swap Message** **Shall** wait **tPRSwapWait** after receiving the **Wait Message** before sending another **PR\_Swap Message**.

### 6.3.12.3 Wait in response to a DR\_Swap Message

The **Wait Message** is used when responding to a **DR\_Swap Message** to indicate that a **Data Role Swap** might be possible in the future. This can occur in any case where the device receiving the **DR\_Swap Message** needs to evaluate the request further. Once it has completed this evaluation one of the Port Partners **Should** initiate the **Data Role Swap** process again by sending a **DR\_Swap Message**.

A Port that receives a **Wait Message** in response to a **DR\_Swap Message** **Shall** wait **tDRSwapWait** after receiving the **Wait Message** before sending another **DR\_Swap Message**.

### 6.3.12.4 Wait in response to a VCONN\_Swap Message

The **Wait Message** is used when responding to a **VCONN\_Swap Message** to indicate that a **VCONN\_Swap** might be possible in the future. This can occur in any case where the device receiving the **VCONN\_Swap Message** needs to evaluate the request further. Once it has completed this evaluation one of the Port Partners **Should** initiate the **VCONN Swap** process again by sending a **VCONN\_Swap Message**.

A Port that receives a **Wait Message** in response to a **VCONN\_Swap Message** **Shall** wait **tVCONNSwapWait** after receiving the **Wait Message** before sending another **VCONN\_Swap Message**.

A Port that is currently the **VCONN Source** **Shall** respond with an **Accept Message** (rather than a **Wait Message**) if the Port Partner's Revision and Version, as reported in the **Revision Message**, is earlier than R3.2 V1.1. A Port Partner supporting an earlier Revision and Version will not expect a **Wait Message** and will generate a **Soft Reset** in response.

### 6.3.12.5 Wait in response to an Enter\_USB Message

The **Wait Message** is used, by the UFP, when responding to an **Enter\_USB Message** to indicate that entering the requested USB Mode might be possible in the future. This can occur, for example, in any case where the UFP needs to **Negotiate** more power to enter the mode. Once the UFP has completed this the DFP **Should** initiate the Enter USB process again by sending an **Enter\_USB Message**.

A DFP that receives a **Wait Message** in response to an **Enter\_USB Message** **Shall** wait **tEnterUSBWait** after receiving the **Wait Message** before sending another **Enter\_USB Message**.

## 6.3.13 Soft Reset Message

A **Soft\_Reset Message** **May** be initiated by either the Source or Sink to its Port Partner requesting a **Soft Reset**. The **Soft\_Reset Message** **Shall** cause a **Soft Reset** of the connected Port Pair (see [Section 6.8.1, "Soft Reset and Protocol Error"](#)). If the **Soft\_Reset Message** fails a **Hard Reset** **Shall** be initiated within **tHardReset** of the last **CRCReceiveTimer** expiring after **nRetryCount** retries have been completed.

A **Soft\_Reset Message** is used to recover from **Protocol Layer** errors; putting the **Message** counters to a known state to regain **Message** synchronization. The **Soft\_Reset Message** has no effect on the Source or Sink; that is the previously **Negotiated** direction. Voltage and current remain unchanged. **Modal Operation** is unaffected by **Soft Reset**. However after a **Soft Reset** has completed, an **Explicit Contract Negotiation** occurs, in order to re-establish PD Communication and to bring state operation for both Port Partners back to either the **PE\_SNK\_Ready** or **PE\_SRC\_Ready** states as appropriate (see [Section 8.3.3.4, "SOP Soft Reset and Protocol Error State Diagrams"](#)).

A **Soft\_Reset Message** **May** be sent by either the Source or Sink when there is a **Message** synchronization error. If the error is not corrected by the **Soft Reset**, **Hard Reset Signaling** **Shall** be issued (see [Section 6.8.3, "Hard Reset"](#)).

A **Soft\_Reset Message** **Shall** be targeted at a specific entity depending on the type of SOP\* Packet used. **Soft\_Reset Messages** sent using SOP Packets **Shall** Soft Reset the Port Partner only. **Soft\_Reset Messages** sent using SOP' Packet/ SOP" Packets **Shall** Soft Reset the corresponding Cable Plug only.

After a VCONN Swap the VCONN Source needs to reset the Cable Plug's Protocol Layer to ensure **MessageID** synchronization. If after a VCONN Swap the VCONN Source wants to communicate with a Cable Plug using SOP' Packets, it **Shall** issue a **Soft\_Reset Message** using a SOP' Packet in order to reset the Cable Plug's Protocol Layer. If

the *VCONN Source* wants to communicate with a *Cable Plug* using *SOP* Packets, it **Shall** issue a *Soft\_Reset Message* using a *SOP* Packet in order to reset the *Cable Plug's Protocol Layer*.

### 6.3.14 Data\_Reset Message

The *Data\_Reset Message* **May** be sent by either the *DFP* or *UFP* and **Shall** reset the USB data connection and exit all *Alternate Modes* with its *Port Partner* while preserving the power on *VBUS*. *USB4<sup>®</sup> Mode* capable ports **Shall** support the *Data\_Reset Message* and other ports **May** support the *Data\_Reset Message*.

The *Data\_Reset Message* **Shall Not** change the existing:

- Power Contract
- Data Roles (i.e., which *Port* is the *DFP* or *UFP*)

The receiver of the *Data\_Reset Message* **Shall** respond by sending an *Accept Message* and then follow the process outlined in the following steps. Neither the sender nor receiver **Shall** initiate a *VCONN Swap* until the *Data Reset* process is complete, and the *Data\_Reset\_Complete Message* has been sent. Following receipt of the *Accept Message*, or *GoodCRC* following the *Accept*, depending which *Port* sends the *Data\_Reset Message*:

- 1) The *DFP* **Shall**:
  - Disconnect the *Port's* [USB 2.0] D+/D- signals.
  - If operating in [USB 3.2] remove the *Port's* Rx Terminations (see [USB 3.2]).
  - If operating in [USB4] drive the *Port's* SBTX to a logic low (see [USB4]).
- 2) Both the *DFP* and *UFP* **Shall** exit all *Alternate Modes* if any.
- 3) Reset the cable:
  - If the *VCONN Source Port* is also the *UFP*, then it **Shall** run the *UFP VCONN Power Cycle* process described in [Section 7.1.15.1, "UFP VCONN Power Cycle"](#).
  - If the *VCONN Source Port* is also the *DFP*, then it **Shall** run the *DFP VCONN Power Cycle* process described in [Section 7.1.15.2, "DFP VCONN Power Cycle"](#).
  - The *DFP* **Shall** exit the *VCONN Power Cycle* process as the *VCONN Source* and be sourcing *VCONN*.
- 4) After *tDataReset* the *DFP* **Shall**:
  - Reconnect the [USB 2.0] D+/D- signals.
  - If the *Port* was operating in [USB 3.2] or [USB4] reapply the *Port's* Rx Terminations (see [USB 3.2]).
- 5) The *Data Reset* process is complete; the *DFP* **Shall** send a *Data\_Reset\_Complete Message* and enter the *USB4<sup>®</sup> Discovery and Entry Flow* (See [USB Type-C 2.4]).

If the *Initiator* of the *Data\_Reset Message* does not receive a **Valid** response within *tSenderResponse* it **Shall** enter the *ErrorRecovery State*.

### 6.3.15 Data\_Reset\_Complete Message

The *Data\_Reset\_Complete Message* **Shall** be sent by the *DFP* to the *UFP* to indicate the completion of the *Data Reset* process (see [Section 6.3.14, "Data\\_Reset Message"](#)).

### 6.3.16 Not\_Supported Message

The *Not\_Supported Message* **Shall** be sent by a *Port* or *Cable Plug* in response to any *Message* it does not support. Returning a *Not\_Supported Message* is assumed in this specification and has not been called out explicitly except in [Section 6.13, "Message Applicability"](#) which defines cases where the *Not\_Supported Message* is returned.



### 6.3.17 Get\_Source\_Cap\_Extended Message

The **Get\_Source\_Cap\_Extended** Message is sent by a *Port* to request additional information about a *Port's Source Capabilities*. The *Port* **Shall** respond by returning a **Source\_Capabilities\_Extended** Message (see [Section 6.5.1, "Source\\_Capabilities\\_Extended Message"](#)).

### 6.3.18 Get\_Status Message

The **Get\_Status** Message is sent by a *Port* using *SOP* to request the *Port Partner's* present status.

The *Port Partner* **Shall** respond by returning a **Status** Message (see [Section 6.5.2, "Status Message"](#)). A *Port* that receives an **Alert** Message (see [Section 6.4.6, "Alert Message"](#)) indicates that the *Source* or *Sink's* Status has changed and **Should** be re-read using a **Get\_Status** Message.

The **Get\_Status** Message **May** also be sent to an *Active Cable* to get its present status using *SOP'/SOP''*.

The *Active Cable* **Shall** respond by returning a **Status** Message (see [Section 6.5.2, "Status Message"](#)).

### 6.3.19 FR\_Swap Message

The **FR\_Swap** Message **Shall** be sent by the *New Source* within **tFRSwapInit** after it has detected a *Fast Role Swap* signal (see [Section 5.8.6.3, "Fast Role Swap Detection"](#) and [Section 6.6.17.3, "tFRSwapInit"](#)). The *Fast Role Swap AMS* is necessary to apply  $R_p$  to the *New Source* and  $R_d$  to the *New Sink* and to re-synchronize the state machines. The **tFRSwapInit** time **Shall** be measured from the time the *Fast Role Swap Request* has been sent for **tFRSwapRx** (max) until the last bit of the **EOP** of the **FR\_Swap** Message has been transmitted by the *PHY Layer*.

The recipient of the **FR\_Swap** Message **Shall** respond by sending an **Accept** Message.

After a successful *Fast Role Swap* the *Port Partners* **Shall** reset their respective *Protocol Layers* (equivalent to a *Soft Reset*): resetting their **MessageIDCounter**, **RetryCounter** and *Protocol Layer* state machines before attempting to establish the *First Explicit Contract*. At this point the *Source* **Shall** also reset its **CapsCounter**.

This ensures that only the *Cable Plug* responds with a **GoodCRC** Message to the **Discover Identity** Command.

Prior to the *Fast Role Swap AMS*, the *New Source* **Shall** have  $R_d$  asserted on the *CC* wire and the *New Sink* **Shall** have  $R_p$  asserted on the *CC* wire.

**Note:** This is an incorrect assignment of  $R_p/R_d$  (since  $R_p$  follows the *Source* and  $R_d$  follows the *Sink* as defined in [\[USB Type-C 2.4\]](#)) that is corrected by the *Fast Role Swap AMS*.

During the *Fast Role Swap AMS*, the *New Source* **Shall** change its *CC wire* resistor from  $R_d$  to  $R_p$  and the *New Sink* **Shall** change its *CC wire* resistor from  $R_p$  to  $R_d$ . The *DFP (Host)*, *UFP (Device)* *Data Roles* and *VCONN Source* **Shall** remain unchanged during the *Fast Role Swap* process.

The *Initial Source* **Should** avoid being the *VCONN Source* (by using the *VCONN Swap* process) whenever not actively communicating with the cable, since it is difficult for the *Initial Source* to maintain *VCONN* power during the *Fast Role Swap* process.

**Note:** A *Fast Role Swap* is a "best effort" solution to a situation where a *PDUSB Device* has lost its external power. This process can occur at any time, even during an *AMS* in which case error handling such as *Hard Reset* or [\[USB Type-C 2.4\]](#) *Error Recovery* will be triggered.

**Note:** During the *Fast Role Swap* process the *Initial Sink* does not disconnect even though *VBUS* drops below **vSafe5V**.

For more information regarding the *Fast Role Swap* process, refer to:

- [Section 7.1.13, "Fast Role Swap"](#)
- [Section 7.2.10, "Fast Role Swap"](#)
- [Section 8.3.3.19.5, "Policy Engine in Source to Sink Fast Role Swap State Diagram"](#)
- [Section 8.3.3.19.6, "Policy Engine in Sink to Source Fast Role Swap State Diagram"](#)

- [Section 9.1.2, "Mapping to USB Device States"](#) for *VBUS* mapping to USB states.

### 6.3.20 Get\_PPS\_Status

The **Get\_PPS\_Status** Message is sent by the *Sink* to request additional information about a *Source's* status. The *Port* **Shall** respond by returning a **PPS\_Status** Message (see [Section 6.5.10, "PPS\\_Status Message"](#)).

### 6.3.21 Get\_Country\_Codes

The **Get\_Country\_Codes** Message is sent by a *Port* to request the alpha-2 country codes its *Port Partner* supports as defined in [\[ISO 3166\]](#). The *Port Partner* **Shall** respond by returning a **Country\_Codes** Message (see [Section 6.5.11, "Country\\_Codes Message"](#)).

### 6.3.22 Get\_Sink\_Cap\_Extended Message

The **Get\_Sink\_Cap\_Extended** (Get Sink Capabilities Extended) Message is sent by a *Port* to request additional information about a *Port's* Sink Capabilities. The *Port* **Shall** respond by returning a **Sink\_Capabilities\_Extended** Message (see [Section 6.5.13, "Sink\\_Capabilities\\_Extended Message"](#)).

### 6.3.23 Get\_Source\_Info Message

The **Get\_Source\_Info** Message is sent by a *Port* to request the type, maximum Capabilities and present Capabilities of the *Port* when it is operating as a *Source*. The *Port* **Shall** respond by returning the **Source\_Info** Message (See [Section 6.4.11, "Source\\_Info Message"](#)).

### 6.3.24 Get\_Revision Message

The **Get\_Revision** Message is sent by a *Port* using *SOP* to request the *Revision* and *Version* of the Power Delivery Specification its *Port Partner* supports.

The *Port Partner* **Shall** respond by returning a *Revision Message* (See [Section 6.4.12, "Revision Message"](#)).

The **Get\_Revision** Message **May** also be sent to a *Cable Plug* to request the *Revision* and *Version* of the Power Delivery Specification it supports using *SOP'/SOP''*.

The *Active Cable* **Shall** respond by returning a *Revision Message* (see [Section 6.4.12, "Revision Message"](#)).

## 6.4 Data Message

A **Data Message** *Shall* consist of a **Message Header** and be followed by one or more **Data Objects**. **Data Messages** are easily identifiable because the **Number of Data Objects** field in the **Message Header** is a non-zero value.

There are many types of **Data Objects** used to compose **Data Messages**. Some examples are:

- **Power Data Object (PDO)** used to expose a **Source Port's power Capabilities** or a **Sink's** power requirements.
- **Request Data Object (RDO)** used by a **Sink Port** to **Negotiate an Explicit Contract**.
- **Vendor Data Object (VDO)** used to convey vendor specific information.
- **BIST Data Object (BDO)** used for **PHY Layer** compliance testing.
- **Battery Status Data Object (BSDO)** used to convey **Battery** status information.
- **Alert Data Object (ADO)** used to indicate events occurring on the **Source** or **Sink**.

The type of **Data Object** being used in a **Data Message** is defined by the **Message Header's Message Type** field and is summarized in **Table 6.6, "Data Message Types"**. The **Sent by** column indicates entities which **May** send the given **Message** (**Source**, **Sink** or **Cable Plug**); entities not listed **Shall Not** issue the corresponding **Message**. The **"Valid Start of Packet"** column indicates the **Messages** which **Shall** only be issued in **SOP Packets** and the **Messages** which **May** be issued in **SOP\* Packets**.

**Table 6.6 Data Message Types**

| Bits 4...0      | Type                       | Sent by                    | Description                                                                         | Valid Start of Packet |
|-----------------|----------------------------|----------------------------|-------------------------------------------------------------------------------------|-----------------------|
| 0_0000          | <b>Reserved</b>            | <b>N/A</b>                 | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. | <b>N/A</b>            |
| 0_0001          | <b>Source_Capabilities</b> | Source or Dual-Role Power  | See <a href="#">Section 6.4.1.5</a>                                                 | SOP only              |
| 0_0010          | <b>Request</b>             | Sink only                  | See <a href="#">Section 6.4.2</a>                                                   | SOP only              |
| 0_0011          | <b>BIST</b>                | Tester, Source or Sink     | See <a href="#">Section 6.4.3</a>                                                   | SOP*                  |
| 0_0100          | <b>Sink_Capabilities</b>   | Sink or Dual-Role Power    | See <a href="#">Section 6.4.2</a>                                                   | SOP only              |
| 0_0101          | <b>Battery_Status</b>      | Source or Sink             | See <a href="#">Section 6.4.5</a>                                                   | SOP only              |
| 0_0110          | <b>Alert</b>               | Source or Sink             | See <a href="#">Section 6.4.6</a>                                                   | SOP only              |
| 0_0111          | <b>Get_Country_Info</b>    | Source or Sink             | See <a href="#">Section 6.4.7</a>                                                   | SOP only              |
| 0_1000          | <b>Enter_USB</b>           | DFP                        | See <a href="#">Section 6.4.8</a>                                                   | SOP*                  |
| 0_1001          | <b>EPR_Request</b>         | Sink                       | See <a href="#">Section 6.4.9</a>                                                   | SOP only              |
| 0_1010          | <b>EPR_Mode</b>            | Source or Sink             | See <a href="#">Section 6.4.10</a>                                                  | SOP only              |
| 0_1011          | <b>Source_Info</b>         | Source                     | See <a href="#">Section 6.4.11</a>                                                  | SOP only              |
| 0_1100          | <b>Revision</b>            | Source, Sink or Cable Plug | See <a href="#">Section 6.4.12</a>                                                  | SOP*                  |
| 0_1101...0_1110 | <b>Reserved</b>            | <b>N/A</b>                 | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. | <b>N/A</b>            |
| 0_1111          | <b>Vendor_Defined</b>      | Source, Sink or Cable Plug | See <a href="#">Section 6.4.4</a>                                                   | SOP*                  |
| 1_0000...1_1111 | <b>Reserved</b>            | <b>N/A</b>                 | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. | <b>N/A</b>            |



## 6.4.1 Capabilities Message

There are two distinct *Capabilities Messages*: one used while in *SPR Mode* and another while in *EPR Mode*. This section defines the *Capabilities Messages* specific to the *SPR Mode* and [Section 6.5.15, "EPR Capabilities Message"](#) defines the *Capabilities Messages* specific to the *EPR Mode*.

### 6.4.1.1 Power Data Objects

Sections [Section 6.4.1.5, "SPR Source Capabilities Message"](#) and [Section 7.1.3, "Types of Sources"](#) describes the *Power Data Objects* (PDOs) used in the construction of a *Capabilities Message* for both *SPR Mode* and *EPR Mode*.

There are three types of *Power Data Objects*. They contain additional information beyond that encoded in the *Message Header* to identify each of the three types of *Power Data Objects*:

- *Fixed Supply* is used to expose well-regulated fixed voltage power supplies.
- *Variable Supply* is used to expose very poorly regulated power supplies.
- *Battery Supply* is used to expose batteries that can be directly connected to *VBUS*.

There are three types of *Augmented Power Data Objects*:

- *SPR PPS* is used to expose a power supply whose output voltage can be programmatically adjusted over the *Advertised voltage* range and limited by the *Source* to a programmable current limit.
- *SPR AVS* and *EPR AVS* are used to expose a power supply whose output voltage can be adjusted over the *Advertised voltage* range but otherwise is equivalent to a *Fixed Supply* (*AVS* does not support a programmable current limit).

*Power Data Objects* are also used to expose additional *Capabilities* that *May* be utilized, such as in the case of a *Power Role Swap*.

A list of one or more *Power Data Objects* **Shall** be sent by the *Source* to convey its *Capabilities*. The *Sink* **May** then request one of these *Capabilities* by returning a *Request Data Object* that contains an index to a *Power Data Object*, to *Negotiate* a mutually agreeable *Explicit Contract*.

Where Maximum and Minimum voltage and current values are given in *PDOs* these **Shall** be taken to be absolute values.

The *Source* and *Sink* **Shall Not Negotiate** a power level that would allow the current to exceed the maximum current supported by their receptacles or the *Attached plug* (see [\[USB Type-C 2.4\]](#)). The *Source* **Shall** limit its offered *Capabilities* to the maximum current supported by its receptacle and *Attached plug*. A *Sink* **Shall** only make a request from any of the *Capabilities* offered by the *Source*. For further details see [Section 4.4, "Cable Type Detection"](#).

*Sources* expose their power *Capabilities* by sending a *Source\_Capabilities Message*. *Sinks* expose their power requirements by sending a *Sink\_Capabilities Message*. Both are composed of several 32-bit *Power Data Objects* (see [Table 6.7, "Power Data Object"](#)).

Table 6.7 Power Data Object

| Bit(s)   | Description                                                                                    |                                             |
|----------|------------------------------------------------------------------------------------------------|---------------------------------------------|
|          | Value                                                                                          | Parameter                                   |
| B31...30 | 00b                                                                                            | <i>Fixed Supply</i> ( $V_{min} = V_{max}$ ) |
|          | 01b                                                                                            | <i>Battery</i>                              |
|          | 10b                                                                                            | <i>Variable Supply</i> (non-Battery)        |
|          | 11b                                                                                            | <i>Augmented Power Data Object (APDO)</i>   |
| B29...0  | Specific <i>Power Capabilities</i> are described by the <i>PDOs</i> in the following sections. |                                             |

The *Augmented Power Data Object* (APDO) is defined to allow support for more than the four *PDO* types by extending the *Power Data Object* field from 2 to 4 bits when the B31...B30 are 11b. The generic APDO structure is shown in [Table 6.8, "Augmented Power Data Object"](#).

Table 6.8 Augmented Power Data Object

| Bit(s)   | Description                                                                       |                                    |
|----------|-----------------------------------------------------------------------------------|------------------------------------|
|          | Value                                                                             | Parameter                          |
| B31...30 | 11b                                                                               | Augmented Power Data Object (APDO) |
| B29...28 | 00b                                                                               | SPR PPS                            |
|          | 01b                                                                               | EPR AVS                            |
|          | 10b                                                                               | SPR AVS                            |
|          | 11b                                                                               | Reserved                           |
| B27...0  | Specific Power Capabilities are described by the APDOs in the following sections. |                                    |

## 6.4.1.2 Source Power Data Objects

This section lists the types of *PDOs* a *Source* can use in an *SPR Capabilities* or *EPR Capabilities* Message.

### 6.4.1.2.1 Fixed Supply Power Data Object

Table 6.9, "Fixed Supply PDO – Source" describes the *Fixed Supply* (00b) *PDO*. See Section 7.1.3, "Types of Sources" for the electrical requirements of the power supply.

Since all *USB Providers* support **vSafe5V**, the required **vSafe5V** *Fixed Supply Power Data Object* is also used to convey additional information that is returned in bits 29...23. All other *Fixed Supply Power Data Objects* **Shall** set bits 29...23 to zero.

For a *Source* offering no *Capabilities*, the **Voltage** field (B19...10) **Shall** be set to 5V and the **Maximum Current** field **Shall** be set to 0mA. This is used in cases such as a *Dual-Role Power* device which offers no *Capabilities* in its default *Power Role* or when external power is required to offer power.

When a *Source* wants a *Sink*, consuming power from *VBUS*, to go to its lowest power state, the **Voltage** field (B19...10) **Shall** be set to 5V and the **Maximum Current** field **Shall** be set to 0mA. This is used in cases where the *Source* wants the *Sink* to draw **pSnkSusp**.

Table 6.9 Fixed Supply PDO – Source

| Bit(s)   | Field                                        | Description                                                     |
|----------|----------------------------------------------|-----------------------------------------------------------------|
| B31...30 | <i>Fixed Supply</i>                          | 00b - <i>Fixed Supply PDO</i>                                   |
| B29      | <b>Dual-Role Power</b>                       | Set to '1' for <i>Dual-Role Power</i> device.                   |
| B28      | <b>USB Suspend Supported</b>                 | Set to '1' if USB suspend is supported.                         |
| B27      | <b>Unconstrained Power</b>                   | Set to '1' if unconstrained power is available.                 |
| B26      | <b>USB Communications Capable</b>            | Set to '1' if capable of <i>USB Communications</i> capable      |
| B25      | <b>Dual-Role Data</b>                        | Set to '1' for a <i>Dual-Role Data</i> device.                  |
| B24      | <b>Unchunked Extended Messages Supported</b> | Set to '1' if <i>Unchunked Extended Messages</i> are supported. |
| B23      | <b>EPR Capable</b>                           | Set to '1' if <i>EPR Capable</i> .                              |
| B22      | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                  |
| B21...20 | <b>Peak Current</b>                          | Peak Current value.                                             |
| B19...10 | <b>Voltage</b>                               | Voltage in 50mV units                                           |
| B9...0   | <b>Maximum Current</b>                       | Maximum current in 10mA units                                   |

#### 6.4.1.2.1.1 Dual-Role Power

The **Dual-Role Power** bit **Shall** be set when the *Port* is *Dual-Role Power* capable i.e., supports the **PR\_Swap** Message.

This is a **Static** capability which **Shall** remain fixed for a given device regardless of the device's present *Power Role*. If the **Dual-Role Power** bit is set to one in the *Source\_Capabilities* Message the **Dual-Role Power** bit in the *Sink\_Capabilities* Message **Shall** also be set to one. If the **Dual-Role Power** bit is set to zero in the *Source\_Capabilities* Message the **Dual-Role Power** bit in the *Sink\_Capabilities* Message **Shall** also be set to zero.

#### 6.4.1.2.1.2 USB Suspend Supported

Prior to an *Explicit Contract* or when the **USB Communications Capable** bit is set to zero, the **USB Suspend Supported** flag is undefined and *Sinks* **Shall** follow the rules for suspend as defined in [USB 2.0], [USB 3.2], [USB4], [USB Type-C 2.4] or [USBBC 1.2]. After an *Explicit Contract* has been *Negotiated*:

- If the **USB Suspend Supported** flag is set, then the *Sink* **Shall** follow the [USB 2.0], [USB 3.2] or [USB4] rules for suspend and resume. A *PDUSB Peripheral* **May** draw up to **pSnkSusp** during suspend; a *PDUSB Hub* **May** draw up to **pHubSusp** during suspend (see Section 7.2.3, "Sink Standby").

- If the **USB Suspend Supported** flag is cleared, then the Sink **Shall Not** apply the **[USB 2.0]**, **[USB 3.2]** or **[USB4]** rules for suspend and **May** continue to draw the **Negotiated** power.

**Note:** When USB is suspended, the USB device state is also suspended.

Sinks **May** indicate to the Source that they would prefer to have the **USB Suspend Supported** flag cleared by setting the No USB Suspend flag in a **Request** Message (see [Section 6.4.2.5, "No USB Suspend"](#)).

#### 6.4.1.2.1.3 Unconstrained Power

The **Unconstrained Power** bit **Shall** be set when an external source of power is available that is sufficient to adequately power the system while charging external devices, or when the device's primary function is to charge external devices.

To set the **Unconstrained Power** bit because of an external source, the external source of power **Should** be either:

- An AC Supply, e.g., a **Charger**, directly connected to the Sink.
- Or, in the case of a **PDUSB Hub**:
  - A PD Source with its **Unconstrained Power** bit set.
  - Multiple PD Sources all with their **Unconstrained Power** bits set.

#### 6.4.1.2.1.4 USB Communications Capable

The **USB Communications Capable** bit **Shall** only be set for Sources capable of communication over the USB data lines (e.g., D+/- or SS Tx/Rx).

#### 6.4.1.2.1.5 Dual-Role Data

The **Dual-Role Data** bit **Shall** be set when the Port is **Dual-Role Data** capable i.e., it supports the **DR\_Swap** Message. This is a **Static** capability which **Shall** remain fixed for a given device regardless of the device's present **Power Role** or **Data Role**. If the **Dual-Role Data** bit is set to one in the **Source\_Capabilities** Message the **Dual-Role Data** bit in the **Sink\_Capabilities** Message **Shall** also be set to one. If the **Dual-Role Data** bit is set to zero in the **Source\_Capabilities** Message the **Dual-Role Data** bit in the **Sink\_Capabilities** Message **Shall** also be set to zero.

#### 6.4.1.2.1.6 Unchunked Extended Messages Supported

The **Unchunked Extended Messages Supported** bit **Shall** be set when the Port can send and receive **Extended Messages** with **Data Size** > **MaxExtendedMsgLegacyLen** bytes in a single, **Unchunked Extended** Message.

#### 6.4.1.2.1.7 EPR Mode Capable

The **EPR Capable** bit is a **Static** bit that **Shall** be set if the Source is designed to supply more than 100W and operate in **EPR Mode**.

When this bit is set, an **EPR Source**:

- Operating in **SPR Mode** **Shall** only send an **EPR\_Source\_Capabilities** Message in response to an **EPR\_Get\_Source\_Cap** Message
- **May** only enter **EPR Mode** when the Cable and the Sink also report that they are **EPR Capable**.

#### 6.4.1.2.1.8 Peak Current

The USB Power Delivery **Fixed Supply** is only required to deliver the amount of current requested in the **Operating Current** field (**IoC**) of an **RDO**. In some usages however, for example computer systems, where there are short bursts of activity, it might be desirable to overload the **Source** for short periods.

For example, when a computer system tries to maintain average power consumption, the higher the peak current, the longer the low current (see [Section 7.2.8, "Sink Peak Current Operation"](#)) period needed to maintain such average power. The **Peak Current** field allows a **Source** to **Advertise** this additional capability. This capability is intended for direct **Port to Port** connections only and **Shall Not** be offered to downstream Sinks via a **Hub**.

Every *Fixed Supply PDO* **Shall** contain a **Peak Current** field. Supplies that want to offer a set of overload *Capabilities* **Shall** Advertise this through the **Peak Current** field in the corresponding *Fixed Supply PDO* (see [Table 6.10, "Fixed Power Source Peak Current Capability"](#)). Supplies that do not support an overload capability **Shall** set these bits to 00b in the corresponding *Fixed Supply PDO*. Supplies that support an extended overload capability specified in the PeakCurrent1...3 fields of the **Source\_Capabilities\_Extended** Message (see [Section 6.5.1, "Source\\_Capabilities\\_Extended Message"](#)) **Shall** also set these bits to 00b. Sinks wishing to utilize these *Extended Capabilities* **Shall** first send the **Get\_Source\_Cap\_Extended** Message to determine what *Capabilities*, if any are supported by the *Source*.

**Table 6.10 Fixed Power Source Peak Current Capability**

| Bits 21...20 | Description                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00           | Peak current equals <i>IoC</i> (default) or look at the <b>Source_Capabilities_Extended</b> Message (send <b>Get_Source_Cap_Extended</b> Message)                                                                                                                                                                                                                              |
| 01           | Overload <i>Capabilities</i> :<br>1. Peak current equals 150% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 97% <i>IoC</i> for 19ms)<br>2. Peak current equals 125% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 97% <i>IoC</i> for 18ms)<br>3. Peak current equals 110% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 90% <i>IoC</i> for 10ms) |
| 10           | Overload <i>Capabilities</i> :<br>1. Peak current equals 200% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 95% <i>IoC</i> for 19ms)<br>2. Peak current equals 150% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 94% <i>IoC</i> for 18ms)<br>3. Peak current equals 125% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 75% <i>IoC</i> for 10ms) |
| 11           | Overload <i>Capabilities</i> :<br>1. Peak current equals 200% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 95% <i>IoC</i> for 19ms)<br>2. Peak current equals 175% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 92% <i>IoC</i> for 18ms)<br>3. Peak current equals 150% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 50% <i>IoC</i> for 10ms) |

#### 6.4.1.2.2

### Variable Supply (non-Battery) Power Data Object

[Table 6.11, "Variable Supply \(non-Battery\) PDO – Source"](#) describes a *Variable Supply (non-Battery)* (10b) *PDO* for a *Source*. See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

The *voltage* fields **Shall** define the range that output *voltage* **Shall** fall within. This does not indicate the *voltage* that will be supplied, except it **Shall** fall within that range. The absolute *voltage*, including any *voltage* variation, **Shall Not** fall below the **Minimum Voltage** field value and **Shall Not** exceed the **Maximum Voltage** field value. The **Minimum Voltage** field value **Shall Not** be less than 80% of the **Maximum Voltage** field value.

**Table 6.11 Variable Supply (non-Battery) PDO – Source**

| Bit(s)   | Field                  | Description                                    |
|----------|------------------------|------------------------------------------------|
| B31...30 | <i>Variable Supply</i> | 01b - <i>Variable Supply (non-Battery) PDO</i> |
| B29...20 | <b>Maximum Voltage</b> | Maximum voltage in 50mV units                  |
| B19...10 | <b>Minimum Voltage</b> | Minimum voltage in 50mV units                  |
| B9...0   | <b>Maximum Current</b> | Maximum current in 10mA units                  |

#### 6.4.1.2.3

### Battery Supply Power Data Object

[Table 6.12, "Battery Supply PDO – Source"](#) describes a *Battery Supply* (01b) *PDO* for a *Source*. See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

The *voltage* fields **Shall** represent the *Battery's* *voltage* range. The *Battery* **Shall** be capable of supplying the Power value over the entire *voltage* range. The absolute *voltage*, including any *voltage* variation, **Shall Not** fall below the **Minimum Voltage** field value and **Shall Not** exceed the **Maximum Voltage** field value.

**Note:** The *Battery Supply* *PDO* uses power instead of current.

The Sink **May** monitor the Battery voltage.

**Table 6.12 Battery Supply PDO – Source**

| Bit(s)   | Field                          | Description                            |
|----------|--------------------------------|----------------------------------------|
| B31...30 | <i>Battery Supply</i>          | 10b - <i>Battery Supply PDO</i>        |
| B29...20 | <i>Maximum Voltage</i>         | Maximum voltage in 50mV units          |
| B19...10 | <i>Minimum Voltage</i>         | Minimum voltage in 50mV units          |
| B9...0   | <i>Maximum Allowable Power</i> | Maximum allowable power in 250mW units |



#### 6.4.1.2.4 Augmented Power Data Object (APDO)

The voltage fields define the output voltage range over which the power supply **Shall** be adjustable in 20mV steps in *SPR PPS Mode* and 100mV steps in both *SPR AVS Mode* and *EPR AVS Mode*. The **Maximum Current** field contains the current the Programmable Power Supply **Shall** be capable of delivering over the *Advertised voltage* range. See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

##### 6.4.1.2.4.1 SPR Programmable Power Supply APDO

[Table 6.13, "SPR Programmable Power Supply APDO – Source"](#) below describes the *SPR PPS* (1100b) *APDO* for a *Source* operating in *SPR Mode* and supplying 5V up to 21V.

**Table 6.13 SPR Programmable Power Supply APDO – Source**

| Bit(s)   | Field                    | Description                                            |
|----------|--------------------------|--------------------------------------------------------|
| B31...30 | <i>APDO</i>              | 11b – <i>Augmented Power Data Object (APDO)</i>        |
| B29...28 | <i>SPR PPS</i>           | 00b – <i>SPR PPS</i>                                   |
| B27      | <b>PPS Power Limited</b> | Set to '1' when <i>PPS Power Limited</i>               |
| B26...25 | <b>Reserved</b>          | <b>Reserved</b> – <b>Shall</b> be set to <b>zero</b> . |
| B24...17 | <b>Maximum Voltage</b>   | Maximum voltage in 100mV increments                    |
| B16      | <b>Reserved</b>          | <b>Reserved</b> – <b>Shall</b> be set to <b>zero</b> . |
| B15...8  | <b>Minimum Voltage</b>   | Minimum voltage in 100mV increments                    |
| B7       | <b>Reserved</b>          | <b>Reserved</b> – <b>Shall</b> be set to <b>zero</b> . |
| B6...0   | <b>Maximum Current</b>   | Maximum current in 50mA increments                     |

The *PPS APDO* is used primarily for *Sink Directed Charge* of a *Battery* in the *Sink*. When applying a current to the *Battery* greater than the cable supports, a high efficiency fixed voltage scaler **May** be used in the *Sink* to reduce the cable current.

##### 6.4.1.2.4.1.1 PPS Power Limited

When the **PPS Power Limited** bit is set, the *SPR PPS Source* **Shall** operate in the same way as if the **PPS Power Limited** bit is clear (see [Section 7.1.4.2, "SPR Programmable Power Supply \(PPS\)"](#) with the below exception:

- May** supply power that exceeds the *Source's* rated *PDP* within the **Optional** operating area in [Figure 7.7, "SPR PPS Constant Power"](#).

When the **PPS Power Limited** bit is cleared, the *SPR PPS Source* **Shall** deliver the **Maximum Current** field value up to the **Maximum Voltage** as *Advertised* in its *APDO*.

The *SPR PPS Source* **Shall Not** reject an *RDO* with an **Operating Current** field value that is less than or equal to the **Maximum Current** field value in the *APDO* even if the requested **Operating Current** field value is greater than the *Source's PDP*/requested Output voltage.

#### 6.4.1.2.4.2

### SPR Adjustable Voltage Supply APDO

[Table 6.14, "SPR Adjustable Voltage Supply APDO – Source"](#) below describes the *SPR AVS* (1110b) *APDO* for a *Source* operating in *SPR Mode* and supplying 9V up to 20V.

**Table 6.14 SPR Adjustable Voltage Supply APDO – Source**

| Bit(s)   | Field                      | Description                                                                                                                                                                                           |
|----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...30 | <i>APDO</i>                | 11b – <i>Augmented Power Data Object (APDO)</i>                                                                                                                                                       |
| B29...28 | <i>SPR AVS</i>             | 10b – <i>SPR AVS</i>                                                                                                                                                                                  |
| B27...26 | <b>Peak Current</b>        | Peak Current (see <a href="#">Table 6.10, "Fixed Power Source Peak Current Capability"</a> )                                                                                                          |
| B25...20 | <b>Reserved</b>            | <b>Reserved</b> – <i>Shall</i> be set to zero.                                                                                                                                                        |
| B19...10 | <b>Maximum Current 15V</b> | For 9V – 15V range: Maximum current in 10mA units equal to the <b>Maximum Current</b> field of the 15V <i>Fixed Supply PDO</i>                                                                        |
| B9...0   | <b>Maximum Current 20V</b> | For 15V – 20V range: Maximum current in 10mA units equal to the <b>Maximum Current</b> field of the 20V <i>Fixed Supply PDO</i> , set to 0 if the maximum voltage in the <i>SPR AVS</i> range is 15V. |

#### 6.4.1.2.4.2.1

### Peak Current

The **Peak Current** field follows the same definition as for the **Peak Current** field (see [Section 6.4.1.2.1.8, "Peak Current"](#) and [Table 6.10, "Fixed Power Source Peak Current Capability"](#)).

#### 6.4.1.2.4.3

### EPR Adjustable Voltage Supply APDO

[Table 6.15, "EPR Adjustable Voltage Supply APDO – Source"](#) below describes the *EPR AVS* (1101b) *APDO* for a *Source* operating in *EPR Mode* and supplying 15V up to 48V.

**Table 6.15 EPR Adjustable Voltage Supply APDO – Source**

| Bit(s)   | Field                                | Description                                                                                    |
|----------|--------------------------------------|------------------------------------------------------------------------------------------------|
| B31...30 | <i>APDO</i>                          | 11b – <i>Augmented Power Data Object (APDO)</i>                                                |
| B29...28 | <i>EPR AVS</i>                       | 01b – <i>EPR AVS</i>                                                                           |
| B27...26 | <b>Peak Current (Source EPR AVS)</b> | Peak Current (see <a href="#">Table 6.16, "EPR AVS Power Source Peak Current Capability"</a> ) |
| B25...17 | <b>Maximum Voltage</b>               | Maximum voltage in 100mV increments                                                            |
| B16      | <b>Reserved</b>                      | <b>Reserved</b> – <i>Shall</i> be set to zero.                                                 |
| B15...8  | <b>Minimum Voltage</b>               | Minimum voltage in 100mV increments                                                            |
| B7...0   | <b>PDP</b>                           | <i>PDP</i> in 1W increments                                                                    |

#### 6.4.1.2.4.3.1

### PDP

✖ The **PDP** field **Shall** contain the *AVS Port's PDP*.

See [Section 10.2.3.3, "Optional Normative Extended Power Range \(EPR\)"](#) and [Figure 10.6, "Valid EPR AVS Operating Region"](#) for more information regarding how *PDP* in the *AVS APDO* relates to maximum available current.

#### 6.4.1.2.4.3.2

### Peak Current

The USB Power Delivery *EPR AVS* is only required to deliver the amount of current requested in the **Operating Current** field (*IoC*) of an *AVS RDO*. In some usages however, for example computer systems, where there are short bursts of activity, it might be desirable to overload the *Source* for short periods.

For example, when a computer system tries to maintain average power consumption, the higher the peak current, the longer the low current period needed to maintain such average power (see [Section 7.2.8, "Sink Peak Current Operation"](#)). The **Peak Current (Source EPR AVS)** field allows a *Source* to *Advertise* this additional capability. This

capability is intended for direct *Port* to *Charger* connections only and **Shall Not** be offered to downstream *Sinks* via a *Hub*.

Every *EPR AVS APDO* **Shall** contain a **Peak Current (Source EPR AVS)** field. Supplies that want to offer a set of overload *Capabilities* **Shall** Advertise this through the **Peak Current (Source EPR AVS)** field in the corresponding *EPR AVS APDO* (see [Table 6.16, "EPR AVS Power Source Peak Current Capability"](#)). Supplies that do not support an overload capability **Shall** set these bits to 00b in the corresponding *EPR AVS APDO*. Supplies that support an extended overload capability specified in the PeakCurrent1...3 fields of the **Source\_Capabilities\_Extended** Message (see [Section 6.5.1, "Source\\_Capabilities\\_Extended Message"](#)) **Shall** set these bits to 00b. *Sinks* wishing to utilize these *Extended Capabilities* **Shall** first send a **Get\_Source\_Cap\_Extended** Message to determine what *Capabilities*, if any are supported by the *Source*.

**Table 6.16 EPR AVS Power Source Peak Current Capability**

| Bits 21...20 | Description                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00           | Peak current equals <i>IoC</i> (default) or look at the <b>Source_Capabilities_Extended</b> Message (send <b>Get_Source_Cap_Extended</b> Message)                                                                                                                                                                                                                              |
| 01           | Overload <i>Capabilities</i> :<br>1. Peak current equals 150% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 97% <i>IoC</i> for 19ms)<br>2. Peak current equals 125% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 97% <i>IoC</i> for 18ms)<br>3. Peak current equals 110% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 90% <i>IoC</i> for 10ms) |
| 10           | Overload <i>Capabilities</i> :<br>1. Peak current equals 200% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 95% <i>IoC</i> for 19ms)<br>2. Peak current equals 150% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 94% <i>IoC</i> for 18ms)<br>3. Peak current equals 125% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 75% <i>IoC</i> for 10ms) |
| 11           | Overload <i>Capabilities</i> :<br>1. Peak current equals 200% <i>IoC</i> for 1ms @ 5% duty cycle (low current equals 95% <i>IoC</i> for 19ms)<br>2. Peak current equals 175% <i>IoC</i> for 2ms @ 10% duty cycle (low current equals 92% <i>IoC</i> for 18ms)<br>3. Peak current equals 150% <i>IoC</i> for 10ms @ 50% duty cycle (low current equals 50% <i>IoC</i> for 10ms) |

### 6.4.1.3 Sink Power Data Objects

This section lists the types of *PDOs* a *Sink* can use in an *SPR* or *EPR Capabilities Message*.

#### 6.4.1.3.1 Sink Fixed Supply Power Data Object

Table 6.17, "*Fixed Supply PDO – Sink*" describes the *Sink Fixed Supply* (00b) *PDO*. See Section 7.1.3, "*Types of Sources*" for the electrical requirements of the power supply. The *Sink* **shall** set the *Voltage* field to its required voltage and the *Operational Current* field to its required operating current. Required operating current is defined as the amount of current a given device needs to be functional. This value could be the maximum current the *Sink* will ever require or could be sufficient to operate the *Sink* in one of its modes of operation.

Since all USB *Consumers* support *vSafe5V*, the required *vSafe5V Fixed Supply Power Data Object* is also used to convey additional information that is returned in bits 29 through 20. All other *Fixed Supply Power Data Objects* **shall** set bits 29...20 to zero.

For a *Sink* requiring no power from the *Source*, the *Voltage* field **shall** be set to 5V and the *Operational Current* field **shall** be set to 0mA.

Table 6.17 Fixed Supply PDO – Sink

| Bit(s)   | Field                                             | Description                                                                                   |                                               |
|----------|---------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------|
| B31...30 | <i>Fixed Supply</i>                               | 00b - <i>Fixed Supply PDO</i>                                                                 |                                               |
| B29      | <i>Dual-Role Power</i>                            | Set to '1' if <i>Dual-Role Power</i> supported                                                |                                               |
| B28      | <i>Higher Capability</i>                          | Set to '1' if <i>Higher Capability</i> supported                                              |                                               |
| B27      | <i>Unconstrained Power</i>                        | Set to '1' if <i>Unconstrained Power</i> supported                                            |                                               |
| B26      | <i>USB Communications Capable</i>                 | Set to '1' if <i>USB Communications Capable</i>                                               |                                               |
| B25      | <i>Dual-Role Data</i>                             | <i>Dual-Role Data</i>                                                                         |                                               |
| B24...23 | <i>Fast Role Swap required USB Type-C Current</i> | <i>Fast Role Swap</i> required <i>USB Type-C</i> current (see also <i>[USB Type-C 2.4]</i> ): |                                               |
|          |                                                   | Value                                                                                         | Description                                   |
|          |                                                   | 00b                                                                                           | <i>Fast Role Swap</i> not supported (default) |
|          |                                                   | 01b                                                                                           | Default <i>USB Port</i>                       |
|          |                                                   | 10b                                                                                           | 1.5A@5V                                       |
|          |                                                   | 11b                                                                                           | 3.0A@5V                                       |
| B22...20 | <i>Reserved</i>                                   | <i>Reserved</i> – <i>Shall</i> be set to zero.                                                |                                               |
| B19...10 | <i>Voltage</i>                                    | Voltage in 50mV units                                                                         |                                               |
| B9...0   | <i>Operational Current</i>                        | Operational current in 10mA units                                                             |                                               |

##### 6.4.1.3.1.1 Dual-Role Power

The *Dual-Role Power* bit **shall** be set when the *Port* is *Dual-Role Power* capable i.e., supports the *PR\_Swap Message*. This is a *Static* capability which **shall** remain fixed for a given device regardless of the device's present *Power Role*. If the *Dual-Role Power* bit is set to one in the *Source\_Capabilities Message* the *Dual-Role Power* bit in the *Sink\_Capabilities Message* **shall** also be set to one. If the *Dual-Role Power* bit is set to zero in the *Source\_Capabilities Message* the *Dual-Role Power* bit in the *Sink\_Capabilities Message* **shall** also be set to zero.

##### 6.4.1.3.1.2 Higher Capability

In the case that the *Sink* needs more than *vSafe5V* (e.g., 15V) to provide full functionality, then the *Higher Capability* bit **shall** be set.

##### 6.4.1.3.1.3 Unconstrained Power

The *Unconstrained Power* bit **shall** be set when an external source of power is available that is sufficient to adequately power the system while charging external devices, or when the device's primary function is to charge external devices.

To set the **Unconstrained Power** bit because of an external source, the external source of power **Should** be either:

- An AC Supply, e.g., a Charger, directly connected to the Sink.
- Or, in the case of a PDUSB Hub:
  - A PD Source with its **Unconstrained Power** bit set.
  - Multiple PD Sources all with their **Unconstrained Power** bits set.

#### 6.4.1.3.1.4 USB Communications Capable

The **USB Communications Capable** bit **Shall** only be set for Sinks capable of communication over the USB data lines (e.g., D+/- or SS Tx/Rx).

#### 6.4.1.3.1.5 Dual-Role Data

The **Dual-Role Data** bit **Shall** be set when the Port is Dual-Role Data capable i.e., it supports the **DR\_Swap** Message. This is a **Static** capability which **Shall** remain fixed for a given device regardless of the device's present Power Role or Data Role. If the **Dual-Role Data** bit is set to one in the **Source\_Capabilities** Message the **Dual-Role Data** bit in the **Sink\_Capabilities** Message **Shall** also be set to one. If the **Dual-Role Data** bit is set to zero in the **Source\_Capabilities** Message the **Dual-Role Data** bit in the **Sink\_Capabilities** Message **Shall** also be set to zero.

#### 6.4.1.3.1.6 Fast Role Swap USB Type-C Current

The **Fast Role Swap required USB Type-C Current** field **Shall** indicate the current level the Sink will require after a Fast Role Swap has been performed.

The **Initial Source** **Shall Not** transmit a Fast Role Swap Request if the **Fast Role Swap required USB Type-C Current** field is set to zero.

Initially when the New Source applies **vSafe5V** it will have  $R_d$  asserted but **Shall** provide the USB Type-C current indicated by the New Sink in this field. If the New Source is not able to supply this level of current, it **Shall Not** perform a Fast Role Swap. When  $R_p$  is asserted by the New Source during the Fast Role Swap AMS (see [Section 6.3.19, "FR\\_Swap Message"](#)), the value of USB Type-C current indicated by  $R_p$  **Shall** be the same or greater than that indicated in the **Fast Role Swap required USB Type-C Current** field.

### 6.4.1.3.2 Variable Supply (non-Battery) Power Data Object

[Table 6.18, "Variable Supply \(non-Battery\) PDO – Sink"](#) describes a Variable Supply (non-Battery) (10b) PDO used by a Sink. See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

The voltage fields **Shall** be set to the output voltage range that the Sink requires to operate. The **Operational Current** field **Shall** be set to the operational current that the Sink requires at the given voltage range. The absolute voltage, including any voltage variation, **Shall Not** fall below the **Minimum Voltage** field value and **Shall Not** exceed the **Maximum Voltage** field value. Required operating current is defined as the amount of current a given device needs to be functional. This value could be the maximum current the Sink will ever require or could be sufficient to operate the Sink in one of its modes of operation.

**Table 6.18 Variable Supply (non-Battery) PDO – Sink**

| Bit(s)   | Field               | Description                             |
|----------|---------------------|-----------------------------------------|
| B31...30 | Variable Supply     | 01b - Variable Supply (non-Battery) PDO |
| B29...20 | Maximum Voltage     | Maximum voltage in 50mV units           |
| B19...10 | Minimum Voltage     | Minimum voltage in 50mV units           |
| B9...0   | Operational Current | Operational current in 10mA units       |

#### 6.4.1.3.3 Battery Supply Power Data Object

[Table 6.19, "Battery Supply PDO – Sink"](#) describes a Battery Supply (01b) PDO used by a Sink. See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

The **voltage** fields **Shall** be set to the output **voltage** range that the *Sink* requires to operate. The **Operational Power** field **Shall** be set to the operational power that the *Sink* requires at the given **voltage** range. The absolute voltage, including any **voltage** variation, **Shall Not** fall below the **Minimum Voltage** field value and **Shall Not** exceed the **Maximum Voltage** field value.

**Note:** Only the *Battery Supply PDO* uses power instead of current.

Required operating power is defined as the amount of power a given device needs to be functional. This value could be the maximum power the *Sink* will ever require or could be sufficient to operate the *Sink* in one of its modes of operation.

**Table 6.19 Battery Supply PDO – Sink**

| Bit(s)   | Field                    | Description                      |
|----------|--------------------------|----------------------------------|
| B31...30 | <i>Battery Supply</i>    | 10b - <i>Battery Supply PDO</i>  |
| B29...20 | <b>Maximum Voltage</b>   | Maximum voltage in 50mV units    |
| B19...10 | <b>Minimum Voltage</b>   | Minimum voltage in 50mV units    |
| B9...0   | <b>Operational Power</b> | Operational Power in 250mW units |

#### 6.4.1.3.4 Augmented Power Data Objects

See [Section 7.1.3, "Types of Sources"](#) for the electrical requirements of the power supply.

The Maximum and Minimum **voltage** fields **Shall** be set to the output **voltage** range that the *Sink* requires to operate.

##### 6.4.1.3.4.1 SPR Programmable Power Supply APDO

[Table 6.20, "SPR Programmable Power Supply APDO – Sink"](#) below describes a *SPR PPS APDO* for a *Sink* operating in *SPR Mode* and consuming 21V or less. The **Maximum Current** field **Shall** be set to the maximum current the *Sink* requires over the **voltage** range. The maximum current is defined as the maximum amount of current the device needs to fully support its function (e.g., *Sink Directed Charge*).

**Table 6.20 SPR Programmable Power Supply APDO – Sink**

| Bit(s)   | Field                  | Description                                     |
|----------|------------------------|-------------------------------------------------|
| B31...30 | <i>APDO</i>            | 11b – <i>Augmented Power Data Object (APDO)</i> |
| B29...28 | <i>SPR PPS</i>         | 00b – <i>SPR PPS</i>                            |
| B27...25 | <b>Reserved</b>        | <b>Reserved – Shall be set to zero.</b>         |
| B24...17 | <b>Maximum Voltage</b> | Maximum voltage in 100mV increments             |
| B16      | <b>Reserved</b>        | <b>Reserved – Shall be set to zero.</b>         |
| B15...8  | <b>Minimum Voltage</b> | Minimum voltage in 100mV increments             |
| B7       | <b>Reserved</b>        | <b>Reserved – Shall be set to zero.</b>         |
| B6...0   | <b>Maximum Current</b> | Maximum current in 50mA increments              |



#### 6.4.1.3.4.2

#### SPR Adjustable Voltage Supply APDO



[Table 6.21, "SPR Adjustable Voltage Supply APDO – Sink"](#) below describes the *SPR AVS* (1110b) *APDO* for a *Sink* operating in *SPR AVS Mode*. The **Maximum Current 15V**/**Maximum Current 20V** fields in the *SPR AVS APDO* for the *Sink* is defined as the maximum current the device needs to fully support its function.

**Table 6.21 SPR Adjustable Voltage Supply APDO – Sink**

| Bit(s)   | Field                      | Description                                                                                                                                                                                           |
|----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...30 | <i>APDO</i>                | 11b – <i>Augmented Power Data Object (APDO)</i>                                                                                                                                                       |
| B29...28 | <i>SPR AVS</i>             | 10b – <i>SPR AVS</i>                                                                                                                                                                                  |
| B27...20 | <b>Reserved</b>            | <b>Reserved – Shall</b> be set to zero.                                                                                                                                                               |
| B19...10 | <b>Maximum Current 15V</b> | For 9V – 15V range: Maximum current in 10mA units equal to the <b>Maximum Current</b> field of the <i>15V Fixed Supply PDO</i>                                                                        |
| B9...0   | <b>Maximum Current 20V</b> | For 15V – 20V range: Maximum Current in 10mA units equal to the <b>Maximum Current</b> field of the <i>20V Fixed Supply PDO</i> , set to 0 if the Maximum voltage in the <i>SPR AVS</i> range is 15V. |

#### 6.4.1.3.4.3

#### EPR Adjustable Voltage Supply APDO

[Table 6.22, "EPR Adjustable Voltage Supply APDO – Sink"](#) below describes a *EPR AVS APDO* for a *Sink* operating in *EPR AVS Mode*. The **PDP** field in the *EPR AVS APDO* for the *Sink* is defined as the *PDP* the device needs to fully support its function.

**Table 6.22 EPR Adjustable Voltage Supply APDO – Sink**

| Bit(s)   | Field                  | Description                                     |
|----------|------------------------|-------------------------------------------------|
| B31...30 | <i>APDO</i>            | 11b – <i>Augmented Power Data Object (APDO)</i> |
| B29...28 | <i>EPR AVS</i>         | 01b – <i>EPR AVS</i>                            |
| B27...26 | <b>Reserved</b>        | <b>Reserved – Shall</b> be set to zero.         |
| B25...17 | <b>Maximum Voltage</b> | Maximum voltage in 100mV increments             |
| B16      | <b>Reserved</b>        | <b>Reserved – Shall</b> be set to zero.         |
| B15...8  | <b>Minimum Voltage</b> | Minimum voltage in 100mV increments             |
| B7...0   | <b>PDP</b>             | <i>PDP</i> in 1W increments                     |

#### 6.4.1.4

#### SPR Capabilities Message Construction

An *SPR Capabilities Message* (**Source Capabilities Message** or **Sink Capabilities Message**) **Shall** have at least one *Power Data Object* for **vSafe5V**. The *SPR Capabilities Message* **Shall** also contain the sending *Port's* information followed by up to 6 additional *Power Data Objects*. *Power Data Objects* in an *SPR Capabilities Message* **Shall** be sent in the following order:

- 1) The **vSafe5V Fixed Supply PDO** **Shall** always be the first (A)PDO.
- 2) The remaining *Fixed Supply PDOs*, if present, **Shall** be sent in voltage order; lowest to highest.
- 3) The *Battery Supply PDOs* if present **Shall** be sent in Minimum voltage order; lowest to highest.
- 4) The *Variable Supply (non-Battery) PDOs*, if present, **Shall** be sent in Minimum voltage order; lowest to highest.
- 5) The *SPR AVS APDO*, if present, **Shall** be sent.
- 6) The *Programmable Power Supply APDOs*, if present, **Shall** be sent in Maximum voltage order, lowest to highest.

**Note:** The *EPR Capabilities Message* construction is defined in [Section 6.5.15.1, "EPR Capabilities Message Construction"](#).

Figure 6.12, "SPR Capabilities Message Construction" describes the construction of an SPR Capabilities Message. The Message will always have at least one Fixed Supply 5V PDO and may have up to six more PDOs depending on the Source Capabilities.

Figure 6.12 SPR Capabilities Message Construction

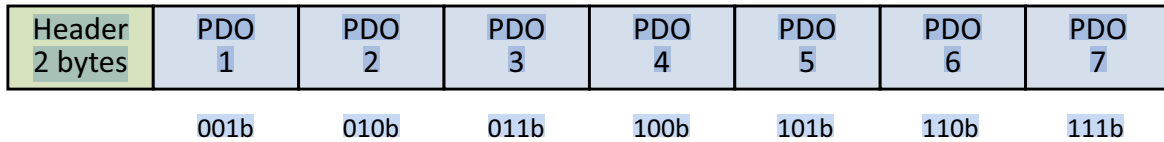


Figure 6.13 Example Capabilities Message with 2 Power Data Objects



In the 27W Source as shown in Figure 6.13, "Example Capabilities Message with 2 Power Data Objects", the **Number of Data Objects** field is 2: **vSafe5V** plus one other voltage.

Power Data Objects (PDO) and Augmented Power Data Objects (APDO) are identified by the Message Header's **Message Type** field. They are used to form SPR Capabilities Messages.

#### 6.4.1.5 SPR Source Capabilities Message

Sources send a **Source\_Capabilities** Message either as part of advertising Port Capabilities, or in response to a **Get\_Source\_Cap** Message. See Section 6.5.15.2, "EPR Source Capabilities Message" for information about EPR Source Capabilities Messages.

Following a **Hard Reset**, a power-on event or plug insertion event, a Source Port **Shall** send a **Source\_Capabilities** Message after every **SourceCapabilityTimer** timeout as an Advertisements that **Shall** be interpreted by the Sink Port on Attachment. The Source **Shall** continue sending a minimum of **nCapsCount** **Source\_Capabilities** Messages until a **GoodCRC** Message is received.

Additionally, a **Source\_Capabilities** Message **Shall** only be sent by a Port in the following cases:

- By the Source Port from the **PE\_SRC\_Ready** state upon a change in its ability to supply power to this Port.
- By a Source Port or Dual-Role Power Port in response to a **Get\_Source\_Cap** Message.
- **Optionally** by a Source Port from the **PE\_SRC\_Ready** state when available power in a multi-Port system changes, even if the Source Capabilities for this Port have not changed.

A Source Port **Shall** report its Capabilities in a series of 32-bit Power Data Objects (see Table 6.7, "Power Data Object") as part of a **Source\_Capabilities** Message (see Figure 6.13, "Example Capabilities Message with 2 Power Data Objects"). Power Data Objects are used to convey a Source Port's Capabilities to provide power including Dual-Role Power ports presently operating as a Sink.

Each Power Data Object **Shall** describe a specific Source capability such as a Battery (e.g., 2.8-4.1V) or a Fixed Supply (e.g., 15V) at a maximum allowable current. The **Number of Data Objects** field in the Message Header **Shall** define the number of Power Data Objects that follow the Message Header in a Data Message. All Sources **Shall** minimally offer one Power Data Object that reports **vSafe5V**. A Source **Shall Not** offer multiple Power Data Objects of the same type (Fixed Supply, Variable Supply, Battery Supply) and the same voltage but **Shall** instead offer one Power Data Object with the highest available current for that Source capability and voltage.

Sinks with Accessory Support do not source *V<sub>BUS</sub>* (see [\[USB Type-C 2.4\]](#)). Sinks with Accessory Support are still considered Sources when sourcing *V<sub>CONN</sub>* to an Accessory even though *V<sub>BUS</sub>* is not applied; in this case they **Shall** Advertise *vSafe5V* with the **Maximum Current** field set to 0mA in the first Power Data Object. The main purpose of this is to enable the Sink with Accessory Support to get into the **PE\_SRC\_Ready** State to enter an *Alternate Mode*.

A Sink in *SPR Mode* **Shall** evaluate every *Source\_Capabilities* Message it receives and **Shall** respond with a *Request* Message. If its power consumption exceeds the Source Capabilities it **Shall** Re-negotiate so as not to exceed the Source's most recently Advertised Capabilities.

A Sink, in *SPR Mode*, in an *Explicit Contract* with a PPS APDO, **Shall** periodically re-request the PPS APDO at least every *tPPSRequest* until either:

- The Sink requests something other than PPS APDO.
- There is a Power Role Swap.
- There is a Hard Reset.
- There is Error Recovery.

A Sink in *EPR Mode* that receives a *Source\_Capabilities* Message in response to a *Get\_Source\_Cap* Message **Shall Not** respond with a *Request* Message. If a Sink in *EPR Mode* receives a *Source\_Capabilities* Message, not in response to a *Get\_Source\_Cap* Message, the Sink **Shall** initiate a Hard Reset.

A Source that has accepted a *Request* Message with a Programmable RDO **Shall** issue *Hard Reset* Signaling if it has not received a *Request* Message with a Programmable RDO within *tPPSTimeout*. The Source **Shall** discontinue this behavior after:

- Receiving a *Request* Message with a Fixed Supply, Variable Supply or Battery Supply RDO.
- There is a Power Role Swap.
- There is a Hard Reset.
- There is Error Recovery.

#### 6.4.1.6 SPR Sink Capabilities Message

Sinks send a *Sink\_Capabilities* Message (see [Section 6.4.2, "Request Message"](#)) in response to a *Get\_Sink\_Cap* Message. See [Section 6.5.15.3, "EPR Sink Capabilities Message"](#) for more information about the Capabilities Message.

A USB Power Delivery capable Sink, upon detecting *vSafe5V* on *V<sub>BUS</sub>* and after a *SinkWaitCapTimer* timeout without seeing a *Source\_Capabilities* Message, **Shall** send a Hard Reset. If the Attached Source is USB Power Delivery capable, it responds by sending *Source\_Capabilities* Messages thus allowing power Negotiations to begin.

A Sink Port **Shall** report power levels it is able to operate at in a series of 32-bit Power Data Objects (see [Section Table 6.7, "Power Data Object"](#)). These are returned as part of a *Sink\_Capabilities* Message in response to a *Get\_Sink\_Cap* Message (see [Figure 6.13, "Example Capabilities Message with 2 Power Data Objects"](#)). This is similar to that used for Source Port Capabilities with equivalent Power Data Objects for Fixed Supply, Variable Supply and Battery Supply as defined in this section. Power Data Objects are used to convey the Sink Port's operational power requirements including Dual-Role Power Ports presently operating as a Source.

Each Power Data Object **Shall** describe a specific Sink operational power level, such as a Battery Supply (e.g., 2.8-4.1V) or a Fixed Supply (e.g., 15V). The **Number of Data Objects** field in the Message Header **Shall** define the number of Power Data Objects that follow the Message Header in a Data Message.

All Sinks **Shall** minimally offer one Power Data Object with a power level at which the Sink can operate. A Sink **Shall Not** offer multiple Power Data Objects of the same type (Fixed Supply, Variable Supply, Battery Supply) and the same voltage but **Shall** instead offer one Power Data Object with the highest available current for that Sink capability and voltage.

All Sinks **Shall** include one Power Data Object that reports *vSafe5V* even if they require additional power to operate fully. In the case where additional power is required for full operation the Higher Capability bit **Shall** be set.

#### 6.4.1.6.1 Use by Dual-Role Power devices

*Dual-Role Power* devices send a **Source\_Capabilities** Message (see [Section 6.4.1.5, "SPR Source Capabilities Message"](#)) as part of advertising *Port Capabilities* when operating in *Source* role. *Dual-Role Power* devices send a **Source\_Capabilities** Message in response to a **Get\_Source\_Cap** Message regardless of their present operating role. Similarly *Dual-Role Power* devices send a **Sink\_Capabilities** Message (see [Section 6.4.1.6, "SPR Sink Capabilities Message"](#)) in response to a **Get\_Sink\_Cap** Message regardless of their present operating role.

#### 6.4.1.6.2 Management of the Power Reserve

This section has been removed. Refer to [Section 8.2.5, "Managing Power Requirements"](#).

## 6.4.2 Request Message

A **Request** Message **Shall** be sent by a Sink to request power during the request phase of an SPR power **Negotiation**. The **Request Data Object** **Shall** be returned by the Sink making a request for power. It **Shall** be sent in response to the most recent **Source\_Capabilities** Message (see [Section 8.3.2.2, "Power Negotiation"](#)) when in **SPR Mode**. A **Request** Message **Shall** return one and only one Sink Request Data Object that **Shall** identify the Power Data Object being requested.

The Source **Shall** respond to a **Request** Message with an **Accept** Message, a **Wait** Message or a **Reject** Message (see [Section 6.9, "Accept, Reject and Wait"](#)).

The **Request** Message includes the requested power level. For example, if the **Source\_Capabilities** Message includes a **Fixed Supply PDO** that offers 9V @ 1.5A and if the Sink only wants 9V @ 0.5A, it will set the **Operating Current** field to 50 (i.e., 10mA \* 50 = 0.5A).

The request uses a different format depending on the kind of power requested.

- The **Fixed Supply Power Data Object** and **Variable Supply Power Data Object** share a common format shown in [Table 6.23, "Fixed and Variable Request Data Object"](#).
- The **Battery Supply Power Data Object** uses the format shown in [Table 6.24, "Battery Request Data Object"](#).
- The **PPS Request Data Object's** format is shown in [Table 6.25, "PPS Request Data Object"](#).
- The **AVS Request Data Object's** format is shown in [Table 6.26, "AVS Request Data Object"](#).

The **Request Data Objects** are also used by the **EPR\_Request** Message when operating in **EPR Mode**. See [Section 6.4.9, "EPR\\_Request Message"](#) for information about the use of the **EPR\_Request** Message.

A Source operating in **EPR Mode** that receives a **Request** Message **Shall** initiate a **Hard Reset**.

**Table 6.23 Fixed and Variable Request Data Object**

| Bit(s)   | Field                                        | Description                                                                                |
|----------|----------------------------------------------|--------------------------------------------------------------------------------------------|
| B31...28 | <b>Object Position</b>                       | Object position (0000b and 1110b...1111b are <b>Reserved</b> and <b>Shall Not</b> be used) |
| B27      | <b>Giveback</b>                              | GiveBack flag = 0 - <b>Deprecated</b> and <b>Shall</b> be set to zero.                     |
| B26      | <b>Capability Mismatch</b>                   | Set to '1' for a <b>Capabilities Mismatch</b>                                              |
| B25      | <b>USB Communications Capable</b>            | Set to '1' if <b>USB Communications Capable</b>                                            |
| B24      | <b>No USB Suspend</b>                        | Set to '1' if requesting <b>No USB Suspend</b>                                             |
| B23      | <b>Unchunked Extended Messages Supported</b> | Set to '1' if <b>Unchunked Extended Messages Supported</b>                                 |
| B22      | <b>EPR Capable</b>                           | Set to '1' if <b>EPR Capable</b>                                                           |
| B21...20 | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                             |
| B19...10 | <b>Operating Current</b>                     | Operating current in 10mA units                                                            |
| B9...0   | <b>Maximum Operating Current</b>             | Maximum Operating current 10mA units                                                       |

Table 6.24 Battery Request Data Object

| Bit(s)   | Field                                        | Description                                                                                |
|----------|----------------------------------------------|--------------------------------------------------------------------------------------------|
| B31...28 | <b>Object Position</b>                       | Object position (0000b and 1110b...1111b are <b>Reserved</b> and <b>Shall Not</b> be used) |
| B27      | <b>Giveback</b>                              | GiveBack flag = 0- <b>Deprecated</b> and <b>Shall</b> be set to zero.                      |
| B26      | <b>Capability Mismatch</b>                   | Set to '1' for a <b>Capabilities Mismatch</b>                                              |
| B25      | <b>USB Communications Capable</b>            | Set to '1' if <b>USB Communications Capable</b>                                            |
| B24      | <b>No USB Suspend</b>                        | Set to '1' if requesting No USB Suspend                                                    |
| B23      | <b>Unchunked Extended Messages Supported</b> | Set to '1' if <b>Unchunked Extended Messages Supported</b>                                 |
| B22      | <b>EPR Capable</b>                           | Set to '1' if <b>EPR Capable</b>                                                           |
| B21...20 | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                             |
| B19...10 | <b>Operating Power</b>                       | Operating Power in 250mW units                                                             |
| B9...0   | <b>Maximum Operating Power</b>               | Maximum Operating Power in 250mW units                                                     |

Table 6.25 PPS Request Data Object

| Bit(s)   | Field                                        | Description                                                                                |
|----------|----------------------------------------------|--------------------------------------------------------------------------------------------|
| B31...28 | <b>Object Position</b>                       | Object position (0000b and 1110b...1111b are <b>Reserved</b> and <b>Shall Not</b> be used) |
| B27      | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                             |
| B26      | <b>Capability Mismatch</b>                   | Set to '1' for a <b>Capabilities Mismatch</b>                                              |
| B25      | <b>USB Communications Capable</b>            | Set to '1' if <b>USB Communications Capable</b>                                            |
| B24      | <b>No USB Suspend</b>                        | Set to '1' if requesting No USB Suspend                                                    |
| B23      | <b>Unchunked Extended Messages Supported</b> | Set to '1' if <b>Unchunked Extended Messages Supported</b>                                 |
| B22      | <b>EPR Capable</b>                           | Set to '1' if <b>EPR Capable</b>                                                           |
| B21      | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                             |
| B20...9  | <b>Output Voltage</b>                        | Output voltage in 20mV units.                                                              |
| B8...7   | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                             |
| B6...0   | <b>Operating Current</b>                     | Operating current 50mA units.                                                              |

Table 6.26 AVS Request Data Object

| Bit(s)   | Field                                        | Description                                                                                                                            |
|----------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| B31...28 | <b>Object Position</b>                       | Object position (0000b and 1110b...1111b are <b>Reserved</b> and <b>Shall Not</b> be used)                                             |
| B27      | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                                                                         |
| B26      | <b>Capability Mismatch</b>                   | Set to '1' for a <b>Capabilities Mismatch</b>                                                                                          |
| B25      | <b>USB Communications Capable</b>            | Set to '1' if <b>USB Communications Capable</b>                                                                                        |
| B24      | <b>No USB Suspend</b>                        | Set to '1' if requesting No USB Suspend                                                                                                |
| B23      | <b>Unchunked Extended Messages Supported</b> | Set to '1' if <b>Unchunked Extended Messages Supported</b>                                                                             |
| B22      | <b>EPR Capable</b>                           | Set to '1' if <b>EPR Capable</b>                                                                                                       |
| B21      | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                                                                         |
| B20...9  | <b>Output Voltage</b>                        | Output voltage in 25mV units, the least two significant bits <b>Shall</b> be set to zero making the effective voltage step size 100mV. |
| B8...7   | <b>Reserved</b>                              | <b>Reserved</b> – <b>Shall</b> be set to zero.                                                                                         |
| B6...0   | <b>Operating Current</b>                     | Operating current 50mA units.                                                                                                          |



### 6.4.2.1 Object Position

The value in the **Object Position** field **Shall** indicate which object in the **Source\_Capabilities** Message or **EPR\_Source\_Capabilities** Message the **RDO** refers to. The value 0001b always indicates the 5V Fixed Supply **PDO** as it is the first object following the **Source\_Capabilities** Message's Message Header or **EPR\_Source\_Capabilities** Message's Extended Message Header. The number 0010b refers to the next **PDO** and so forth.

The **Object Position** field values 0001b...0111b **Shall** only be used to refer to **SPR (A)PDOs**. **SPR (A)PDOs** **May** be requested by either a **Request** or an **EPR\_Request** Message. Object positions 1000b...1011b **Shall** only be used to refer to **EPR (A)PDOs**. **EPR (A)PDOs** **Shall** only be requested by an **EPR\_Request** Message. If the **Object Position** field in a **Request** Message contains a value greater than 0111b, the Source **Shall** send **Hard Reset** Signaling.

### 6.4.2.2 GiveBack Flag (Deprecated)

The **Giveback** flag has been **Deprecated** and **Shall** be set to zero.

### 6.4.2.3 Capability Mismatch

A **Capabilities** Mismatch occurs when the Source cannot satisfy the Sink's power requirements based on the Source Capabilities it has offered. In this case the Sink **Shall** make a **Valid** request from the offered Source Capabilities and **Shall** set the **Capability Mismatch** bit (see [Section 8.2.5.2, "Power Capability Mismatch"](#)). When a Capabilities Mismatch condition does not exist, the Sink **Shall Not** set the **Capability Mismatch** bit.

When a Sink returns a Request Data Object with the **Capability Mismatch** bit set in response to a Source Capabilities Message, it indicates that it wants more power than the Source is currently offering. This can be due to either a specific **voltage** that is not being offered or there is not sufficient current for the **voltages** that are being offered.

Sources whose **Port Reported PDP** is less than their **Port Present PDP** (see [Section 6.4.11, "Source\\_Info Message"](#)) **Shall** respond to the Requests with the **Capability Mismatch** bit set as follows. The Source within **tCapabilitiesMismatchResponse** of the **PS\_RDY** Message **Shall** send a new Source Capabilities Message that offers either:

- 1) The set of Source Capabilities to minimally satisfy the Sink's requirements based on what it actually requires for full operation by evaluating the:
  - a) **Sink\_Capabilities\_Extended** Message (if supported by the Sink) and/or
  - b) **Sink\_Capabilities** or **EPR\_Sink\_Capabilities** Message.
- 2) The set of Source Capabilities the Source can supply at this time based on the **Port Present PDP**.

To prevent looping, Sources **Should Not** send a new Source Capabilities Message in response to subsequent Request Message with the **Capability Mismatch** flag set until its **Port Present PDP** changes.

Once a Guaranteed Capability Source that has responded to a **Capability Mismatch**, it **Shall Not** subsequently send out another Source Capabilities Message at a lower **PDP** unless the power required by the Sink (as indicated in its Sink Capabilities Message or **Sink\_Capabilities\_Extended** Message) has also been reduced. Sources wishing to manage their power **May** periodically check the Sink Capabilities Message or **Sink\_Capabilities\_Extended** Message to determine whether these have changed.

**Note:** A Source Capabilities Message refers to a **Source\_Capabilities** Message or an **EPR\_Source\_Capabilities** Message, and a Sink Capabilities Message refers to a **Sink\_Capabilities** Message or **EPR\_Sink\_Capabilities** Message, Request refers to a **Request** Message or **EPR\_Request** depending on operating mode.

In this context a **Valid** Request Message means the following:

- The **Object Position** field **Shall** contain a reference to an object that was present in the last received Source Capabilities Message.
- The **Operating Current/Operating Power** field **Shall** contain a value which is less than or equal to the maximum current/power offered by the selected (A)PDO the Source Capabilities Message.

#### 6.4.2.4 USB Communications Capable

\* The **USB Communications Capable** flag **Shall** be set to one when the *Sink* has USB data lines and is capable of communicating using either **[USB 2.0]**, **[USB 3.2]** or **[USB4]** protocols. The **USB Communications Capable** flag **Shall** be set to zero when the *Sink* does not have USB data lines or is otherwise incapable of communicating using either **[USB 2.0]**, **[USB 3.2]** or **[USB4]** protocols. This is used by the *Source* to determine operation in certain cases such as USB suspend. If the **USB Communications Capable** flag has been set to zero by a *Sink*, then the *Source* needs to be aware that USB Suspend rules cannot be observed by the *Sink*.

#### 6.4.2.5 No USB Suspend

The **No USB Suspend** flag **May** be set by the *Sink* to indicate to the *Source* that this device is requesting to continue its **Explicit Contract** during USB Suspend. *Sinks* setting this flag typically have functionality that can use power for purposes other than **USB Communication** e.g., for charging a *Battery*.

The *Source* uses this flag to evaluate whether it **Should** re-issue the **Source\_Capabilities** Message with the **USB Suspend Supported** flag cleared.

#### 6.4.2.6 Unchunked Extended Messages Supported

The **Unchunked Extended Messages Supported** bit **Shall** be set when the *Port* can send and receive **Extended Messages** with **Data Size** > **MaxExtendedMsgLegacyLen** bytes in a single, **Unchunked Extended Message**.

#### 6.4.2.7 EPR Mode Capable

The **EPR Capable** bit **Shall** indicate whether or not the *Sink* is capable of operating in **EPR Mode**. When the *Sink's* ability to operate in **EPR Mode** changes, it **Shall** send a new **Request** Message with the updated **EPR Capable** bit set in the **RDO**.

#### 6.4.2.8 Operating Current

The **Operating Current** field in the **Request Data Object** **Shall** be set to the highest current the *Sink* will draw during the **Explicit Contract**. A new **Request** Message or **EPR\_Request** Message, with an updated **Operating Current** value, **Shall** be issued whenever the *Sink's* power needs change.

The **Operating Current** field in the **SPR Programmable Request Data Object** is used in addition by the *Sink* to request the *Source* for the **Current Limit** level it needs. When the request is accepted the *Source's* output current supplied into any load **Shall** be less than or equal to the **Operating Current** value. When the *Sink* attempts to consume more current, the *Source* **Shall** reduce the output voltage so as not to exceed the **Operating Current** value.

The **Operating Current** field in the **AVS Request Data Object** **Shall** be set to the highest current the *Sink* will draw during the **Explicit Contract**.

**Note:** A *Source* in **AVS Mode**, unlike the *SPR Source* in **PPS Mode**, does not support current limit; the *Sink* is responsible not to take more current than it requested.

A new **Request** / **EPR\_Request** Message, with an updated **Operating Current** value, **Shall** be issued whenever the *Sink's* power needs change.

The value in the **Operating Current** field **Shall Not** exceed the value in the **Maximum Current** field of the **Source\_Capabilities** Message. For **EPR AVS**, the **Operating Current** field **Shall Not** exceed the **PDP** / Output voltage rounded down to the nearest 50 mA.

This field **Shall** apply to the **Fixed Supply**, **Variable Supply**, **Programmable** and **AVS RDOs**.

#### 6.4.2.9 Maximum Operating Current

The **Maximum Operating Current** field has been functionally **Deprecated**. In order to maintain backward compatibility with *Sources* that may try to interpret the **Maximum Operating Current** field in the **Request** Message or **EPR\_Request** Message, the field **Shall** be set equal to the value of the **Operating Current** field. To ensure backward compatibility, the *Source* **Should Ignore** this field.

This field **shall** apply to the *Fixed Supply* and *Variable Supply* RDO in *SPR Mode* and the *Fixed Supply* RDO in *EPR Capable*.

#### 6.4.2.10 Operating Power

The **Operating Power** field in the *Request Data Object* **shall** be set to the highest power the *Sink* will draw throughout the *Explicit Contract*.

This field **shall** apply to the *Battery Supply* RDO.

#### 6.4.2.11 Maximum Operating Power

The **Maximum Operating Power** field has been functionally **Deprecated**. In order to maintain backward compatibility with *Sources* that may try to interpret the **Maximum Operating Power** field in the *Request Message*, the field **shall** be set equal to the value of the **Operating Power** field. To ensure backward compatibility, the *Source* **should ignore** this field.

This field **shall** apply to the *Battery Supply* RDO.

#### 6.4.2.12 Output Voltage

The Output Voltage field in the *Programmable* and *AVS Request Data Objects* **shall** be set by the *Sink* to the voltage the *Sink* requires as measured at the *Source's* output connector. The Output Voltage field **shall** be greater than or equal to the Minimum Voltage field and less than or equal to the Maximum Voltage field in the *Programmable Power Supply* and *AVS APDOs*, respectively.

This field **shall** apply to the *Programmable RDO* and *AVS RDO*.

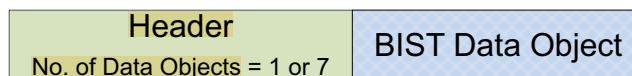
### 6.4.3 BIST Message

The **BIST Message** is sent to request the *Port* to enter a *PHY Layer* test mode (see [Section 5.9, "Built in Self-Test \(BIST\)"](#)) that performs one of the following functions:

- Enter a *Continuous BIST Mode* to send a continuous stream of test data to the *Tester*.
- Enter and leave a *Shared Capacity Group* test mode.

The *Message* format is as shown in [Figure 6.14, "BIST Message"](#).

Figure 6.14 BIST Message



All Ports **shall** be able to be a *Unit Under Test (UUT)* only when operating at *vSafe5V*. All of the following *BIST Modes* **shall** be supported:

- Process reception of a **BIST Carrier Mode** *BIST Data Object* that **shall** result in the generation of the appropriate carrier signal.
- Process reception of a **BIST Test Data** *BIST Data Object* that **shall** result in the *Message* being **Ignored**.

*UUTs* with Ports constituting a *Shared Capacity Group* (see [\[USB Type-C 2.4\]](#)) **shall** support the following *BIST Mode*:

- Process reception of a **BIST Shared Test Mode Entry** *BIST Data Object* that **shall** cause the *UUT* to enter *BIST Shared Capacity Test Mode*; a mode in which the *UUT* offers its full *Source Capabilities* on every *Port* in the *Shared Capacity Group*.

- Process reception of a **BIST Shared Test Mode Exit** BIST Data Object that **Shall** cause the UUT to exit the **BIST Shared Capacity Test Mode**.

When a Port receives a **BIST** Message BIST Data Object for a BIST Mode when not operating at **vSafe5V**, the **BIST** Message **Shall** be **Ignored**.

When a Port receives a **BIST** Message BIST Data Object for a BIST Mode it does not support the **BIST** Message **Shall** be **Ignored**.

When a Port or Cable Plug receives a **BIST** Message BIST Data Object for a Continuous BIST Mode the Port or Cable Plug enters the requested BIST Mode and **Shall** remain in that BIST Mode for **tBISTContMode** and then **Shall** return to normal operation (see [Section 6.6.7.2, "BISTContModeTimer"](#)).

The usage model of the PHY Layer BIST Modes generally assumes that some controlling agent will request a test of its Port Partner.

In [Section 8.3.2.15, "Built in Self-Test \(BIST\)"](#) there is a sequence description of the test sequences used for compliance testing.

The fields in the BIST Data Object are defined in the [Table 6.27, "BIST Data Object"](#).

**Table 6.27 BIST Data Object**

| Bit(s)   | Value         | Parameter                          | Description                                           | Reference                         | Applicability                           |
|----------|---------------|------------------------------------|-------------------------------------------------------|-----------------------------------|-----------------------------------------|
| B31...28 | 0000b...0100b | <b>Reserved</b>                    | <b>Shall Not</b> be used                              | <a href="#">Section 1.4.2</a>     | -                                       |
|          | 0101b         | <b>BIST Carrier Mode</b>           | Request Transmitter to enter BIST Carrier Mode        | <a href="#">Section 6.4.3.1</a>   | Mandatory                               |
|          | 0110b...0111b | <b>Reserved</b>                    | <b>Shall Not</b> be used                              | <a href="#">Section 1.4.2</a>     | -                                       |
|          | 1000b         | <b>BIST Test Data</b>              | Sends a Test Frame.                                   | <a href="#">Section 6.4.3.2</a>   | Mandatory                               |
|          | 1001b         | <b>BIST Shared Test Mode Entry</b> | Requests UUT to enter BIST Shared Capacity Test Mode. | <a href="#">Section 6.4.3.3.1</a> | Mandatory for UUTs with shared capacity |
|          | 1010b         | <b>BIST Shared Test Mode Exit</b>  | Requests UUT to exit BIST Shared Capacity Test Mode.  | <a href="#">Section 6.4.3.3.2</a> | Mandatory for UUTs with shared capacity |
|          | 1011b...1111b | <b>Reserved</b>                    | <b>Shall Not</b> be used                              | <a href="#">Section 1.4.2</a>     | -                                       |
| B27...0  |               | <b>Reserved</b>                    | <b>Shall</b> be set to zero.                          | <a href="#">Section 1.4.2</a>     | -                                       |

### 6.4.3.1 BIST Carrier Mode

Upon receipt of a **BIST** Message, with a **BIST Carrier Mode** BIST Data Object, the UUT **Shall** send out a continuous string of BMC encoded alternating "1"s and "0"s.

The UUT **Shall** exit the Continuous BIST Mode within **tBISTContMode** of this Continuous BIST Mode being enabled (see [Section 6.6.7.2, "BISTContModeTimer"](#)).

### 6.4.3.2 BIST Test Data Mode

Upon receipt of a **BIST** Message, with a **BIST Test Data** BIST Data Object, the UUT **Shall** return a **GoodCRC** Message and **Shall** enter **BIST Test Data Mode** in which it sends no further Messages except for **GoodCRC** Messages in response to received Messages. See [Section 5.9.2, "BIST Test Data Mode"](#) for the definition of the Test Frame.

The test **Shall** be ended by sending **Hard Reset Signaling** to reset the UUT.

### 6.4.3.3 BIST Shared Capacity Test Mode

A **Shared Capacity Group** of Ports share a common power source that is not capable of simultaneously powering all the ports to their full **Source Capabilities** (see [USB Type-C 2.4]). The **BIST Shared Capacity Test Mode** **Shall** only be implemented by ports in a **Shared Capacity Group**.

The **UUT Shared Capacity Group** of Ports **Shall** contain one or more Ports, designated as Master Ports, that recognize both the **BIST Shared Test Mode Entry** BIST Data Object and the **BIST Shared Test Mode Exit** BIST Data Object.

#### 6.4.3.3.1 BIST Shared Test Mode Entry

When any **master** Port in a **Shared Capacity Group** receives a **BIST** Message with a **BIST Shared Test Mode Entry** BIST Data Object, while in the **PE\_SRC\_Ready** State, the UUT **Shall** enter a compliance test mode where the maximum **Source Capabilities** are always offered on every **Port**, regardless of the availability of shared power i.e., all shared power management is disabled.

Ports in the **Shared Capacity Group** that are not Master Ports **Shall Not** enter compliance mode on receiving the **BIST Shared Test Mode Entry** BIST Data Object.

Upon receipt of a **BIST** Message, with a **BIST Shared Test Mode Entry** BIST Data Object, the UUT **Shall** return a **GoodCRC** Message and **Shall** enter the **BIST Shared Capacity Test Mode**.

On entering this mode, the UUT **Shall** send a new **Source\_Capabilities** Message from each Port in the **Shared Capacity Group** within **tBISTSharedTestMode**. The Tester will not exceed the shared capacity during this mode.

#### 6.4.3.3.2 BIST Shared Test Mode Exit

Upon receipt of a **BIST** Message, with a **BIST Shared Test Mode Exit** BIST Data Object, the UUT **Shall** return a **GoodCRC** Message and **Shall** exit the **BIST Shared Capacity Test Mode**. If any other Message, aside from a **BIST** Message, with a **BIST Shared Test Mode Exit** BIST Data Object, is received while in **BIST Shared Capacity Test Mode** this **Shall Not** cause the UUT to exit the **BIST Shared Capacity Test Mode**.

On exiting the mode, the UUT **May** send a new **Source\_Capabilities** Message to each Port in the **Shared Capacity Group** or the UUT **May** perform **ErrorRecovery** on each Port.

Ports in the **Shared Capacity Group** that are not Master Ports **Shall Not** exit compliance mode on receiving the **BIST Shared Test Mode Exit** BIST Data Object.

Ports in the **Shared Capacity Group** that are not Master Ports **Should Not** exit compliance mode on receiving the **BIST Shared Test Mode Exit** BIST Data Object.

- The UUT **Shall** exit **BIST Shared Capacity Test Mode** when It is powered off.
- The UUT **Shall** remain in **BIST Shared Capacity Test Mode** for any PD event (except when a **BIST Shared Test Mode Exit** BIST Data Object, is received); specifically the UUT **Shall** remain in **BIST Shared Capacity Test Mode** when any of the following PD events occurs:
  - Hard Reset
  - Cable Reset
  - Soft Reset
  - Data Role Swap
  - Power Role Swap
  - Fast Role Swap
  - VCONN Swap.
- The UUT **May** leave **BIST Shared Capacity Test Mode** if the Tester makes a request that exceeds the **Capabilities** of the UUT.



## 6.4.4 Vendor Defined Message

✖ The **Vendor\_Defined** Message (VDM) is provided to allow vendors to exchange information outside of that defined by this specification.

A **Vendor\_Defined** Message **Shall** consist of at least one *Vendor Data Object (VDO)*, the VDM Header, and **May** contain up to a maximum of six additional VDOs.

To ensure vendor uniqueness of **Vendor\_Defined** Messages, all **Vendor\_Defined** Messages **Shall** contain a **Valid** USB Standard or Vendor ID (SVID) allocated by USB-IF in the VDM Header.

Two types of **Vendor\_Defined** Messages are defined: *Structured VDMs* and *Unstructured VDMs*. A *Structured VDM* defines an extensible structure designed to support *Modal Operation*. An *Unstructured VDM* does not define any structure and Messages **May** be created in any manner that the vendor chooses.

**Vendor\_Defined** Messages **Shall Not** be used for direct power *Negotiation*. They **May** however be used to alter *Local Policy*, affecting what is offered or consumed via the normal PD Messages.

The Message format **Shall** be as shown in [Figure 6.15, "Vendor Defined Message"](#).

Figure 6.15 Vendor Defined Message



The VDM Header **Shall** be the first 4-byte object in a Vendor Defined Message. The VDM Header provides *Command* space to allow vendors to customize Messages for their own purposes. Additionally, vendors **May** make use of the *Commands* in a *Structured VDM*.

The fields in the VDM Header for an *Unstructured VDM*, when the **VDM Type** Bit is set to zero, **Shall** be as defined in [Table 6.28, "Unstructured VDM Header"](#). The fields in the VDM Header for a *Structured VDM*, when the **VDM Type** Bit is set to one **Shall** be as defined in [Table 6.29, "Structured VDM Header"](#).

Both *Unstructured VDMs* and *Structured VDMs* **Shall** only be sent and received after an *Explicit Contract* has been established. The only exception to this is the **Discover Identity** Command which **May** be sent by *Source* when a *Default Contract* or an *Implicit Contract* (in place after *Attach*, a *Power Role Swap* or *Fast Role Swap*) is in place in order to discover *Cable Capabilities* (see [Section 8.3.3.25.3, "Source Startup Structured VDM Discover Identity of a Cable Plug State Diagram"](#)).

### 6.4.4.1 Unstructured VDM

✖ The *Unstructured VDM* does not define the contents of bits B14...0 in the VDM Header. Their definition and use are the sole responsibility of the vendor indicated by the VID. The *Port Partners* and *Cable Plugs* **Shall** exit any states entered using an *Unstructured VDM* when a *Hard Reset* appears on PD.

The following rules apply to the use of *Unstructured VDM Messages*:

- *Unstructured VDMs* **Shall** only be used when an *Explicit Contract* is in place.
- Prior to establishing an *Explicit Contract* *Unstructured VDMs* **Shall Not** be sent and **Shall** be **Ignored** if received.
- Only the DFP **Shall** be an *Initiator* of *Unstructured VDMs*.
- Only the UFP or a *Cable Plug* **Shall** be a *Responder* to *Unstructured VDM*.
- *Unstructured VDMs* **Shall Not** be initiated or responded to under any other circumstances.



- *Unstructured VDMs* **Shall** only be used during *Modal Operation* in the context of an *Active Mode* i.e., only after the *UFP* has **Ack'd** the *Enter Mode* Command can *Unstructured VDMs* be sent or received. The *Active Mode* and the associated *Unstructured VDMs* **Shall** use the same *SVID*.
- *Unstructured VDMs* **May** be used with *SOP\* Packets*.
- When a *DFP* or *UFP* does not support *Unstructured VDMs* or does not recognize the *VID* it **Shall** return a *Not\_Supported* Message.

[Table 6.28, "Unstructured VDM Header"](#) illustrates the *VDM Header* bits.

**Table 6.28 Unstructured VDM Header**

| Bit(s)   | Parameter                | Description                                                           |
|----------|--------------------------|-----------------------------------------------------------------------|
| B31...16 | <i>Vendor ID (VID)</i>   | Unique 16-bit unsigned integer. Assigned by the USB-IF to the Vendor. |
| B15      | <i>VDM Type</i>          | 0 = <i>Unstructured VDM</i>                                           |
| B14...0  | Available for Vendor Use | Content of this field is defined by the vendor.                       |

#### 6.4.4.1.1 USB Vendor ID

The *Vendor ID (VID)* field **Shall** contain the 16-bit *Vendor ID* value assigned to the vendor by the USB-IF (*VID*). No other value **Shall** be present in this field.

#### 6.4.4.1.2 VDM Type

✖ The *VDM Type* field **Shall** be set to zero indicating that this is an *Unstructured VDM*.

#### 6.4.4.2 Structured VDM

✖ Setting the *VDM Type* field to 1 (*Structured VDM*) defines the use of bits B14...0 in the *Structured VDM Header*. The fields in the *Structured VDM Header* are defined in [Table 6.29, "Structured VDM Header"](#).

The following rules apply to the use of *Structured VDM Messages*:

- *Structured VDMs* **Shall** only be used when an *Explicit Contract* is in place with the following exception:
  - ✖ Prior to establishing the *First Explicit Contract*, a Source **May** issue *Discover Identity* Messages, to a Cable Plug using *SOP' Packets*, as an Initiator (see [Section 8.3.3.25.3, "Source Startup Structured VDM Discover Identity of a Cable Plug State Diagram"](#)).
- Either Port **May** be an Initiator of *Structured VDMs* except for the *Enter Mode* and *Exit Mode* Commands which **Shall** only be initiated by the *DFP*.
- A Cable Plug **Shall** only be a Responder to *Structured VDMs*.
- *Structured VDMs* **Shall Not** be initiated or responded to under any other circumstances.
- When a *DFP* or *UFP* does not support *Structured VDMs* any *Structured VDMs* received **Shall** return a *Not\_Supported* Message.
- When using any of the *SVID Specific Commands* in the *Structured VDM Header* (*VDM Header* b4...0 - value 16 - 31) the Responder **Shall NAK** Messages where the *SVID* in the *VDM Header* is not recognized as an *SVID* that uses *SVID Specific Commands* or the use of *SVID Specific Commands* is not supported for the *SVID*.
- ✖ When a Cable Plug does not support *Structured VDMs* any *Structured VDMs* received **Shall** be Ignored.

A DFP, UFP or Cable Plug which supports *Structured VDMs* and receiving a *Structured VDM* for a *SVID* that it does not recognize **Shall** reply with a **NAK** Command.

**Table 6.29 Structured VDM Header**

| Bit(s)              | Field                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...16            | Standard or Vendor ID (SVID)   | Unique 16-bit unsigned integer, assigned by the USB-IF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| B15                 | VDM Type                       | 1 = Structured VDM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| B14...13            | Structured VDM Version (Major) | Version Number (Major) of the Structured VDM (not this specification Version): <ul style="list-style-type: none"><li>Version 1.0 = 00b (Deprecated and Shall Not be used)</li><li>Version 2.x = 01b</li><li>Values 2-3 are Reserved and Shall Not be used</li></ul>                                                                                                                                                                                                                                                                                                  |
| B12...11            | Structured VDM Version (Minor) | For Commands 0...15 Version Number (Minor) of the Structured VDM <ul style="list-style-type: none"><li>Version 2.0 = 00b (Used for ports implemented prior to USB PD Revision 3.1, Version 1.6)</li><li>Version 2.1 = 01b (Used for ports implemented starting with USB PD Revision 3.1, Version 1.6)</li><li>All other Values are Reserved and Shall Not be used</li><li>SVID Specific Commands (16...31) defined by the SVID.</li></ul>                                                                                                                            |
| B10...8             | Object Position                | For the Enter Mode, Exit Mode, and Attention Commands (Requests/Responses): <ul style="list-style-type: none"><li>000b = Reserved and Shall Not be used.</li><li>001b...110b = Index into the list of VDOs to identify the desired Alternate Mode VDO</li><li>111b = Exit all Active Modes (equivalent of a power on reset). Shall only be used with the Command.</li></ul> Commands 0...3, 7...15: <ul style="list-style-type: none"><li>000b</li><li>001b...111b = Reserved and Shall Not be used.</li></ul> SVID Specific Commands (16...31) defined by the SVID. |
| B7...6              | Command Type                   | 00b = REQ (Request from Initiator Port)<br>01b = ACK (Acknowledge Response from Responder Port)<br>10b = NAK (Negative Acknowledge Response from Responder Port)<br>11b = BUSY (Busy Response from Responder Port)                                                                                                                                                                                                                                                                                                                                                   |
| B5                  | Reserved                       | Shall be set to zero and Shall be Ignored                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| B4...0 <sup>1</sup> | Command                        | 0 = Reserved and Shall Not be used.<br>1 = Discover Identity<br>2 = Discover SVIDs<br>3 = Discover Modes<br>4 = Enter Mode<br>5 = Exit Mode<br>6 = Attention<br>7-15 = Reserved and Shall Not be used.<br>16...31 = SVID Specific Commands                                                                                                                                                                                                                                                                                                                           |

1) In the case where a SID is used the modes are defined by a standard. When a VID is used the modes are defined by the Vendor.

[Section Table 6.30, "Structured VDM Commands"](#) shows the *Commands*, which *SVID* to use with each *Command* and the *SOP\** values which **Shall** be used.

**Table 6.30 Structured VDM Commands**

| Command                       | VDM Header SVID Field                     | SOP* used                                                |
|-------------------------------|-------------------------------------------|----------------------------------------------------------|
| <i>Discover Identity</i>      | <b>Shall</b> only use the <b>PD SID</b> . | <b>Shall</b> only use <i>SOP/SOP'</i> .                  |
| <i>Discover SVIDs</i>         | <b>Shall</b> only use the <b>PD SID</b> . | <b>Shall</b> only use <i>SOP/SOP'</i> .                  |
| <i>Discover Modes</i>         | <b>Valid</b> with any <i>SVID</i> .       | <b>Shall</b> only use <i>SOP/SOP'</i> .                  |
| <i>Enter Mode</i>             | <b>Valid</b> with any <i>SVID</i> .       | <b>Valid</b> with <i>SOP*</i> .                          |
| <i>Exit Mode</i>              | <b>Valid</b> with any <i>SVID</i> .       | <b>Valid</b> with <i>SOP*</i> .                          |
| <i>Attention</i>              | <b>Valid</b> with any <i>SVID</i> .       | <b>Valid</b> with <b>SOP*</b> .                          |
| <i>SVID Specific Commands</i> | <b>Valid</b> with any <i>SVID</i> .       | <b>Valid</b> with <i>SOP*</i> (defined by <i>SVID</i> ). |

#### 6.4.4.2.1 SVID

The **Standard or Vendor ID (SVID)** field **Shall** contain either a 16-bit USB Standard ID value (*SID*) or the 16-bit assigned to the vendor by the USB-IF (*VID*). No other value **Shall** be present in this field.

[Section Table 6.31, "SVID Values"](#) lists specific *SVID* values referenced by this specification.

**Table 6.31 SVID Values**

| Parameter       | Value  | Description                                            |
|-----------------|--------|--------------------------------------------------------|
| <b>PD SID</b>   | 0xFF00 | Standard ID allocated to this specification by USB-IF. |
| <b>DPTC SID</b> | 0xFF01 | Standard ID allocated to <b>[DPTC2.1]</b> by USB-IF.   |

#### 6.4.4.2.2 VDM Type

The **VDM Type** field **Shall** be set to one indicating that this is a *Structured VDM*.

#### 6.4.4.2.3 Structured VDM Version

The **Structured VDM Version (Major)/Structured VDM Version (Minor)** fields indicate the level of functionality supported in the *Structured VDM* part of the specification. This is not the same *Version* as the *Version* of this specification. The **Structured VDM Version (Major)** **Shall** be set to 01b to indicate Version 2.x with the **Structured VDM Version (Minor)** field set as appropriate based on whether the *Port* is implemented to USB PD Revision 3.1, Version 1.6 (or newer) or a prior *Version*.

To ensure interoperability with existing *PDUSB* products, *PDUSB* products **Shall** support every *Structured VDM Version* number starting from Version 1.0.

On receipt of a *VDM Header* with a higher *Version* number than it supports, a *Port* or *Cable Plug* **Shall** respond using the highest *Version* number it supports. On receipt of a *VDM Header* with a lower *Version* number than it supports, a *Port* or *Cable Plug* **Shall** respond using the same *Version* number it received.

The **Structured VDM Version (Major)/Structured VDM Version (Minor)** fields of the *Discover Identity* Command sent and received during the *Discovery Process* **Shall** be used to determine the lowest common *Structured VDM Version* supported by the *Port Partners* or *Cable Plug* and **Shall** continue to operate using this *Specification Revision* until they are *Detached*. After discovering the *Structured VDM Version*, the **Structured VDM Version (Major)/Structured VDM Version (Minor)** fields **Shall** match the agreed common *Structured VDM Version*.



The **Object Position** field **Shall** be used by the **Enter Mode** and **Exit Mode** Commands. The **Discover Modes** Command returns a list of zero to six VDOs, each of which describes an **Alternate Mode**. The value in **Object Position** field is an index into that list that indicates which VDO (e.g., **Alternate Mode**) in the list the **Enter Mode** and **Exit Mode** Command refers to. The **Object Position** **Shall** start with one for the first **Alternate Mode** in the list. If the SVID is a VID, the content of the VDO for the **Alternate Mode** **Shall** be defined by the vendor. If the **Standard or Vendor ID (SVID)** is a SID, the value **Shall** be assigned, by the USB-IF, to the given Standard. The VDO's content **May** be as simple as a numeric value or as complex as bit mapped description of **Capabilities** of the **Alternate Mode**. In all cases, the **Responder** is responsible for deciphering the contents to know whether or not it supports the **Alternate Mode** at the Object Position.

This field **Shall** be set to zero in the Request or Response (**REQ**, **ACK**, **NAK** or **BUSY**) when not required by the specification of the individual Command.

## 6.4.4.2.5 Command Type

### 6.4.4.2.5.1 Commands other than Attention

This **Command Type** field **Shall** be used to indicate the type of Command request/response being sent.

An Initiator **Shall** set the **Command Type** field to **REQ** to indicate that this is a Command request from an Initiator.

If Structured VDMs are supported, then the responses are as follows:

- "Responder **ACK**" is the normal return and **Shall** be sent to indicate that the Command request was received and handled normally.
- "Responder **NAK**" **Shall** be returned when the Command request:
  - Has an **Invalid** parameter (e.g., **Invalid SVID** or **Alternate Mode**).
  - Cannot be acted upon because the configuration is not correct (e.g., an **Alternate Mode** which has a dependency on another **Alternate Mode** or a request to exit an **Alternate Mode** which is not an **Active Mode**).
  - Is an Unrecognized Message.
  - The handling of "Responder **NAK**" is left up to the Initiator.
- "Responder **BUSY**" **Shall** be sent in the response to a VDM when the Responder is unable to respond to the Command request immediately, but the Command request **May** be retried. The Initiator **Shall** wait **tVDMBusy** after a "Responder **BUSY**" response is received before retrying the Command request.

### 6.4.4.2.5.2 Attention Command

This **Command Type** field **Shall** be used to indicate the type of Command request being sent. An Initiator **Shall** set the field to **REQ** to indicate that this is a Command request from an Initiator. If Structured VDMs are supported, then no response **Shall** be made to an **Attention** Command.

## 6.4.4.2.6 Command

### 6.4.4.2.6.1 Commands other than Attention

The **Command** field contains the value for the VDM Command being sent. The Commands explicitly listed in the **Command** field are used to identify devices and manage their operational Modes. There is a further range of Command values left for the vendor to use to manage additional extensions.

A Structured VDM Command consists of a Command request and a Command response (**ACK**, **NAK** or **BUSY**). A Structured VDM Command is deemed to be completed (and if applicable, the transition to the requested functionality is made) when the **GoodCRC** Message has been successfully received by the Responder in reply to its Command response.

If *Structured VDMs* are supported, but the *Structured VDM Command* request is an *Unrecognized Message*, it **Shall** be *NAKed* (see [Table 6.32, "Commands and Responses"](#)).

#### 6.4.4.2.6.2 Attention Command

The **Command** field contains the value for the *VDM Command* being sent (**Attention**). The **Attention Command** **May** be used by the *Initiator* to notify the *Responder* that it requires service.

A *Structured VDM Attention Command* consists of a *Command* request but no *Command* response. A *Structured VDM Attention Command* is deemed to be completed when the **GoodCRC** *Message* has been successfully received by the *Initiator* in reply to its **Attention Command** request.

If *Structured VDMs* are supported, but the *Structured VDM Attention Command* request is an *Unrecognized Message* it **Shall** be **Ignored** (see [Table 6.32, "Commands and Responses"](#)).

#### 6.4.4.3 Use of Commands

The *VDM Header* for a *Structured VDM Message* defines *Commands* used to retrieve a list of *SVIDs* the device supports, to discover the *Modes* associated with each *SVID*, and to enter/exit the *Modes*. The *Commands* include:

- **Discover Identity**
-  **Discover SVIDs**
-  **Discover Modes**
-  **Enter Mode**
-  **Exit Mode**
-  **Attention**

Additional *Command* space is also **Reserved** for Standard and Vendor use and for future extensions.

The *Command AMSS* use the terms *Initiator* and *Responder* to identify messaging roles the ports are taking on relative to each other. This role is independent of the *Port's* power capability (*Provider*, *Consumer* etc.) or its present **Power Role** (*Source* or *Sink*). The *Initiator* is the *Port* sending the initial *Command* request and the *Responder* is the *Port* replying with the *Command* response. See [Section 6.4.4.4, "Command Processes"](#).

All *Ports* that support *Modes* **Shall** support the **Discover Identity**, **Discover SVIDs**, the **Discover Modes**, the **Enter Mode** and **Exit Mode** *Commands*.

[Table 6.32, "Commands and Responses"](#) details the responses a *Responder* **May** issue to each *Command* request. Responses not listed for a given *Command* **Shall Not** be sent by a *Responder*. A **NAK** response **Should** be taken as an indication not to retry that particular *Command*.

Table 6.32 Commands and Responses

| Command                  | Allowed Response                                                                                          | Reference                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Discover Identity</b> | <b>ACK, NAK, BUSY</b>                                                                                     | <a href="#">Section 6.4.4.3.1</a>                                                                                     |
| <b>Discover SVIDs</b>    |  <b>ACK, NAK, BUSY</b> | <a href="#">Section 6.4.4.3.2</a>                                                                                     |
| <b>Discover Modes</b>    |  <b>ACK, NAK, BUSY</b> | <a href="#">Section 6.4.4.3.3</a>                                                                                     |
| <b>Enter Mode</b>        |  <b>ACK, NAK</b>       | <a href="#">Section 6.4.4.3.4</a>                                                                                     |
| <b>Exit Mode</b>         |  <b>ACK, NAK</b>       | <a href="#">Section 6.4.4.3.5</a>  |
| <b>Attention</b>         | None                                                                                                      | <a href="#">Section 6.4.4.3.6</a>  |

Examples of *Command* usage can be found in [Appendix C, "VDM Command Examples"](#).



The **Discover Identity** Command is provided to enable an *Initiator* to identify its *Port Partner* and for an *Initiator* (*VCONN Source*) to identify the *Responder* (*Cable Plug* or *VPD*). The **Discover Identity** Command is also used to determine whether a *Cable Plug* or *VPD* is PD-Capable by looking for a **GoodCRC** Message Response.

The **Discover Identity** Command **Shall** only be sent to *SOP* when there is an *Explicit Contract*.

The **Discover Identity** Command **Shall** be used to determine whether a given *Cable Plug* or *VPD* is PD Capable (see [Section 8.3.3.21.1, "Initiator Structured VDM Discover Identity State Diagram"](#) and [Section 8.3.3.25.3, "Source Startup Structured VDM Discover Identity of a Cable Plug State Diagram"](#)). In this case a **Discover Identity** Command request sent to *SOP* **Shall Not** cause a *Soft Reset* if a **GoodCRC** Message response is not returned since this can indicate a non-PD Capable cable or *VPD*.

**Note:** A *Cable Plug* or *VPD* will not be ready for PD Communication until *tVCONNStable* after *VCONN* has been applied (see [\[USB Type-C 2.4\]](#)).

During *Cable Plug* or *VPD* discovery, when there is an *Explicit Contract*, **Discover Identity** Commands are sent at a rate defined by the **DiscoverIdentityTimer** (see [Section 6.6.15, "DiscoverIdentityTimer"](#)) up to a maximum of **nDiscoverIdentityCount** times (see [Section 6.7.5, "Discover Identity Counter"](#)).

A PD-Capable *Cable Plug* or *VPD* **Shall** return a **Discover Identity** Command **ACK** in response to a **Discover Identity** Command request sent to *SOP*.

The **Discover Identity** Command **Shall** be used to determine the identity and/or **Capabilities** of the *Port Partner*. The following products **Shall** return a **Discover Identity** Command **ACK** in response to a **Discover Identity** Command request sent to *SOP*:

- A PD-Capable *UFP* that supports *Modal Operation*.
- A PD-Capable product that has multiple *DFPs*.
- A PD-Capable **[USB4]** product.

The *SVID* in the **Discover Identity** Command request **Shall** be set to the PD SID (see [Section Table 6.31, "SVID Values"](#)).

The **Number of Data Objects** field in the *Message Header* in the **Discover Identity** Command request **Shall** be set to 1 since the **Discover Identity** Command request **Shall Not** contain any *VDOs*.

The **Discover Identity** Command **ACK** sent back by the *Responder* **Shall** contain an *ID Header VDO*, a *Cert Stat VDO*, a *Product VDO* and the *Product Type VDOs* defined by the *Product Type* as shown in [Figure 6.16, "Discover Identity Command response"](#). This specification defines the following *Product Type VDOs*:

- *Passive Cable VDO* (see [Section 6.4.4.3.1.6, "Passive Cable VDO"](#))
- *Active Cable VDOs* (see [Section 6.4.4.3.1.7, "Active Cable VDOs"](#))
- *VCONN Powered USB Device (VPD) VDO* (see [Section 6.4.4.3.1.9, "VCONN Powered USB Device VDO"](#))
- *UFP VDO* (see [Section 6.4.4.3.1.4, "UFP VDO"](#))
- *DFP VDO* (see [Section 6.4.4.3.1.5, "DFP VDO"](#))

No *VDOs* other than those defined in this specification **Shall** be sent as part of the **Discover Identity** Command response. Where there is no *Product Type VDO* defined for a specific *Product Type*, no *VDOs* **Shall** be sent as part of the **Discover Identity** Command response. Any additional *VDOs* received by the *Initiator* **Shall** be **Ignored**.



**Figure 6.16 Discover Identity Command response**

| Header<br>No. of Data Objects = 4-7 <sup>1</sup> | VDM Header | ID Header VDO | Cert Stat VDO | Product VDO | 0..3 <sup>2</sup> Product Type VDO(s) |
|--------------------------------------------------|------------|---------------|---------------|-------------|---------------------------------------|
|--------------------------------------------------|------------|---------------|---------------|-------------|---------------------------------------|

1. Only Data objects defined in this specification can be sent as part of the *Discover Identity* Command.
2. The following sections define the number and content of the VDOs for each Product Type:

The **Number of Data Objects** field in the *Message Header* in the *Discover Identity* Command **NAK** and **BUSY** responses **Shall** be set to 1 since they **Shall Not** contain any VDOs.

If the product is a DRD both a *Product Type (UFP)* and a *Product Type (DFP)* are declared in the ID Header. These products **Shall** return *Product Type VDOs* for both *UFP* and *DFP* beginning with the *UFP VDO*, then by a 32-bit Pad Object (defined as all '0's'), followed by the *DFP VDO* as shown in [Figure 6.17, "Discover Identity Command response for a DRD"](#).

**Figure 6.17 Discover Identity Command response for a DRD**

| Header<br>No. of Data Objects = 7 | VDM Header | ID Header VDO | Cert Stat VDO | Product VDO | Product Type VDO(s) |     |     |
|-----------------------------------|------------|---------------|---------------|-------------|---------------------|-----|-----|
|                                   |            |               |               |             | UFP                 | Pad | DFP |

#### 6.4.4.3.1.1 ID Header VDO

The *ID Header VDO* contains information corresponding to the Power Delivery Product. The fields in the *ID Header VDO* **Shall** be as defined in [Section Table 6.33, "ID Header VDO"](#).

**Table 6.33 ID Header VDO**

| Bit(s)   | Description                                                                                                                                                                                                                                                                                                                                                                                                                      | Reference                             |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| B31      | <b>USB Communications Capable as USB Host</b> <ul style="list-style-type: none"> <li><b>Shall</b> be set to one if the product is capable of enumerating <i>USB Devices</i>.</li> <li><b>Shall</b> be set to zero otherwise.</li> </ul>                                                                                                                                                                                          | <a href="#">Section 6.4.4.3.1.1.1</a> |
| B30      | <b>USB Communications Capable as a USB Device</b> <ul style="list-style-type: none"> <li><b>Shall</b> be set to one if the product is capable of being enumerated as a <i>USB Device</i>.</li> <li><b>Shall</b> be set to zero otherwise</li> </ul>                                                                                                                                                                              | <a href="#">Section 6.4.4.3.1.1.2</a> |
| B29...27 | <b>SOP Product Type (UFP)</b> <ul style="list-style-type: none"> <li>000b – Not a <i>UFP</i></li> <li>001b – <i>PDUSB Hub</i></li> <li>010b – <i>PDUSB Peripheral</i></li> <li>011b – <i>PSD</i></li> <li>100b...111b – <b>Reserved, Shall Not</b> be used.</li> </ul>                                                                                                                                                           | <a href="#">Section 6.4.4.3.1.1.3</a> |
|          | <b>SOP' Product Type (Cable Plug/VPD)</b> <ul style="list-style-type: none"> <li>000b – Not a <i>Cable Plug/VPD</i></li> <li>001b...010b – <b>Reserved, Shall Not</b> be used.</li> <li>011b – <i>Passive Cable</i></li> <li>100b – <i>Active Cable</i></li> <li>101b – <b>Reserved, Shall Not</b> be used.</li> <li>110b – <i>VCONN Powered USB Device (VPD)</i></li> <li>111b – <b>Reserved, Shall Not</b> be used.</li> </ul> |                                       |

Table 6.33 ID Header VDO (Continued)

| Bit(s)           | Description                                                                                                                                                                                                                                                                                                  | Reference                                                                                                               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| B26 <sup>*</sup> | <b>Modal Operation Supported</b>                                                                                                                                                                                                                                                                             | <a href="#">Section 6.4.4.3.1.1.4</a>                                                                                   |
|                  | <ul style="list-style-type: none"> <li><b>Shall</b> be set to one if the product (<i>UFP/Cable Plug</i>) is capable of supporting <i>Modal Operation (Alternate Modes)</i>.</li> <li><b>Shall</b> be set to zero otherwise.</li> </ul>                                                                       |                                                                                                                         |
| B25...23         | <b>SOP - Product Type (DFP)</b> <ul style="list-style-type: none"> <li>000b – Not a <i>DFP</i></li> <li>001b – <i>PDUSB Hub</i></li> <li>010b – <i>PDUSB Host</i></li> <li>011b – Power Brick</li> <li>100b...111b – <b>Reserved, Shall Not</b> be used.</li> </ul> <b>SOP: Reserved, Shall Not</b> be used. | <a href="#">Section 6.4.4.3.1.1.6</a>                                                                                   |
| B22...21         | <b>Connector Type</b> <ul style="list-style-type: none"> <li>00b – <b>Reserved</b>, for compatibility with legacy systems.</li> <li>01b – <b>Reserved, Shall Not</b> be used.</li> <li>10b – <i>USB Type-C</i> Receptacle</li> <li>11b – <i>USB Type-C</i> Plug</li> </ul>                                   | <a href="#">Section 6.4.4.3.1.1.7</a>                                                                                   |
| B20...16         | <b>Reserved, Shall Not</b> be used.                                                                                                                                                                                                                                                                          |                                                                                                                         |
| B15...0          | <b>USB Vendor ID</b>                                                                                                                                                                                                                                                                                         | <a href="#">Section 6.4.4.3.1.1.8</a><br><a href="#">[USB 2.0]</a> / <a href="#">[USB 3.2]</a> / <a href="#">[USB4]</a> |

#### 6.4.4.3.1.1.1 USB Communications Capable as a USB Host

The **USB Communications Capable as a USB Host** field is used to indicate whether or not the *Port* has a *USB Host* Capability.

#### 6.4.4.3.1.1.2 USB Communications Capable as a USB Device

The **USB Communications Capable as a USB Device** field is used to indicate whether or not the *Port* has a *USB Device* Capability.

#### 6.4.4.3.1.1.3 Product Type (UFP)

The **SOP Product Type (UFP)** field indicates the type of Product when in *UFP Data Role*, whether a *VDO* will be returned and if so the type of *VDO* to be returned. The *Product Type* indicated in the **SOP Product Type (UFP)** field **Shall** be the closest categorization of the main functionality of the Product in *UFP Data Role* or "Undefined" when there is no suitable category for the product. For DRD Products this field **Shall** always indicate the *Product Type* when in *UFP* role regardless of the present *Data Role*. [Table 6.34, "Product Types \(UFP\)"](#) defines the *Product Type VDOs* which **Shall** be returned.

Table 6.34 Product Types (UFP)

| Product Type           | Description                                                                                    | Product Type VDO            | Reference                             |
|------------------------|------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------|
| Undefined              | <b>Shall</b> be used when this is not a <i>UFP</i> .                                           | None                        |                                       |
| <sup>*</sup> PDUSB Hub | <b>Shall</b> be used when the Product is a <i>PDUSB Hub</i> .                                  | <i>UFP VDO</i> <sup>*</sup> | <a href="#">Section 6.4.4.3.1.1.4</a> |
| PDUSB Peripheral       | <b>Shall</b> be used when the Product is a <i>PDUSB Device</i> other than a <i>PDUSB Hub</i> . | <i>UFP VDO</i>              | <a href="#">Section 6.4.4.3.1.1.4</a> |
| PSD                    | <b>Shall</b> be used when the Product is a PSD, e.g., power bank. <sup>*</sup>                 | None                        |                                       |

#### 6.4.4.3.1.1.4

#### Product Type (Cable Plug)

The **SOP' Product Type (Cable Plug/VPD)** field indicates the type of Product when the Product is a **Cable Plug** or **VPD**, whether a **VDO** will be returned and if so the type of **VDO** to be returned. [Table 6.35, "Product Types \(Cable Plug/VPD\)"](#) defines the **Product Type VDOs** which **Shall** be returned.

**Table 6.35 Product Types (Cable Plug/VPD)**

| Product Type             | Description                                                                                              | Product Type VDO  | Reference                           |
|--------------------------|----------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------|
| Undefined                | <b>Shall</b> be used where no other <b>Product Type</b> value is appropriate.                            | None              |                                     |
| Active Cable             | <b>Shall</b> be used when the Product is a cable that incorporates signal conditioning circuits.         | Active Cable VDO  | <a href="#">Section 6.4.4.3.1.7</a> |
| Passive Cable            | <b>Shall</b> be used when the Product is a cable that does not incorporate signal conditioning circuits. | Passive Cable VDO | <a href="#">Section 6.4.4.3.1.6</a> |
| VCONN Powered USB Device | <b>Shall</b> be used when the Product is a <b>PDUSB VCONN Powered USB Device</b> .                       | VPD VDO           | <a href="#">Section 6.4.4.3.1.9</a> |

#### 6.4.4.3.1.1.5

#### Modal Operation Supported

The **Modal Operation Supported** bit is used to indicate whether or not the Product (either a **Cable Plug** or a device that can operate in the **UFP** role) is capable of supporting Modes. The **Modal Operation Supported** bit does not describe a **DFP's Alternate Mode Controller** functionality.

A product that supports **Modal Operation** **Shall** respond to the **Discover SVIDs Command** with a list of **SVIDs** for all of the Modes it is capable of supporting whether or not those Modes can currently be entered.

#### 6.4.4.3.1.1.6

#### Product Type (DFP)

The **SOP - Product Type (DFP)** field indicates the type of Product when in **DFP Data Role**, whether a **VDO** will be returned and if so the type of **VDO** to be returned. The **Product Type** indicated in the **SOP - Product Type (DFP)** field **Shall** be the closest categorization of the main functionality of the Product in **DFP Data Role** or "Undefined" when there is no suitable category for the product. For **DRD** Products this field **Shall** always indicate the **Product Type** when in **DFP** role regardless of the present **Data Role**. [Table 6.36, "Product Types \(DFP\)"](#) defines the **Product Type VDOs** which **Shall** be returned.

In **SOP' Communication (Cable Plugs and VPDs)** this bit field is **Reserved** and **Shall** be set to zero.

**Table 6.36 Product Types (DFP)**

| Product Type | Description                                                                                                                                             | Product Type VDO | Reference                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------|
| Undefined    | <b>Shall</b> be used where no other <b>Product Type</b> value is appropriate.                                                                           | None             |                                     |
| PDUSB Hub    | <b>Shall</b> be used when the Product is a <b>PDUSB Hub</b> .                                                                                           | DFP VDO          | <a href="#">Section 6.4.4.3.1.7</a> |
| PDUSB Host   | <b>Shall</b> be used when the Product is a <b>PDUSB Host</b> or a <b>PDUSB host</b> that supports one or more <b>Alternate Modes</b> as an <b>AMC</b> . | DFP VDO          | <a href="#">Section 6.4.4.3.1.6</a> |
| Charger      | <b>Shall</b> be used when the Product is a <b>Charger</b> .                                                                                             | DFP VDO          | <a href="#">Section 6.4.4.3.1.9</a> |

#### 6.4.4.3.1.1.7

#### Connector Type Field

The **Connector Type** field (B22...21) **Shall** contain a value identifying it as either a **USB Type-C** receptacle or a **USB Type-C** plug.

#### 6.4.4.3.1.1.8

#### Vendor ID

Manufacturers **Shall** set the **USB Vendor ID** field to the value of the *Vendor ID* assigned to them by USB-IF. For *USB Devices* or *Hubs* which support *USB Communications* the **USB Vendor ID** field **Shall** be identical to the Vendor ID field defined in the product's *USB Device Descriptor* (see [\[USB 2.0\]](#) and [\[USB 3.2\]](#)).

#### 6.4.4.3.1.2

#### Cert Stat VDO

The *Cert Stat VDO* **Shall** contain the XID assigned by USB-IF to the product before certification in binary format. The fields in the *Cert Stat VDO* **Shall** be as defined in [Table 6.37, "Cert Stat VDO"](#).

Table 6.37 Cert Stat VDO

| Bit(s)  | Description                  | Reference          |
|---------|------------------------------|--------------------|
| B31...0 | 32-bit unsigned integer, XID | Assigned by USB-IF |

#### 6.4.4.3.1.3

#### Product VDO

The *Product VDO* contains identity information relating to the product. The fields in the *Product VDO* **Shall** be as defined in [Table 6.38, "Product VDO"](#).

Table 6.38 Product VDO

| Bit(s)   | Description                             | Reference                                             |
|----------|-----------------------------------------|-------------------------------------------------------|
| B31...16 | 16-bit unsigned integer, USB Product ID | <a href="#">[USB 2.0]</a> / <a href="#">[USB 3.2]</a> |
| B15...0  | 16-bit unsigned integer, bcdDevice      | <a href="#">[USB 2.0]</a> / <a href="#">[USB 3.2]</a> |

Manufacturers **Should** set the USB Product ID field to a unique value identifying the product and **Should** set the bcdDevice field to a version number relevant to the release version of the product.

#### 6.4.4.3.1.4

#### UFP VDO

The *UFP VDO* defined in this section **Shall** be returned by Ports capable of operating as a *UFP* including traditional USB peripherals, *USB Hub's* upstream Port and DRD capable host Ports. The *UFP VDO* defined in this section **Shall** be sent when the *Product Type* (UFP) field in the *ID Header VDO* is given as a PDUSB Peripheral or PDUSB Hub. [Table 6.39, "UFP VDO"](#) defines the *UFP VDO* that **Shall** be sent based on the *Product Type*.

A [\[USB4\]](#) UFP **Shall** support the Structured VDM **Discover Identity** Command.

Table 6.39 UFP VDO

| Bit(s)   | Description                    | Reference                                                                                                                                                                                 |                                                           |
|----------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| B31...29 | <b>UFP VDO Version</b>         | Version Number of the VDO (not this specification Version): <ul style="list-style-type: none"><li>Version 1.3 = 011b</li></ul> Values 100b...111b are <b>Reserved, Shall Not</b> be used. |                                                           |
| B28      | <b>Reserved</b>                | <b>Shall</b> be set to zero.                                                                                                                                                              |                                                           |
| B27...24 | <b>Device Capability</b>       | <b>Bit</b>                                                                                                                                                                                | <b>Description</b>                                        |
|          |                                | 0                                                                                                                                                                                         | <a href="#">[USB 2.0]</a> Device Capable                  |
|          |                                | 1                                                                                                                                                                                         | <a href="#">[USB 2.0]</a> Device Capable (Billboard only) |
|          |                                | 2                                                                                                                                                                                         | <a href="#">[USB 3.2]</a> Device Capable                  |
| B23...22 | <b>Connector Type (Legacy)</b> | 3                                                                                                                                                                                         | <a href="#">[USB4]</a> Device Capable                     |
|          |                                | <b>Deprecated, Shall</b> be set to 00b.                                                                                                                                                   |                                                           |
| B21...11 | <b>Reserved</b>                | <b>Shall</b> be set to zero.                                                                                                                                                              |                                                           |

Table 6.39 UFP VDO (Continued)

| Bit(s)  | Description                                                                                                  | Reference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
|---------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------|---|---------------------------------|---|--------------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------|
| B10...8 | VCONN Power                                                                                                  | <p>When the VCONN Required field is set to “Yes” the VCONN Power Field indicates the VCONN power needed by the AMA for full functionality:</p> <ul style="list-style-type: none"><li>• 000b = 1W</li><li>• 001b = 1.5W</li><li>• 010b = 2W</li><li>• 011b = 3W</li><li>• 100b = 4W</li><li>• 101b = 5W</li><li>• 110b = 6W</li></ul> <p>111b = Reserved, Shall Not be used.</p> <p>When the VCONN Required field is set to “No” the VCONN Power field is Reserved and Shall be set to zero.</p> |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| B7      | VCONN Required                                                                                               | <p>Indicates whether the AMA requires VCONN in order to function.</p> <ul style="list-style-type: none"><li>• 0 = No</li><li>• 1 = Yes</li></ul> <p>When the Alternate Modes field indicates no modes are supported, the VCONN Required field is Reserved and Shall be set to zero.</p>                                                                                                                                                                                                         |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| B6      | VBUS Required                                                                                                | <p>Indicates whether the AMA requires VBUS in order to function.</p> <ul style="list-style-type: none"><li>• 0 = Yes</li><li>• 1 = No</li></ul> <p>When the Alternate Modes field indicates no modes are supported, the VBUS Required field is Reserved and Shall be set to zero.</p>                                                                                                                                                                                                           |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| B5...3  | Alternate Modes                                                                                              | <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Supports [TBT3] Alternate Mode.</td></tr><tr><td>1</td><td>Supports Alternate Modes that reconfigure the signals on the [USB Type-C 2.4] connector – except for [TBT3].</td></tr><tr><td>2</td><td>Supports Alternate Modes that do not reconfigure the signals on the [USB Type-C 2.4] connector.</td></tr></table>                                                                                                          | Bit                             | Description | 0 | Supports [TBT3] Alternate Mode. | 1 | Supports Alternate Modes that reconfigure the signals on the [USB Type-C 2.4] connector – except for [TBT3]. | 2 | Supports Alternate Modes that do not reconfigure the signals on the [USB Type-C 2.4] connector. |
|         |                                                                                                              | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description                     |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
|         |                                                                                                              | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Supports [TBT3] Alternate Mode. |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| 1       | Supports Alternate Modes that reconfigure the signals on the [USB Type-C 2.4] connector – except for [TBT3]. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| 2       | Supports Alternate Modes that do not reconfigure the signals on the [USB Type-C 2.4] connector.              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
| B2...0  | USB Highest Speed                                                                                            | <ul style="list-style-type: none"><li>• 000b = [USB 2.0] only, no SuperSpeed support</li><li>• 001b = [USB 3.2] Gen1</li><li>• 010b = [USB 3.2]/[USB4] Gen2</li><li>• 011b = [USB4] Gen3</li><li>• 100b = [USB4] Gen4</li><li>• 101b...111b = Reserved and Shall be set to zero.</li></ul>                                                                                                                                                                                                      |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |
|         |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |             |   |                                 |   |                                                                                                              |   |                                                                                                 |

#### 6.4.4.3.1.4.1 VDO Version Field

The **UFP VDO Version** field contains a **VDO Version** for this **VDM Version** number. This field indicates the expected content for the **UFP VDOs**.

#### 6.4.4.3.1.4.2 Device Capability Field

The **Device Capability** bit-field describes the **UFP's Capabilities** when operating as either a **PDUSB Device** or **PDUSB Hub**.

The bits in the bit-field **Shall** be non-zero when the corresponding *USB Device* speed is supported and **Shall** be set to zero when the corresponding *USB Device* speed is not supported.

[USB 2.0] "Device capable" and "Device capable Billboard only" (bits 0 and 1) **Shall Not** be simultaneously set.

#### 6.4.4.3.1.4.3 Connector Type Field

The **Connector Type (Legacy)** field was previously used for the *UFP VDO's* Connector Type. **Shall** be set to 00b by the *Cable Plug* and **Shall** be **Ignored** by the receiver. The receiver can find this information in the **Connector Type** field in the *ID Header VDO* ([Section 6.4.4.3.1.1.7, "Connector Type Field"](#)).

#### 6.4.4.3.1.4.4 VCONN Power Field

When the **VCONN Required** field indicates that *VCONN* is required the **VCONN Power** field **Shall** indicate how much power an *AMA* needs in order to fully operate. When the **VCONN Required** field is set to "No" the **VCONN Power** field is **Reserved** and **Shall** be set to zero.

#### 6.4.4.3.1.4.5 VCONN Required Field

The **VCONN Required** field **Shall** indicate whether *VCONN* is needed for the *AMA* to operate. The **VCONN Required** field **Shall** only be used if the **Alternate Modes** field indicates that an *Alternate Mode* is supported. If no *Alternate Modes* are supported, this field is **Reserved** and **Shall** be set to zero.

#### 6.4.4.3.1.4.6 VBUS Required Field

The **VBUS Required** field **Shall** indicate whether *VBUS* is needed for the *AMA* to operate. The **VBUS** required field **Shall** only be used if the **Alternate Modes** field indicates that an *Alternate Mode* is supported. If no *Alternate Modes* are supported, this field is **Reserved** and **Shall** be set to zero.

#### 6.4.4.3.1.4.7 Alternate Modes Field

The **Alternate Modes** field **Shall** be used to identify all the types of *Alternate Modes*, if any, a device supports.

#### 6.4.4.3.1.4.8 USB Highest Speed Field

The **USB Highest Speed** field **Shall** indicate the *Port's* highest speed capability. The *DFP* **Shall** consider all values indicated in this field that are higher than the highest value that the *DFP* recognizes as being **Valid** and functionally compatible with the highest speed that the *DFP* supports.



### 6.4.4.3.1.5 DFP VDO

The **DFP VDO** **Shall** be returned by Ports capable of operating as a **DFP**; including those implemented by *Hosts*, *Hubs* and *Power Bricks*. The **DFP VDO** **Shall** be returned when the *Product Type (DFP)* field in the *ID Header VDO* is given as *Power Brick*, *PDUSB Host* or *PDUSB Hub*. [Table 6.40, "DFP VDO"](#) defines the **DFP VDO** that **Shall** be sent.

Table 6.40 DFP VDO

| Bit(s)   | Field                          | Description                                                                                                                                                                                        |                               |
|----------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| B31...29 | <b>DFP VDO Version</b>         | Version Number of the VDO (not this specification Version): <ul style="list-style-type: none"><li>Version 1.2 = 010b</li></ul> Values 011b...111b are <b>Reserved</b> and <b>Shall Not</b> be used |                               |
| B28...27 | <b>Reserved</b>                | <b>Shall</b> be set to zero.                                                                                                                                                                       |                               |
| B26...24 | <b>Host Capability</b>         | Bit                                                                                                                                                                                                | Description                   |
|          |                                | 0                                                                                                                                                                                                  | <b>[USB 2.0]</b> Host Capable |
|          |                                | 1                                                                                                                                                                                                  | <b>[USB 3.2]</b> Host Capable |
|          |                                | 2                                                                                                                                                                                                  | <b>[USB4]</b> Host Capable    |
| B23...22 | <b>Connector Type (Legacy)</b> | <b>Shall</b> be set to 00b.                                                                                                                                                                        |                               |
| B21...5  | <b>Reserved</b>                | <b>Shall</b> be set to zero.                                                                                                                                                                       |                               |
| B4...0   | <b>Port Number</b>             | Unique <b>Port</b> number to identify a specific <b>Port</b> on a multi- <b>Port</b> device.                                                                                                       |                               |

#### 6.4.4.3.1.5.1 VDO Version Field

The **DFP VDO Version** field **Shall** contain a **VDO Version** for this **VDM Version** number. This field indicates the expected content for the **DFP VDO**.

#### 6.4.4.3.1.5.2 Host Capability Field

The **Host Capability** bit-field **Shall** describe whether the **DFP** can operate as a **PDUSB Host** and the **DFP's Capabilities** when operating as a **PDUSB Host**.

Power Bricks and **PDUSB Hubs** **Shall** set the **Host Capability** bits to zero.

#### 6.4.4.3.1.5.3 Connector Type Field

The **Connector Type (Legacy)** field was previously used for the **UFP VDO's** Connector Type. **Shall** be set to 00b by the **Cable Plug** and **Shall** be **Ignored** by the receiver. The receiver can find this information in the **Connector Type** field in the **ID Header VDO** ([Section 6.4.4.3.1.1.7, "Connector Type Field"](#)).

#### 6.4.4.3.1.5.4 Port Number Field

The **Port Number** field **Shall** be a **Static** unique number that unambiguously identifies each **[USB Type-C 2.4]** **DFP**, including **DRPs**, on the device.

**Note:** This number is independent of the USB **Port** number.

#### 6.4.4.3.1.6

#### Passive Cable VDO

The *Passive Cable VDO* defined in this section **shall** be sent when the *Product Type* is given as *Passive Cable*. [Table 6.41, "Passive Cable VDO"](#) defines the *Cable VDO* which **shall** be sent.

A *Passive Cable* has a USB Plug on each end at least one of which is a *Cable Plug* supporting *SOP' Communication*. A *Passive Cable* **shall not** incorporate data bus signal conditioning circuits and hence has no concept of Super Speed Directionality. A *Passive Cable* **shall** include a *VBUS* wire and **shall** only respond to *SOP' Communication*. *Passive Cables* **shall** support the Structured VDM *Discover Identity* Command and **shall** return the *Passive Cable VDO* in a *Discover Identity* Command *ACK* as shown in [Table 6.41, "Passive Cable VDO"](#).

Table 6.41 Passive Cable VDO

| Bit(s)   | Field                                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...28 | <i>HW Version</i>                                                                              | 0000b...1111b assigned by the <i>VID</i> owner                                                                                                                                                                                                                                                                                                                                                                                                                                |
| B27...24 | <i>Firmware Version</i>                                                                        | 0000b...1111b assigned by the <i>VID</i> owner                                                                                                                                                                                                                                                                                                                                                                                                                                |
| B23...21 | <i>VDO Version</i>                                                                             | Version Number of the <i>VDO</i> (not this specification Version): <ul style="list-style-type: none"> <li>Version 1.0 = 000b</li> </ul> Values 001b...111b are <b>Reserved</b> and <b>shall not</b> be used.                                                                                                                                                                                                                                                                  |
| B20      | <b>Reserved</b>                                                                                | <b>shall</b> be set to zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| B19...18 | <i>USB Type-C plug to USB Type-C/Captive (Passive Cable)</i>                                   | <ul style="list-style-type: none"> <li>00b = <b>Reserved</b> and <b>shall not</b> be used</li> <li>01b = <b>Reserved</b> and <b>shall not</b> be used</li> <li>10b = <i>USB Type-C</i></li> <li>11b = <i>Captive</i></li> </ul>                                                                                                                                                                                                                                               |
| B17      | <i>EPR Capable (Passive Cable)</i>                                                             | <ul style="list-style-type: none"> <li>0b – Cable is not <i>EPR Capable</i></li> <li>1b = Cable is <i>EPR Capable</i></li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| B16...13 | <i>Cable Latency (Passive Cable)</i>                                                           | <ul style="list-style-type: none"> <li>0000b – <b>Reserved</b> and <b>shall not</b> be used</li> <li>0001b – &lt;10ns (~1m)</li> <li>0010b – 10ns to 20ns (~2m)</li> <li>0011b – 20ns to 30ns (~3m)</li> <li>0100b – 30ns to 40ns (~4m)</li> <li>0101b – 40ns to 50ns (~5m)</li> <li>0110b – 50ns to 60ns (~6m)</li> <li>0111b – 60ns to 70ns (~7m)</li> <li>1000b – &gt; 70ns (&gt;~7m)</li> </ul> <b>Note:</b> 1001b ....1111b <b>Reserved</b> and <b>shall not</b> be used |
| B12...11 | <i>Cable Termination Type (Passive Cable)</i>                                                  | <ul style="list-style-type: none"> <li>00b = <i>VCONN</i> not required. <i>Cable Plugs</i> that only support <i>Discover Identity</i> Commands <b>shall</b> set these bits to 00b.</li> <li>01b = <i>VCONN</i> required</li> <li>10b...11b = <b>Reserved</b> and <b>shall not</b> be used</li> </ul>                                                                                                                                                                          |
| B10...9  | <i>Maximum VBUS Voltage (Passive Cable)</i>                                                    | Maximum Cable <i>VBUS</i> Voltage <sup>2</sup> : <ul style="list-style-type: none"> <li>00b – 20V</li> <li>01b – 30V<sup>1</sup> (<b>Deprecated</b>)</li> <li>10b – 40V<sup>1</sup> (<b>Deprecated</b>)</li> <li>11b – 50V</li> </ul>                                                                                                                                                                                                                                         |
| B8...7   | <b>Reserved</b>                                                                                | <b>shall</b> be set to zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1)       | Values no longer allowed. When present the field <b>shall</b> be interpreted as if it was 00b. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2)       | <i>EPR Sinks</i> with a captive cable <b>shall</b> report 50V.                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Table 6.41 Passive Cable VDO (Continued)

| Bit(s) | Field                                                                                          | Description                                                                                                                                                                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B6...5 | <b>V<sub>BUS</sub> Current Handling Capability (Passive Cable)</b>                             | <ul style="list-style-type: none"> <li>00b = <b>Reserved</b> and <b>Shall Not</b> be used</li> <li>01b = 3A</li> <li>10b = 5A</li> <li>11b = <b>Reserved</b> and <b>Shall Not</b> be used</li> </ul>                                                                                                                               |
| B4...3 | <b>Reserved</b>                                                                                | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                       |
| B2...0 | <b>USB Highest Speed (Passive Cable)</b>                                                       | <ul style="list-style-type: none"> <li>000b = <b>[USB 2.0]</b> only, no SuperSpeed support</li> <li>001b = <b>[USB 3.2]</b> Gen1</li> <li>010b = <b>[USB 3.2]/[USB4]</b> Gen2</li> <li>011b = <b>[USB4]</b> Gen3</li> <li>100b = <b>[USB4]</b> Gen4</li> <li>101b...111b = <b>Reserved</b> and <b>Shall Not</b> be used</li> </ul> |
| 1)     | Values no longer allowed. When present the field <b>Shall</b> be interpreted as if it was 00b. |                                                                                                                                                                                                                                                                                                                                    |
| 2)     | <b>EPR Sinks</b> with a captive cable <b>Shall</b> report 50V.                                 |                                                                                                                                                                                                                                                                                                                                    |

#### 6.4.4.3.1.6.1 HW Version Field

The **HW Version** (B31...28) contains a HW **version** assigned by the **VID** owner.

#### 6.4.4.3.1.6.2 FW Version Field

The **Firmware Version** field (B27...24) contains a FW **version** assigned by the **VID** owner.

#### 6.4.4.3.1.6.3 VDO Version Field

The **VDO Version** field (B23...20) contains a **VDO Version** for this **VDM Version** number. This field indicates the expected content for this **VDO**.

#### 6.4.4.3.1.6.4 USB Type-C plug to USB Type-C/Captive Field

The **USB Type-C plug to USB Type-C/Captive (Passive Cable)** field (B19...18) **Shall** contain a value indicating whether the opposite end from the **USB Type-C** plug is another **USB Type-C** plug (i.e., a detachable Standard **USB Type-C** Cable Assembly) or is a Captive Cable Assembly.

#### 6.4.4.3.1.6.5 EPR Mode Capable

The **EPR Capable (Passive Cable)** bit is a **Static** bit which **Shall** only be set when the cable is specifically designed for safe operation when carrying up to 48 volts at 5 amps.

#### 6.4.4.3.1.6.6 Cable Latency Field

The **Cable Latency (Passive Cable)** field (B16...13) **Shall** contain a value corresponding to the signal latency through the cable which can be used as an approximation for its length.

#### 6.4.4.3.1.6.7 Cable Termination Type Field

The **Cable Termination Type (Passive Cable)** field (B12...11) **Shall** contain a value indicating whether the **Passive Cable** needs **VCONN** only initially in order to support the **Discover Identity Command**, after which it can be removed, or the **Passive Cable** needs **VCONN** to be continuously applied in order to power some feature of the **Cable Plug**.

#### 6.4.4.3.1.6.8 Maximum V<sub>BUS</sub> Voltage Field

The **Maximum V<sub>BUS</sub> Voltage (Passive Cable)** field (B10...9) **Shall** contain the maximum **voltage** that **Shall** be **Negotiated** using a **Fixed Supply** over the cable as part of an **Explicit Contract** where the maximum **voltage** that **Shall** be applied to the cable is **vSrcNew** max + **vSrcValid** max. For example, when the **Maximum V<sub>BUS</sub> Voltage (Passive Cable)** field is 20V, a **Fixed Supply** of 20V can be **Negotiated** as part of an **Explicit Contract** where the absolute maximum **voltage** that can be applied to the cable is 21.55V. Similarly, when the **Maximum V<sub>BUS</sub> Voltage (Passive**

**Cable**) field is 50V, a *Fixed Supply* of 48V can be *Negotiated* as part of an *Explicit Contract* where the absolute maximum voltage that can be applied to the cable is 50.9V. **Maximum VBUS Voltage (Passive Cable)** field values of 01b and 10b (formerly 30V and 40V) **Shall** be treated if they were 00b (20V).

#### 6.4.4.3.1.6.9 VBUS Current Handling Capability Field

The **VBUS Current Handling Capability (Passive Cable)** field (B6...5) **Shall** indicate whether the cable is capable of carrying 3A or 5A.

#### 6.4.4.3.1.6.10 USB Highest Speed Field

The **USB Highest Speed (Passive Cable)** field (B2...0) **Shall** indicate the highest rate the cable supports. The *DFP* **Shall** consider all values indicated in this field that are higher than the highest value that the *DFP* recognizes as being **Valid** and functionally compatible with the highest speed that the *DFP* supports.

An *Active Cable* has a USB Plug on each end at least one of which is a *Cable Plug* supporting *SOP' Communication*. An *Active Cable* **shall** incorporate data bus signal conditioning circuits and **may** have a concept of Super Speed Directionality on its Super Speed wires. An *Active Cable* **may** include a *V<sub>BUS</sub>* wire.

An *Active Cable*:

- **shall** respond to *SOP' Communication*.
- **may** respond to *SOP'' Communication*.
- **shall** support the Structured VDM *Discover Identity* Command.
- In the *Discover Identity* Command **ACK**:
  - **shall** set the *Product Type* in the *ID Header VDO* to *Active Cable*.
  - **shall** return the *Active Cable VDOs* defined in [Table 6.42, "Active Cable VDO1"](#) and [Table 6.43, "Active Cable VDO2"](#).

**Table 6.42 Active Cable VDO1**

| Bit(s)   | Field                                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...28 | <i>HW Version</i>                                                                              | 0000b...1111b assigned by the <i>VID</i> owner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| B27...24 | <i>Firmware Version</i>                                                                        | 0000b...1111b assigned by the <i>VID</i> owner                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| B23...21 | <i>VDO Version</i>                                                                             | Version Number of the <i>VDO</i> (not this specification <i>Version</i> ): <ul style="list-style-type: none"> <li>• Version 1.0 = 000b</li> </ul> Values 001b...111b are <b>Reserved</b> and <b>shall Not</b> be used.                                                                                                                                                                                                                                                                                                                                                                                                             |
| B20      | <b>Reserved</b>                                                                                | <b>shall</b> be set to zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| B19...18 | <i>USB Type-C plug to USB Type-C/Captive</i>                                                   | <ul style="list-style-type: none"> <li>• 00b = <b>Reserved</b> and <b>shall Not</b> be used</li> <li>• 01b = <b>Reserved</b> and <b>shall Not</b> be used</li> <li>• 10b = <i>USB Type-C</i></li> <li>• 11b = <i>Captive</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |
| B17      | <i>EPR Capable (Active Cable)</i>                                                              | <ul style="list-style-type: none"> <li>• 0b – Cable is not <i>EPR Capable</i></li> <li>• 1b = Cable is <i>EPR Capable</i></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| B16...13 | <i>Cable Latency</i>                                                                           | <ul style="list-style-type: none"> <li>• 0000b – <b>Reserved</b> and <b>shall Not</b> be used</li> <li>• 0001b – &lt;10ns (~1m)</li> <li>• 0010b – 10ns to 20ns (~2m)</li> <li>• 0011b – 20ns to 30ns (~3m)</li> <li>• 0100b – 30ns to 40ns (~4m)</li> <li>• 0101b – 40ns to 50ns (~5m)</li> <li>• 0110b – 50ns to 60ns (~6m)</li> <li>• 0111b – 60ns to 70ns (~7m)</li> <li>• 1000b – 1000ns (~100m)</li> <li>• 1001b – 2000ns (~200m)</li> <li>• 1010b – 3000ns (~300m)</li> <li>• 1001b ....1111b <b>Reserved</b> and <b>shall Not</b> be used</li> </ul> <b>Note:</b> Includes latency of electronics in <i>Active Cable</i> . |
| B12...11 | <i>Cable Termination Type (Active Cable)</i>                                                   | <ul style="list-style-type: none"> <li>• 00b...01b = <b>Reserved</b> and <b>shall Not</b> be used</li> <li>• 10b = One end Active, one end passive, <i>V<sub>CONN</sub></i> required</li> <li>• 11b = Both ends Active, <i>V<sub>CONN</sub></i> required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                |
| 1)       | Values no longer allowed. When present the field <b>shall</b> be interpreted as if it was 00b. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2)       | <i>EPR Sinks</i> with a captive cable <b>shall</b> report 50V.                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

Table 6.42 Active Cable VDO1 (Continued)

| Bit(s)                                                                                            | Field                                                                  | Description                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B10...9                                                                                           | <b>Maximum <math>V_{BUS}</math> Voltage (Active Cable)</b>             | Maximum Cable $V_{BUS}$ voltage <sup>2</sup> :<br><ul style="list-style-type: none"> <li>00b – 20V</li> <li>01b – 30V<sup>1</sup> (<i>Deprecated</i>)</li> <li>10b – 40V<sup>1</sup> (<i>Deprecated</i>)</li> <li>11b – 50V</li> </ul>                                                                                                                               |
| B8...7                                                                                            | <b>Reserved</b>                                                        | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                                                         |
| B8                                                                                                | <b>SBU Supported</b>                                                   | <ul style="list-style-type: none"> <li>0 = SBU connections supported</li> <li>1 = SBU connections are not supported</li> </ul>                                                                                                                                                                                                                                       |
| B7                                                                                                | <b>SBU Type</b>                                                        | When SBU Supported = 1 this bit <b>Shall</b> be <i>Ignored</i><br>When SBU Supported = 0: <ul style="list-style-type: none"> <li>0 = SBU is passive</li> <li>1 = SBU is active</li> </ul>                                                                                                                                                                            |
| B6...5                                                                                            | <b><math>V_{BUS}</math> Current Handling Capability (Active Cable)</b> | When <b><math>V_{BUS}</math> Through Cable</b> is “No”, this field <b>Shall</b> be <i>Ignored</i> .<br>When <b><math>V_{BUS}</math> Through Cable</b> is “Yes”: <ul style="list-style-type: none"> <li>00b = <b>Reserved</b> and <b>Shall Not</b> be used</li> <li>01b = 3A</li> <li>10b = 5A</li> <li>11b = <b>Reserved</b> and <b>Shall Not</b> be used</li> </ul> |
| B4                                                                                                | <b><math>V_{BUS}</math> Through Cable</b>                              | <ul style="list-style-type: none"> <li>0 = No</li> <li>1 = Yes</li> </ul>                                                                                                                                                                                                                                                                                            |
| B3                                                                                                | <b>SOP” Controller Present</b>                                         | <ul style="list-style-type: none"> <li>0 = No SOP” controller present</li> <li>1 = SOP” controller present</li> </ul>                                                                                                                                                                                                                                                |
| B2...0                                                                                            | <b>USB Highest Speed (Active Cable)</b>                                | <ul style="list-style-type: none"> <li>000b = <b>[USB 2.0]</b> only, no SuperSpeed support</li> <li>001b = <b>[USB 3.2]</b> Gen1</li> <li>010b = <b>[USB 3.2]/[USB4]</b> Gen2</li> <li>011b = <b>[USB4]</b> Gen3</li> <li>100b = <b>[USB4]</b> Gen4</li> <li>101b...111b = <b>Reserved</b> and <b>Shall Not</b> be used</li> </ul>                                   |
| 1) Values no longer allowed. When present the field <b>Shall</b> be interpreted as if it was 00b. |                                                                        |                                                                                                                                                                                                                                                                                                                                                                      |
| 2) <i>EPR Sinks</i> with a captive cable <b>Shall</b> report 50V.                                 |                                                                        |                                                                                                                                                                                                                                                                                                                                                                      |

Table 6.43 Active Cable VDO2

| Bit(s)   | Field                                | Description                                                                                                                                                                                                                                                             |
|----------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...24 | <b>Maximum Operating Temperature</b> | The maximum internal operating temperature in °C. It might or might not reflect the plug’s skin temperature.                                                                                                                                                            |
| B23...16 | <b>Shutdown Temperature</b>          | The temperature, in °C, at which the cable will go into thermal shutdown so as not to exceed the allowable plug skin temperature.                                                                                                                                       |
| B15      | <b>Reserved</b>                      | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                            |
| B14...12 | <b>U3/CLd Power</b>                  | <ul style="list-style-type: none"> <li>000b: &gt;10mW</li> <li>001b: 5-10mW</li> <li>010b: 1-5mW</li> <li>011b: 0.5-1mW</li> <li>100b: 0.2-0.5mW</li> <li>101b: 50-200μW</li> <li>110b: &lt;50μW</li> <li>111b: <b>Reserved</b> and <b>Shall Not</b> be used</li> </ul> |



Table 6.43 Active Cable VDO2 (Continued)

| Bit(s) | Field                                                                                                        | Description                                                                                                                                                                                                                                          |
|--------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B11    | <i>U3 to U0 transition mode</i>                                                                              | <ul style="list-style-type: none"> <li>0b: U3 to U0 direct</li> <li>1b: U3 to U0 through U3S</li> </ul>                                                                                                                                              |
| B10    | <i>Physical connection</i>  | <ul style="list-style-type: none"> <li>0b = Copper</li> <li>1b = Optical</li> </ul>                                                                                                                                                                  |
| B9     | <i>Active element</i>                                                                                        | <ul style="list-style-type: none"> <li>0b = Active Re-driver</li> <li>1b = Active Re-timer</li> </ul>                                                                                                                                                |
| B8     | <i>USB4 Supported</i>                                                                                        | <ul style="list-style-type: none"> <li>0b = <i>[USB4]</i> supported</li> <li>1b = <i>[USB4]</i> not supported</li> </ul>                                                                                                                             |
| B7...6 | <i>USB 2.0 Hub Hops Consumed</i>                                                                             | Number of <i>[USB 2.0]</i> 'hub hops' cable consumes.<br><b>Shall</b> be set to zero if USB 2.0 not supported.                                                                                                                                       |
| B5     | <i>USB 2.0 Supported</i>    | <ul style="list-style-type: none"> <li>0b = <i>[USB 2.0]</i> supported</li> <li>1b = <i>[USB 2.0]</i> not supported</li> </ul>                                                                                                                       |
| B4     | <i>USB 3.2 Supported</i>                                                                                     | <ul style="list-style-type: none"> <li>0b = <i>[USB 3.2]</i> SuperSpeed supported</li> <li>1b = <i>[USB 3.2]</i> SuperSpeed not supported</li> </ul>                                                                                                 |
| B3     | <i>USB Lanes Supported</i>  | <ul style="list-style-type: none"> <li>0b = One lane</li> <li>1b = Two lanes</li> </ul>                                                                                                                                                              |
| B2     | <i>Optically Isolated Active Cable</i>                                                                       | <ul style="list-style-type: none"> <li>0b = No</li> <li>1b = Yes</li> </ul>                                                                                                                                                                          |
| B1     | <i>USB4 Asymmetric Mode Supported</i>                                                                        | <ul style="list-style-type: none"> <li>0b = No</li> <li>1b = Yes</li> </ul> <b>Shall</b> be set to zero if asymmetry is not supported.                                                                                                               |
| B0     | <i>USB Gen</i>                                                                                               | <ul style="list-style-type: none"> <li>0b = Gen 1</li> <li> 1b = Gen 2 or higher</li> </ul> <b>Note:</b> See VDO1 USB Highest Speed for details of Gen supported. |

#### 6.4.4.3.1.7.1 HW Version Field

The **HW Version** field (B31...28) contains a HW version assigned by the VID owner.

#### 6.4.4.3.1.7.2 FW Version Field

The **Firmware Version** field (B27...24) contains a FW version assigned by the VID owner.

#### 6.4.4.3.1.7.3 VDO Version Field

The **VDO Version** field (B23...20) contains a VDO Version for this VDM Version number. This field indicates the expected content for the Active Cable VDOs.

#### 6.4.4.3.1.7.4 Connector Type Field

The **USB Type-C plug to USB Type-C/Captive** field (B19...18) **Shall** contain a value indicating whether the opposite end from the USB Type-C plug is another USB Type-C plug (i.e., a detachable Standard USB Type-C Cable Assembly) or is a Captive Cable Assembly.

#### 6.4.4.3.1.7.5 EPR Mode Capable

The **EPR Capable (Active Cable)** is a **Static** bit which **Shall** only be set when the cable is specifically designed for safe operation when carrying up to 48 volts at 5 amps.

#### 6.4.4.3.1.7.6 Cable Latency Field

The **Cable Latency** field (B16...13) **Shall** contain a value corresponding to the signal latency through the cable which can be used as an approximation for its length.

#### 6.4.4.3.1.7.7

#### Cable Termination Type Field

The **Cable Termination Type (Active Cable)** field (B12...11) **shall** contain a value corresponding to whether the *Active Cable* has one or two *Cable Plugs* requiring power from *VCONN*.

#### 6.4.4.3.1.7.8

#### Maximum VBUS Voltage Field

The **Maximum VBUS Voltage (Active Cable)** field (B10...9) **shall** contain the maximum voltage that **shall** be *Negotiated* as part of an *Explicit Contract* where the maximum voltage that **shall** be applied to the cable is *vSrcNew* max + *vSrcValid* max. When this field is set to 20V, the cable will safely carry a Programmable Power Supply *APDO* of 20V where the absolute maximum voltage that can be applied to the cable is 21.55V. Similarly, when the **Maximum VBUS Voltage (Active Cable)** field is 50V, a *Fixed Supply* of 48V can be *Negotiated* as part of an *Explicit Contract* where the absolute maximum voltage that can be applied to the cable is 50.9V. **Maximum VBUS Voltage (Active Cable)** field values of 01b and 10b (formerly 30V and 40V) **shall** be treated if they were 00b (20V).

#### 6.4.4.3.1.7.9

#### SBU Supported Field

The **SBU Supported** field (B8) **shall** indicate whether the cable supports the SBUs in the cable.

#### 6.4.4.3.1.7.10

#### SBU Type Field

The **SBU Type** field (B7) **shall** indicate whether the SBUs are passive or active (e.g., digital).

#### 6.4.4.3.1.7.11

#### VBUS Current Handling Capability Field

The **VBUS Current Handling Capability (Active Cable)** field (B6...5) **shall** indicate whether the cable is capable of carrying 3A or 5A. The **VBUS Current Handling Capability (Active Cable)** field **shall** only be **Valid** when the **VBUS Current Handling Capability (Active Cable)** field indicates an end-to-end *VBUS* wire.

#### 6.4.4.3.1.7.12

#### VBUS Through Cable Field

The **VBUS Through Cable** field (B4) **shall** indicate whether the cable contains an end-to-end *VBUS* wire.

#### 6.4.4.3.1.7.13

#### SOP" Controller Present Field

The **SOP" Controller Present** field (B3) **shall** indicate whether one of the *Cable Plugs* is capable of *SOP"* Communication in addition to the *Normative SOP'* Communication.

#### 6.4.4.3.1.7.14

#### USB Highest Speed Field

The **USB Highest Speed (Active Cable)** field (B2...0) **shall** indicate the highest rate the cable supports. The *DFP* **shall** consider all values indicated in this field that are higher than the highest value that the *DFP* recognizes as being **Valid** and functionally compatible with the highest speed that the *DFP* supports.

#### 6.4.4.3.1.7.15

#### Maximum Operating Temperature Field

**Maximum Operating Temperature** field (B31...24) **shall** report the maximum allowable operating temperature inside the plug in °C.

#### 6.4.4.3.1.7.16

#### Shutdown Temperature Field

**Shutdown Temperature** field (B23...16) **shall** indicate the temperature inside the plug, in °C, at which the plug will shut down its active signaling components. When this temperature is reached, it will be reported in the *Active Cable Status* Message through the Thermal Shutdown bit.

#### 6.4.4.3.1.7.17

#### U3/CLd Power Field

The **U3/CLd Power** field (B14...12) **shall** indicate the power the cable consumes while in *[USB 3.2]* U3 or *[USB4]* CLd.

#### 6.4.4.3.1.7.18

#### U3 to U0 Transition Mode Field

The **U3 to U0 transition mode** field (B11) **shall** indicate which U3 to U0 mode the cable supports. This does not include the power in U3S if supported.

#### 6.4.4.3.1.7.19

#### Physical Connection Field

The **Physical connection** field (B10) **Shall** indicate the cable's construction, whether the connection between the active elements is copper or optical.

#### 6.4.4.3.1.7.20

#### Active element Field

The **Active element** field (B9) **Shall** indicate the cable's active element, whether the active element is a re-timer or a re-driver.

#### 6.4.4.3.1.7.21

#### USB4<sup>+</sup> Supported Field

The **USB4<sup>+</sup> Supported** field (B8) **Shall** indicate whether or not the cable supports [USB4] operation.

#### 6.4.4.3.1.7.22

#### USB 2.0 Hub Hops Consumed field

The **USB 2.0 Hub Hops Consumed** field (B7...6) **Shall** indicate the number of USB 2.0 'hub hops' that are lost due to the transmission time of the cable.

#### 6.4.4.3.1.7.23

#### USB 2.0 Supported Field

The **USB 2.0 Supported** field (B5) **Shall** indicate whether or not the cable supports [USB 2.0] only signaling.

#### 6.4.4.3.1.7.24

#### USB 3.2 Supported Field

The **USB 3.2 Supported** field (B4) **Shall**, indicate whether or not the cable supports [USB 3.2] SuperSpeed signaling.

#### 6.4.4.3.1.7.25

#### USB Lanes Supported Field

The **USB Lanes Supported** field (B3) **Shall** indicate whether the cable supports one or two lanes of [USB 3.2] SuperSpeed signaling.

#### 6.4.4.3.1.7.26

#### Optically Isolated Active Cable Field

The **Optically Isolated Active Cable** field (B2) **Shall** indicate whether this cable is an optically isolated *Active Cable* or not (as defined in [USB Type-C 2.4]). Optically Isolated *Active Cables* **Shall** have a re-timer or linear re-driver (LRD) as the active element and do not support [USB 2.0] or carry *VBUS*.

#### 6.4.4.3.1.7.27

#### USB4<sup>+</sup> Asymmetric Mode Supported Field

The **USB4<sup>+</sup> Asymmetric Mode Supported** field (B1) **Shall** indicate that the *Active Cable* supports asymmetric mode as defined in [USB4] and [USB Type-C 2.4].

#### 6.4.4.3.1.7.28

#### USB Gen Field

The **USB Gen** field (B0) **Shall** indicate the signaling Gen the cable supports. Gen 1 **Shall** only be used by [USB 3.2] cables as indicated by the USB 3.2 Supported field. Gen 2 or higher **May** be used by either [USB 3.2] or [USB4] cables as indicated by their respective supported fields. When Gen 2 or higher is indicated the **USB Highest Speed (Active Cable)** field in VDO1 **Shall** indicate the actual Gen supported.

#### 6.4.4.3.1.8

#### Alternate Mode Adapter VDO

The *Alternate Mode Adapter (AMA) VDO* has been **Deprecated**. *PDUSB Devices* which support one or more *Alternate Modes* **Shall** set an appropriate *Product Type (UFP)*, and **Shall** set the **Modal Operation Supported** bit to '1'.

#### 6.4.4.3.1.9

#### VCONN Powered USB Device VDO

The *VCONN Powered USB Device* (VPD) VDO defined in this section **Shall** be sent when the *Product Type* is given as *VCONN Powered USB Device*. [Table 6.44, "VPD VDO"](#) defines the VPD VDO which **Shall** be sent.

**Table 6.44 VPD VDO**

| Bit(s)                                                                                            | Field                                 | Description                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...28                                                                                          | <b>HW Version</b>                     | 0000b...1111b assigned by the VID owner                                                                                                                                                                                                                                        |
| B27...24                                                                                          | <b>Firmware Version</b>               | 0000b...1111b assigned by the VID owner                                                                                                                                                                                                                                        |
| B23...21                                                                                          | <b>VDO Version</b>                    | Version Number of the VDO (not this specification Version): <ul style="list-style-type: none"> <li>Version 1.0 = 000b</li> <li>Values 001b...111b are <b>Reserved</b> and <b>Shall Not</b> be used.</li> </ul>                                                                 |
| B20...17                                                                                          | <b>Reserved</b>                       | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                   |
| B16...15                                                                                          | <b>Maximum VBUS Voltage</b>           | Maximum VPD VBUS Voltage: <ul style="list-style-type: none"> <li>00b – 20V</li> <li>01b – 30V<sup>1</sup> (<b>Deprecated</b>)</li> <li>10b – 40V<sup>1</sup> (<b>Deprecated</b>)</li> <li>11b – 50V<sup>1</sup> (<b>Deprecated</b>)</li> </ul>                                 |
| B14                                                                                               | <b>Charge Through Current Support</b> | Charge Through Current Support bit=1b: <ul style="list-style-type: none"> <li>0b - 3A capable.</li> <li>1b - 5A capable</li> </ul> Charge Through Current Support bit = 0b: <ul style="list-style-type: none"> <li><b>Reserved</b> and <b>Shall</b> be set to zero.</li> </ul> |
| B13                                                                                               | <b>Reserved</b>                       | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                   |
| B12...7                                                                                           | <b>VBUS Impedance</b>                 | Charge Through Current Support bit = 1b: VBUS impedance through the VPD in 2 mΩ increments. Values less than 10 mΩ are <b>Reserved</b> and <b>Shall Not</b> be used.<br>Charge Through Current Support bit = 0b: <b>Reserved</b> and <b>Shall</b> be set to zero.              |
| B6...1                                                                                            | <b>Ground Impedance</b>               | Charge Through Current Support bit = 1b: Ground impedance through the VPD in 1 mΩ increments. Values less than 10 mΩ are <b>Reserved</b> and <b>Shall Not</b> be used.<br>Charge Through Current Support bit = 0b: <b>Shall</b> be set to zero.                                |
| B0                                                                                                | <b>Charge Through Support</b>         | <ul style="list-style-type: none"> <li>1b – the VPD supports Charge Through</li> <li>0b – the VPD does not support Charge Through</li> </ul>                                                                                                                                   |
| 1) Values no longer allowed. When present the field <b>Shall</b> be interpreted as if it was 00b. |                                       |                                                                                                                                                                                                                                                                                |

#### 6.4.4.3.1.9.1

#### HW Version Field

The **HW Version** field (B31...28) contains a HW version assigned by the VID owner.

#### 6.4.4.3.1.9.2

#### FW Version Field

The **Firmware Version** field (B27...24) contains a FW version assigned by the VID owner.

#### 6.4.4.3.1.9.3

#### VDO Version Field

The **VDO Version** field (B23...20) contains a VDO Version for this VDM Version number. This field indicates the expected content for this VDO.

#### 6.4.4.3.1.9.4

#### Maximum VBUS Voltage Field

The **Maximum VBUS Voltage** field (B16...15) **Shall** contain the maximum voltage that a Sink **Shall Negotiate** through the VPD Charge Through Port as part of an Explicit Contract.

**Note:** The maximum voltage that will be applied to the cable is **vSrcNew** max + **vSrcValid** max. For example, when the **Maximum VBUS Voltage** field is 20V, a Fixed Supply of 20V can be **Negotiated** as part of an

*Explicit Contract* where the absolute maximum voltage that can be applied to the cable is 21.55V.

**Maximum VBUS Voltage** field values of 01b and 10b (formerly 30V and 40V) **Shall** be treated if they were 00b (20V).

#### 6.4.4.3.1.9.5 **VBUS Impedance Field**

The **VBUS Impedance** field (B12...7) **Shall** contain the impedance the *VPD* adds in series between the *Source* and the *Sink*. The *Sink* **Shall** take this value into account when requesting current so as to not to exceed the **VBUS IR Drop** limit of 0.5V between the *Source* and itself. If the *Sink* can tolerate a larger *IR Drop* on **VBUS** it **May** do so.

#### 6.4.4.3.1.9.6 **Ground Impedance Field**

The **Ground Impedance** field (B6...1) **Shall** contain the impedance the *VPD* adds in series between the *Source* and the *Sink*. The *Sink* **Shall** take this value into account when requesting current so as to not to exceed the **Ground IR Drop** limit of 0.25V between the *Source* and itself.

#### 6.4.4.3.1.9.7 **Charge Through Field**

The **Firmware Version** field (B0) **Shall** be set to 1b when the *VPD* supports *Charge Through* and 0b otherwise.

### 6.4.4.3.2 Discover SVIDs

The **Discover SVIDs** Command is used by an *Initiator* to determine the SVIDs for which a *Responder* has Modes. The **Discover SVIDs** Command is used in conjunction with the **Discover Modes** Command in the *Discovery Process* to determine which Modes a device supports. The list of SVIDs is always terminated with one or two 0x0000 SVIDs.

The SVID in the **Discover SVIDs** Command **Shall** be set to the PD SID (see "[Table 6.31, "SVID Values"](#)") by both the *Initiator* and the *Responder* for this Command.

The **Number of Data Objects** field in the *Message Header* in the **Discover SVIDs** Command request **Shall** be set to 1 since the **Discover SVIDs** Command request **Shall Not** contain any VDOs.

The **Discover SVIDs** Command **ACK** sent back by the *Responder **Shall** contain one or more SVIDs. The SVIDs are returned 2 per VDO (see "[Table 6.45, "Discover SVIDs Responder VDO"](#)"). If there are an odd number of supported SVIDs, the **Discover SVIDs** Command is returned ending with a SVID value of 0x0000 in the last part of the last VDO. If there are an even number of supported SVIDs, the **Discover SVIDs** Command is returned ending with an additional VDO containing two SVIDs with values of 0x0000. A *Responder* **Shall** only return SVIDs for which a **Discover Modes** Command request for that SVID will return at least one *Alternate Mode*.*

A *Responder* that does not support any SVIDs **Shall** return a **NAK**.

The **Number of Data Objects** field in the *Message Header* in the **Discover SVIDs** Command **NAK** and **BUSY** responses **Shall** be set to 1 since they **Shall Not** contain any VDOs.

If the *Responder* supports 12 or more SVIDs then the **Discover SVIDs** Command **Shall** be executed multiple times until a **Discover SVIDs** VDO is returned ending either with a SVID value of 0x0000 in the last part of the last VDO or with a VDO containing two SVIDs with values of 0x0000. Each **Discover SVID ACK** Message, other than the one containing the terminating 0x0000 SVID, **Shall** convey 12 SVIDs. The *Responder* **Shall** restart the list of SVIDs each time a **Discover Identity** Command request is received from the *Initiator*.

**Note:** Since a *Cable Plug* does not retry Messages if the **GoodCRC** Message from the *Initiator* becomes corrupted the *Cable Plug* will consider the **Discover SVIDs** Command **ACK** unsent and will send the same list of SVIDs again.

[Figure 6.18, "Example Discover SVIDs response with 3 SVIDs"](#) shows an example response to the **Discover SVIDs** Command request with two VDOs containing three SVIDs. [Figure 6.19, "Example Discover SVIDs response with 4 SVIDs"](#) shows an example response with two VDOs containing four SVIDs followed by an empty VDO to terminate the response. [Figure 6.20, "Example Discover SVIDs response with 12 SVIDs followed by an empty response"](#) shows an example response with six VDOs containing twelve SVIDs followed by an additional request that returns an empty VDO indicating there are no more SVIDs to return.

**Table 6.45 Discover SVIDs Responder VDO**

| Bit(s)   | Field    | Description                                                                                                                                         |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...16 | SVID n   | 16-bit unsigned integer, assigned by the USB-IF or 0x0000 if this is the last VDO and the <i>Responder</i> supports an even number of SVIDs.        |
| B15...0  | SVID n+1 | 16-bit unsigned integer, assigned by the USB-IF or 0x0000 if this is the last VDO and the <i>Responder</i> supports an odd or even number of SVIDs. |

**Figure 6.18 Example Discover SVIDs response with 3 SVIDs**

| Header<br>No. of Data Objects = 3 | VDM Header | VDO 1               |                    | VDO 2               |                    |
|-----------------------------------|------------|---------------------|--------------------|---------------------|--------------------|
|                                   |            | SVID 0<br>(B31..16) | SVID 1<br>(B15..0) | SVID 2<br>(B31..16) | 0x0000<br>(B15..0) |



**Figure 6.19 Example Discover SVIDs response with 4 SVIDs**

| Header<br>No. of Data Objects = 4 | VDM Header | VDO 1               |                    | VDO 2               |                    | VDO 3               |                    |
|-----------------------------------|------------|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|
|                                   |            | SVID 0<br>(B31..16) | SVID 1<br>(B15..0) | SVID 2<br>(B31..16) | SVID 3<br>(B15..0) | 0x0000<br>(B31..16) | 0x0000<br>(B15..0) |

**Figure 6.20 Example Discover SVIDs response with 12 SVIDs followed by an empty response**

| Header<br>No. of Data Objects = 7 | VDM Header | VDO 1               |                    | VDO 2               |                    | VDO 3               |                    | VDO 4               |                    | VDO 5               |                    | VDO 6                |                     |
|-----------------------------------|------------|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|---------------------|--------------------|----------------------|---------------------|
|                                   |            | SVID 0<br>(B31..16) | SVID 1<br>(B15..0) | SVID 2<br>(B31..16) | SVID 3<br>(B15..0) | SVID 4<br>(B31..16) | SVID 5<br>(B15..0) | SVID 6<br>(B31..16) | SVID 7<br>(B15..0) | SVID 8<br>(B31..16) | SVID 9<br>(B15..0) | SVID 10<br>(B31..16) | SVID 11<br>(B15..0) |

| Header<br>No. of Data Objects = 2 | VDM Header | VDO 1               |                    |
|-----------------------------------|------------|---------------------|--------------------|
|                                   |            | 0x0000<br>(B31..16) | 0x0000<br>(B15..0) |

### 6.4.4.3.3 Discover Modes

The **Discover Modes** Command is used by an *Initiator* to determine the Modes a *Responder* supports for a given *SVID*.

The *SVID* in the **Discover Modes** Command **Shall** be set to the *SVID* for which Modes are being requested by both the *Initiator* and the *Responder* for this Command.

The **Number of Data Objects** field in the *Message Header* in the **Discover Modes** Command request **Shall** be set to 1 since the **Discover Modes** Command request **Shall Not** contain any VDOs.

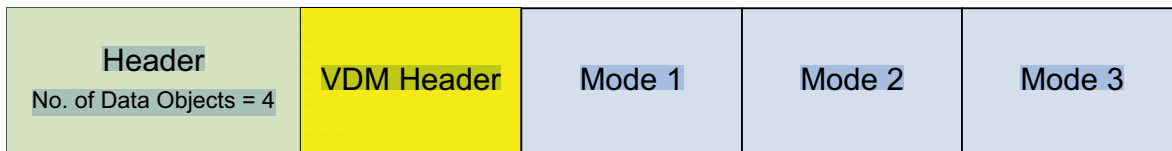
The **Discover Modes** Command **ACK** sent back by the *Responder **Shall** contain one or more Modes. The **Discover Modes** Command **ACK** **Shall** contain a *Message Header* with the **Number of Data Objects** field set to a value of 2 to 7 (the actual value is the number of *Alternate Mode* objects plus one). If the ID is a *VID*, the structure and content of the VDO is left to the Vendor. If the ID is a *SID*, the structure and content of the VDO is defined by the relevant standard's body.*

A *Responder* that does not support any Modes **Shall** return a **NAK**.

The **Number of Data Objects** field in the *Message Header* in the **Discover Modes** Command **NAK** and **BUSY** responses **Shall** be set to 1 since they **Shall Not** contain any VDOs.

Figure 6.21, "Example Discover Modes response for a given SVID with 3 Modes" shows an example of a **Discover Modes** Command response from a *Responder* which supports three Modes for a given *SVID*.

Figure 6.21 Example Discover Modes response for a given SVID with 3 Modes



### 6.4.4.3.4 Enter Mode Command

The **Enter Mode** Command is used by an *Initiator* (*DFP*) to command a *Responder* (*UFP* or *Cable Plug*) to enter a specified *Alternate Mode* of operation. Only a *DFP* **Shall** initiate the **Enter Mode** Process which it starts after it has successfully completed the *Discovery Process*.

The value in the Object Position field in the *VDM Header* **Shall** indicate to which *Alternate Mode* in the **Discover Modes** Command the VDO refers (see Figure 6.21, "Example Discover Modes response for a given SVID with 3 Modes"). The value 1 always indicates the first *Alternate Mode* as it is the first object following the *VDM Header*. The value 2 refers to the next *Alternate Mode* and so forth.

The **Number of Data Objects** field in the *Message Header* in the Command request **Shall** be set to either 1 or 2 since the **Enter Mode** Command request **Shall Not** contain more than 1 VDO. When a VDO is included in an **Enter Mode** Command request the contents of the 32-bit VDO is defined by the *Alternate Mode*.

The **Number of Data Objects** field in the Command response **Shall** be set to 1 since an **Enter Mode** Command response (**ACK**, **NAK**) **Shall Not** contain any VDOs.

Before entering a *Alternate Mode*, by sending the **Enter Mode** Command request that requires the reconfiguring of any pins on entry to that *Alternate Mode*, the *Initiator* **Shall** ensure that those pins being reconfigured are placed into the *USB Safe State*. Before entering an *Alternate Mode* that requires the reconfiguring of any pins, the *Responder* **Shall** ensure that those pins being reconfigured are placed into either *USB operation* or the *USB Safe State*.

A device **May** support multiple Modes with one or more active at any point in time. Any interactions between them are the responsibility of the Standard or Vendor. Where there are multiple *Active Modes* at the same time *Modal Operation* **Shall** start on entry to the first *Alternate Mode*.

On receiving an **Enter Mode** Command requests the *Responder* **shall** respond with either an **ACK** or a **NAK** response. The *Responder* is not allowed to return a **BUSY** response. The value in the Object Position field of the **Enter Mode** Command response **shall** contain the same value as the received **Enter Mode** Command request.

If the *Responder* responds to the **Enter Mode** Command request with an **ACK**, the *Responder* **shall** enter the **Alternate Mode** before sending the **ACK**. The *Initiator* **shall** enter the **Alternate Mode** on reception of the **ACK**. Successful transmission of the **Message** confirms to the *Responder* that the *Initiator* will enter an **Active Mode**.

See [Figure 8.111, "DFP to UFP Enter Mode"](#) for more details.

If the *Responder* responds to the **Enter Mode** Command request with a **NAK**, the **Alternate Mode** is not entered. If not presently in *Modal Operation* the *Initiator* **shall** return to USB operation. If not presently in *Modal Operation* the *Responder* **shall** remain in either USB operation or the **USB Safe State**.

If the *Initiator* fails to receive a response within **tVDMWaitModeEntry** it **shall Not** enter the **Alternate Mode** but return to USB operation.

[Figure 6.22, "Successful Enter Mode sequence"](#) shows the sequence of events during the transition between USB operation and entering an **Alternate Mode**. It illustrates when the *Responder's* **Alternate Mode** changes and when the *Initiator's* **Alternate Mode** changes. [Figure 6.23, "Unsuccessful Enter Mode sequence due to NAK"](#) illustrates that when the *Responder* returns a **NAK** the transition to an **Alternate Mode** do not take place and the *Responder* and *Initiator* remain in their default USB roles.

 **Figure 6.22 Successful Enter Mode sequence**

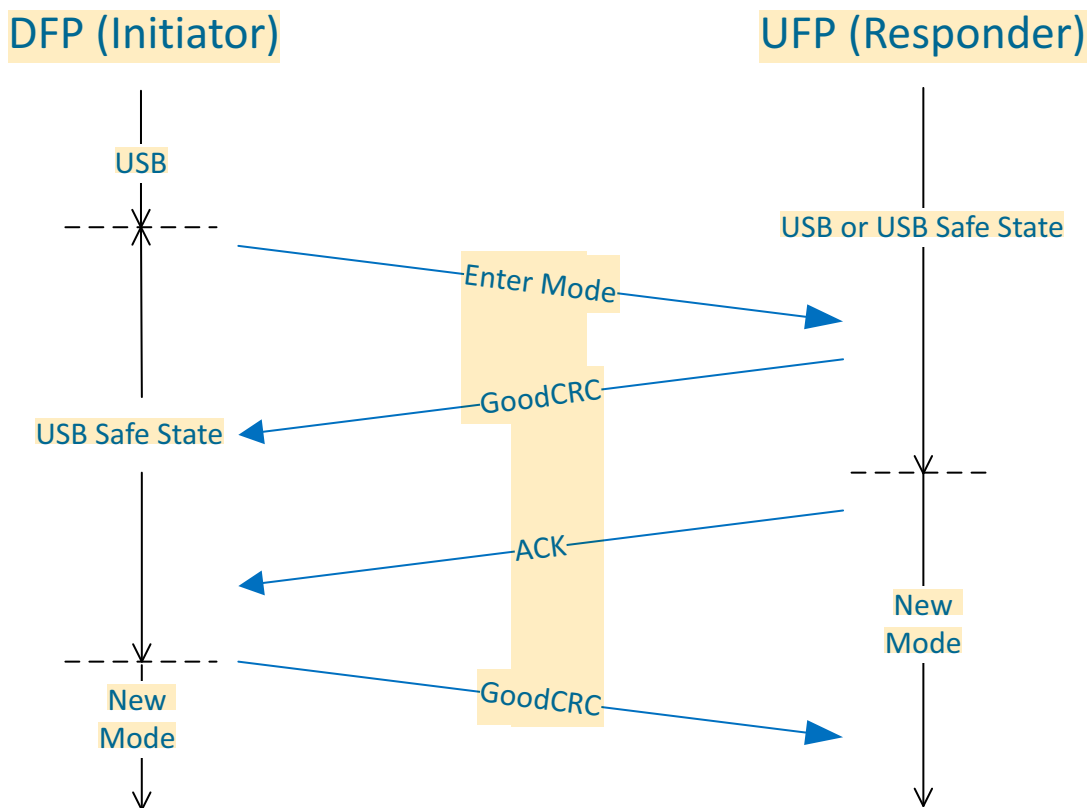
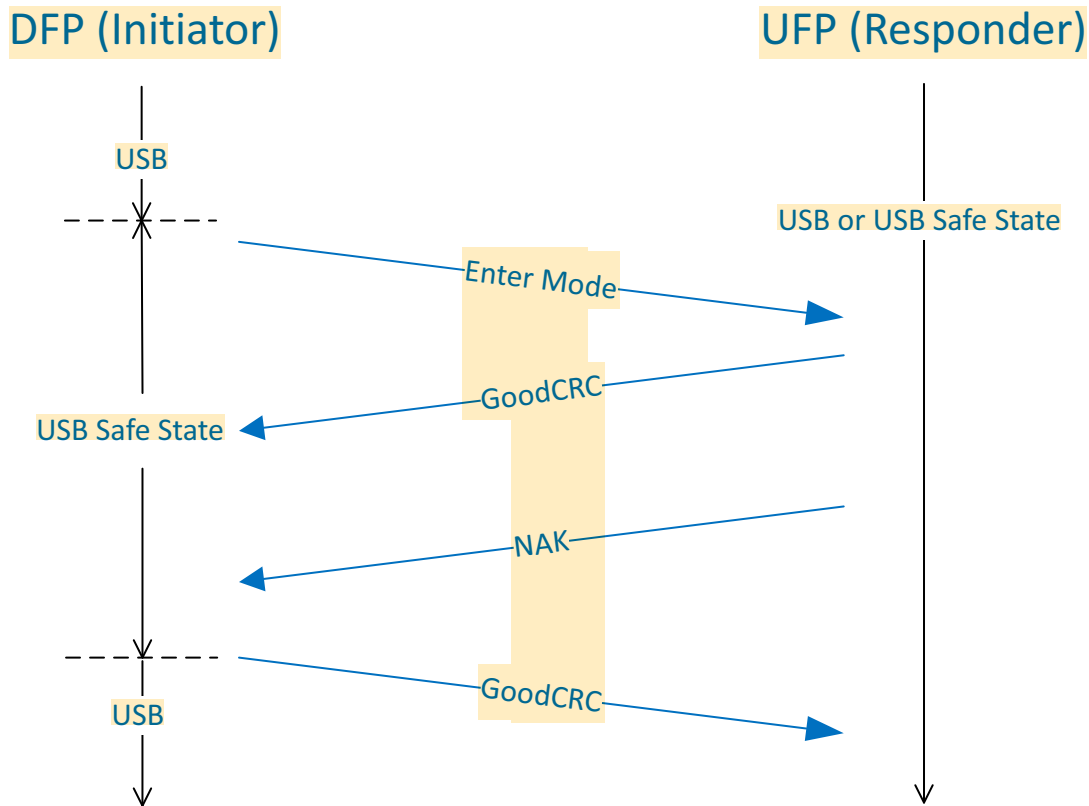


Figure 6.23 Unsuccessful Enter Mode sequence due to NAK



Once the *Alternate Mode* is entered, the device **shall** remain in that *Active Mode* until the *Exit Mode Command* is successful (see [Section 6.4.4.3.5, "Exit Mode Command"](#)).

The following events **shall** also cause the *Port Partners* and *Cable Plug(s)* to exit all *Active Modes*:

- A PD Hard Reset.
- [Error Recovery](#).
- The *Port Partners* or *Cable Plug(s)* are *Detached*.
- A *Cable Reset* (only exits the *Cable Plug's Active Modes*).
- A *Data Reset* (removing power briefly resets all the *Active Modes* in the *Cable Plug*).

The *Initiator shall* return to USB Operation within *tVDMExitMode* of a *disconnect*, of *Hard Reset Signaling* being detected or *Error Recovery*.

The *Responder shall* return to either USB operation or *USB Safe State* within *tVDMExitMode* of a *disconnect*, of *Hard Reset Signaling* being detected or *Error Recovery*.

A *DR\_Swap* Message **shall Not** be sent during *Modal Operation* between the *Port Partners* (see [Section 6.3.9, "DR\\_Swap Message"](#)).

#### 6.4.4.3.5 Exit Mode Command

The *Exit Mode Command* is used by an *Initiator (DFP)* to command a *Responder (UFP or Cable Plug)* to exit its *Active Mode* and return to normal USB operation. Only the *DFP shall* initiate the *Exit Mode Process*.

The value in the Object Position field **shall** indicate to which *Alternate Mode* in the *Discover Modes Command* the VDO refers (see [Figure 6.21, "Example Discover Modes response for a given SVID with 3 Modes"](#)) and **shall** have been used previously in an *Enter Mode Command* request for an *Active Mode*. The value 1 always indicates the first

**Alternate Mode** as it is the first object following the **VDM Header**. The value 2 refers to the next **Alternate Mode** and so forth. A value of 111b in the Object Position field **Shall** indicate that all **Active Modes** **Shall** be exited.

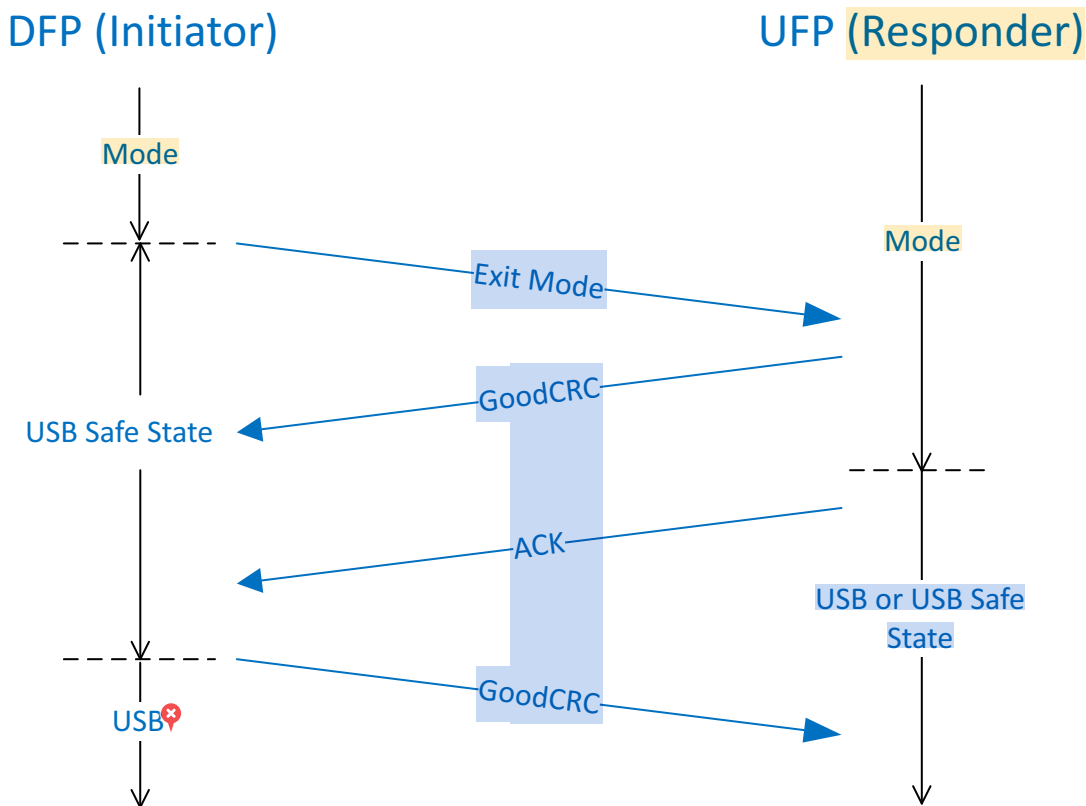
The **Number of Data Objects** field in both the **Command** request and **Command** response (**ACK**, **NAK**) **Shall** be set to 1 since an **Exit Mode** **Command** **Shall Not** contain any VDOs.

The **Responder** **Shall** exit its **Active Mode** before sending the response **Message**. The **Initiator** **Shall** exit its **Active Mode** when it receives the **ACK**. The **Responder** **Shall Not** return a **BUSY** acknowledgment and **Shall** only return a **NAK** acknowledgment to a request not containing an **Active Mode** (i.e., **Invalid** object position). An **Initiator** which fails to receive an **ACK** within **tVDMWaitModeExit** or receives a **NAK** or **BUSY** response **Shall** exit its **Active Mode**.

See [Figure 8.112, "DFP to UFP Exit Mode"](#) for more details.

[Figure 6.24, "Exit Mode sequence"](#) shows the sequence of events during the transition between exiting an **Active Mode** and USB operation. It illustrates when the **Responder's Alternate Mode** changes and when the **Initiator's Alternate Mode** changes.

**Figure 6.24 Exit Mode sequence**



#### 6.4.4.3.6 Attention

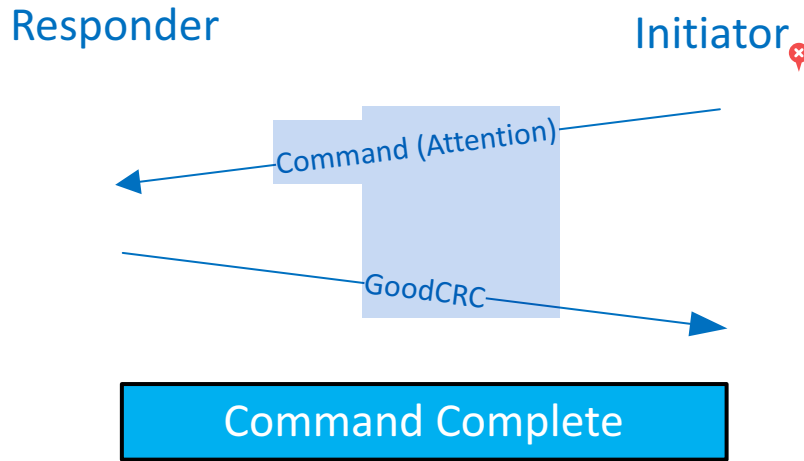
The **Attention** **Command** **May** be used by the **Initiator** to notify the **Responder** that it requires service.

The value in the Object Position field **Shall** indicate to which **Alternate Mode** in the **Discover Modes** **Command** the VDO refers (see [Figure 6.21, "Example Discover Modes response for a given SVID with 3 Modes"](#)) and **Shall** have been used previously in an **Enter Mode** **Command** request for an **Active Mode**. The value 1 always indicates the first **Alternate Mode** as it is the first object following the **VDM Header**. The value 2 refers to the next **Alternate Mode** and so forth. A value of 000b or 111b in the Object Position field **Shall Not** be used by the **Attention** **Command**.

The **Number of Data Objects** field in the *Message Header* **Shall** be set to 1 or 2 since the **Attention Command** **Shall Not** contain more than 1 VDO. When a VDO is included in an **Attention Command** the contents of the 32-bit VDO is defined by the *Alternate Mode*.

[Figure 6.24, "Exit Mode sequence"](#) shows the sequence of events when an **Attention Command** is received.

**Figure 6.25 Attention Command request/response sequence**

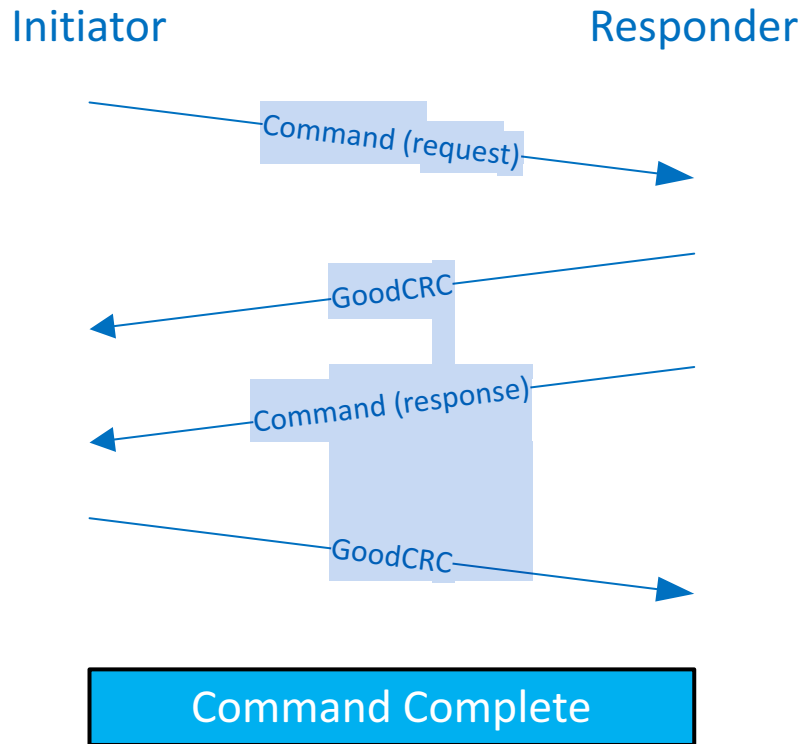


#### 6.4.4.4 Command Processes

The *Message flow of Commands* during a Process is a query followed by a response. Every *Command* request sent has to be responded to with a **GoodCRC** Message. The **GoodCRC** Message only indicates the *Command* request was received correctly; it does not mean that the *Responder* understood or even supports a particular *SVID*. [Figure 6.26, "Command request/response sequence"](#) shows the request/response sequence including the **GoodCRC** Messages.



Figure 6.26 Command request/response sequence



In order for the *Initiator* to know that the *Command* request was actually consumed, it needs an **acknowledgment** from the *Responder*. There are three responses that indicate the *Responder* received and processed the *Command* request:

- **ACK**
- **NAK**
- **BUSY**

The *Responder* **Shall** complete:

- **Enter Mode** requests within *tVDMEnterMode*.
- **Exit Mode** requests within *tVDMExitMode*.
- Other requests within *tVDMReceiverResponse*.

An *Initiator* not receiving a response within the following times **Shall** timeout and return to either the *PE\_SRC\_Ready* or *PE\_SNK\_Ready* state (as appropriate):

- **Enter Mode** requests within *tVDMWaitModeEntry*.
- **Exit Mode** requests within *tVDMWaitModeExit*.
- Other requests within *tVDMSenderResponse*.

The *Responder* **Shall** respond with:

- **ACK** if it recognizes the *SVID* and can process it at this time.
- **NAK**:
  - if it recognizes the *SVID* but cannot process the *Command* request

- or if it does not recognize the *SVID*
- or if it does not support the *Command*
- or if a VDO contains a field which is **Invalid**.
- **BUSY** if it recognizes the *SVID* and the *Command* but cannot process the *Command* request at this time.

The **ACK**, **NAK** or **BUSY** response **Shall** contain the same *SVID* as the *Command* request.

#### 6.4.4.4.1 Discovery Process

The *Initiator* (usually the *DFP*) always begins the *Discovery Process*. The *Discovery Process* has two phases. In the first phase, the **Discover SVIDs** *Command* request is sent by the *Initiator* to get the list of *SVIDs* the *Responder* supports. In the second phase, the *Initiator* sends a **Discover Modes** *Command* request for each *SVID* supported by both the *Initiator* and *Responder*.

#### 6.4.4.4.2 Enter Vendor Mode / Exit Vendor Mode Processes

The result of the *Discovery Process* is that both the *Initiator* and *Responder* identify the Modes they mutually support. The *Initiator* (*DFP*), upon finding a suitable *Alternate Mode*, uses the **Enter Mode** *Command* to enable the *Alternate Mode*.

The *Responder* (*UFP* or *Cable Plug*) and *Initiator* continue using the *Active Mode* until the *Active Mode* is exited.

In a managed termination, using the **Exit Mode** *Command*, the *Active Mode* **Shall** be exited in a controlled manner as described in [Section 6.4.4.3.5, "Exit Mode Command"](#).

In an unmanaged termination, triggered by:

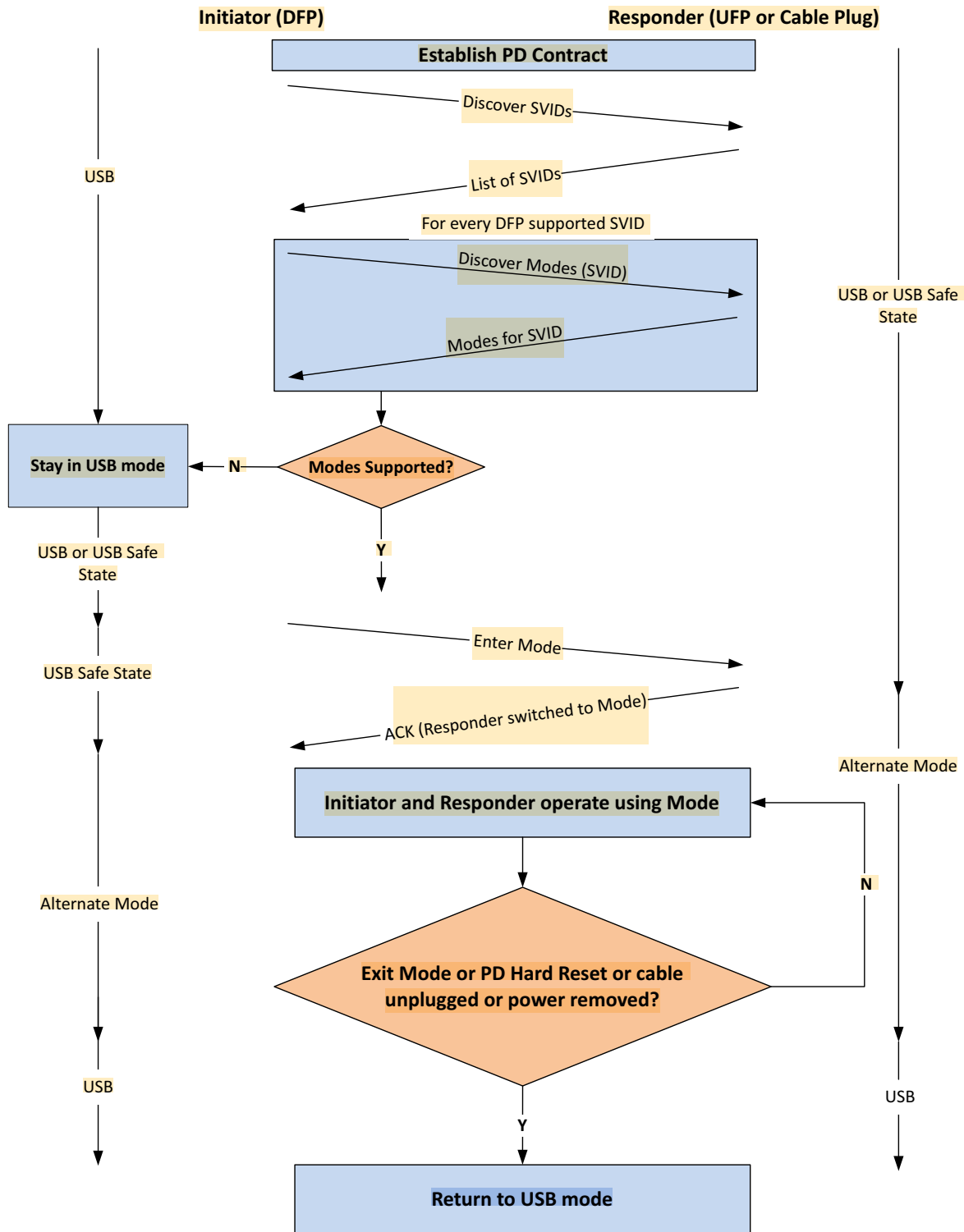
- A Power Delivery *Hard Reset* (i.e. **Hard Reset** *Signaling* sent by either *Port Partner*) or
- By cable *Detach* (device unplugged) or
- By **Error Recovery**

the *Active Mode* **Shall** still be exited but there **Shall Not** be a transition through the *USB Safe State*.

In both the managed and unmanaged terminations, the *Initiator* and *Responder* return to USB operation as defined in [\[USB Type-C 2.4\]](#) following an exit from an *Alternate Mode*.

The overall *Message* flow is illustrated in [Figure 6.27, "Enter/Exit Mode Process"](#).

### Figure 6.27 Enter/Exit Mode Process



#### 6.4.4.5 VDM Message Timing and Normal PD Messages

The timing and interspersing of *VDMs* between regular PD *Messages* **Shall** be done without perturbing the PD *AMSs*. This requirement **Shall** apply to both *Unstructured VDMs* and *Structured VDMs*.

The use of *Structured VDMs* by an *Initiator* **Shall Not** interfere with the normal *PD Message* timing requirements nor **Shall** either the *Initiator* or *Responder* interrupt a *PD AMS* (e.g., *Negotiation*, *Power Role Swap*, *Data Role Swap* etc.). The use of *Unstructured VDMs* **Shall Not** interfere with normal *PD Message* timing.

### 6.4.5 Battery\_Status Message

The *Battery\_Status* Message **Shall** be sent in response to a *Get\_Battery\_Status* Message. The *Battery\_Status* Message contains one Battery Status Data Object (BSDO) for one of the Batteries it supports as reported by *Number of Batteries/Battery Slots* field in the *Source\_Capabilities\_Extended* Message. The returned BSDO **Shall** correspond to the *Battery* requested in the *Battery\_Status\_Ref* field contained in the *Get\_Battery\_Status* Message.

The *Battery\_Status* Message returns a BSDO whose format **Shall** be as shown in *Figure 6.28, "Battery\_Status Message"* and *Table 6.46, "Battery Status Data Object (BSDO)"*. The *Number of Data Objects* field in the *Battery\_Status* Message **Shall** be set to 1.

Figure 6.28 Battery\_Status Message

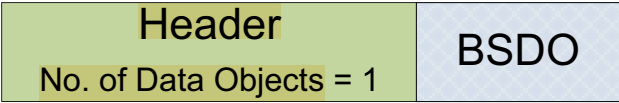


Table 6.46 Battery Status Data Object (BSDO)

| Bit(s)   | Field                    | Description                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31...16 | Battery Present Capacity | Battery's State of Charge (SoC) in 0.1 WH increments<br>Note: 0xFFFF = Battery's SOC unknown |                                                                                                                                                                                                                                                                                                                                                                                                               |
| B15...8  | Battery Info             | Bit                                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                   |
|          |                          | 0                                                                                            | Invalid Battery Reference<br>Invalid Battery reference                                                                                                                                                                                                                                                                                                                                                        |
|          |                          | 1                                                                                            | Battery Present<br>Battery is present when set                                                                                                                                                                                                                                                                                                                                                                |
|          |                          | 3...2                                                                                        | Battery Charging Status<br>When Battery Present is '1' Shall contain the Battery charging status: <ul style="list-style-type: none"><li>00b: Battery is Charging.</li><li>01b: Battery is Discharging.</li><li>10b: Battery is Idle.</li><li>11b: Reserved, Shall Not be used.</li></ul> When Battery Present is '0': <ul style="list-style-type: none"><li>11b...00b: Reserved, Shall Not be used.</li></ul> |
| B7...0   | Reserved                 | 7...4                                                                                        | Reserved, Shall Not be used.<br>Shall be set to zero                                                                                                                                                                                                                                                                                                                                                          |

#### 6.4.5.1 Battery Present Capacity

The *Battery Present Capacity* field **Shall** return either the *Battery's* State of Charge (SoC) in tenths of WH or indicate that the *Battery's* present State of Charge (SOC) is unknown.

## 6.4.5.2 Battery Info

✖ The **Battery Info** field **Shall** be used to report additional information about the **Battery's** present status. The **Battery Info** field's bits **Shall** reflect the present conditions under which the **Battery** is operating in the systems.

### 6.4.5.2.1 Invalid Battery Reference

The **Invalid Battery Reference** bit **Shall** be set when the **Get\_Battery\_Status** Message contains a reference to a **Battery** or **Battery Slot** (see [Section 6.5.1.13, "Number of Batteries/Battery Slots Field"](#)) that does not exist.

### 6.4.5.2.2 ⚠ Battery Present

✖ The **Battery Present** bit **Shall** be set whenever the **Battery** is present. It **Shall** always be set for Batteries that are not Hot Swappable Batteries. For Hot Swappable Batteries, the **Battery Present** bit **Shall** indicate whether the **Battery** is *Attached* or *Detached*.

### 6.4.5.2.3 Battery Charging Status

The **Battery Charging Status** bits indicate whether the **Battery** is being charged, discharged or is idle (neither charging nor discharging). These bits **Shall** be set when the **Battery Present** bit is set. Otherwise, when the **Battery Present** bit is zero the **Battery Charging Status** bits **Shall** also be zero.

## 6.4.6 Alert Message

The **Alert Message** is provided to allow *Port Partners* to inform each other when there is a status change event. Some of the events are critical such as *OCP*, *OVP* and *OTP*, while others are **informational** such as change in a *Battery's* status from charging to neither charging nor discharging.

The **Alert Message** **Shall** only be sent when the *Source* or *Sink* detects a status change.

The **Alert Message** **Shall** contain exactly one Alert Data Object (ADO) and the format **Shall** be as shown in [Figure 6.29, "Alert Message"](#) and [Table 6.47, "Alert Data Object \(ADO\)"](#).

Figure 6.29 Alert Message

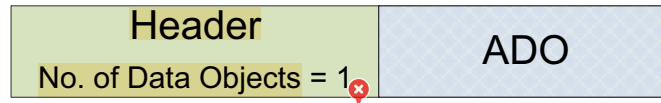


Table 6.47 Alert Data Object (ADO)

| Bit(s)   | Field                          | Description                                                                                                                                                                                    |                                                                                                                                |
|----------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|          |                                | Bit                                                                                                                                                                                            | Description                                                                                                                    |
| B31...24 | <b>Type of Alert</b>           | 0                                                                                                                                                                                              | <b>Reserved</b> and <b>Shall</b> be set to zero.                                                                               |
|          |                                | 1                                                                                                                                                                                              | <b>Battery Status Change Event</b><br>Battery Status Change Event ( <i>Attach/Detach/charging/discharging/idle</i> )           |
|          |                                | 2                                                                                                                                                                                              | <b>OCP Event</b><br>OCP event when set ( <i>Source</i> only, for <i>Sink</i> <b>Reserved</b> and <b>Shall</b> be set to zero). |
|          |                                | 3                                                                                                                                                                                              | <b>OTP Event</b><br>OTP event when set                                                                                         |
|          |                                | 4                                                                                                                                                                                              | <b>Operating Condition Change</b><br>Operating Condition Change when set                                                       |
|          |                                | 5                                                                                                                                                                                              | <b>Source Input Change Event</b><br>Source Input Change Event when set                                                         |
|          |                                | 6                                                                                                                                                                                              | <b>OVP Event</b><br>OVP event when set                                                                                         |
|          |                                | 7                                                                                                                                                                                              | <b>Extended Alert Event</b><br>Extended Alert Event when set                                                                   |
| B23...20 | <b>Fixed Batteries</b>         | When <b>Battery Status Change Event</b> bit set indicates which <i>Fixed Batteries</i> have had a status change. B20 corresponds to <i>Battery</i> 0 and B23 corresponds to <i>Battery</i> 3.  |                                                                                                                                |
| B19...16 | <b>Hot Swappable Batteries</b> | When <b>Battery Status Change Event</b> bit set indicates which Hot Swappable Batteries have had a status change. B16 corresponds to <i>Battery</i> 4 and B19 corresponds to <i>Battery</i> 7. |                                                                                                                                |



Table 6.47 Alert Data Object (ADO) (Continued)

| Bit(s)  | Field                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B15...4 | <b>Reserved</b>                  | <b>Shall</b> be set to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| B3...0  | <b>Extended Alert Event Type</b> | <p>When the <b>Extended Alert Event</b> bit in the <b>Type of Alert</b> field equals '1', then the <b>Extended Alert Event Type</b> field indicates the event which has occurred:</p> <ul style="list-style-type: none"> <li>0 = <b>Reserved</b>.</li> <li>1 = Power state change (<i>DFP</i> only)</li> <li>2 = Power button press (<i>UFP</i> only)</li> <li>3 = Power button release (<i>UFP</i> only)</li> <li>4 = Controller initiated wake e.g., Wake on LAN (<i>UFP</i> only)</li> <li>5-15 = <b>Reserved</b></li> </ul> <p>When the <b>Extended Alert Event</b> bit in the <b>Type of Alert</b> field equals '0', then the <b>Extended Alert Event Type</b> field is <b>Reserved</b> and <b>Shall</b> be set to zero.</p> |

### 6.4.6.1 Type of Alert

The **Type of Alert** field **Shall** be used to report *Source* or *Sink* status changes. Only one **Alert Message** **Shall** be generated for each Event or Change; however multiple Type of Alert bits **May** be set in one **Alert Message**. Once the **Alert Message** has been sent the **Type of Alert** field **Shall** be cleared.

A **Get\_Battery\_Status** Message **Should** be sent in response to a *Battery* status change in an **Alert Message** to get the details of the change.

A **Get\_Status** Message **Should** be sent in response to a non-*Battery* status change in an **Alert Message** from to get the details of the change.

#### 6.4.6.1.1 Battery Status Change

The **Battery Status Change Event** bit **Shall** be set when any *Battery's* power state changes between charging, discharging, neither. For Hot Swappable Batteries, it **Shall** also be set when a *Battery* is *Attached* or *Detached*.

#### 6.4.6.1.2 Over-Current Protection Event

The **OCPEvent** bit **Shall** be set when a *Source* detects its output current exceeds its limits triggering its protection circuitry. This bit is **Reserved** for a *Sink*.

#### 6.4.6.1.3 Over-Temperature Protection Event

The **OTPEvent** bit **Shall** be set when a *Source* or *Sink* shuts down due to over-temperature triggering its protection circuitry.

#### 6.4.6.1.4 Operating Condition Change

The **Operating Condition Change** bit **Shall** be set when a *Source* or *Sink* detects its Operating Condition enters or exits either the 'warning' or 'over temperature' temperature states.

The **Operating Condition Change** bit **Shall** be set when the *Source* operating in the Programmable Power Supply mode detects it has changed its operating condition between *Constant Voltage (CV)* and *Current Limit (CL)*.

#### 6.4.6.1.5 Source Input Change Event

The **Source Input Change Event** bit **Shall** be set when the *Source/Sink's* input changes. For example, when the AC input is removed, and the *Source/Sink* continues to be powered from one or more of its batteries or when AC returns and the *Source/Sink* transitions from *Battery* to AC operation or when the *Source/Sink* changes operation from one (or more) *Battery* to another (or more) *Battery*.

#### 6.4.6.1.6 Over-Voltage Protection Event

The **OVP Event** bit **Shall** be set when the *Sink* detects its output voltage exceeds its limits triggering its protection circuitry.

The **OVP Event** bit **May** be set when the *Source* detects its output voltage exceeds its limits triggering its protection circuitry.

#### 6.4.6.1.7 Extended Alert Event

The **Extended Alert Event** bit **Shall** be set when the event is defined as an Extended Alert Type.

#### 6.4.6.2 Fixed Batteries

The **Fixed Batteries** field indicates which *Fixed Batteries* have had a status change. B20 corresponds to *Battery 0* and B23 corresponds to *Battery 3*.

Once the **Alert Message** has been sent the **Fixed Batteries** field **Shall** be cleared.

#### 6.4.6.3 Hot Swappable Batteries

The **Hot Swappable Batteries** field indicates which Hot Swappable Batteries have had a status change. B16 corresponds to *Battery 0* and B19 corresponds to *Battery 3*.

Once the **Alert Message** has been sent the **Hot Swappable Batteries** field **Shall** be cleared.

#### 6.4.6.4 Extended Alert Event Types

The **Extended Alert Event Type** field provides extensions to the available types for the **Alert Message**. If the **Extended Alert Event Type** bit is not set, then the Extended Alert Event Type is **Reserved** and **Shall** be set to zero.

##### 6.4.6.4.1 Power State Change

The Power state change event value **May** be set when the *DFP* transitions into a new power state. The new power state **Shall** be communicated via the Power state change byte in the **Status Message**. This **Message Should** be sent by the host in response to any system power state change.

##### 6.4.6.4.2 Power Button Press

The Power button press event value **May** be set when the power button on the *UFP* is pressed. The press and release events are separated into two different events so that devices that respond differently to a long button press will see a long button press. On the host-side, the power button press event typically initiates the same behavior as a power button press of the **host's** power button.

##### 6.4.6.4.3 Power Button Release

If a Power button press event was sent, then the Power button release event value **Shall** be sent by the *UFP* following the Power button press event. If a physical power button press initiated the Power button press event, then the Power button release event **Should** be sent when the physical button is released.

##### 6.4.6.4.4 Controller Initiated Wake

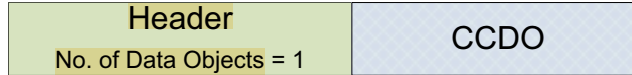
The Controller initiated wake is used to communicate a wake event from the *UFP* to the *DPF* such as Wake on **LAN** from a NIC or another controller. This event **doesn't** need the press/release form of the Power button press, because it only needs to communicate the presence of the event, and not the timing.

## 6.4.7 Get\_Country\_Info Message

The **Get\_Country\_Info** Message **Shall** be sent by a **Port** to get country specific information from its **Port Partner** using the country's Alpha-2 Country Code defined by **[ISO 3166]**. The **Port Partner** responds with a **Country\_Info** Message that contains the country specific information. The **Get\_Country\_Info** Message **Shall** be as shown in **Figure 6.30, "Get\_Country\_Info Message"** and **Table 6.48, "Country Code Data Object (CCDO)"**.

For example, if the request is for China information, then the Country Code Data Object (CCDO) would be CCDO [31:0] = 434E0000h for "CN" country code.

**Figure 6.30 Get\_Country\_Info Message**



**Table 6.48 Country Code Data Object (CCDO)**

| Bit(s)   | Description                                                               |
|----------|---------------------------------------------------------------------------|
| B31...24 | First character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b>  |
| B23...16 | Second character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b> |
| B15...0  | <b>Reserved, Shall</b> be set to zero.                                    |

## 6.4.8 Enter\_USB Message

The **Enter\_USB** Message **Shall** be sent by the *DFP* to its *UFP Port Partner* and to the *Cable Plug(s)* of an *Active Cable*, when in an *Explicit Contract*, to enter a specified *USB Mode* of operation. The recipient of the Message **Shall** respond by sending an **Accept** Message, a **Wait** Message or a **Reject** Message (see [Section 6.9, "Accept, Reject and Wait"](#)).

When entering **[USB4]** operation, the **Enter\_USB** Message **Shall** be sent by a **[USB4]** *PDUSB Hub's DFP(s)* or **[USB4]** *PDUSB Host's DFP(s)* within **tEnterUSB**:

- following a PD Connection.
- after a *Data Reset* to enter **[USB4]** operation is completed.
- after a *Data Role Swap* is completed.

The **Enter\_USB** Message **May** be sent by a *PDUSB Hub's DFP(s)* or *PDUSB Host's DFP(s)* within **tEnterUSB** following a PD Connection or after a *Data Reset* to enter **[USB 3.2]** or **[USB 2.0]** operation.

The **Enter\_USB** Message **Shall** be used by a *PDUSB Hub's DFP(s)* to speculatively train the USB links or enter **[DPTC2.1]** or **[TBT3]** *Alternate Modes* prior to the presence of a host. In this case, the **Host Present** bit **Shall** be cleared. When the Host is *Connected* the **Enter\_USB** Message **Shall** be resent with the **Host Present** bit set. The **Enter\_USB** Message's Enter USB Data Object (EUDO), received from the Root Hub when the *USB Host* is connected, **Shall** be propagated down through the *Hub* tree.

See **[USB Type-C 2.4]** USB4<sup>®</sup> *Hub* Connection Requirements.

The **Enter\_USB** Message **Shall** be as shown in [Figure 6.31, "Enter\\_USB Message"](#) and [Table 6.49, "Enter\\_USB Data Object \(EUDO\)"](#).

Figure 6.31 Enter\_USB Message

|                         |      |
|-------------------------|------|
| Header                  | EUDO |
| No. of Data Objects = 1 |      |

Table 6.49 Enter\_USB Data Object (EUDO)

| Bit(s)   | Field                             | Description                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B31      | <b>Reserved</b>                   | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                        |
| B30...28 | <b>USB Mode</b> <sup>1</sup>      | <ul style="list-style-type: none"> <li>000b:</li> <li>001b:</li> <li>010b:</li> <li>111b...011b: <b>Reserved, Shall Not</b> be used.</li> </ul>                                                                                                                                                                                     |
| B27      | <b>Reserved</b>                   | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                        |
| B26      | <b>USB4 DRD</b> <sup>2</sup>      | <ul style="list-style-type: none"> <li>0b: Not capable of operating as a <b>[USB4]</b> Device</li> <li>1b: Capable of operating as a <b>[USB4]</b> Device</li> </ul>                                                                                                                                                                |
| B25      | <b>USB3 DRD</b> <sup>2</sup>      | <ul style="list-style-type: none"> <li>0b: Not capable of operating as a <b>[USB 3.2]</b> Device</li> <li>1b: Capable of operating as a <b>[USB 3.2]</b> Device</li> </ul>                                                                                                                                                          |
| B24      | <b>Reserved</b>                   | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                        |
| B23...21 | <b>Cable Speed</b> <sup>2,3</sup> | <ul style="list-style-type: none"> <li>000b: <b>[USB 2.0]</b> only, no SuperSpeed support</li> <li>001b: <b>[USB 3.2]</b> Gen1</li> <li>010b: <b>[USB 3.2]</b> Gen2 and <b>[USB4]</b> Gen2</li> <li>011b: <b>[USB4]</b> Gen3</li> <li>100b: <b>[USB4]</b> Gen4</li> <li>101b...111b: <b>Reserved, Shall Not</b> be used.</li> </ul> |
| B20...19 | <b>Cable Type</b> <sup>2,3</sup>  | <ul style="list-style-type: none"> <li>00b: Passive</li> <li>01b: Active Re-timer</li> <li>10b: Active Re-driver</li> <li>11b: Optically Isolated</li> </ul>                                                                                                                                                                        |
| B18...17 | <b>Cable Current</b> <sup>2</sup> | <ul style="list-style-type: none"> <li>00b = <b>V<sub>BUS</sub></b> is not supported</li> <li>01b = <b>Reserved</b></li> <li>10b = 3A</li> <li>11b = 5A</li> </ul>                                                                                                                                                                  |
| B16      | <b>PCIe Support</b> <sup>2</sup>  | <b>[USB4]</b> PCIe tunneling supported by the host                                                                                                                                                                                                                                                                                  |
| B15      | <b>DP Support</b> <sup>2</sup>    | <b>[USB4]</b> DP tunneling supported by the host                                                                                                                                                                                                                                                                                    |
| B14      | <b>TBT Support</b> <sup>2</sup>   | <b>[TBT3]</b> is supported by the host's <b>USB4<sup>®</sup></b> Connection Manager                                                                                                                                                                                                                                                 |
| B13      | <b>Host Present</b> <sup>2</sup>  | <p>A <b>Host</b> is present at the top of the USB tree.</p> <p>When this bit is set <b>PCIe Support</b>, <b>DP Support</b> and <b>TBT Support</b> represent the <b>Host's Capabilities</b> that <b>Shall</b> be propagated down the <b>Hub</b> tree.</p>                                                                            |
| B12...0  | <b>Reserved</b>                   | <b>Shall</b> be set to zero.                                                                                                                                                                                                                                                                                                        |

1) Entry into **[USB 3.2]** and **[USB4]** include entry into **[USB 2.0]**.

2) **Shall** be **Ignored** when received by a **Cable Plug** (e.g., **SOP'** or **SOP''**).

3) The **DFP Shall** interpret the **Cable Plug's** reported capability as defined in **[USB Type-C 2.4]** in the USB4 Discovery and Entry Section.

#### 6.4.8.1 USB Mode Field

The **USB Mode** field **Shall** be used by the *DFP* to direct the *USB Mode* the *Port Partner* is to enter.

#### 6.4.8.2 USB4® DRD Field

The **USB4 DRD** field **Shall** be set when the *Host DFP* is capable of operating as a **[USB4]** Device. A **[USB4]** *Host DFP* that sets the **USB4 DRD** field **Shall** also be capable of operating as a **[USB 2.0]** Device.

#### 6.4.8.3 USB3 DRD Field

The **USB3 DRD** field **Shall** be set when the *Host DFP* is capable of operating as a **[USB 3.2]** Device. A **[USB 3.2]** *Host DFP* that sets the **USB3 DRD** field **Shall** also be capable of operating as a **[USB 2.0]** Device.

#### 6.4.8.4 Cable Speed Field

The **Cable Speed** field **Shall** be used to indicate the **cable's** maximum speed. The value is read from the *Cable Plug* and interpreted by the *DFP* as defined by **[USB Type-C 2.4]** in the USB4 Discovery and Entry Section.

#### 6.4.8.5 Cable Type Field

The **Cable Type** field **Shall** be used to indicate whether the cable is passive or active. Further if the cable is active, it indicates the type of active circuits in the cable and if the cable is optically isolated. The value is read from the *Cable Plug* and interpreted by the *DFP* as defined by **[USB Type-C 2.4]** in the USB4 Discovery and Entry Section.

#### 6.4.8.6 Cable Current Field

The **Cable Current** field **Shall** be used to indicate the **cable's** current carrying capability. The value is reported by the *Cable Plug* in the **VBUS Current Handling Capability (Passive Cable)/VBUS Current Handling Capability (Active Cable)** field.

#### 6.4.8.7 PCIe Support Field

The **PCIe Support** field **Shall** be set when the *Host DFP* is capable of tunneling PCIe over **[USB4]**.

The **PCIe Support** field **May** be set speculatively when the *Hub's DFP* is capable of tunneling PCIe over **[USB4]**.

#### 6.4.8.8 DP Support Field

The **DP Support** field **Shall** be set when the *Host DFP* is capable of tunneling DP over **[USB4]**.

The **DP Support** field **May** be set speculatively when the *Hub's DFP* is capable of tunneling DP over **[USB4]**.

#### 6.4.8.9 TBT Support Field

The **TBT Support** field **Shall** be set when the *Host DFP* is capable of tunneling Thunderbolt™ over **[USB4]** and that the Connection Manager (CM) supports discovery and configuration of Thunderbolt devices connected to the *DFP* of **[USB4]** Hubs.

The **TBT Support** field **May** be set speculatively when the *Hub's DFP* is capable of tunneling Thunderbolt over **[USB4]**.

#### 6.4.8.10 Host Present Field

The **Host Present** field **Shall** be set to indicate that a *Host* is present upstream.



## 6.4.9 EPR\_Request Message

An **EPR\_Request** Message **Shall** be sent by a Sink, operating in *EPR Mode*, to request power, typically during the request phase of a power *Negotiation*. The **EPR\_Request** Message **Shall** be sent in response to the most recent **EPR\_Source\_Capabilities** Message. The **EPR\_Request** Message **Shall** return a Sink Request Data Object (RDO) that **Shall** identify the Power Data Object being requested followed by a copy of the Power Data Object being requested.

**Note:** The requested Power Data Object **May** be either an EPR (A)PDO or SPR (A)PDO.

The **EPR\_Request** Message **Shall** be as shown in [Figure 6.32, "EPR\\_Request Message"](#).

**Figure 6.32 EPR\_Request Message**

| Header                  | RDO | Copy of PDO |
|-------------------------|-----|-------------|
| No. of Data Objects = 2 |     |             |

The Source **Shall** verify the PDO in the **EPR\_Request** Message exactly matches the PDO in the latest **EPR\_Source\_Capabilities** Message pointed to by the Object Position field in the RDO.

The Source **Shall** respond to an **EPR\_Request** Message in the same manner as it responds to a **Request** Message with an **Accept** Message, or a **Reject** Message (see [Section 6.9, "Accept, Reject and Wait"](#)). The Explicit Contract Negotiation process for EPR is the same as the process for SPR Mode except that the **Source\_Capabilities** Message is replaced by the **EPR\_Source\_Capabilities** and the **Request** Message is replaced by the **EPR\_Request** Message.

An EPR Source operating in SPR Mode that receives a **EPR\_Request** Message **Shall** initiate a Hard Reset.

The RDO takes a different form depending on the kind of power requested. The PDO and APDO formats are detailed in [Section 6.4.2, "Request Message"](#).

## 6.4.10 EPR\_Mode Message

The **EPR\_Mode** Message is used to enter, acknowledge, and exit the *EPR Mode*. The Action field is used to describe the action that is to be taken by the recipient of the **EPR\_Mode** Message. The Data field provides additional information for the Message recipient in the *EPR Mode* Data Object (ERMDO).

The **EPR\_Mode** Message **shall** be as shown in [Figure 6.33, "EPR Mode DO Message"](#) and [Table 6.50, "EPR Mode Data Object \(ERMDO\)"](#).

Figure 6.33 EPR Mode DO Message

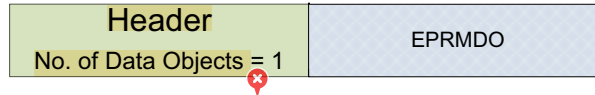


Table 6.50 EPR Mode Data Object (EPRMDO)

| Bit(s)   | Field           | Description                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                               |
|----------|-----------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| B31...24 | Action          | Value                       | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Sent By                                                                                       |
|          |                 | 0x00                        | <b>Reserved</b> and <b>Shall Not</b> be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                               |
|          |                 | 0x01                        | <b>Enter</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  Sink only |
|          |                 | 0x02                        | <b>Enter Acknowledged</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Source only                                                                                   |
|          |                 | 0x03                        | <b>Enter Succeeded</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Source only                                                                                   |
|          |                 | 0x04                        | <b>Enter Failed</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Source only                                                                                   |
|          |                 | 0x05                        | <b>Exit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Sink or Source                                                                                |
|          |                 | 0x06...0xFF                 | <b>Reserved</b> and <b>Shall Not</b> be used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                               |
| B23...16 | Data            | Action Field                | Data Field Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                               |
|          |                 | <b>Enter</b>                | <b>Shall</b> be set to the <i>EPR Sink Operational PDP</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                               |
|          |                 | <b>Enter Acknowledged</b>   | <b>Shall</b> be set to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                               |
|          |                 | <b>Enter Succeeded</b>      | <b>Shall</b> be set to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                               |
|          |                 | <b>Enter Failed</b>         | <b>Shall</b> be one of the following values: <ul style="list-style-type: none"><li>0x00 - Unknown cause</li><li>0x01 - Cable not <i>EPR Capable</i></li><li>0x02 –Source failed to become  <i>VCONN Source</i>.</li><li>0x03 – <i>EPR Capable</i> bit not set in <i>RDO</i>.</li><li>0x04 – Source unable to enter <i>EPR Mode</i><sup>1</sup>.</li><li> 0x05 - <i>EPR Capable</i> bit not set in <i>PDO</i>.</li></ul> All other values are <b>Reserved</b> and <b>Shall Not</b> be used |                                                                                               |
|          |                 | <b>Exit</b>                 | <b>Shall</b> be set to zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                               |
| B15...0  | <b>Reserved</b> | <b>Shall</b> be set to zero |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                               |

1) The Sink **May** retry entering *EPR Mode* after receiving this **Enter Failed** response.

### 6.4.10.1 Process to enter EPR Mode

An *EPR Source* **shall** enter *EPR Mode* upon request by an *EPR Sink* connected with an *EPR Cable* when able to offer the *Source Capabilities* as defined in the *Power Rules* (See [Table 10.12, "EPR Source Capabilities based on the Port Maximum PDP and using an EPR Capable Cable"](#) and [Table 10.13, "EPR Source Capabilities when Port Present PDP is less than Port Maximum PDP and using an EPR-capable cable"](#)).

For *Port Partners* to successfully enter *EPR Mode*, the following conditions must be met:

- The Sink **shall** request entry into the *EPR Mode*.

- The Source **Shall** verify the cable is *EPR Capable*.
- A Sink **Shall Not** be Connected to the Source through a *Charge Through VPD (CT-VPD)*.
- The Source and Sink **Shall** already be in an *SPR Explicit Contract*.
- The *EPR Capable* bit **Shall** be set in the *Fixed Supply 5V PDO*.
- The *EPR Capable* bit **Shall** have been set in the *RDO* in the last *Request Message* received by the Source.

To verify the cable is *EPR Capable*, the *EPR Source* **Shall** have already done the following (see [Section 6.6.21.4, "tEPRSourceCableDiscovery"](#)):

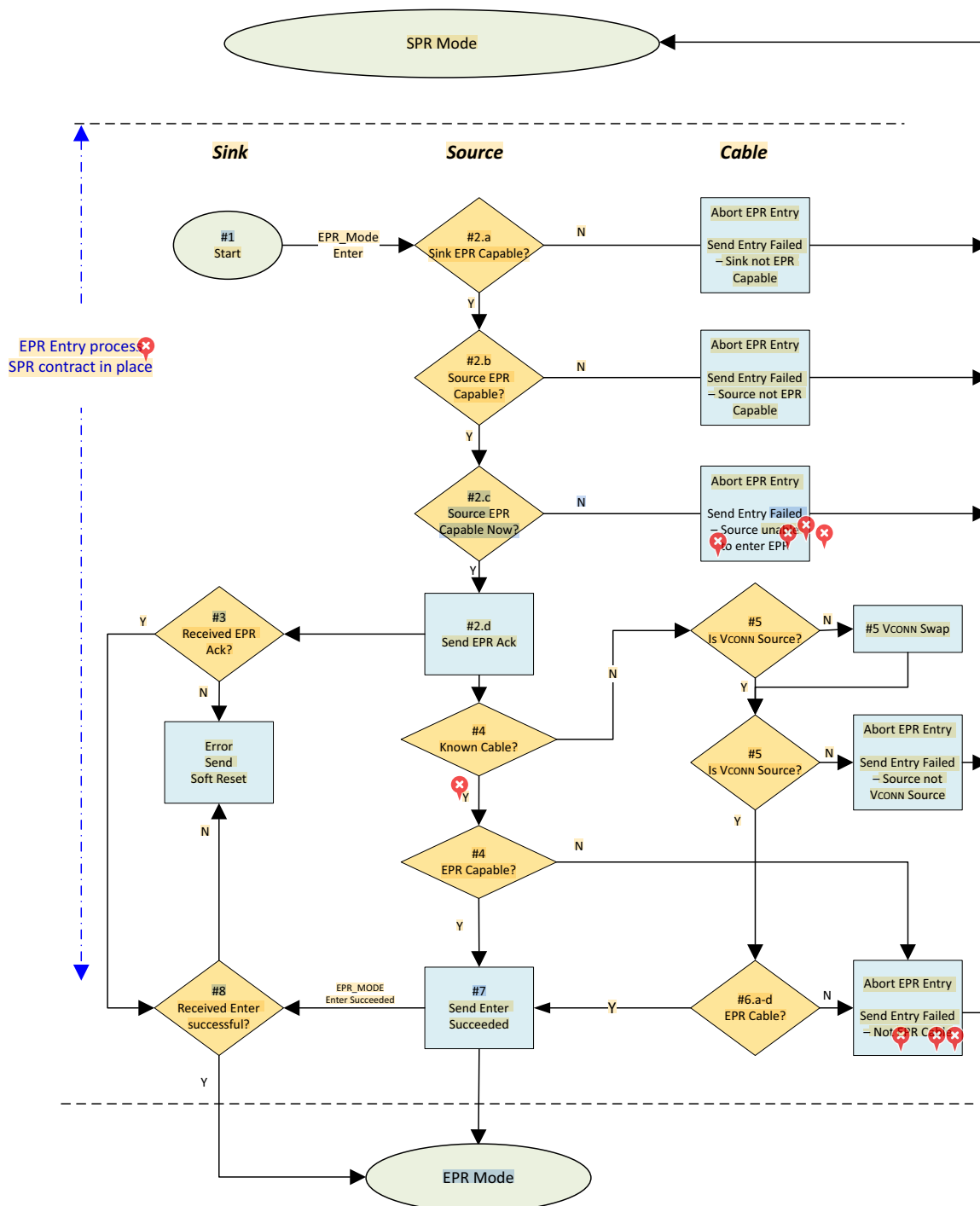
- Discover the cable prior to entering its *First Explicit Contract*
- Alternatively, within *tEPRSourceCableDiscovery* of entry into the *First Explicit Contract*
  - If it is the *VCONN Source*, discover the cable.
  - If not the *VCONN Source*, do a *VCONN Swap* then discover the cable.

and can verify the cable is *EPR Capable* by completing steps 5 and 6 in the entry process in [Figure 6.34, "Illustration of process to enter EPR Mode"](#).

The *EPR Mode* entry process is a *Non-interruptible multi-Message AMS*. An illustration of this *AMS* is shown in [Figure 6.34, "Illustration of process to enter EPR Mode"](#).

**Note:** [Figure 6.34, "Illustration of process to enter EPR Mode"](#) is not **Normative** but is **Informative** only.

**Figure 6.34 Illustration of process to enter EPR Mode**



The entry process ***Shall*** follow these steps in order:

- 1) The Sink **Shall** send the **EPR\_Mode** Message with the Action field set to 1 (**Enter**) and the Data field set to its Operational PDP. If the EPR Source receives an **EPR\_Mode** Message with the Action field not set to **Enter** it **Shall** initiate a **Soft Reset**.
- 2) The Source **Shall** do the following:

- a) Verify the **EPR Capable** bit was set in the most recent *RDO*. If not set, the *Source* **Shall** do the following:
    - i) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and the Data field set to 3 ("EPR Mode Capable bit not set in the *RDO*").
    - ii) Abort the *EPR Mode* entry process and remain in the existing SPR *Explicit Contract*.
  - b) Verify the **EPR Capable** bit was set in the most recent 5V *Fixed Supply* *PDO*. If not set, the *Source* **Shall** do the following:
    - i) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and the Data field set to 5 ("EPR Mode Capable bit not set in the *Fixed Supply* 5V *PDO*").
    - ii) Abort the *EPR Mode* entry process and remain in the existing SPR *Explicit Contract*.
  - c) Verify the *Source* is still able to support *EPR Mode*. If not, the *Source* **Shall** do the following:
    - i) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and Data field set to 4 ("Unable at this time").
    - ii) Abort the *EPR Mode* entry process and remain in the existing SPR *Explicit Contract*. The *Sink* **May** at some time in the future send another request to enter *EPR Mode*.
  - d) Send an **EPR\_Mode** Message with the Action field set to 2 (**Enter Acknowledged**).
- 3) If the *Sink* receives any Message, other than an **EPR\_Mode** Message with the Action Field set to 2, the *Sink* **Shall** initiate a *Soft Reset*.
  - 4) When the *EPR Source* has used the **Discover Identity** Command to determine and remembers the *Cable Capabilities* or the *EPR Source* is connected with a captive cable:
    - a) If the cable is *EPR Capable* it **Should** go directly to Step 7, but **May** continue to Step 5.
    - b) If the cable is not *EPR Capable* it **Shall** do the following:
    - c) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and the Data field set to 1 ("Cable not EPR capable").
    - d) Abort the *EPR Mode* entry process and remain in the existing SPR *Explicit Contract*.
  - 5) If the *Source* is not the *VCONN Source*, it **Shall** send a **VCONN\_Swap** Message
    - a) If the *Source* fails to become the *VCONN Source*, it **Shall**:
      - i) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and the Data field set to 2 (not *VCONN Source*).
      - ii) Abort the *EPR Mode* entry process and remain in the existing SPR *Explicit Contract*.
  - 6) The *Source* **Shall** use the **Discover Identity** Command to read the cable's e-Marker and verify the following:
    - a) Cable VDO - **Maximum VBUS Voltage (Passive Cable)/Maximum VBUS Voltage (Active Cable)** field is 11b (50V)
    - b) Cable VDO - **VBUS Current Handling Capability (Passive Cable)/VBUS Current Handling Capability (Active Cable)** field is 10b (5A)
    - c) Cable VDO - **EPR Capable (Passive Cable)/EPR Capable (Active Cable)** field is 1b (*EPR Capable*)
    - d) If the cable fails to respond to the **Discover Identity** Command or is not *EPR Capable*, the *Source* **Shall** do the following:
      - i) Send an **EPR\_Mode** Message with the Action field set to 4 (**Enter Failed**) and the Data field to 1 ("Cable not EPR capable").

- ii) Abort the *EPR Mode* entry process and remain in the existing *SPR Explicit Contract*.
- 7) The Source **Shall** send the *EPR\_Mode* Message with the Action field set to 3 (*Enter Succeeded*) and **Shall** enter *EPR Mode*.
- 8) If the Sink receives an *EPR\_Mode* Message with the Action field set to 3 (*Enter Succeeded*) it **Shall** enter *EPR Mode*, otherwise it **Shall** initiate a *Soft Reset*.

If the *EPR Mode* entry process successfully completes within *tEnterEPR* of the last bit of the *GoodCRC* Message sent in response to the *EPR\_Mode* Message with the Action field set to 1 (*Enter*), the Source **Shall** send an *EPR\_Source\_Capabilities* Message within *tFirstSourceCap*.

If the *EPR Mode* entry process has not been aborted or does not complete within *tEnterEPR* of the last bit of the *GoodCRC* Message sent in response to the *EPR\_Mode* Message with the Action field set to 1 (*Enter*), the Sink **Shall** initiate a *Soft Reset*.

## 6.4.10.2 Operation in EPR Mode

While operating in *EPR Mode*, the Source **Shall** only send *EPR\_Source\_Capabilities* Messages to Advertise its power Capabilities and the Sink **Shall** only respond with *EPR\_Request* Messages to Negotiate Explicit Contracts. The *EPR\_Request* Message **May** be for either an *SPR* or *EPR (A)PDO*.

If the Source sends a *Source\_Capabilities* Message, that is not in response to a Sink *Get\_Source\_Cap* Message, the Sink **Shall** initiate a *Hard Reset*. If the Sink sends a *Request* Message, the Source **Shall** initiate a *Hard Reset*.

The Source **Shall** monitor the CC communications path to ensure that there is periodic traffic. The Sink **Shall** send an *EPR\_KeepAlive* Message when it has not sent any Messages for more than *tSinkEPRKeepAlive* to ensure there is timely periodic traffic. If there is no traffic for more than *tSourceEPRKeepAlive*, the Source **Shall** initiate a *Hard Reset*.

## 6.4.10.3 Exiting EPR Mode

### 6.4.10.3.1 Commanded Exit

While in *EPR Mode*, either the Source or Sink **May** exit *EPR Mode* by sending an *EPR\_Mode* Message with the Action field set to 5 (*Exit*).

The ports **Shall** be in an *Explicit Contract* with an *SPR (A)PDO* prior to the *EPR Mode* exit process by either:

- The Source sending an *EPR\_Source\_Capabilities* Message with no *EPR (A)PDO*s (e.g., only *SPR (A)PDO*s) or
- The Sink negotiating a new *Explicit Contract* with bit 31 in the *RDO* set to zero (e.g., only *SPR (A)PDO*s).

The process to exit *EPR Mode* is a *Non-interruptible multi-Message AMS* and **Shall** follow these steps in order:

- 1) The Port Partners **Shall** be in an *Explicit Contract* with an *SPR (A)PDO*.
- 2) Either the Source or Sink **Shall** send an *EPR\_Mode* Message with the Action field set to 5 (*Exit*) to exit the *EPR Mode*
- 3) The Source **Shall** send a *Source\_Capabilities* Message within *tFirstSourceCap* of the *GoodCRC* Message in response to the *EPR\_Mode* Message with the Action field set to 5 (*Exit*).
- 4) If the Sink does not receive a *Source\_Capabilities* Message within *tTypeCSinkWaitCap* of the last bit of the *GoodCRC* Message in response to the *EPR\_Mode* Message with the Action field set to 5 (*Exit*), Sink **Shall** initiate a *Hard Reset*.

### 6.4.10.3.2 Implicit Exit

*EPR Mode* **Shall** be exited as the side-effect of the *Power Role Swap* and *Fast Role Swap* processes. This is because at the end of these processes *V<sub>BUS</sub>* will be at *vSafe5V* and the Ports will be in an *Implicit Contract*. The New Source will then send a *Source\_Capabilities* Message (not an *EPR\_Source\_Capabilities* Message) to begin the process of



negotiating an SPR *Explicit Contract*. Once an SPR *Explicit Contract* is entered, the *Source* and *Sink* can then enter *EPR Mode* if needed.

### 6.4.10.3.3 Exits due to errors

Other critical errors can occur while in *EPR Mode*; these errors **shall** result in *Hard Reset* being initiated by the *Port* that detects the error. Some of these errors include:

- An *EPR\_Mode* Message with the Action field set to 5 (**Exit**) to exit *EPR Mode* is received by a *Port* in an *Explicit Contract* with an EPR (A)PDO.
- The *Sink* receives an *EPR\_Source\_Capabilities* Message with an EPR (A)PDO in any of the first seven object positions.
- The (A)PDO in the *EPR\_Request* Message does not match the (A)PDO in the latest *EPR\_Source\_Capabilities* Message pointed to by the Object Position field in the RDO.
- The *Source* receives a *Request* Message.
- The *Sink* receives a *Source\_Capabilities* Message not in response to a *Get\_Source\_Cap* Message.

## 6.4.11 Source\_Info Message

The **Source\_Info** Message **Shall** be sent in response to a **Get\_Source\_Info** Message. The **Source\_Info** Message contains one Source Information Data Object (SIDO).

The **Source\_Info** Message returns a SIDO whose format **Shall** be as shown in [Figure 6.35, "Source\\_Info Message"](#) and [Table 6.51, "Source\\_Info Data Object \(SIDO\)"](#). The **Number of Data Objects** field in the **Source\_Info** Message **Shall** be set to 1.

The **Port Maximum PDP**, **Port Present PDP**, **Port Reported PDP** and the **Port Type** are used to identify **Capabilities** of a **Source Port**.

Figure 6.35 Source\_Info Message

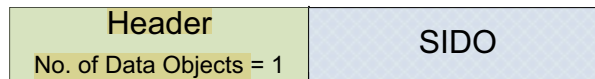


Table 6.51 Source\_Info Data Object (SIDO)

| Bit(s)           | Field                    | Description                                                                                                                      |
|------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| B31 <sup>x</sup> | <b>Port Type</b>         | <ul style="list-style-type: none"><li>0 = <i>Managed Capability Port</i></li><li>1 = <i>Guaranteed Capability Port</i></li></ul> |
| B30...24         | <b>Reserved</b>          | <b>Shall</b> be set to zero                                                                                                      |
| B23...16         | <b>Port Maximum PDP</b>  | Power the <b>Port</b> is designed to supply                                                                                      |
| B15...8          | <b>Port Present PDP</b>  | Power the <b>Port</b> is presently capable of supplying                                                                          |
| B7...0           | <b>Port Reported PDP</b> | Power the <b>Port</b> is actually advertising                                                                                    |

### 6.4.11.1 Port Type Field

**Port Type** is a **Static** field that **Shall** be used to indicate whether the amount of power the **Port** can provide is fixed or can change dynamically.

For **Ports** that are part of a **Shared Capacity Group**, the **Port Type** field **Shall** be set to *Managed Capability Port*.

For **Ports** not part of a **Shared Capacity Group**, the **Port Type** field **May** be set to either *Managed Capability Port* or *Guaranteed Capability Port*.

### 6.4.11.2 Port Maximum PDP Field

**Port Maximum PDP** is a **Static** field that **Shall** report the integer portion of the **PDP Rating** of the **Port**. A *Guaranteed Capability Port* (as indicated by the **Port Type** field being set to '1') **Shall** always be capable of supplying this amount of power. A *Managed Capability Port* (as indicated by the **Port Type** field being set to '0') **Shall** be able to offer this amount of power at some time.

The **Port Maximum PDP** **Shall** be the same as the larger of the **SPR** *Source PDP Rating* and the *EPR* *Source PDP Rating* in the **Source\_Capabilities\_Extended** Message.

### 6.4.11.3 Port Present PDP Field

The **Port Present PDP** field **Shall** indicate the integer part of the amount of power the **Port** is presently capable of supplying including limitations due to *Cable Capabilities* or abnormal operating conditions (e.g., elevated temperature, low input voltage, etc.).

A *Guaranteed Capability Port* **Shall** always set its **Port Present PDP** to be the same as its **Port Maximum PDP** or the highest possible value when limited.

A *Managed Capability Port* that is part of a *Shared Capacity Group* **Shall** set its **Port Present PDP** to Shared Port Power Available as defined in **[USB Type-C 2.4]** or to a lower value when limited.

A *Managed Capability Port* that is part of an *Assured Capacity Group* **Shall** set its **Port Present PDP** to the **Port Maximum PDP** or the highest value possible when limited.

#### 6.4.11.4 Port Reported PDP Field

The **Port Reported PDP** field **Shall** track the amount of power the *Port* is offering in its **Source\_Capabilities** Message or **EPR\_Source\_Capabilities** Message. The **Port Reported PDP** field **May** be dynamic or **Static** depending on the *Port's* other characteristics such as Managed/Guaranteed Capability, SPR/EPR **Mode**, its power policy etc.

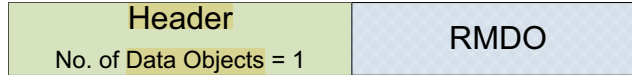
**Note:** The **Port Reported PDP** field is computed as the integer part of, the largest of the products of the voltage times current of the **Fixed Supply** PDOs returned in the **Source\_Capabilities** Message or **EPR\_Source\_Capabilities** Messages.

## 6.4.12 Revision Message

The **Revision Message** *shall* be sent in response to the **Get\_Revision Message** sent by the *Port Partner*. This *Message* is used to identify the highest **Revision** the *Port* is capable of operating at. The **Revision Message** contains one **Revision Message Data Object (RMDO)**.

The **Revision Message** returns an RMDO whose format *shall* be as shown in [Figure 6.36, "Revision Message Data Object"](#) and [Table 6.52, "Revision Message Data Object \(RMDO\)"](#). The **Number of Data Objects** field in the **Revision Message** *shall* be set to 1.

**Figure 6.36 Revision Message Data Object**



**Table 6.52 Revision Message Data Object (RMDO)**

| Bit(s)   | Description                            |
|----------|----------------------------------------|
| B31...28 | Revision.major                         |
| B27...24 | Revision.minor                         |
| B23...20 | Version.major                          |
| B19...16 | Version.minor                          |
| B15...0  | Reserved, <i>shall</i> be set to zero. |

E.g., for Revision 3.2, Version 1.1 the fields would be the following:

- Revision.major = 0011b
- Revision.minor = 0010b
- Version.major = 0001b
- Version.minor = 0001b

## 6.5 Extended Message

An *Extended Message* **Shall** contain an *Extended Message Header* (indicated by the **Extended** field in the *Message Header* being set) and be followed by zero or more data bytes. Additional bytes that might be added to existing *Messages* in future *Revision* of this specification **Shall** be **Ignored**.

The format of the *Extended Message* is defined by the *Message Header's* **Message Type** field and is summarized in [Table 6.53, "Extended Message Types"](#). The Sent by column indicates entities which **May** send the given *Message* (*Source*, *Sink* or *Cable Plug*); entities not listed **Shall Not** issue the corresponding *Message*. The "Valid Start of Packet" column indicates the *Messages* which **Shall** only be issued in *SOP Packets* and the *Messages* which **May** be issued in *SOP\* Packets*.

**Table 6.53 Extended Message Types**

| Bits<br>4...0       | Type                                         | Sent by                    | Description                                                                         | Valid<br>Start of<br>Packet |
|---------------------|----------------------------------------------|----------------------------|-------------------------------------------------------------------------------------|-----------------------------|
| 0 0000              | <b>Reserved</b>                              |                            | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                             |
| 0 0001              | <a href="#">Source_Capabilities_Extended</a> | Source or Dual-Role Power  | See <a href="#">Section 6.5.1</a>                                                   | SOP only                    |
| 0 0010              | <b>Status</b>                                | Source, Sink or Cable Plug | See <a href="#">Section 6.5.2</a>                                                   | SOP*                        |
| 0 0011              | <a href="#">Get_Battery_Cap</a>              | Source or Sink             | See <a href="#">Section 6.5.3</a>                                                   | SOP only                    |
| 0 0100              | <a href="#">Get_Battery_Status</a>           | Source or Sink             | See <a href="#">Section 6.5.4</a>                                                   |                             |
| 0 0101              | <a href="#">Battery_Capabilities</a>         | Source or Sink             | See <a href="#">Section 6.5.5</a>                                                   | SOP only                    |
| 0 0110              | <a href="#">Get_Manufacturer_Info</a>        | Source or Sink             | See <a href="#">Section 6.5.6</a>                                                   | SOP*                        |
| 0 0111              | <a href="#">Manufacturer_Info</a>            | Source, Sink or Cable Plug | See <a href="#">Section 6.5.7</a>                                                   | SOP*                        |
| 0 1000              | <a href="#">Security_Request</a>             | Source or Sink             | See <a href="#">Section 6.5.8.1</a>                                                 | SOP*                        |
| 0 1001              | <a href="#">Security_Response</a>            | Source, Sink or Cable Plug | See <a href="#">Section 6.5.8.2</a>                                                 | SOP*                        |
| 0 1010              | <a href="#">Firmware_Update_Request</a>      | Source or Sink             | See <a href="#">Section 6.5.9.1</a>                                                 | SOP*                        |
| 0 1011              | <a href="#">Firmware_Update_Response</a>     | Source, Sink or Cable Plug | See <a href="#">Section 6.5.9.2</a>                                                 | SOP*                        |
| 0 1100              | <a href="#">PS_Status</a>                    | Source                     | See <a href="#">Section 6.5.10</a>                                                  | SOP only                    |
| 0 1101              | <a href="#">Country_Info</a>                 | Source or Sink             | See <a href="#">Section 6.5.12</a>                                                  | SOP only                    |
| 0 1110              | <a href="#">Country_Codes</a>                | Source or Sink             | See <a href="#">Section 6.5.11</a>                                                  | SOP only                    |
| 0 1111              | <a href="#">Sink_Capabilities_Extended</a>   | Sink or Dual-Role Power    | See <a href="#">Section 6.5.13</a>                                                  | SOP only                    |
| 1 0000              | <a href="#">Extended_Control</a>             | Source or Sink             | See <a href="#">Section 6.5.14</a>                                                  | SOP only                    |
| 1 0001              | <a href="#">EPR_Source_Capabilities</a>      | Source or Dual-Role Power  | See <a href="#">Section 6.5.15.2</a>                                                | SOP only                    |
| 1 0010              | <a href="#">EPR_Sink_Capabilities</a>        | Sink or Dual-Role Power    | See <a href="#">Section 6.5.15.3</a>                                                | SOP only                    |
| 1 0011...<br>1 1101 | <b>Reserved</b>                              |                            | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                             |
| 1 1110              | <a href="#">Vendor_Defined_Extended</a>      | Source, Sink or Cable Plug | See <a href="#">Section 6.5.16</a>                                                  | SOP*                        |
| 1 1111              | <b>Reserved</b>                              |                            | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                             |

## 6.5.1 Source\_Capabilities\_Extended Message

The **Source\_Capabilities\_Extended** Message **Should** be sent in response to a **Get\_Source\_Cap\_Extended** Message. The **Source\_Capabilities\_Extended** Message enables a *Source* or a *DRP* to inform the *Sink* about its **Capabilities** as a *Source*.

The **Source\_Capabilities\_Extended** Message **Shall** return a 25-byte Source Capabilities Extended Data Block (SCEDB) whose format **Shall** be as shown in [Figure 6.37, "Source\\_Capabilities\\_Extended Message"](#) and [Table 6.54, "Source Capabilities Extended Data Block \(SCEDB\)"](#).

Figure 6.37 Source\_Capabilities\_Extended Message

| Extended Header | SCEDB                |
|-----------------|----------------------|
| Data Size = 25  | (25-byte Data Block) |

Table 6.54 Source Capabilities Extended Data Block (SCEDB)

| Offset | Field                                        | Description                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                    |
|--------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>VID</b>                                   | Vendor ID (assigned by the USB-IF)                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                    |
| 2      | <b>PID</b>                                   | Product ID (assigned by the manufacturer)                                                                                                                                                                                                                                                                            |                                                                                                                                                                                    |
| 4      | <b>XID</b>                                   | Value provided by the USB-IF assigned to the product                                                                                                                                                                                                                                                                 |                                                                                                                                                                                    |
| 8      | <b>FW Version</b>                            | Firmware version number                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                    |
| 9      | <b>HW Version</b>                            | Hardware version number                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                    |
| 10     | <b>Voltage Regulation</b>                    | <b>Bit</b>                                                                                                                                                                                                                                                                                                           | <b>Description</b>                                                                                                                                                                 |
|        |                                              | 1...0                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"><li>00b: 150mA/μs Load Step (default)</li><li>01b: 500mA/μs Load Step</li><li>11b...10b: <b>Reserved</b> and <b>Shall Not</b> be used.</li></ul> |
|        |                                              | 2                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>0b: 25% <i>IoC</i> (default)</li><li>1b: 90% <i>IoC</i></li></ul>                                                                            |
|        |                                              | 3...7                                                                                                                                                                                                                                                                                                                | <b>Reserved</b> and <b>Shall Not</b> be used                                                                                                                                       |
| 11     | <b>Holdup Time</b>                           | Output will stay with regulated limits for this number of milliseconds after removal of the AC from the input. <ul style="list-style-type: none"><li>0x00 = feature not supported</li></ul> <b>Note:</b> A value of at least 3ms <b>Should</b> be used (see <a href="#">Section 7.1.12.2, "Holdup Time Field"</a> ). |                                                                                                                                                                                    |
| 12     | <b>Compliance</b>                            | Compliance in <b>SPR Mode</b> :                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                    |
|        |                                              | <b>Bit</b>                                                                                                                                                                                                                                                                                                           | <b>Description</b>                                                                                                                                                                 |
|        |                                              | 0                                                                                                                                                                                                                                                                                                                    | <i>LPS</i> compliant when set                                                                                                                                                      |
|        |                                              | 1                                                                                                                                                                                                                                                                                                                    | PS1 compliant when set                                                                                                                                                             |
|        |                                              | 2                                                                                                                                                                                                                                                                                                                    | PS2 compliant when set                                                                                                                                                             |
| 3...7  | <b>Reserved</b> and <b>Shall Not</b> be used |                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                    |
| 13     | <b>Touch Current</b>                         | <b>Bit</b>                                                                                                                                                                                                                                                                                                           | <b>Description</b>                                                                                                                                                                 |
|        |                                              | 0                                                                                                                                                                                                                                                                                                                    | Low touch <b>current</b> EPS when set                                                                                                                                              |
|        |                                              | 1                                                                                                                                                                                                                                                                                                                    | Ground pin <b>supported</b> when set                                                                                                                                               |
|        |                                              | 2                                                                                                                                                                                                                                                                                                                    | Ground pin intended for protective earth when set                                                                                                                                  |
| 3...7  | <b>Reserved</b> and <b>Shall Not</b> be used |                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                    |



Table 6.54 Source Capabilities Extended Data Block (SCEDB) (Continued)

| Offset | Field                                     | Description                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                           |
|--------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14     | <i>Peak Current1</i>                      | Bit                                                                                                                                                                                                                                                            | Description                                                                                                                                                                                                               |
|        |                                           | 0...4                                                                                                                                                                                                                                                          | Percent overload in 10% increments Values higher than 25 (11001b) are clipped to 250%.                                                                                                                                    |
|        |                                           | 5...10                                                                                                                                                                                                                                                         | Overload period in 20ms                                                                                                                                                                                                   |
|        |                                           | 11...14                                                                                                                                                                                                                                                        | Duty cycle in 5% increments                                                                                                                                                                                               |
|        |                                           | 15                                                                                                                                                                                                                                                             | <i>V<sub>BUS</sub></i> voltage droop                                                                                                                                                                                      |
| 16     | <i>Peak Current2</i>                      | Bit                                                                                                                                                                                                                                                            | Description                                                                                                                                                                                                               |
|        |                                           | 0...4                                                                                                                                                                                                                                                          | Percent overload in 10% increments Values higher than 25 (11001b) are clipped to 250%.                                                                                                                                    |
|        |                                           | 5...10                                                                                                                                                                                                                                                         | Overload period in 20ms                                                                                                                                                                                                   |
|        |                                           | 11...14                                                                                                                                                                                                                                                        | Duty cycle in 5% increments                                                                                                                                                                                               |
|        |                                           | 15                                                                                                                                                                                                                                                             | <i>V<sub>BUS</sub></i> voltage droop                                                                                                                                                                                      |
| 18     | <i>Peak Current3</i>                      | Bit                                                                                                                                                                                                                                                            | Description                                                                                                                                                                                                               |
|        |                                           | 0...4                                                                                                                                                                                                                                                          | Percent overload in 10% increments Values higher than 25 (11001b) are clipped to 250%.                                                                                                                                    |
|        |                                           | 5...10                                                                                                                                                                                                                                                         | Overload period in 20ms                                                                                                                                                                                                   |
|        |                                           | 11...14                                                                                                                                                                                                                                                        | Duty cycle in 5% increments                                                                                                                                                                                               |
|        |                                           | 15                                                                                                                                                                                                                                                             | <i>V<sub>BUS</sub></i> voltage droop                                                                                                                                                                                      |
| 20     | <i>Touch Temp</i>                         | Temperature conforms to: <ul style="list-style-type: none"> <li>0 = <i>[IEC 60950-1]</i> (default)</li> <li>1 = <i>[IEC 62368-1]</i> TS1</li> <li>2 = <i>[IEC 62368-1]</i> TS2</li> </ul> Note: All other values <b>Reserved</b> and <b>Shall Not</b> be used. |                                                                                                                                                                                                                           |
| 21     | <i>Source Inputs</i>                      | Bit                                                                                                                                                                                                                                                            | Description                                                                                                                                                                                                               |
|        |                                           | 0                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>0b: No external supply</li> <li>1b: External supply present</li> </ul>                                                                                                             |
|        |                                           | 1                                                                                                                                                                                                                                                              | If bit 0 is set: <ul style="list-style-type: none"> <li>0b: External supply is constrained.</li> <li>1b: External supply is unconstrained.</li> </ul> If bit 0 is not set <b>Reserved</b> and <b>Shall</b> be set to zero |
|        |                                           | 2                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>0b: No internal <i>Battery</i></li> <li>1b: Internal <i>Battery</i> present</li> </ul>                                                                                             |
|        |                                           | 3...7                                                                                                                                                                                                                                                          | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                                                                                           |
| 22     | <i>Number of Batteries/ Battery Slots</i> | Upper Nibble = Number of Hot Swappable <i>Battery Slots</i> (0...4)<br>Lower Nibble = Number of <i>Fixed Batteries</i> (0...4)                                                                                                                                 |                                                                                                                                                                                                                           |
| 23     | <i>SPR Source PDP Rating</i>              | 0...6: <i>Source PDP Rating</i> ( <i>EPR Source's PDP Rating</i> when operating in <i>SPR Mode</i> ).<br>7: <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                    |                                                                                                                                                                                                                           |
| 24     | <i>EPR Source PDP Rating</i>              | 0...7: <i>EPR Source PDP Rating</i>                                                                                                                                                                                                                            |                                                                                                                                                                                                                           |

### 6.5.1.1 Vendor ID (VID) Field

The **VID** field **Shall** contain the 16-bit *Vendor ID (VID)* assigned to the *Source's* vendor by the USB-IF. If the vendor does not have a *VID*, the **VID** field **Shall** be set to 0xFFFF. *Devices* that have a USB data interface **Shall** report the same *VID* as the *idVendor* in the Standard Device Descriptor (see *[USB 2.0]* and *[USB 3.2]*).

### 6.5.1.2 Product ID (PID) Field

The **PID** field **shall** contain the 16-bit Product ID (PID) assigned by the *Source's* vendor. *Devices* that have a USB data interface **shall** report the same PID as the idProduct in the Standard Device Descriptor (see [\[USB 2.0\]](#) and [\[USB 3.2\]](#)).

### 6.5.1.3 XID Field

The **XID** field **shall** contain the 32-bit XID provided by the USB-IF to the vendor who in turns assigns it to a product. If the vendor does not have an XID, then it **shall** return zero in this field (see [\[USB 2.0\]](#) and [\[USB 3.2\]](#)).

### 6.5.1.4 Firmware Version Field

The **FW Version** field **shall** contain an 8-bit firmware version number assigned to the device by the vendor.

### 6.5.1.5 Hardware Version Field

The **HW Version** field **shall** contain an 8-bit hardware version number assigned to the device by the vendor.

### 6.5.1.6 Voltage Regulation Field

The **Voltage Regulation** field contains bits covering Load Step Slew Rate and Magnitude.

See [Section 7.1.12.1, "Voltage Regulation Field"](#) for further details.

#### 6.5.1.6.1 Load Step Slew Rate

The *Source* **shall** report its load step response capability in bits 0...1 of the **Voltage Regulation** bit field.

#### 6.5.1.6.2 Load Step Magnitude

The *Source* **shall** report its load step magnitude rate as a percentage of *IoC* in bit 2 of the **Voltage Regulation** field.

### 6.5.1.7 Holdup Time Field

The **Holdup Time** field **shall** contain the *Source's* holdup time (see [Section 7.1.12.2, "Holdup Time Field"](#)).

### 6.5.1.8 Compliance Field

The **Compliance** is **Static** and **shall** contain the standards the *Source* is compliant with in SPR (see [Section 7.1.12.3, "Compliance Field"](#)).

### 6.5.1.9 Touch Current Field

The **Touch Current** field reports whether the *Source* meets certain leakage current levels and if it has a ground pin.

A *Source* **shall** set the **Touch Current** bit (bit 0) when their leakage current is less than 65µA rms when *Source's* maximum capability is less than or equal to 30W, or when their leakage current is less than 100 µA rms when its power capability is between 30W and 100W. The total combined leakage current **shall** be measured in accordance with [\[IEC 60950-1\]](#) when tested at 250VAC rms at 50 Hz.

A *Source* with a ground pin **shall** set the Ground pin bit (bit 1).

A *Source* whose Ground pin is intended to be connected to a protective earth **shall** set both bit1 and bit 2.

### 6.5.1.10 Peak Current Field

The **Peak Current1/Peak Current2/Peak Current3** fields **shall** contain the combinations of Peak Current that the *Source* supports (see [Section 7.1.12.4, "Peak Current"](#)).

Peak Current provides a means for *Source* report its ability to provide current in excess of the *Negotiated* amount for short periods. The Peak Current descriptor defines up to three combinations of % overload, duration and duty

cycle defined as **Peak Current1**, **Peak Current2** and **Peak Current3** that the *Source* supports. A *Source* **May** offer no Peak Current capability. A *Source* **Shall** populate unused Peak Current bit fields with zero.

The Bit Fields within **Peak Current1**, **Peak Current2** and **Peak Current3** contain the following subfields:

- Percentage Overload
  - **Shall** be the maximum peak current reported in 10% increments as a percentage of the **Negotiated** operating current (*IoC*) offered by the *Source*. Values higher than 25 (11001b) are clipped to 250%.
- Overload Period
  - **Shall** be the minimum rolling average time window in 20ms increments, where a value of 20ms is recommended.
-  **Duty Cycle**
  - **Shall** be the maximum percentage of overload period reported in 5% increments. The values **Should** be 5%, 10% and 50% for PeakCurrent1, PeakCurrent2, and PeakCurrent3, respectively.
- **V<sub>BUS</sub> Droop**
  - **Shall** be set to one to indicate there is an additional 5% voltage droop on **V<sub>BUS</sub>** when the overload conditions occur as defined by **vSrcPeak**. However, it is recommended that the *Source* **Should** provide **V<sub>BUS</sub>** in the range of **vSrcNew** when overload conditions occur and set this bit to zero.

### 6.5.1.11 Touch Temp Field

The **Touch Temp** field **Shall** report the IEC standard used to determine the surface temperature of the *Source's* enclosure. Safety limits for the *Source's* touch temperature are set in applicable product safety standards (e.g., **[IEC 60950-1]** or **[IEC 62368-1]**). The *Source* **May** report when its touch temperature performance conforms to the TS1 or TS2 limits described in **[IEC 62368-1]**.

### 6.5.1.12 Source Inputs Field

The **Source Inputs** field **Shall** identify the possible inputs that provide power to the *Source*:

- When bit 0 is set, the *Source* can be sourced by an external power supply.
- When bits 0 and 1 are set, the *Source* can be sourced by an external power supply which is assumed to be effectively "infinite" i.e., it won't run down over time.
- When bit 2 is set the *Source* can be sourced by an internal *Battery*.

**Note:** Some *Sources* are only powered by a *Battery* (e.g., an automobile) rather than the more common AC Supply.

**Note:** Bit 2 **May** be set independently of bits 0 and 1.

### 6.5.1.13 Number of Batteries/Battery Slots Field

The **Number of Batteries/Battery Slots** field **Shall** report the number of *Fixed Batteries* and Hot Swappable *Battery Slots* the *Source* supports. This field **Shall** independently report the number of *Battery Slots* and the number of *Fixed Batteries*.

A *Source* **Shall** have no more than 4 *Fixed Batteries* and no more than 4 *Battery Slots*.

*Fixed Batteries* **Shall** be numbered consecutively from 0 to 3. The number assigned to a given *Fixed Battery* **Shall Not** change between *Attach* and *Detach*.

*Battery Slots* **Shall** be numbered consecutively from 4 to 7. The number assigned to a given *Battery Slot* **Shall Not** change between *Attach* and *Detach*.

#### 6.5.1.14 SPR Source PDP Rating Field

For an *SPR Source* the *SPR Source PDP Rating* field **Shall** report the integer portion of the *PDP Rating* of the *Port*.

For an *EPR Source*, the *SPR Source PDP Rating* field **Shall** report the integer portion of the maximum amount of power that the *Port* is designed to deliver in SPR Mode.

The *SPR Source PDP Rating* field that is reported **Shall** be **Static**.

#### 6.5.1.15 EPR Source PDP Rating Field

For an *EPR Source* the *EPR Source PDP Rating* field **Shall** report the integer portion of the *PDP Rating* of the *Port*.

For an *SPR Source* this field **Shall** be set to zero.

The *EPR Source PDP Rating* field that is reported **Shall** be **Static**.

## 6.5.2 Status Message

The **Status Message** **Shall** be sent in response to a **Get\_Status Message**. The content of the **Status Message** depends on the target of the **Get\_Status Message**. When sent to **SOP** the **Status Message** returns the status of the **Port's Port Partner**. When sent to **SOP'** or **SOP''** the **Status Message** returns the status of one of the **Active Cable's Cable Plugs**.

### 6.5.2.1 SOP Status Message

A **Status Message**, sent in response to **Get\_Status Message** to **SOP**, enables a **Port** to inform its **Port Partner** about the present status of the **Source** or **Sink**. Typically, a **Get\_Status Message** will be sent by the **Port** after receipt of an **Alert Message**. Some of the reported events are critical such as **OCP**, **OVP** and **OTP**, while others are **informational** such as change in a **Battery's** status from charging to neither charging nor discharging.

The **Status Message** returns a 7-byte Status Data Block (SDB) whose format **Shall** be as shown in [Figure 6.38, "SOP Status Message"](#) and [Table 6.55, "SOP Status Data Block \(SDB\)"](#).

Figure 6.38 SOP Status Message



Table 6.55 SOP Status Data Block (SDB)

| Offset (Byte) | Field                 | Description                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                   |
|---------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Internal Temp         |                                                                                                                                                                                                                                                                                                                                                            | Source or Sink's internal temperature in °C <ul style="list-style-type: none"><li>0 = feature not supported</li><li>1 = temperature is less than 2°C.</li><li>2-255 = temperature in °C</li></ul> |
| 1             | Present Input         | Bit                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                                                                                       |
|               |                       | 0                                                                                                                                                                                                                                                                                                                                                          | Reserved and Shall be set to zero                                                                                                                                                                 |
|               |                       | 1                                                                                                                                                                                                                                                                                                                                                          | External Power when set                                                                                                                                                                           |
|               |                       | 2                                                                                                                                                                                                                                                                                                                                                          | External Power AC/DC (Valid when Bit 1 set) <ul style="list-style-type: none"><li>0: DC</li><li>1: AC</li></ul>                                                                                   |
|               |                       |                                                                                                                                                                                                                                                                                                                                                            | Reserved when Bit 1 is zero                                                                                                                                                                       |
|               |                       | 3                                                                                                                                                                                                                                                                                                                                                          | Internal Power from Battery when set                                                                                                                                                              |
|               |                       | 4                                                                                                                                                                                                                                                                                                                                                          | Internal Power from non-Battery power source when set                                                                                                                                             |
|               |                       | 5...7                                                                                                                                                                                                                                                                                                                                                      | Reserved and Shall be set to zero                                                                                                                                                                 |
| 2             | Present Battery Input | When Present Input field bit 3 set Shall contain the bit corresponding to the Battery or Batteries providing power: <ul style="list-style-type: none"><li>Upper nibble = Hot Swappable Battery (b7...4)</li><li>Lower nibble = Fixed Battery (b3...0)</li></ul> When Present Input field bit 3 is not set this field is Reserved and Shall be set to zero. |                                                                                                                                                                                                   |

Table 6.55 SOP Status Data Block (SDB) (Continued)

| Offset (Byte) | Field              | Description |                                                                                                                                                 |                                                          |
|---------------|--------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 3             | Event Flags        | Bit         | Flag                                                                                                                                            | Description                                              |
|               |                    | 0           |                                                                                                                                                 | <b>Reserved</b> and <b>Shall</b> be set to zero          |
|               |                    | 1           | OCP Event                                                                                                                                       | OCP event when set                                       |
|               |                    | 2           | OTP Event                                                                                                                                       | OTP event when set                                       |
|               |                    | 3           | OVP Event                                                                                                                                       | OVP event when set                                       |
|               |                    | 4           | CL/CV Mode                                                                                                                                      | In PPS Mode only: CL mode when set, CV mode when cleared |
|               |                    | 5...7       |                                                                                                                                                 | <b>Reserved</b> and <b>Shall</b> be set to zero          |
| 4             | Temperature Status | Bit         | Description                                                                                                                                     |                                                          |
|               |                    | 0           | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                 |                                                          |
|               |                    | 1...2       | <ul style="list-style-type: none"> <li>00 – Not Supported.</li> <li>01 – Normal</li> <li>10 – Warning</li> <li>11 – Over temperature</li> </ul> |                                                          |
|               |                    | 3...7       | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                 |                                                          |
| 5             | Power Status       | Bit         | Description                                                                                                                                     |                                                          |
|               |                    | 0           | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                 |                                                          |
|               |                    | 1           | Source power limited due to cable supported current                                                                                             |                                                          |
|               |                    | 2           | Source power limited due to insufficient power available while sourcing other ports                                                             |                                                          |
|               |                    | 3           | Source power limited due to insufficient external power                                                                                         |                                                          |
|               |                    | 4           | Source power limited due to Event Flags in place (Event Flags must also be set)                                                                 |                                                          |
|               |                    | 5           | Source power limited due to temperature                                                                                                         |                                                          |
|               |                    | 6...7       | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                 |                                                          |

Table 6.55 SOP Status Data Block (SDB) (Continued)

| Offset (Byte) | Field                     | Description |                                                            |
|---------------|---------------------------|-------------|------------------------------------------------------------|
| 6             | <b>Power State Change</b> | Bit         | Description                                                |
|               |                           | 0...2       | <b>New Power State</b>                                     |
|               |                           |             | Value      Description                                     |
|               |                           |             | 0      Status not supported                                |
|               |                           |             | 1      S0                                                  |
|               |                           |             | 2      Modern Standby                                      |
|               |                           |             | 3      S3                                                  |
|               |                           |             | 4      S4                                                  |
|               |                           |             | 5      S5 (Off with battery, wake events supported)        |
|               |                           |             | 6      G3 (Off with no battery, wake events not supported) |
|               |                           |             | 7 <b>Reserved</b> and <b>Shall</b> be set to zero          |
|               |                           | 3...5       | <b>New Power State indicator</b>                           |
|               |                           |             | Value      Description                                     |
|               |                           |             | 0      Off LED                                             |
|               |                           |             | 1      On LED                                              |
|               |                           |             | 2      Blinking LED                                        |
|               |                           |             | 3      Breathing LED                                       |
|               |                           | 4...7       | <b>Reserved</b> and <b>Shall</b> be set to zero            |
|               |                           | 6...7       | <b>Reserved</b> and <b>Shall</b> be set to zero            |

### 6.5.2.1.1 Internal Temp Field

The **Internal Temp** field reports the instantaneous temperature of a portion of the *Source* or *Sink*.

### 6.5.2.1.2 Present Input Field

The **Present Input** field indicates which supplies are presently powering the *Source* or *Sink*.

The following bits are defined:

- Bit 1: indicates that an external power source is present.
- Bit 2: indicates whether the external unconstrained power source is AC or DC.
- Bit 3: indicates that power is being provided from *Battery*.
- Bit 4: indicates an alternative internal source of power that is not a *Battery*.

### 6.5.2.1.3 Present Battery Input Field

The **Present Battery Input** field indicates which *Battery* or Batteries are presently supplying power to the *Source* or *Sink*. The **Present Battery Input** field is only **Valid** when the **Present Input** field indicates that there is Internal Power from *Battery*.

The upper nibble of the field indicates which Hot Swappable *Battery*/Batteries are supplying power with bit 4 in upper nibble corresponding to *Battery* 4 and bit 7 in the upper nibble corresponding to *Battery* 7 (see [Section 6.5.3, "Get Battery Cap Message"](#) and [Section 6.5.4, "Get Battery Status Message"](#)).

The lower nibble of the field indicates which *Fixed Battery*/Batteries are supplying power with bit 0 in lower nibble corresponding to *Battery* 0 and bit 3 in the lower nibble corresponding to *Battery* 3 (see [Section 6.5.3, "Get Battery Cap Message"](#) and [Section 6.5.4, "Get Battery Status Message"](#)).



#### 6.5.2.1.4 Event Flags Field

The **Event Flags** field returns event flags. The *OTP*, *OVP* and *OCF* event flags **shall** be set when there is an event and **shall** only be cleared when read with the *Get\_Status* Message.

When the **OTP Event** flag is set the **Temperature Status** field **shall** also be set to over temperature.

The **CL/CV Mode** flag is only **Valid** when operating as a Programmable Power Supply and **shall** be **Ignored** otherwise. When the *Source* is operating as a Programmable Power Supply the **CL/CV Mode** flag **shall** be set when operating in **Current Limit** mode (**CL**) and **shall** be cleared when operating in **Constant Voltage** mode (**CV**).

#### 6.5.2.1.5 Temperature Status Field

The **Temperature Status** field returns the current temperature status of the device either: normal, warning or over temperature. When the **Temperature Status** field is set to over temperature the **OTP Event** flag **shall** also be set.

#### 6.5.2.1.6 Power Status Field

The **Power Status** field indicates the current status of a *Source*. A non-zero return of the field indicates *Advertised Source* power is being reduced for either:

- The cable does not support the full *Source* current.
- The *Source* is supplying power to other ports and is unable to provide its full power.
- The external power to the *Source* is insufficient to support full power.
- An Event has occurred that is causing the *Source* to reduce its *Advertised* power.

A *Sink* **shall** set this field to zero.

### 6.5.2.1.7 Power state change

The **Power State Change** field contains two status bytes; the *New Power State* and *New Power State indicator* status bytes.

#### 6.5.2.1.7.1 New power state

The *New Power State* status byte indicates a power state change to one of the specified power states. Any device that supports the ACPI standard system power states **shall** use the ACPI states. For devices that do not support the ACPI power states, the following mapping **should** be used:

- High power (on) state -> S0
- Sleep state -> S3
- Low power (off) state -> S5 or G3

#### 6.5.2.1.7.2 New power state indicator

The *New Power State indicator* status byte defines the host's desired indicator for the specified power state. This indicator allows several possibilities for predefined behaviors that the host can specify to indicate its system power state to the user via the downstream device. The *New Power State indicator* is a "best effort" indicator. If the device cannot provide the requested indicator, then it provides the best indicator that it can. If a Breathing indicator cannot be provided, then a Blinking indicator **should** be provided. If a Blinking indicator cannot be provided, then a constant on indicator **should** be provided.

*New Power State indicator*s in decreasing precedence:

- Breathing
- Blinking
- Constant on
- No indicator

### 6.5.2.2 SOP'/SOP'' Status Message

A **Status Message**, sent in response to a **Get\_Status Message** to **SOP'** or **SOP''**, enables a *Source* or *Sink* to get the present status of the **Cable's Cable Plug(s)**. Typically, a **Get\_Status Message** will be used by the *USB Host* and/or *USB Device* to manage the **Cable's Cable Plug(s)** temperature. The **Status Message** returns a 2-byte Status Data Block (SDB) whose format **Shall** be as shown in [Figure 6.39, "SOP'/SOP'' Status Message"](#) and [Table 6.56, "'SOP'/SOP'' Status Data Block \(SPDB\)'"](#).

Passive *Cable Plugs* **Shall Not** indicate Thermal Shutdown.

Figure 6.39 SOP'/SOP'' Status Message

|                                  |                        |
|----------------------------------|------------------------|
| Extended Header<br>Data Size = 2 | SPDB<br>(2-byte block) |
|----------------------------------|------------------------|

Table 6.56 "SOP'/SOP'' Status Data Block (SPDB)"

| Offset (Byte) | Field                | Value                                                                                      | Description                                                                                                                                                                                              |                                                 |
|---------------|----------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 0             | <b>Internal Temp</b> | Unsigned  | <i>Cable Plug's</i> internal temperature in °C. <ul style="list-style-type: none"><li>0 = feature not supported</li><li>1 = temperature is less than 2°C.</li><li>2...255 = temperature in °C.</li></ul> |                                                 |
| 1             | <b>Flags</b>         | Bit Field                                                                                  | Bit                                                                                                                                                                                                      | Description                                     |
|               |                      |                                                                                            | 0                                                                                                                                                                                                        | Thermal Shutdown                                |
|               |                      |                                                                                            | <b>1...7</b>                                                                                                                                                                                             | <b>Reserved</b> and <b>Shall</b> be set to zero |

#### 6.5.2.2.1 Internal Temp Field

The **Internal Temp** field reports the instantaneous temperature of the plug in °C. The internal temperature **Shall** be monotonic. The *Cable Plug* **Shall** report its internal temperature every **tACTempUpdate**.

#### 6.5.2.2.2 Thermal Shutdown Field

The **Flags** flag **Shall** also be set when the **plug's** internal temperature exceeds the Internal Maximum Temperature reported in the *Active Cable* VDO. Once this bit has been set, it **Shall** remain set and the plug **Shall** remain in Thermal Shutdown until there is a *Hard Reset* or the *Active Cable's* power is removed. The Thermal Shutdown flag **Shall Not** be cleared by a *Cable Reset*.

### 6.5.3 Get\_Battery\_Cap Message

The **Get\_Battery\_Cap** (Get Battery Capabilities) Message is used to request the capability of a Battery present in its Port Partner. The Port **Shall** respond by returning a **Battery\_Capabilities** Message (see [Section 6.5.5, "Battery Capabilities Message"](#)) containing a Battery Capabilities Data Block (BCDB) for the targeted Battery.

The **Get\_Battery\_Cap** Message contains a 1-byte Get Battery Cap Data Block (GBCDB), whose format **Shall** be as shown in [Figure 6.40, "Get\\_Battery\\_Cap Message"](#) and [Table 6.57, "Get Battery Cap Data Block \(GBCDB\)"](#). This block defines for which Battery the request is being made.

The **Data Size** field in the **Get\_Battery\_Cap** Message **Shall** be set to 1.

Figure 6.40 Get\_Battery\_Cap Message

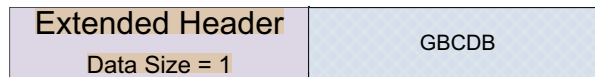


Table 6.57 Get Battery Cap Data Block (GBCDB)

| Offset | Field           | Description                                                                                                                                                                                                                                                               |
|--------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Battery Cap Ref | Number of the Battery indexed from zero: <ul style="list-style-type: none"><li>Values 0...3 represent the Fixed Batteries.</li><li>Values 4...7 represent the Hot Swappable Batteries.</li><li>Values 8...255 are <b>Reserved</b> and <b>Shall Not</b> be used.</li></ul> |

### 6.5.4 Get\_Battery\_Status Message

The **Get\_Battery\_Status** (Get Battery Status) Message is used to request the status of a Battery present in its Port Partner. The Port **Shall** respond by returning a **Battery\_Status** Message (see [Section 6.4.5, "Battery Status Message"](#)) containing a Battery Status Data Object (BSDO) for the targeted Battery.

The **Get\_Battery\_Status** Message contains a 1-byte Get Battery Status Data Block (GBSDB) whose format **Shall** be as shown in [Figure 6.41, "Get\\_Battery\\_Status Message"](#) and [Table 6.58, "Get Battery Status Data Block \(GBSDB\)"](#). This block contains details of the requested Battery. The **Data Size** field in the **Get\_Battery\_Status** Message **Shall** be set to 1.

Figure 6.41 Get\_Battery\_Status Message

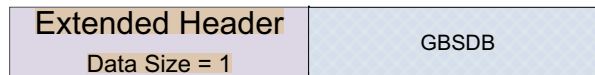


Table 6.58 Get Battery Status Data Block (GBSDB)

| Offset | Field              | Description                                                                                                                                                                                                                                                               |
|--------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | Battery Status Ref | Number of the Battery indexed from zero: <ul style="list-style-type: none"><li>Values 0...3 represent the Fixed Batteries.</li><li>Values 4...7 represent the Hot Swappable Batteries.</li><li>Values 8...255 are <b>Reserved</b> and <b>Shall Not</b> be used.</li></ul> |

## 6.5.5 Battery\_Capabilities Message

The **Battery\_Capabilities** Message is sent in response to a **Get\_Battery\_Cap** Message. The **Battery\_Capabilities** Message contains one Battery Capability Data Block (BCDB) for one of the Batteries its supports as reported by **Number of Batteries/Battery Slots** field in the **Source\_Capabilities\_Extended** Message. The returned BCDB **Shall** correspond to the **Battery** requested in the **Battery Cap Ref** field contained in the **Get\_Battery\_Cap** Message.

The **Battery\_Capabilities** Message returns a 9-byte BCDB whose format **Shall** be as shown in [Figure 6.42](#), "[Battery\\_Capabilities Message](#)" and [Table 6.59](#), "[Battery Capability Data Block \(BCDB\)](#)".

Figure 6.42 Battery\_Capabilities Message

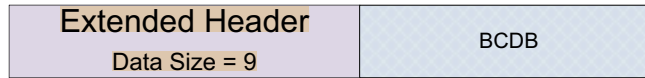


Table 6.59 Battery Capability Data Block (BCDB)"

| Offset (Byte) | Field                             | Description                                                                                                                                                                                                                 |                           |                                     |             |   |                           |                                     |       |     |          |
|---------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------|-------------|---|---------------------------|-------------------------------------|-------|-----|----------|
| 0             | VID                               | Vendor ID (assigned by the USB-IF)                                                                                                                                                                                          |                           |                                     |             |   |                           |                                     |       |     |          |
| 2             | PID                               | Product ID (assigned by the manufacturer)                                                                                                                                                                                   |                           |                                     |             |   |                           |                                     |       |     |          |
| 4             | Battery Design Capacity           | Battery's design capacity in 0.1 WH<br>Note: <ul style="list-style-type: none"><li>0x0000 = Battery not present</li><li>0xFFFF = design capacity unknown</li></ul>                                                          |                           |                                     |             |   |                           |                                     |       |     |          |
| 6             | Battery Last Full Charge Capacity | Battery's last full charge capacity in 0.1 WH<br>Note: <ul style="list-style-type: none"><li>0x0000 = Battery not present</li><li>0xFFFF = last full charge capacity unknown</li></ul>                                      |                           |                                     |             |   |                           |                                     |       |     |          |
| 8             | Battery Type                      | <table><tr><th>Bit</th><th>Field</th><th>Description</th></tr><tr><td>0</td><td>Invalid Battery Reference</td><td>Invalid Battery reference when set.</td></tr><tr><td>1...7</td><td>---</td><td>Reserved</td></tr></table> | Bit                       | Field                               | Description | 0 | Invalid Battery Reference | Invalid Battery reference when set. | 1...7 | --- | Reserved |
|               |                                   | Bit                                                                                                                                                                                                                         | Field                     | Description                         |             |   |                           |                                     |       |     |          |
|               |                                   | 0                                                                                                                                                                                                                           | Invalid Battery Reference | Invalid Battery reference when set. |             |   |                           |                                     |       |     |          |
| 1...7         | ---                               | Reserved                                                                                                                                                                                                                    |                           |                                     |             |   |                           |                                     |       |     |          |

### 6.5.5.1 Vendor ID (VID)

The **VID** field **Shall** contain the manufacturer **VID** associated with the **Battery**, as assigned by the USB-IF, or 0xFFFF in the case that no such **VID** exists.

If the **Battery Cap Ref** field in the **Get\_Battery\_Cap** Message is **Invalid**, the **VID** field **Shall** be 0xFFFF.

### 6.5.5.2 Product ID (PID)

The following rules apply to the **PID** field. When the **VID**:

-  Belongs to the *Battery* vendor the **PID** field **Shall** contain the *Battery's* 16-bit product identifier designated by the *Battery* vendor.
- Belongs to the *Device* vendor the **PID** field **Shall** contain the *Battery's* 16-bit product identifier designated by the *Device* vendor.
- Is 0xFFFF the **PID** field **Shall** be set to 0x0000.

### 6.5.5.3 Battery Design Capacity Field

The **Battery Design Capacity** field **Shall** return the *Battery's* design capacity in tenths of WH. If the *Battery* is Hot Swappable and is not present, the **Battery Design Capacity** field **Shall** be set to zero. If the *Battery* is unable to report its Design Capacity, the **Battery Design Capacity** field **Shall** be set to 0xFFFF.

### 6.5.5.4 Battery Last Full Charge Capacity Field

The **Battery Last Full Charge Capacity** field **Shall** contain the *Battery's* last full charge capacity in tenths of WH. If the *Battery* is Hot Swappable and is not present, the **Battery Last Full Charge Capacity** field **Shall** be set to zero. If the *Battery* is unable to report its Design Capacity, the **Battery Last Full Charge Capacity** field **Shall** be set to 0xFFFF.

### 6.5.5.5 Battery Type Field

The **Battery Type** field is used to report additional information about the *Battery's Capabilities*.

#### 6.5.5.5.1 Invalid Battery Reference

The **Invalid Battery Reference** bit **Shall** be set when the *Get\_Battery\_Cap* Message contains a reference to a *Battery* that does not exist.

## 6.5.6 Get\_Manufacturer\_Info Message

The **Get\_Manufacturer\_Info** (Get Manufacturer Info) Message is sent by a *Port* to request manufacturer specific information relating to its *Port Partner*, *Cable Plug* or of a *Battery* behind a *Port*. The *Port* **shall** respond by returning a **Manufacturer\_Info** Message ([Section 6.5.7, "Manufacturer\\_Info Message"](#)) containing a Manufacturer Info Data Block (MIDB). Support for this feature by the *Cable Plug* is **Optional Normative**.

The **Get\_Manufacturer\_Info** Message contains a 2-byte Get Manufacturer Info Data Block (GMIDB). This block defines whether it is the *Device* or *Battery* manufacturer information being requested and for which *Battery* the request is being made.

The **Get\_Manufacturer\_Info** Message returns a GMIDB whose format **shall** be as shown in [Figure 6.43, "Get\\_Manufacturer\\_Info Message"](#) and [Table 6.60, "Get\\_Manufacturer\\_Info Data Block \(GMIDB\)"](#).

Figure 6.43 Get\_Manufacturer\_Info Message

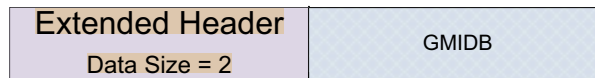


Table 6.60 Get Manufacturer Info Data Block (GMIDB)

| Offset | Field                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <i>Manufacturer Info Target</i> | <ul style="list-style-type: none"><li>0: <i>Port/Cable Plug</i></li><li>1: <i>Battery</i></li><li>255...2: <b>Reserved</b> and <b>Shall Not</b> be used.</li></ul>                                                                                                                                                                                                                                                                                                                              |
| 1      | <i>Manufacturer Info Ref</i>    | <p>If the <b>Manufacturer Info Target</b> field is <i>Battery</i> (01b) the <b>Manufacturer Info Ref</b> field <b>shall</b> contain the <i>Battery</i> number reference which is the number of the <i>Battery</i> indexed from zero:</p> <ul style="list-style-type: none"><li>Values 0...3 represent the <i>Fixed Batteries</i>.</li><li>Values 4...7 represent the <i>Hot Swappable Batteries</i>.</li></ul> <p>Otherwise, this field is <b>Reserved</b> and <b>Shall</b> be set to zero.</p> |



## 6.5.7 Manufacturer\_Info Message

The **Manufacturer\_Info** Message **Shall** be sent in response to a **Get\_Manufacturer\_Info** Message. The **Manufacturer\_Info** Message contains the USB VID and the Vendor's PID to identify the device or Battery and the device or Battery's manufacturer byte array in a variable length Data Block of up to **MaxExtendedMsgLegacyLen**.

The **Manufacturer\_Info** Message returns a Manufacturer Info Data Block (MIDB) whose format **Shall** be as shown in [Figure 6.44, "Manufacturer\\_Info Message"](#) and [Table 6.61, "Manufacturer Info Data Block \(MIDB\)"](#).

Figure 6.44 Manufacturer\_Info Message

| Extended Header   | MIDB |
|-------------------|------|
| Data Size = 5..26 |      |

Table 6.61 Manufacturer Info Data Block (MIDB)

| Offset | Field                      | Description                                                                                                                                                                                                                                                                                   |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | <b>VID</b>                 | Vendor ID (assigned by the USB-IF)                                                                                                                                                                                                                                                            |
| 2      | <b>PID</b>                 | Product ID (assigned by the manufacturer)                                                                                                                                                                                                                                                     |
| 4      | <b>Manufacturer String</b> | Vendor defined null terminated string of 0...21 characters. If the <b>Manufacturer Info Target</b> field or <b>Manufacturer Info Ref</b> field in the <b>Get_Manufacturer_Info</b> Message is unrecognized the field <b>Shall</b> return a null terminated ASCII text string "Not Supported". |

### 6.5.7.1 Vendor ID (VID)

If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message is associated with this **Port/Cable Plug**, the **VID** field **Shall** contain:

- The manufacturer's VID associated with the **Port/Cable Plug**, as defined by the USB-IF, or
- 0xFFFF in the case that the vendor does not have a VID.

If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message is associated with a **Device** that has a USB data interface, the **Device** **Shall** report the same VID as the idVendor in the Standard Device Descriptor (see [\[USB 2.0\]](#) and [\[USB 3.2\]](#)).

If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message is associated with a **Battery**, the **VID** field **Shall** contain:

- The manufacturer VID associated with the **Battery** specified, as defined by the USB-IF, or
- 0xFFFF in the case that the vendor does not have a VID.

If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message:

- Is **Invalid**, this **VID** field **Shall** be 0xFFFF.
- Is **Battery** (01b) and the **Manufacturer Info Ref** field is **Invalid**, the **VID** field **Shall** be 0xFFFF.

### 6.5.7.2 Product ID (PID)

✖ If the **VID** is 0xFFFF, the **PID** field **Shall** contain 0x0000.

Otherwise:

- If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message is associated with this **Port/Cable Plug**, the PID field **Shall** contain the device's 16-bit product identifier designated by the device vendor.
- If the in **Manufacturer Info Target** field in the **Get\_Manufacturer\_Info** Message is associated with a **Battery**:
  - And the VID belongs to the *Battery* vendor, the **PID** field **Shall** contain the **Battery's** 16-bit product identifier designated by the *Battery* vendor.
  - And the VID belongs to the *Device* vendor, the **PID** field **Shall** contain the **Battery's** 16-bit product identifier designated by the *Device* vendor.

### 6.5.7.3 Manufacturer String

The **Manufacturer String** field **Shall** contain the device's or *Battery's* manufacturer string as defined by the vendor.

If the **Manufacturer Info Target** field or **Manufacturer Info Ref** field in the **Get\_Manufacturer\_Info** Message is unrecognized the field **Shall** return a null terminated ASCII text string "Not Supported".

## 6.5.8 Security Messages

The authentication process between *Port Partners* or a *Port* and *Cable Plug* is fully described in [\[USBTypeCAuthentication 1.0\]](#). This specification describes two *Extended Messages* used by the authentication process when applied to PD.

In the authentication process described in [\[USBTypeCAuthentication 1.0\]](#) there are three basic exchanges that serve to:

- Get the *Port* or *Cable Plug's* certificates.
- Get the *Port* or *Cable Plug's* digest.
- Challenge the *Port Partner* or *Cable Plug*.

Certificates are used to convey information, attested to by a signer, which attests to the *Port Partner's* or *Cable Plug's* authenticity. The *Port's* or *Cable Plug's* certificates are needed when a *Port* encounters a *Port Partner* or *Cable Plug* it has not been *Attached* to before. To minimize calculations after the initial *Attachment*, a *Port* can also use a digest consisting of hashes of the certificates rather than the certificates themselves. Once the *Port* has the certificates and has calculated the hashes, it stores the hashes and uses the digest in future exchanges. After the *Port* gets the certificates or digest, it challenges its *Port Partner* or the *Cable Plug* to detect replay attacks.

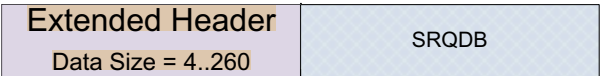
For further details refer to [\[USBTypeCAuthentication 1.0\]](#).

### 6.5.8.1 Security\_Request

The *Security\_Request* Message is used by a *Port* to pass a security data structure to its *Port Partner* or a *Cable Plug*.

The *Security\_Request* Message contains a Security Request Data Block (SRQDB) whose format **shall** be as shown in [Figure 6.45, "Security\\_Request Message"](#). The contents of the SRQDB and its use are defined in [\[USBTypeCAuthentication 1.0\]](#).

Figure 6.45 Security\_Request Message

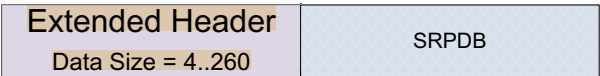


### 6.5.8.2 Security\_Response

The *Security\_Response* Message is used by a *Port* or *Cable Plug* to pass a security data structure to the *Port* that sent the *Security\_Request* Message.

The *Security\_Response* Message contains a Security Response Data Block (SRPDB) whose format **shall** be as shown in [Figure 6.46, "Security\\_Response Message"](#). The contents of the SRPDB and its use are defined in [\[USBTypeCAuthentication 1.0\]](#).

Figure 6.46 Security\_Response Message



# 6.5.9 Firmware Update Messages

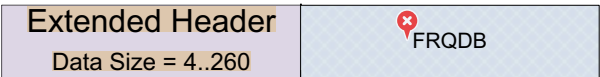
The firmware update process between *Port Partners* or a *Port* and *Cable Plug* is fully described in [\[USBPDFirmwareUpdate 1.0\]](#). This specification describes two *Extended Messages* used by the firmware update process when applied to PD.

## 6.5.9.1 Firmware\_Update\_Request

The *Firmware\_Update\_Request* Message is used by a *Port* to pass a firmware update data structure to its *Port Partner* or a *Cable Plug*.

The *Firmware\_Update\_Request* Message contains a Firmware Update Request Data Block (FRQDB) whose format **Shall** be as shown in [Figure 6.47, "Firmware\\_Update\\_Request Message"](#). The contents of the FRQDB and its use are defined in [\[USBPDFirmwareUpdate 1.0\]](#).

Figure 6.47 Firmware\_Update\_Request Message

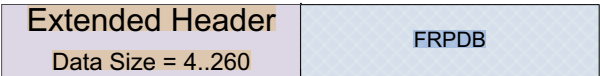


## 6.5.9.2 Firmware\_Update\_Response

The *Firmware\_Update\_Response* Message is used by a *Port* or *Cable Plug* to pass a firmware update data structure to the *Port* that sent the *Firmware\_Update\_Request* Message.

The *Firmware\_Update\_Response* Message contains a Firmware Update Response Data Block (FRPDB) whose format **Shall** be as shown in [Figure 6.48, "Firmware\\_Update\\_Response Message"](#). The contents of the FRPDB and its use are defined in [\[USBPDFirmwareUpdate 1.0\]](#).

Figure 6.48 Firmware\_Update\_Response Message



## 6.5.10 PPS\_Status Message

The **PPS\_Status** Message **Shall** be sent in response to a **Get\_PPS\_Status** Message. The **PPS\_Status** Message enables a *Sink* to query the *Source* to get additional information about its operational state. The **Get\_PPS\_Status** Message and the **PPS\_Status** Message **Shall** only be supported when the **Alert** Message is also supported.

The **PPS\_Status** Message **Shall** return a 4-byte PPS Status Data Block (PPSSDB) whose format **Shall** be as shown in [Figure 6.49, "PPS\\_Status Message"](#) and [Table 6.62, "PPS Status Data Block \(PPSSDB\)"](#).

Figure 6.49 PPS\_Status Message

| Extended Header | PPSSDB              |
|-----------------|---------------------|
| Data Size = 4   | (4-byte Data Block) |

Table 6.62 PPS Status Data Block (PPSSDB)

| Offset (Byte) | Field                  | Description |                                                                                                                                                                          |
|---------------|------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | <b>Output Voltage</b>  | 2           | <i>Source's</i> output voltage in 20mV units.<br>When set to 0xFFFF, the <i>Source</i> does not support this field.                                                      |
| 2             | <b>Output Current</b>  | 1           | <i>Source's</i> output current in 50mA units.<br>When set to 0xFF, the <i>Source</i> does not support this field.                                                        |
| 3             | <b>Real Time Flags</b> | Bit         | Description                                                                                                                                                              |
|               |                        | 0           | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                                          |
|               |                        | 1...2       | <b>PTF</b> <ul style="list-style-type: none"><li>PTF: 00 – Not Supported</li><li>PTF: 01 – Normal</li><li>PTF: 10 – Warning</li><li>PTF: 11 – Over temperature</li></ul> |
|               |                        | 3           | <b>OMF</b><br>OMF (Operating Mode Flag) is set when operating in <b>Current Limit</b> mode and cleared when operating in <b>Constant Voltage</b> mode.                   |
|               |                        | 4...7       | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                                          |

### 6.5.10.1 Output Voltage Field

The **Output Voltage** field **Shall** return the *Source's* output voltage at the time of the request. The output voltage is measured either at the *Source's* receptacle or, if the *Source* has a captive cable, where the voltage is applied to the cable.

The measurement accuracy **Shall** be +/-3% rounded to the nearest 20mV in *SPR PPS Mode*.

If the *Source* does not support the **Output Voltage** field, the field **Shall** be set to 0xFFFF.

### 6.5.10.2 Output Current Field

The **Output Current** field **Shall** return the *Source's* output current at the time of the request measured at the *Source's* receptacle.

The measurement accuracy **Shall** be +/-150mA.

If the *Source* does not support the **Output Current** field, the field **Shall** be set to 0xFF.

### 6.5.10.3 Real Time Flags Field



Real Time flags provide a real-time indication of the *Source's* operating state:

- The **PTF** (Present Temperature Flag) **Shall** provide a real-time indication of the *Source's* internal thermal status. If the PTF is not supported, it will be set to **zero**.
  - **Normal** indicates that the *Source* is operating within its normal thermal envelope.
  - **Warning** indicates that the *Source* is over-heating but is not in imminent danger of shutting down.
  - **Over Temperature** indicates that the *Source* is over heated and will shut down soon or has already shutdown and has sent the **OTP Event** flag in an **Alert Message**.
- The **OMF** (Operating Mode Flag) **Shall** provide a real-time indication of the *SPR PPS Source's* operating mode. When set, the *Source* is operating in **Current Limit** mode; when cleared it is operating **Constant Voltage** mode. This bit **Shall** be set to zero when not in *SPR PPS Mode*.

# 6.5.11 Country\_Codes Message

The **Country\_Codes** Message **Shall** be sent in response to a **Get\_Country\_Codes** Message. The **Country\_Codes** Message enables a **Port** to query its **Port Partner** to get a list of alpha-2 country codes as defined in **[ISO 3166]** for which the **Port Partner** has country specific information.

The **Country\_Codes** Message **Shall** contain a 4...26-byte Country Code Data Block (CCDB) whose format **Shall** be as shown in **Figure 6.50, "Country\_Codes Message"** and **Table 6.63, "Country Codes Data Block (CCDB)"**.

Figure 6.50 Country\_Codes Message



Table 6.63 Country Codes Data Block (CCDB)

| Offset              | Field        | Description                            |                              |                                                                           |
|---------------------|--------------|----------------------------------------|------------------------------|---------------------------------------------------------------------------|
| 0                   | Length       | Number of country codes in the Message |                              |                                                                           |
| 1                   | Reserved     | Shall be set to zero.                  |                              |                                                                           |
| 2...<br>Length * 2n | Country Code | Offset                                 | Field                        | Description                                                               |
|                     |              | 2                                      | 1 <sup>st</sup> Country Code | First character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b>  |
|                     |              | 3                                      |                              | Second character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b> |
|                     |              | 4                                      | 2 <sup>nd</sup> Country Code | First character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b>  |
|                     |              | 5                                      |                              | Second character of the Alpha-2 Country Code defined by <b>[ISO 3166]</b> |
|                     |              | ...                                    | ...                          |                                                                           |
|                     |              | Length * 2n                            | n <sup>th</sup> Country Code |                                                                           |

## 6.5.11.1 Country Code Field

The **Country Code** field **Shall** contain **Length** Country Codes in the Alpha-2 Country Code defined by **[ISO 3166]**.



## 6.5.12 Country\_Info Message

The **Country\_Info** Message **Shall** be sent in response to a **Get\_Country\_Info** Message. The **Country\_Info** Message enables a *Port* to get additional country specific information from its *Port Partner*.

The **Country\_Info** Message **Shall** contain a 4...26-byte Country Info Data Block (CIDB) whose format **Shall** be as shown in [Figure 6.51, "Country\\_Info Message"](#) and [Table 6.64, "Country Info Data Block \(CIDB\)"](#).

Figure 6.51 Country\_Info Message



Table 6.64 Country Info Data Block (CIDB)

| Offset | Field                 | Size                                                                                                       |
|--------|-----------------------|------------------------------------------------------------------------------------------------------------|
| 0      | Country Code          | First character of the Alpha-2 Country Code received in the corresponding <b>Get_Country_Info</b> Message. |
| 1      |                       | Second character of the Alpha-2 Country Code received in the corresponding <b>Get_Country_Info</b> Message |
| 2...3  | Reserved              | <b>Shall</b> be set to zero.                                                                               |
| 4      | Country Specific Data | 1...22 bytes of content defined by the country's authority.                                                |

### 6.5.12.1 Country Code Field

The **Country Code** field **Shall** contain the Alpha-2 Country Code received in the corresponding **Get\_Country\_Info** Message.

### 6.5.12.2 Country Specific Data Field

The **Country Specific Data** field **Shall** contain content defined by and formatted in a manner determined by an official agency of the country indicated in the Country Code field.

If the Country Code field in the **Get\_Country\_Info** Message is unrecognized then Country Specific Data field **Shall** return the null terminated ASCII text string "Unsupported Code".

## 6.5.13 Sink\_Capabilities\_Extended Message

The *Sink\_Capabilities\_Extended* Message **Shall** be sent in response to a *Get\_Sink\_Cap\_Extended* Message. The *Sink\_Capabilities\_Extended* Message enables a Sink or a DRP to inform the Source about its *Capabilities* as a Sink.

The *Sink\_Capabilities\_Extended* Message **Shall** return a 24-byte Sink Capabilities Extended Data Block (SKEDB) whose format **Shall** be as shown in [Figure 6.52, "Sink\\_Capabilities\\_Extended Message"](#) and [Table 6.65, "Sink Capabilities Extended Data Block \(SKEDB\)"](#).

Figure 6.52 Sink\_Capabilities\_Extended Message



Table 6.65 Sink Capabilities Extended Data Block (SKEDB)

| Offset (Byte) | Field                     | Size (Bytes) | Type      | Description                                                                                                                                                                         |                                                                                                                                                                            |
|---------------|---------------------------|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | VID                       | 2            | Numeric   | Vendor ID (assigned by the USB-IF)                                                                                                                                                  |                                                                                                                                                                            |
| 2             | PID                       | 2            | Numeric   | Product ID (assigned by the manufacturer)                                                                                                                                           |                                                                                                                                                                            |
| 4             | XID                       | 4            | Numeric   | Value provided by the USB-IF assigned to the product                                                                                                                                |                                                                                                                                                                            |
| 8             | FW Version                | 1            | Numeric   | Firmware version number                                                                                                                                                             |                                                                                                                                                                            |
| 9             | HW Version                | 1            | Numeric   | Hardware version number                                                                                                                                                             |                                                                                                                                                                            |
| 10            | SKEDB Version             | 1            | Numeric   | SKEDB Version (not the specification Version): <ul style="list-style-type: none"><li>Version 1.0 = 1</li></ul> Values 0 and 2-255 are <b>Reserved</b> and <b>Shall Not</b> be used. |                                                                                                                                                                            |
| 11            | Load Step                 | 1            | Bit Field | Bit                                                                                                                                                                                 | Description                                                                                                                                                                |
|               |                           |              |           | 0...1                                                                                                                                                                               | <ul style="list-style-type: none"><li>00b: 150mA/μs Load Step (default)</li><li>01b: 500mA/μs Load Step</li></ul> 11b...10b: <b>Reserved</b> and <b>Shall Not</b> be used. |
|               |                           |              |           | 2...7                                                                                                                                                                               | <b>Reserved</b> and <b>Shall</b> be set to zero                                                                                                                            |
| 12            | Sink Load Characteristics | 2            | Bit Field | Bit                                                                                                                                                                                 | Description                                                                                                                                                                |
|               |                           |              |           | 0...4                                                                                                                                                                               | Percent overload in 10% increments.<br>Values higher than 25 (11001b) are clipped to 250%. 00000b is the default.                                                          |
|               |                           |              |           | 5...10                                                                                                                                                                              | Overload period in 20ms when bits 0...4 non-zero.                                                                                                                          |
|               |                           |              |           | 1...14                                                                                                                                                                              | Duty cycle in 5% increments when bits 0...4 are non-zero.                                                                                                                  |
|               |                           |              |           | 15                                                                                                                                                                                  | Can tolerate VBUS voltage droop                                                                                                                                            |
| 14            | Compliance                | 1            | Bit Field | Bit                                                                                                                                                                                 | Description                                                                                                                                                                |
|               |                           |              |           | 0                                                                                                                                                                                   | Requires LPS Source when set                                                                                                                                               |
|               |                           |              |           | 1                                                                                                                                                                                   | Requires PS1 Source when set                                                                                                                                               |
|               |                           |              |           | 2                                                                                                                                                                                   | Requires PS2 Source when set                                                                                                                                               |
|               |                           |              |           | 3...7                                                                                                                                                                               | Reserved and Shall be set to zero                                                                                                                                          |

Table 6.65 Sink Capabilities Extended Data Block (SKEDB) (Continued)

| Offset (Byte) | Field                    | Size (Bytes) | Type      | Description                                                                                                                                                                                                                               |                                                                                                                                             |
|---------------|--------------------------|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 15            | Touch Temp               | 1            | Value     | Temperature conforms to: <ul style="list-style-type: none"><li>0 = Not applicable</li><li>1 = [IEC 60950-1] (default)</li><li>2 = [IEC 62368-1] TS1</li><li>3 = [IEC 62368-1] TS2</li></ul> <b>Note:</b> All other values <b>Reserved</b> |                                                                                                                                             |
| 16            | Battery Info             | 1            | Byte      | Upper Nibble = Number of Hot Swappable Battery Slots (0...4)<br>Lower Nibble = Number of Fixed Batteries (0...4)                                                                                                                          |                                                                                                                                             |
| 17            | Sink Modes               | 1            | Bit Field | Bit                                                                                                                                                                                                                                       | Description                                                                                                                                 |
|               |                          |              |           | 0                                                                                                                                                                                                                                         | PPS charging supported                                                                                                                      |
|               |                          |              |           | 1                                                                                                                                                                                                                                         | VBUS powered                                                                                                                                |
|               |                          |              |           | 2                                                                                                                                                                                                                                         | AC Supply powered                                                                                                                           |
|               |                          |              |           | 3                                                                                                                                                                                                                                         | Battery powered                                                                                                                             |
|               |                          |              |           | 4                                                                                                                                                                                                                                         | Battery essentially unlimited                                                                                                               |
|               |                          |              |           | 5                                                                                                                                                                                                                                         | AVS Support                                                                                                                                 |
|               |                          |              |           | 6...7                                                                                                                                                                                                                                     | Reserved and Shall be set to zero                                                                                                           |
| 18            | SPR Sink Minimum PDP     | 1            | Byte      | Bit                                                                                                                                                                                                                                       | Description                                                                                                                                 |
|               |                          |              |           | 0...6                                                                                                                                                                                                                                     | The PDP of the Source that the Sink requires to operate at its lowest functionality without consuming power from its Battery if it has one. |
|               |                          |              |           | 7                                                                                                                                                                                                                                         | Reserved and Shall be set to zero                                                                                                           |
| 19            | SPR Sink Operational PDP | 1            | Byte      | Bit                                                                                                                                                                                                                                       | Description                                                                                                                                 |
|               |                          |              |           | 0...6                                                                                                                                                                                                                                     | The PDP of the Source that the Sink requires to operate at its normal functionality.                                                        |
|               |                          |              |           | 7                                                                                                                                                                                                                                         | Reserved and Shall be set to zero                                                                                                           |
| 20            | SPR Sink Maximum PDP     | 1            | Byte      | Bit                                                                                                                                                                                                                                       | Description                                                                                                                                 |
|               |                          |              |           | 0...6                                                                                                                                                                                                                                     | The maximum PDP the Sink will ever request.                                                                                                 |
|               |                          |              |           | 7                                                                                                                                                                                                                                         | Reserved and Shall be set to zero                                                                                                           |
| 21            | EPR Sink Minimum PDP     | 1            | Byte      | The PDP of the Source that the EPR Sink requires to operate at its lowest functionality without consuming power from its Battery if it has one.                                                                                           |                                                                                                                                             |
| 22            | EPR Sink Operational PDP | 1            | Byte      | The PDP of the Source that the EPR Sink requires to operate at its normal functionality.                                                                                                                                                  |                                                                                                                                             |
| 23            | EPR Sink Maximum PDP     | 1            | Byte      | The maximum PDP that the EPR Sink will ever request.                                                                                                                                                                                      |                                                                                                                                             |

### 6.5.13.1 Vendor ID (VID) Field

The **VID** field **Shall** contain the 16-bit *Vendor ID (VID)* assigned to the Sink's vendor by the USB-IF. If the vendor does not have a *VID*, the **VID** field **Shall** be set to 0xFFFF. Devices that have a USB data interface **Shall** report the same *VID* as the *idVendor* in the Standard Device Descriptor (see [USB 2.0] and [USB 3.2]).

### 6.5.13.2 Product ID (PID) Field

The **PID** field **Shall** contain the 16-bit *Product ID (PID)* assigned by the Sink's vendor. Devices that have a USB data interface **Shall** report the same *PID* as the *idProduct* in the Standard Device Descriptor (see [USB 2.0] and [USB 3.2]).

### 6.5.13.3 XID Field

The **XID** field **shall** contain the 32-bit XID provided by the USB-IF to the vendor who in turns assigns it to a product. If the vendor does not have an XID, then it **shall** return zero in this field (see [\[USB 2.0\]](#) and [\[USB 3.2\]](#)).

### 6.5.13.4 Firmware Version Field

The **FW Version** field **shall** contain an 8-bit firmware version number assigned to the device by the vendor.

### 6.5.13.5 Hardware Version Field

The **HW Version** field **shall** contain an 8-bit hardware version number assigned to the device by the vendor.

### 6.5.13.6 SKEDB Version Field

The **SKEDB Version** field contains the version level of the SKEDB. Currently only Version 1 is defined.

### 6.5.13.7 Load Step Field

The **Load Step** field contains bits indicating the Load Step Slew Rate and Magnitude that this *Sink* prefers. See [Section 7.1.12.1, "Voltage Regulation Field"](#) for further details.

### 6.5.13.8 Sink Load Characteristics Field

The *Sink* **shall** report its preferred load characteristics in the **Sink Load Characteristics** field. Regardless of this value, in operation its load **shall Not** exceed the *Capabilities* reported in the **Source\_Capabilities\_Extended Message**.

### 6.5.13.9 Compliance Field

The **Compliance** field **shall** contain the types of *Sources* the *Sink* has been tested and certified with (see [Section 7.1.12.3, "Compliance Field"](#)).

### 6.5.13.10 Touch Temp

The **Touch Temp** field **shall** report the IEC standard used to determine the surface temperature of the *Sink's* enclosure. Safety limits for the *Sink's* touch temperature are set in applicable product safety standards (e.g., [\[IEC 60950-1\]](#) or [\[IEC 62368-1\]](#)). The *Sink* **May** report when its touch temperature performance conforms to the TS1 or TS2 limits described in [\[IEC 62368-1\]](#).

### 6.5.13.11 Battery Info

The **Battery Info** field **shall** report the number of *Fixed Batteries* and *Hot Swappable Battery* slots the *Sink* supports. This field **shall** independently report the number of *Battery Slots* and the number of *Fixed Batteries*. The information reported in the **Battery Info** field **shall** match that reported in the **Number of Batteries/Battery Slots** field of the **Source\_Capabilities\_Extended Message**.

A *Sink* **shall** have no more than 4 *Fixed Batteries* and no more than 4 *Battery Slots*.

*Fixed Batteries* **shall** be numbered consecutively from 0 to 3. The number assigned to a given *Fixed Battery* **shall Not** change between *Attach* and *Detach*.

*Battery Slots* **shall** be numbered consecutively from 4 to 7. The number assigned to a given *Battery Slot* **shall Not** change between *Attach* and *Detach*.

### 6.5.13.12 Sink Modes

\* The **Sink Modes** bit field **shall** identify the charging **Capabilities** and the power sources that can be used by the **Sink**. When bit 0 is set, the **Sink** has the ability to use a **PPS Source** for fast charging.

The source of power a **Sink** can use:

- When bit 1 is set, the **Sink** has the ability to be sourced by **V<sub>BUS</sub>**.
- When bit 2 is set, the **Sink** has the ability to be sourced by an **AC Supply**.
- When bit 3 is set, the **Sink** has the ability to be sourced by a **Battery**.
- When bit 4 is set, the **Sink** has the ability to be sourced by a **Battery** with essentially infinite energy (e.g., a car battery).

Bits 1-4 **may** be set independently of one another. The combination indicates what sources of power the **Sink** can utilize. For example, some **Sinks** are only powered by a **Battery** (e.g., an automobile battery) rather than the more common **AC Supply** and some **Sinks** are only powered from **V<sub>BUS</sub>** or **V<sub>CONN</sub>**.

When bit 5 is set, the **Sink** has the ability to support **AVS**.

### 6.5.13.13 SPR Sink Minimum PDP

The **SPR Sink Minimum PDP** field **shall** contain the minimum power **Source PDP** needed by the **Sink**, rounded up to the next integer, to operate at its lowest level of functionality without requiring power from its **Battery** if present. **Battery** charging may be an opportunistic feature, however this **PDP** **should** be designed for basic functionality, not for charging. The **SPR Sink Minimum PDP** field **shall** be less than or equal to the **SPR Sink Operational PDP**. The value is used by the **Source** to determine whether or not it has sufficient power to minimally support the **Attached Sink**. If the **Sink** is **EPR Capable** and is unable to operate at **PDPs** less than 100W, it **shall** set this field to the minimum power to sustain PD communication.

If the **Sink** is self-powered, such that it doesn't need power from a **Source**, then it **shall** set this field to zero.

The **SPR Sink Minimum PDP** is used to indicate to **Shared Capacity Chargers** the power that **should** be delivered to the **Sink** to guarantee at least basic functionality for the end user.

Possible examples of **SPR Sink Minimum PDP** could be:

- The minimum power a wireless **Charger** would require in order to detect, and deliver the minimum required amount of power to the attached device.
- The power required to have basic functionality by a **Batteryless Sink**,
- On a device with a **Battery**, it can power the minimum functionality of the device

### 6.5.13.14 SPR Sink Operational PDP

The **SPR Sink Operational PDP** field **shall** contain the **Source PDP** that the manufacturer recommends for the normal functionality of the **Sink**, rounded up to the next integer. This corresponds to the **PDP Rating of Sources** that the **Sink** is designed to operate with (See [Section 10.3.2, "Normative Sink Rules"](#)). The **SPR Sink Operational PDP** field **shall** be sufficient to operate all the **Sink's** functional modes normally AND charge the **Sink's Battery** if present. For **Sinks** with a **Battery(s)**, the **SPR Sink Operational PDP** field **shall** correspond to the **PDP Rating** of the **Charger** shipped with the **Sink** or the recommended **Charger's PDP Rating**. If the **Sink** is **EPR Capable** and is unable to operate at **PDPs** less than 100W, it **shall** set the **SPR Sink Minimum PDP** field to the minimum power to sustain PD communication.

If the **Sink** is self-powered, such that it doesn't need power from a **Source**, then it **shall** set this field to zero.

The **SPR Sink Operational PDP** is used to indicate to **Shared Capacity Chargers** that at this power level the user is not expected to receive any performance warning related to the power being supplied to the **Sink**.

### 6.5.13.15 SPR Sink Maximum PDP

The **SPR Sink Maximum PDP** field **Shall** contain the highest PDP the Sink will ever request under any operating condition, rounded up to the next integer, including charging its Battery if present. The **SPR Sink Maximum PDP** field **Shall Not** be less than the **SPR Sink Operational PDP** field, but **May** be the same. The value is used by the Source to determine the maximum amount of power it has to budget for the Attached Sink. If the Sink is EPR Capable and is unable to operate at PDPs less than 100W, it **Shall** set this field to the minimum power to sustain PD communication.

If the Sink is self-powered, such that it doesn't need power from a Source, then it **Shall** set this field to zero.

### 6.5.13.16 EPR Sink Minimum PDP

The **EPR Sink Minimum PDP** field **Shall** contain the Source PDP needed by an EPR Sink, rounded up to the next integer, to operate at its lowest level of functionality without requiring power from its Battery, if present. Battery charging may be an opportunistic feature, however this PDP **Should** be designed for basic functionality, not for charging. The **EPR Sink Minimum PDP** field **Shall** be less than or equal to the **EPR Sink Operational PDP** field value. The value is used by the Source to determine whether or not it has sufficient power to minimally support the Attached Sink. If the Sink is not EPR Capable, or if the Sink is self-powered, such that it doesn't need power from a Source, this field **Shall** be set to zero.

The **EPR Sink Minimum PDP** is used to indicate to Shared Capacity Chargers the power that **Should** be delivered to the Sink to guarantee at least basic functionality for the end user.

Possible examples of **EPR Sink Minimum PDP** could be:

- The power required to have basic functionality by a Batteryless Sink,
- On a device with a Battery, it can power the minimum functionality of the device.

**Note:** **EPR Sink Minimum PDP** can be the same as its **SPR Sink Minimum PDP**.

### 6.5.13.17 EPR Sink Operational PDP

The **EPR Sink Operational PDP** field **Shall** contain the Source PDP that the manufacturer recommends for the normal functionality of the Sink, rounded up to the next integer. This corresponds to the PDP Rating of EPR Sources that the Sink is designed to operate with (See [Section 10.3.2, "Normative Sink Rules"](#)). The **EPR Sink Operational PDP** **Shall** be sufficient to operate all the Sink's functional modes normally AND charge the Sink's Battery if present. For Sinks with a Battery(s), it **Shall** correspond to the PDP Rating of the Charger shipped with the EPR Sink or the recommended Charger's PDP Rating. If the Sink is not EPR Capable, or if the Sink is self-powered, such that it doesn't need power from a Source, this field **Shall** be set to zero.

The **EPR Sink Operational PDP** is used to indicate to Shared Capacity Chargers that at this power level the user is not expected to receive any performance warning related to the power being supplied to the Sink.

### 6.5.13.18 EPR Sink Maximum PDP

The **EPR Sink Maximum PDP** field **Shall** be highest PDP the EPR Sink will ever request under any operating condition, rounded up to the next integer, including charging its Battery if present. The **EPR Sink Maximum PDP** field **Shall Not** be less than the **EPR Sink Operational PDP**, but **May** be the same. The value is used by the Source to determine the maximum amount of power it has to budget for the Attached Sink. If the Sink is not EPR Capable, or if the Sink is self-powered, such that it doesn't need power from a Source, this field **Shall** be set to zero.



## 6.5.14 Extended\_Control Message

The **Extended\_Control** Message extends the **Control Message** space. The **Extended\_Control** Message includes one byte of data. The **Extended\_Control** Message **Shall** be as shown in [Figure 6.53, "Extended\\_Control Message"](#) and [Table 6.66, "Extended Control Data Block \(ECDB\)"](#).

Figure 6.53 Extended\_Control Message

|                                  |                        |
|----------------------------------|------------------------|
| Extended Header<br>Data Size = 2 | ECDB<br>(2-byte block) |
|----------------------------------|------------------------|

Table 6.66 Extended Control Data Block (ECDB)

| Offset | Field | Value        | Description                                |
|--------|-------|--------------|--------------------------------------------|
| 0      | Type  | Unsigned Int | Extended Control Message Type              |
| 1      | Data  | Byte         | <b>Shall</b> be set to zero when not used. |

The **Extended\_Control** Message types are specified in the Type field of the ECDB and are listed in [Table 6.67, "Extended Control Message Types"](#). The Sent by column indicates entities which **May** send the given Message (Source, Sink or Cable Plug); entities not listed **Shall Not** issue the corresponding Message. The "Valid Start of Packet" column indicates the Messages which **Shall** only be issued in SOP Packets.

Table 6.67 Extended Control Message Types

| Type    | Data     | Message Type                       | Sent by       | Description                                                                         | Valid Start of Packet |
|---------|----------|------------------------------------|---------------|-------------------------------------------------------------------------------------|-----------------------|
| 0       |          | <b>Reserved</b>                    |               | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                       |
| 1       | Not used | <a href="#">EPR_Get_Source_Cap</a> | Sink or DRP   | See <a href="#">Section 6.5.14.1</a>                                                | SOP only              |
| 2       | Not used | <a href="#">EPR_Get_Sink_Cap</a>   | Source or DRP | See <a href="#">Section 6.5.14.2</a>                                                | SOP only              |
| 3       | Not used | <a href="#">EPR_KeepAlive</a>      | Sink          | See <a href="#">Section 6.5.14.3</a>                                                | SOP only              |
| 4       | Not Used | <a href="#">EPR_KeepAlive_Ack</a>  | Source        | See <a href="#">Section 6.5.14.4</a>                                                | SOP only              |
| 5...255 |          | <b>Reserved</b>                    |               | All values not explicitly defined are <b>Reserved</b> and <b>Shall Not</b> be used. |                       |

### 6.5.14.1 EPR\_Get\_Source\_Cap Message

The **EPR\_Get\_Source\_Cap** (EPR Get Source Capabilities) Message **Shall** only be sent by a Port capable of operating as a Sink and that supports EPR Mode to request the Source Capabilities and Dual-Role Power capability of its Port Partner. A Port that can operate as an EPR Source **Shall** respond by returning an **EPR\_Source\_Capabilities** Message (see [Section 6.5.15.2, "EPR\\_Source\\_Capabilities Message"](#)). A Port that does not support EPR Mode as a Source **Shall** return the **Not\_Supported** Message.

An EPR Capable Sink Port that is operating in SPR Mode **Shall** treat the **EPR\_Source\_Capabilities** Message as informational only and **Shall Not** respond with an **EPR\_Request** Message.

### 6.5.14.2 EPR\_Get\_Sink\_Cap Message

The **EPR\_Get\_Sink\_Cap** (EPR Get Sink Capabilities) Message **Shall** only be sent by a Port capable of operating as a Source and that supports EPR Mode to request the Sink Capabilities and Dual-Role Power capability of its Port Partner. A Port that is EPR Capable operating as a Sink **Shall** respond by returning an **EPR\_Sink\_Capabilities** Message (see [Section 6.5.15.3, "EPR\\_Sink\\_Capabilities Message"](#)). A Port that does not support EPR Mode as a Sink **Shall** return the **Not\_Supported** Message.



### 6.5.14.3 EPR\_KeepAlive Message

\* The **EPR\_KeepAlive** Message **May** be sent by a Sink operating in *EPR Mode* to meet the requirement for periodic traffic. The Source operating on *EPR Mode* responds by returning an **EPR\_KeepAlive\_Ack** Message to the Sink. See [Section 6.4.9, "EPR\\_Request Message"](#) for additional information.

### 6.5.14.4 EPR\_KeepAlive\_Ack Message

The **EPR\_KeepAlive\_Ack** Message **Shall** be sent by a Source operating in *EPR Mode* in response to an **EPR\_KeepAlive** Message. See [Section 6.4.9, "EPR\\_Request Message"](#) for additional information.\*

### 6.5.15 EPR Capabilities Message

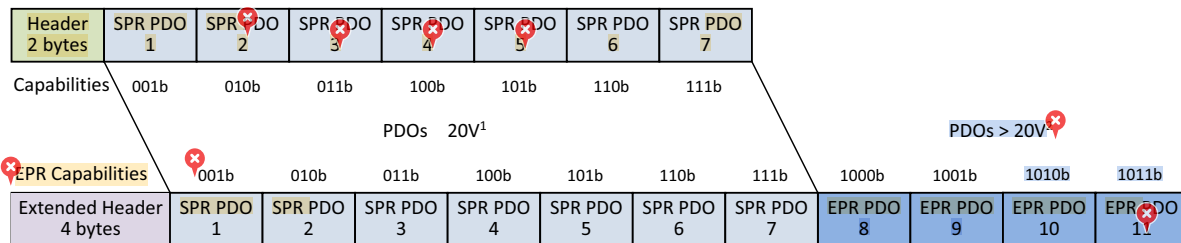
\* The **EPR Capabilities** Message is an **Extended Capabilities Message** made of **Power Data Objects (PDO)** defined in [Section 6.4.1, "Capabilities Message"](#). It is used to form **EPR\_Source\_Capabilities** Messages and **EPR\_Sink\_Capabilities** Messages. Sources expose their EPR power **Capabilities** by sending an **EPR\_Source\_Capabilities** Message. Sinks expose their EPR power requirements by returning an **EPR\_Sink\_Capabilities** Message when requested. Both are composed of a number of 32-bit **Power Data Objects** (see [Table 6.7, "Power Data Object"](#)).

An **EPR Capabilities** Message **Shall** have a **5V Fixed Supply PDO** containing the sending **Port's** information in the first object position followed by up to 10 additional **PDOs**.

#### 6.5.15.1 EPR Capabilities Message Construction

The **EPR Capabilities** Messages (**EPR\_Source\_Capabilities** and **EPR\_Sink\_Capabilities**) are **Extended Messages** with the first seven positions filled with the same **SPR (A)PDOs** returned by the **SPR Capabilities** Messages (**Source\_Capabilities** and **Sink\_Capabilities**) followed by the **EPR (A)PDOs** starting in the eighth position. See [Figure 6.54, "Mapping SPR Capabilities to EPR Capabilities"](#).

Figure 6.54 Mapping SPR Capabilities to EPR Capabilities



<sup>1)</sup> See [Section 10 "Power Rules"](#) for rules, on which SPR (A)PDOs are allowed to be used for a given PDP.\*

<sup>2)</sup> See [Section 10 "Power Rules"](#) for rules, on which EPR (A)PDOs are allowed to be used for a given PDP.

**Power Data Objects** in the **EPR Capabilities** Messages **Shall** be sent in the following order:

- 1) The **SPR (A)PDOs** as reported in the **SPR Capabilities** Message. The **Number of Data Objects** field in the **Message Header** of the **EPR Capabilities** Message is the same as the **Number of Data Objects** field in the **Message Header** of the **SPR Capabilities** Message.
- 2) If the **SPR Capabilities** Message contains fewer than 7 **PDOs**, the unused **Data Objects** **Shall** be zero filled.
- 3) The **EPR (A)PDOs** as defined in [Section 6.4.1, "Capabilities Message"](#) **Shall** start at **Data Object** position 8 and **Shall** be sent in the following order:
  - a) **Fixed Supply PDOs** that offer 28V, 36V or 48V, if present, **Shall** be sent in **voltage** order; lowest to highest.
  - b) One **EPR 4VS APDO** **Shall** be sent.\*

### 6.5.15.2 EPR\_Source\_Capabilities Message

The **EPR\_Source\_Capabilities** is an *EPR Capabilities Message* containing a list of *Power Data Objects* that the *EPR Source* is capable of supplying. It is sent by an *EPR Source* in order to convey its *Capabilities* to a *Sink*. An *EPR Source* **Shall** send the **EPR\_Source\_Capabilities Message**:

- When entering *EPR Mode*
- While in *EPR Modes* when its *Capabilities* change
- In response to an **EPR\_Get\_Source\_Cap** Message
- After a *Soft Reset* while in *EPR Mode*

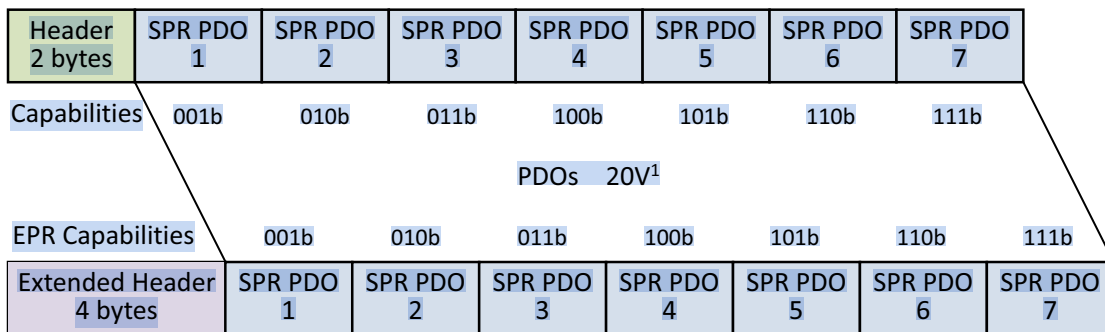
An *EPR Sink* operating in *EPR Mode* **Shall** evaluate every **EPR\_Source\_Capabilities Message** it receives and **Shall** respond with a **EPR\_Request** Message. If its power consumption exceeds the *Source Capabilities*, it **Shall Re-negotiate** so as not to exceed the *Source's* most recently *Advertised Source Capabilities*.

While operating in *SPR Mode*, an *EPR Sink* receiving an **EPR\_Source\_Capabilities Message** in response to an **EPR\_Get\_Source\_Cap** Messages **Shall Not** respond with an **EPR\_Request** Message.

The (A)PDOs in an **EPR\_Source\_Capabilities Message** **Shall** only be requested using the **EPR\_Request** Message and only when in *EPR Mode*.

When *Source* wants to exit *EPR Mode*, if not already in power contract with an *SPR (A)PDO*, it **Shall** send an **EPR\_Source\_Capabilities Message** with no *EPR (A)PDOs* (i.e. seven *SPR (A)PDOs* including any zero padded ones). See [Figure 6.55, "EPR Source Capabilities message with no EPR PDOs"](#).

Figure 6.55 EPR\_Source\_Capabilities message with no EPR PDOs



<sup>1)</sup> See [Section 10 "Power Rules"](#) for rules, on which *SPR (A)PDOs* are allowed to be used for a given *PDP*.

### 6.5.15.3 EPR\_Sink\_Capabilities Message

The **EPR\_Sink\_Capabilities** is an *EPR Capabilities Message* that contains a list of *Power Data Objects* that the *EPR Sink* requires to operate. It is sent by an *EPR Sink* in order to convey its power requirements to an *EPR Source*. The *EPR Sink* **Shall** only send the **EPR\_Sink\_Capabilities Message** in response to an **EPR\_Get\_Sink\_Cap** Message.

## 6.5.16 Vendor\_Defined\_Extended Message

The **Vendor\_Defined\_Extended Message** (VDEM) is provided to allow vendors to exchange information outside of that defined by this specification using the **Extended Message** format.

A **Vendor\_Defined\_Extended Message** **Shall** consist of at least one *Vendor Data Object*, the *VDM Header*, and **May** contain up to a maximum of 256 additional data bytes.

To ensure vendor uniqueness of **Vendor\_Defined\_Extended Messages**, all **Vendor\_Defined\_Extended Messages** **Shall** contain a **Valid** USB Standard or Vendor ID (SVID) allocated by USB-IF in the *VDM Header*.

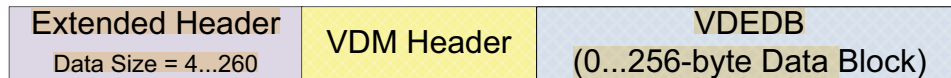
A VDEM does not define any structure and Messages **May** be created in any manner that the vendor chooses.

**Vendor\_Defined\_Extended Messages** **Shall Not** be used for direct power **Negotiation**. They **May** however be used to alter *Local Policy*, affecting what is offered or consumed via the normal PD Messages. For example, a **Vendor\_Defined\_Extended Message** could be used to enable the *Source* to offer additional power via a **Source\_Capabilities Message**.

**Vendor\_Defined\_Extended Messages** **Shall Not** be used where equivalent functionality is contained in the PD Specification e.g., authentication or firmware update.

The Message format **Shall** be as shown in [Figure 6.56, "Vendor\\_Defined\\_Extended Message"](#).

**Figure 6.56 Vendor\_Defined\_Extended Message**



The *VDM Header* **Shall** be the first 4-bytes in a *Vendor Defined Extended Message*. The *VDM Header* provides **Command** space to allow vendors to customize Messages for their own purposes.

The *VDM Header* in the VDEM **Shall** follow the *Unstructured VDM Header* format as defined in [Section 6.4.4.1, "Unstructured VDM"](#).

VDEMs **Shall** only be sent and received after an *Explicit Contract* has been established.

A VDEM **AMS** **Shall Not** interrupt any other PD **AMS**.

The VDEM does not define the contents of bits B14...0 in the *VDM Header*. Their definition and use are the sole responsibility of the vendor indicated by the SVID. The *Port Partners* and *Cable Plugs* **Shall** exit any states entered using a VDEM according to the rules defined in [Section 6.4.4.3.4, "Enter Mode Command"](#).

The following rules apply to the use of VDEM Messages:

- **VDEMs** **Shall Not** be initiated or responded to under any other circumstances than the following:
  - **VDEMs** **Shall** only be used when an *Explicit Contract* is in place.
  - Prior to establishing an *Explicit Contract* VDEMs **Shall Not** be sent and **Shall** be **Ignored** if received.
  - *Cable Plugs* **Shall Not** initiate VDEMs.

-  **VDEMs *Shall*** only be used during *Modal Operation* in the context of an *Active Mode* i.e., only after the *UFP* has **Ack'ed** the ***Enter Mode*** Command can *VDEMs* be sent or received. The *Active Mode* and the associated *VDEMs* ***Shall*** use the same *SVID*.
- ***VDEMs May*** be used with *SOP\* Packets*.
-  When a *DFP* or *UFP* does not support *VDEMs* or does not recognize the *VID* it ***Shall*** return a ***Not\_Supported*** Message.

**Note:** Usage of *VDEMs* with *Chunking* is not recommended since this is less efficient than using *Unstructured VDMs*.

## 6.6 Timers

All the following timers are defined in terms of bits on the bus regardless of where they are implemented in terms of the logical architecture. This is to ensure a fixed reference for the starting and stopping of timers. It is left to the implementer to ensure that this timing is observed in a real system.

### 6.6.1 CRCReceiveTimer

The **CRCReceiveTimer** **Shall** be used by the sender's **Protocol Layer** to ensure that a **Message** has not been lost. Failure to receive an acknowledgment of a **Message** (a **GoodCRC** **Message**) whether caused by a bad **GoodCRC** **Message** on the receiving end or by a garbled **Message** within **tReceive** is detected when the **CRCReceiveTimer** expires.

The sender's **Protocol Layer** response when a **CRCReceiveTimer** expires **Shall** be to retry **nRetryCount** times.

**Note:** *Cable Plugs* do not retry **Messages** and large **Extended Messages** that are not **Chunked** are not retried (see [Section 6.7.2, "Retry Counter"](#)).

Sending of the Preamble corresponding to the retried **Message** **Shall** start within **tRetry** of the **CRCReceiveTimer** expiring.

The **CRCReceiveTimer** **Shall** be started when the last bit of the **Message EOP** has been transmitted by the **PHY Layer**. The **CRCReceiveTimer** **Shall** be stopped when the last bit of the **EOP** corresponding to the **GoodCRC** **Message** has been received by the **PHY Layer**.

The **Protocol Layer** receiving a **Message** **Shall** respond with a **GoodCRC** **Message** within **tTransmit** in order to ensure that the sender's **CRCReceiveTimer** does not expire. The **tTransmit** time **Shall** be measured from when the last bit of the **Message EOP** has been received by the **PHY Layer** until the first bit of the Preamble of the **GoodCRC** **Message** has been transmitted by the **PHY Layer**.

### 6.6.2 SenderResponseTimer

The **SenderResponseTimer** **Shall** be used by the sender's **Policy Engine** to ensure that a **Message** requesting a response (e.g., **Get\_Source\_Cap** **Message**) is responded to within a bounded time of **tSenderResponse**. Failure to receive the expected response is detected when the **SenderResponseTimer** expires.

For **Extended Messages** received as **Chunks**, the **SenderResponseTimer** will also be started and stopped by the **Chunking Rx State Machine**. See [Section 8.3.3.1.1, "SenderResponseTimer State Diagram"](#) for more details of the **SenderResponseTimer** operation.

The **Policy Engine's** response when the **SenderResponseTimer** expires **Shall** be dependent on the **Message** sent (see [Section 8.3, "Policy Engine"](#)).

The **SenderResponseTimer** **Shall** be started from the time the last bit of the **GoodCRC** **Message EOP**, corresponding to the **Message** requesting a response, has been received by the **PHY Layer**.

The **SenderResponseTimer** **Shall** be stopped when the last bit of the **EOP** of the **GoodCRC** **Message**, corresponding to the expected response **Message**, has been transmitted by the **PHY Layer**.

The receiver of a **Message** requiring a response **Shall** respond within **tReceiverResponse** in order to ensure that the sender's **SenderResponseTimer** does not expire.

The **tReceiverResponse** time **Shall** be measured from the time the last bit of the **GoodCRC** **Message EOP**, corresponding to the expected request **Message**, has been transmitted by the **PHY Layer** until the first bit of the response **Message** Preamble has been transmitted by the **PHY Layer**.

### 6.6.3 Capability Timers

**Sources** and **Sinks** use **Capability Timers** to determine **Attachment** of a PD Capable device. By periodically sending or requesting **Capabilities**, it is possible to determine PD device **Attachment** when a response is received.

### 6.6.3.1 SourceCapabilityTimer

Prior to the *First Explicit Contract* a Source **Shall** use the *SourceCapabilityTimer* to periodically send out a *Source\_Capabilities* Message every *tTypeCSendSourceCap* while:

- The Port is Attached.
- The Source is not in an active connection with a PD Sink Port.

Whenever there is a *SourceCapabilityTimer* timeout the Source **Shall** send a *Source\_Capabilities* Message. It **Shall** then re-initialize and restart the *SourceCapabilityTimer*. The *SourceCapabilityTimer* **Shall** be stopped when the last bit of the *EOP* corresponding to the *GoodCRC* Message has been received by the *PHY* Layer since a PD connection has been established. At this point, the Source waits for a *Request* Message or a response timeout.

**Note:** The Source can also stop sending *Source\_Capabilities* Message after *nCapsCount* Messages have been sent without a *GoodCRC* Message response (see [Section 6.7.4, "Capabilities Counter"](#)).

See [Section 8.3.3.2, "Policy Engine Source Port State Diagram"](#) for more details of when *Source\_Capabilities* Messages are transmitted.

### 6.6.3.2 SinkWaitCapTimer

The Sink **Shall** support the *SinkWaitCapTimer*.

While in a *Default Contract* or an *Implicit Contract* when a Sink observes an absence of *Source\_Capabilities* Messages, after *VBUS* is present, for a duration of *tTypeCSinkWaitCap* the Sink **May** issue *Hard Reset* Signaling in order to restart the sending of *Source\_Capabilities* Messages by the Source (see [Section 6.7.4, "Capabilities Counter"](#)) or continue to operate at *USB Type-C* current.

When a Sink, entering *EPR Mode*, observes an absence of *EPR\_Source\_Capabilities* Messages, after the *GoodCRC* Message acknowledging the *EPR\_Mode* Message with the Action field set to 3 (*Enter Succeeded*), for a duration of *tTypeCSinkWaitCap* the Sink **Shall** issue *Hard Reset* Signaling in order to exit *EPR Mode* (see [Section 6.4.10, "EPR\\_Mode Message"](#)).

When a Sink, exiting *EPR Mode*, observes an absence of *Source\_Capabilities* Messages, after the *GoodCRC* Message acknowledging the *EPR\_Mode* Message with the Action field set to 5 (*Exit*), for a duration of *tTypeCSinkWaitCap* the Sink **Shall** issue *Hard Reset* Signaling in order to restart the sending of *Source\_Capabilities* Messages by the Source (see [Section 6.7.4, "Capabilities Counter"](#)).

See [Section 8.3.3.3, "Policy Engine Sink Port State Diagram"](#) for more details of when the *SinkWaitCapTimer* is run.

### 6.6.3.3 tFirstSourceCap

After Port Partners are Attached or after a *Hard Reset* or after a *Power Role Swap* or after a *Fast Role Swap* a Source **Shall** send its first *Source\_Capabilities* Message within *tFirstSourceCap* of *VBUS* reaching *vSafe5V*.

After *Soft Reset*, a Source **Shall** send its first *Source\_Capabilities* Message within *tFirstSourceCap* after last bit of the *GoodCRC* Message *EOP* corresponding to *Accept* Message.

This ensures that the Sink receives a *Source\_Capabilities* Message before the Sink's *SinkWaitCapTimer* expires.

A Source entering *EPR Mode* **Shall** send its first *EPR\_Source\_Capabilities* Message within *tFirstSourceCap* of the *GoodCRC* Message acknowledging the *EPR\_Mode* Message with the Action field set to 3 (*Enter Succeeded*).

A Source exiting *EPR Mode* **Shall** send its first *Source\_Capabilities* Message within *tFirstSourceCap* of the *GoodCRC* Message acknowledging the *EPR\_Mode* Message with the Action field set to 5 (*Exit*).



## 6.6.4 Wait Timers and Times

### 6.6.4.1 SinkRequestTimer

The **SinkRequestTimer** is used to ensure that the time before the next *Sink Request Message*, after a **Wait Message** has been received from the *Source* in response to a *Sink Request Message*, is a minimum of **tSinkRequest** min (see [Section 6.3.12, "Wait Message"](#)).

The **SinkRequestTimer** **Shall** be started when the **EOP** of a **Wait Message** has been received and **Shall** be stopped if any other *Message* is received or during a *Hard Reset*.

The *Sink* **Shall** wait at least **tSinkRequest**, after receiving the **EOP** of a **Wait Message** sent in response to a *Sink Request Message*, before sending a new *Request Message*. Whenever there is a **SinkRequestTimer** timeout the *Sink* **May** send a *Request Message*. It **Shall** then re-initialize and restart the **SinkRequestTimer**.

### 6.6.4.2 tPRSwapWait

The time before the next **PR\_Swap Message**, after a **Wait Message** has been received in response to a **PR\_Swap Message** is a minimum of **tPRSwapWait** min (see [Section 6.3.12, "Wait Message"](#)). The *Port* **Shall** wait at least **tPRSwapWait** after receiving the **EOP** of a **Wait Message** sent in response to a **PR\_Swap Message**, before sending a new **PR\_Swap Message**.

### 6.6.4.3 tDRSwapWait

The time before the next **DR\_Swap Message**, after a **Wait Message** has been received in response to a **DR\_Swap Message** is a minimum of **tDRSwapWait** min (see [Section 6.3.12, "Wait Message"](#)). The *Port* **Shall** wait at least **tDRSwapWait** after receiving the **EOP** of a **Wait Message** sent in response to a **DR\_Swap Message**, before sending a new **DR\_Swap Message**.

### 6.6.4.4 tVCONNSwapWait

The time before the next **VCONN\_Swap Message**, after a **Wait Message** has been received in response to a **VCONN\_Swap Message** is a minimum of **tVCONNSwapWait** min (see [Section 6.3.12, "Wait Message"](#)). The *Port* **Shall** wait at least **tVCONNSwapWait** after receiving the **EOP** of a **Wait Message** sent in response to a **VCONN\_Swap Message**, before sending a new **VCONN\_Swap Message**.

### 6.6.4.5 tVCONNSwapDelayDFP

The time delay for *DFP* after losing *VCONN Source* role due to an incoming *VCONN Swap* request from *UFP* and before sending the next **VCONN\_Swap Message**. The *DFP* **Shall** wait at least **tVCONNSwapDelayDFP** after sending the **EOP** of the **GoodCRC Message** in response to **PS\_RDY Message** received at the end of the previous *VCONN Swap AMS*.

### 6.6.4.6 tVCONNSwapDelayUFP

The time delay for *UFP* after losing *VCONN Source* role due to an incoming *VCONN Swap* request from *DFP* and before sending the next **VCONN\_Swap Message**. The *UFP* **Shall** wait at least **tVCONNSwapDelayUFP** after sending the **EOP** of the **GoodCRC Message** in response to **PS\_RDY Message** received at the end of the previous *VCONN Swap AMS*.

### 6.6.4.7 tEnterUSBWait

The time before the next **Enter\_USB Message**, after a **Wait Message** has been received in response to a **Enter\_USB Message** is a minimum of **tEnterUSBWait** min (see [Section 6.3.12, "Wait Message"](#)). The *DFP* **Shall** wait at least **tEnterUSBWait** after receiving the **EOP** of a **Wait Message** sent in response to an **Enter\_USB Message**, before sending a new **Enter\_USB Message**.

## 6.6.5 Power Supply Timers

See [Section 7.3, "Transitions"](#) for diagrams showing the usage of the timers in this section.



### 6.6.5.1 PSTransitionTimer

The **PSTransitionTimer** is used by the Policy Engine to timeout on a **PS\_RDY** Message. It is started when a request for new Source Capabilities has been accepted and will timeout after **tPSTransition** if a **PS\_RDY** Message has not been received. This condition leads to a Hard Reset and a return to USB Default Operation. The **PSTransitionTimer** relates to the time taken for the Source to transition from one voltage, or current level, to another (see [Section 7.1, "Source Requirements"](#)).

The **PSTransitionTimer** **Shall** be started when the last bit of the **GoodCRC** Message **EOP**, corresponding to an **Accept** Message, has been transmitted by the **PHY** Layer. The **PSTransitionTimer** **Shall** be stopped when the last bit of the **GoodCRC** Message **EOP**, corresponding to the **PS\_RDY** Message, has been transmitted by the **PHY** Layer.

### 6.6.5.2 PSSourceOffTimer

#### 6.6.5.2.1 Use during Power Role Swap

The **PSSourceOffTimer** is used by the Policy Engine in Dual-Role Power Device that is currently acting as a Sink to timeout on a **PS\_RDY** Message during a Power Role Swap **AMS**. This condition leads to USB Type-C Error Recovery.

If a **PR\_Swap** Message request has been sent by the Dual-Role Power Device currently acting as a Source the Sink can respond with an **Accept** Message. When the last bit of the **GoodCRC** Message **EOP**, corresponding to this transmitted **Accept** Message, is received by the Sink's **PHY** Layer, then the **PSSourceOffTimer** **Shall** be started.

If a **PR\_Swap** Message request has been sent by the Dual-Role Power Device currently acting as a Sink the Source can respond with an **Accept** Message. When the last bit of the **GoodCRC** Message **EOP**, corresponding to this received **Accept** Message, is transmitted by the Sink's **PHY** Layer, then the **PSSourceOffTimer** **Shall** be started.

The **PSSourceOffTimer** **Shall** be stopped when the last bit of the **GoodCRC** Message **EOP**, corresponding to the received **PS\_RDY** Message, is transmitted by the **PHY** Layer.

The **PSSourceOffTimer** relates to the time taken for the remote Dual-Role Power Device to stop supplying power (see also [Section 7.3.2.1, "Sink Requested Power Role Swap"](#) and [Section 7.3.2.2, "Source Requested Power Role Swap"](#)). The timer **Shall** time out if a **PS\_RDY** Message has not been received from the remote Dual-Role Power Device within **tPSSourceOff** indicating this has occurred.

#### 6.6.5.2.2 Use during Fast Role Swap

The **PSSourceOffTimer** is used by the Policy Engine in Dual-Role Power Device that is the **Initial Sink** (currently providing **vSafe5V**) to timeout on a **PS\_RDY** Message during a Fast Role Swap **AMS**. This condition leads to USB Type-C Error Recovery.

When the **FR\_Swap** Message request has been sent by the **Initial Sink**, the **Initial Source** **Shall** respond with an **Accept** Message. When the last bit of the **GoodCRC** Message **EOP**, corresponding to this **Accept** Message is received by the **Initial Sink's** **PHY** Layer, then the **PSSourceOffTimer** **Shall** be started.

The **PSSourceOffTimer** **Shall** be stopped when

the last bit of the **GoodCRC** Message **EOP**, corresponding to the received **PS\_RDY** Message, is transmitted by the **PHY** Layer.

The **PSSourceOffTimer** relates to the time taken for the **Initial Source** to stop supplying power and for **V<sub>BUS</sub>** to revert to **vSafe5V** (see also [Section 7.2.10, "Fast Role Swap"](#) and [Section 7.3.4, "Transitions Caused by Fast Role Swap"](#)). The timer **Shall** time out if a **PS\_RDY** Message has not been received from the **Initial Source** within **tPSSourceOff** indicating this has occurred.

### 6.6.5.3 PSSourceOnTimer

#### 6.6.5.3.1 Use during Power Role Swap

The **PSSourceOnTimer** is used by the Policy Engine in Dual-Role Power Device that has just stopped sourcing power and is waiting to start sinking power to timeout on a **PS\_RDY** Message during a Power Role Swap. This condition leads to USB Type-C Error Recovery.

The **PSSourceOnTimer** *Shall* be started when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the transmitted **PS\_RDY** Message, is received by the **PHY** Layer.
- The **PSSourceOnTimer** *Shall* be stopped when:
- The last bit of the **GoodCRC** Message **EOP**, corresponding to the received **PS\_RDY** Message, is transmitted by the **PHY** Layer.

The **PSSourceOnTimer** relates to the time taken for the remote *Dual-Role Power Device* to start sourcing power (see also [Section 7.3.2.1, "Sink Requested Power Role Swap"](#) and [Section 7.3.2.2, "Source Requested Power Role Swap"](#)) and will time out if a **PS\_RDY** Message indicating this has not been received within **tPSSourceOn**.

### 6.6.5.3.2 Use during Fast Role Swap

The **PSSourceOnTimer** is used by the *Policy Engine* in *Dual-Role Power Device* that has just stopped sourcing power and is waiting to start sinking power to timeout on a **PS\_RDY** Message during a *Fast Role Swap*. This condition leads to *USB Type-C Error Recovery*.

The **PSSourceOnTimer** *Shall* be started when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the transmitted **PS\_RDY** Message, is received by the **PHY** Layer.

The **PSSourceOnTimer** *Shall* be stopped when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the received **PS\_RDY** Message, is transmitted by the **PHY** Layer.

The **PSSourceOnTimer** relates to the time taken for the remote *Dual-Role Power Device* to start sourcing power (see also [Section 7.2.10, "Fast Role Swap"](#) and [Section 7.3.4, "Transitions Caused by Fast Role Swap"](#)) and will time out if a **PS\_RDY** Message indicating this has not been received within **tPSSourceOn** ❗

### 6.6.6 NoResponseTimer

❗ The **NoResponseTimer** is used by the *Policy Engine* in a *Source* to determine that its *Port Partner* is not responding after a *Hard Reset*. When the **NoResponseTimer** times out, the *Policy Engine* *Shall* issue up to **nHardResetCount** additional *Hard Resets* before determining that the *Port Partner* is non-responsive to *USB Power Delivery* messaging.

If the *Source* fails to receive a **GoodCRC** Message in response to a **Source\_Capabilities** Message within **tNoResponse** of:

- The last bit of a **Hard Reset** Signaling being sent by the **PHY** Layer if the **Hard Reset** Signaling was initiated by the *Sink*.
- The last bit of a **Hard Reset** Signaling being received by the **PHY** Layer if the **Hard Reset** Signaling was initiated by the *Source*.

Then the *Source* *Shall* issue additional *Hard Resets* up to **nHardResetCount** times (see [Section 6.8.3, "Hard Reset"](#)).

For a non-responsive device, the *Policy Engine* in a *Source* *May* either decide to continue sending **Source\_Capabilities** Messages or to go to non-*USB Power Delivery* operation and cease sending **Source\_Capabilities** Messages.

### 6.6.7 BIST Timers

#### ❗ 6.6.7.1 tBISTCarrierMode

**tBISTCarrierMode** is used to define the maximum time that a *UUT* has to enter *BIST Carrier Mode* when requested by a *Tester*.

A UUT **Shall** enter *BIST Carrier Mode* within **tBISTCarrierMode** of the last bit of the **GoodCRC** Message **EOP**, corresponding to the received the **BIST** Message used to initiate the test, being transmitted by the **PHY** Layer. In *BIST Carrier Mode* when transmitting a continuous carrier signal transmission **Shall** start as soon as the UUT enters **BIST Mode**.

### 6.6.7.2 BISTContModeTimer

The **BISTContModeTimer** is used by a UUT to ensure that a *Continuous BIST Mode* (i.e., *BIST Carrier Mode*) is exited in a timely fashion. A UUT that has been put into a *Continuous BIST Mode* **Shall** return to normal operation (either **PE\_SRC\_Transition\_to\_default**, **PE\_SNK\_Transition\_to\_default**, or **PE\_CBL\_Ready**) within **tBISTContMode** of starting to transmit a continuous carrier signal.

### 6.6.7.3 tBISTSharedTestMode

**tBISTSharedTestMode** is used to define the maximum time that a UUT has to enter *BIST Shared Capacity Test Mode* when requested by a *Tester*.

A UUT **Shall** enter *BIST Shared Capacity Test Mode* and send a new **Source\_Capabilities** Message from all Ports within the **Shared Capacity Group** within **tBISTSharedTestMode** of the last bit of the **GoodCRC** Message **EOP**, corresponding to the received the **BIST** Message used to initiate the test, being transmitted by the **PHY** Layer. 

## 6.6.8 Power Role Swap Timers

### 6.6.8.1 SwapSourceStartTimer

The **SwapSourceStartTimer** **Shall** be used by the **New Source**, after a *Power Role Swap* or *Fast Role Swap*, to ensure that it does not send **Source\_Capabilities** Message before the **New Sink** is ready to receive the **Source\_Capabilities** Message. The **New Source** **Shall Not** send the **Source\_Capabilities** Message earlier than **tSwapSourceStart** after the last bit of the **EOP** of **GoodCRC** Message sent in response to the **PS\_RDY** Message sent by the **New Source** indicating that its power supply is ready. The **Sink** **Shall** be ready to receive a **Source\_Capabilities** Message **tSwapSinkReady** after having sent the last bit of the **EOP** of **GoodCRC** Message sent in response to the **PS\_RDY** Message sent by the **New Source** indicating that its power supply is ready.

## 6.6.9 Soft Reset Timers

### 6.6.9.1 tSoftReset

A failure to see a **GoodCRC** Message in response to any Message within **tReceive** (after **nRetryCount** retries), when a *Port Pair* is *Connected*, is indicative of a communications failure. This **Shall** cause the *Source* or *Sink* to send a **Soft\_Reset** Message, transmission of which **Shall** be completed within **tSoftReset** of the **CRCReceiveTimer** expiring.

### 6.6.9.2 tProtErrSoftReset

If the *Protocol Error* occurs that causes the *Source* or *Sink* to send a **Soft\_Reset** Message, the transmission of the **Soft\_Reset** Message **Shall** be completed within **tProtErrSoftReset** of the **EOP** of the **GoodCRC** sent in response to the Message that caused the *Protocol Error*.

## 6.6.10 Data Reset Timers

### 6.6.10.1 VCONNDISchargeTimer

The **VCONNDISchargeTimer** is used by the *Policy Engine* in the *DFP* to ensure the *UFP* actively discharges **VCONN** in a timely manner to ensure the cable will restore **Ra**. Once the *UFP* has discharged **VCONN** below **vRaReconnect** (see [\[USB Type-C 2.4\]](#)) it sends a **PS\_RDY** Message (see also [Section 7.1.15, "VCONN Power Cycle"](#)).

If the *DFP* does not receive a **PS\_RDY** Message from the *UFP* within **tVCONNSourceDischarge** of the last bit of the **GoodCRC** acknowledging the **Accept** Message in response to the **Data\_Reset** Message, the **VCONNDISchargeTimer** will time out and the *Policy Engine* **Shall** enter the **ErrorRecovery** State. 

### 6.6.10.2 tDataReset

The *DFP* **shall** complete the *Data\_Reset* process (as defined in [Section 6.3.14, "Data\\_Reset Message"](#)) within *tDataReset* of the last bit of the *GoodCRC* Message *EOP*, corresponding to the *Accept* Message, being transmitted by the *PHY* Layer.

### 6.6.10.3 DataResetFailTimer

The *DataResetFailTimer* **shall** be used by the *DFP's* Policy Engine to ensure the *Data\_Reset* process completes within *tDataResetFail* of the last bit of the *GoodCRC* acknowledging the *Accept* Message in response to the *Data\_Reset* Message. If the *DFP's* *DataResetFailTimer* expires, the *DFP* **shall** enter the *ErrorRecovery* State.

### 6.6.10.4 DataResetFailUFPTimer

The *DataResetFailUFPTimer* **shall** be used by the *UFP's* Policy Engine to ensure the *Data\_Reset* process completes within *tDataResetFailUFP* of the last bit of the *GoodCRC* acknowledging the *Accept* Message in response to the *Data\_Reset* Message. If the *UFP's* *DataResetFailUFPTimer* expires, the *UFP* **shall** enter the *ErrorRecovery* State.

## 6.6.11 Hard Reset Timers

### 6.6.11.1 HardResetCompleteTimer

The *HardResetCompleteTimer* is used by the *Protocol Layer* in the case where it has asked the *PHY* Layer to send *Hard\_Reset* Signaling and the *PHY* Layer is unable to send the *Signaling* within a reasonable time due to a *non-Idle* channel. If the *PHY* Layer does not indicate that the *Hard\_Reset* Signaling has been sent within *tHardResetComplete* of the *Protocol Layer* requesting transmission, then the *Protocol Layer* **shall** inform the *Policy Engine* that the *Hard\_Reset* Signaling has been sent in order to ensure the power supply is reset in a timely fashion.

### 6.6.11.2 PSHardResetTimer

The *PSHardResetTimer* is used by the *Policy Engine* in a *Source* to ensure that the *Sink* has had sufficient time to process *Hard\_Reset* Signaling before turning off its power supply to *VBUS*.

When a *Hard\_Reset* occurs the *Source*, stops driving *VCONN*, removes *R<sub>p</sub>* from the *CC* pin and starts to transition the *VBUS* voltage to *vSafe0V* either:

- *tPSHardReset* after the last bit of the *Hard\_Reset* Signaling has been received from the *Sink* or
- *tPSHardReset* after the last bit of the *Hard\_Reset* Signaling has been sent by the *Source*.

See [Section 7.1.5, "Response to Hard Resets"](#).

### 6.6.11.3 tDRSwapHardReset

If a *DR\_Swap* Message is received during *Modal Operation* then a *Hard\_Reset* **shall** be initiated by the recipient of the unexpected *DR\_Swap* Message; *Hard\_Reset* Signaling **shall** be generated within *tDRSwapHardReset* of the *EOP* of the *GoodCRC* sent in response to the *DR\_Swap* Message.

### 6.6.11.4 tProtErrHardReset

If a *Protocol Error* occurs that directly leads to a *Hard\_Reset*, the transmission of the *Hard\_Reset* Signaling **shall** be completed within *tProtErrHardReset* of the *EOP* of the *GoodCRC* sent in response to the Message that caused the *Protocol Error*.

## 6.6.12 Structured VDM Timers

### 6.6.12.1 VDMResponseTimer

The *VDMResponseTimer* **shall** be used by the *Initiator's* Policy Engine to ensure that a *Structured VDM Command* request needing a response (e.g. *Discover Identity* Command request) is responded to within a bounded time of *tVDMSenderResponse*. The *VDMResponseTimer* **shall** be applied to all *Structured VDM Commands* except the



**Enter Mode** and **Exit Mode** Commands which have their own timers (**VDMModeEntryTimer** and **VDMModeExitTimer** respectively). Failure to receive the expected response is detected when the **VDMResponseTimer** expires.

The Policy Engine's response when the **VDMResponseTimer** expires **Shall** be dependent on the Message sent (see [Section 8.3, "Policy Engine"](#)).

The **VDMResponseTimer** **Shall** be started from the time the last bit of the **GoodCRC** Message **EOP**, corresponding to the VDM Command requesting a response, has been received by the **PHY** Layer. The **VDMResponseTimer** **Shall** be stopped when the last bit of the **EOP** of the **GoodCRC** Message, corresponding to the expected VDM Command response, has been transmitted by the **PHY** Layer.

The receiver of a Message requiring a response **Shall** respond within **tVDMReceiverResponse** in order to ensure that the sender's **VDMResponseTimer** does not expire.

The **tVDMReceiverResponse** time **Shall** be measured from the time the last bit of the Message **EOP** has been transmitted by the **PHY** Layer until the first bit of the response Message Preamble has been transmitted by the **PHY** Layer.

### 6.6.12.2 VDMModeEntryTimer

The **VDMModeEntryTimer** **Shall** be used by the Initiator's Policy Engine to ensure that the response to a Structured VDM **Enter Mode** Command request (**ACK** or **NAK** with **ACK** indicating that the requested **Alternate Mode** has been entered) arrives within a bounded time of **tVDMWaitModeEntry**. Failure to receive the expected response is detected when the **VDMModeEntryTimer** expires.

The Policy Engine's response when the **VDMModeEntryTimer** expires is to inform the Device Policy Manager (see [Section 8.3.3.23.1, "DFP Structured VDM Mode Entry State Diagram"](#)).

The **VDMModeEntryTimer** **Shall** be started from the time the last bit of the **EOP** of the **GoodCRC** Message, corresponding to the VDM Command request, has been received by the **PHY** Layer. The **VDMModeEntryTimer** **Shall** be stopped when the last bit of the **EOP** of the **GoodCRC** Message, corresponding to the expected Structured VDM Command response (**ACK**, **NAK** or **BUSY**), has been transmitted by the **PHY** Layer.

The receiver of a Message requiring a response **Shall** respond within **tVDMEnterMode** in order to ensure that the sender's **VDMModeEntryTimer** does not expire.

The **tVDMEnterMode** time **Shall** be measured from the time the last bit of the **EOP** of the **GoodCRC** Message, corresponding to VDM Command Request, has been transmitted by the **PHY** Layer until the first bit of the response Message Preamble has been transmitted by the **PHY** Layer.

### 6.6.12.3 VDMModeExitTimer

The **VDMModeExitTimer** **Shall** be used by the Initiator's Policy Engine to ensure that the **ACK** response to a Structured VDM **Exit Mode** Command, indicating that the requested **Alternate Mode** has been exited, arrives within a bounded time of **tVDMWaitModeExit**. Failure to receive the expected response is detected when the **VDMModeExitTimer** expires.

The Policy Engine's response when the **VDMModeExitTimer** expires is to inform the Device Policy Manager (see [Section 8.3.3.23.2, "DFP Structured VDM Mode Exit State Diagram"](#)).

The **VDMModeExitTimer** **Shall** be started from the time the last bit of the **GoodCRC** Message **EOP**, corresponding to the VDM Command requesting a response, has been received by the **PHY** Layer. The **VDMModeExitTimer** **Shall** be stopped when the last bit of the **GoodCRC** Message **EOP**, corresponding to the expected Structured VDM Command response **ACK**, has been transmitted by the **PHY** Layer.

The receiver of a Message requiring a response **Shall** respond within **tVDMExitMode** in order to ensure that the sender's **VDMModeExitTimer** does not expire.

The **tVDMExitMode** time **Shall** be measured from the time the last bit of the Message **EOP** has been received by the **PHY** Layer until the first bit of the response Message Preamble has been transmitted by the **PHY** Layer.

#### 6.6.12.4 tVDMBusy

✖ The Initiator **Shall** wait at least **tVDMBusy**, after receiving a **BUSY** Command response, before repeating the Structured VDM request again.

#### 6.6.13 VCONN Timers

##### ✖ 6.6.13.1 VCONNOnTimer

The **VCONNOnTimer** is used during a VCONN Swap.

The **VCONNOnTimer** **Shall** be started when:

- The last bit of **GoodCRC** Message **EOP**, corresponding to the **Accept** Message, is transmitted or received by the **PHY** Layer.

The **VCONNOnTimer** **Shall** be stopped when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the **PS\_RDY** Message, is transmitted by the **PHY** Layer.

Prior to sending the **PS\_RDY** Message, the Port **Shall** have turned VCONN On.

#### 6.6.13.2 tVCONNSourceOff

The **tVCONNSourceOff** time applies during a VCONN Swap. The initial VCONN Source **Shall** cease sourcing VCONN within **tVCONNSourceOff** of the last bit of the **GoodCRC** Message **EOP**, corresponding to the **PS\_RDY** Message, being transmitted by the **PHY** Layer.

#### 6.6.14 tCableMessage

Ports compliant with **Revision 3.x** of the specification **Shall Not** wait **tCableMessage** before sending an **SOP' Packet** or **SOP'' Packet** even when communicating using **[USBPD 2.0]** with a Cable Plug. This specification defines **Collision Avoidance** mechanisms that obviate the need for this time.

Cable Plugs **Shall** only wait **tCableMessage** before sending an **SOP' Packet** or **SOP'' Packet** when operating at **[USBPD 2.0]**. When operating at Revisions higher than **[USBPD 2.0]** Cable Plugs **Shall Not** wait **tCableMessage** before sending an **SOP' Packet** or **SOP'' Packet**.

#### 6.6.15 DiscoverIdentityTimer

The **DiscoverIdentityTimer** is used prior to or during an **Explicit Contract** when discovering whether a Cable Plug is PD Capable using **SOP'**. When performing **Cable Discovery** during an **Explicit Contract** the **Discover Identity** Command request **Shall** be sent every **tDiscoverIdentity**. No more than **nDiscoverIdentityCount** **Discover Identity** Messages without a **GoodCRC** Message response **Shall** be sent. If no **GoodCRC** Message response is received after **nDiscoverIdentityCount** **Discover Identity** Command requests have been sent by a Port, the Port **Shall Not** send any further **SOP'/SOP''** Messages.

#### 6.6.16 Collision Avoidance Timers

##### 6.6.16.1 SinkTxTimer

The **SinkTxTimer** is used by the Protocol Layer in a Source to allow the Sink to complete its transmission before initiating an AMS.

The Source **Shall** wait a minimum of **tSinkTx** after changing  $R_p$  from **SinkTxOK** to **SinkTxNG** before initiating an AMS by sending a Message.

A Sink **Shall** only initiate an AMS when it has determined that  $R_p$  is set to **SinkTxOK**.

## 6.6.16.2 tSrcHoldsBus

✖ If a transition into the *PE\_SRC\_Ready* state will result in an immediate transition out of the *PE\_SRC\_Ready* state within *tSrcHoldsBus* e.g. it is due to a *Protocol Error* that has not resulted in a *Soft Reset*, then the notifications of the end of *AMS* and first *Message* in an *AMS* **May** Not be sent to avoid changing the  $R_p$  value unnecessarily. ✖

## 6.6.17 Fast Role Swap Timers

### 6.6.17.1 tFRSwap5V

The *tFRSwap5V* time **Shall** be measured from:

- The later of:
  - ✖ The last bit of the *GoodCRC* Message *EOP*, corresponding to the *Accept* Message or
  - *VBUS* being within *vSafe5V*.
- Until the first bit of the response *PS\_RDY* Message Preamble has been transmitted by the *PHY* Layer.

During a *Fast Role Swap*, the *Initial Source* **Shall** start the *PS\_RDY* Message within *tFRSwap5V* after both:

- The *Initial Source* has sent the *Accept* Message, and
- *VBUS* is at or below *vSafe5V*.

### 6.6.17.2 tFRSwapComplete

During a fast-role swap, the *Initial Sink* **Shall** respond with a the *PS\_RDY* Message within *tFRSwapComplete* after it has received the *PS\_RDY* Message from the *Initial Source*. The *tFRSwapComplete* time **Shall** be measured from the time the last bit of the *GoodCRC* Message *EOP*, corresponding to the *PS\_RDY* Message, has been transmitted by the *PHY* Layer until the first bit of the response *PS\_RDY* Message Preamble has been transmitted by the *PHY* Layer.

### 6.6.17.3 tFRSwapInit

That last bit of the *EOP* of the *FR\_Swap* Message **Shall** be transmitted by the *New Source* no later than *tFRSwapInit* after the *Fast Role Swap Request* has been detected (see [Section 5.8.6.3, "Fast Role Swap Detection"](#)).

## 6.6.18 Chunking Timers

### 6.6.18.1 ChunkingNotSupportedTimer

The *ChunkingNotSupportedTimer* is used by a *Source* or *Sink* which does not support multi-chunk *Chunking* but has received a *Message Chunk*.

The *ChunkingNotSupportedTimer* **Shall** be started when:

- The last bit of the *GoodCRC* Message *EOP*, corresponding to a *Message Chunk* of a multi-chunk *Message*, is transmitted by the *PHY* Layer. The *Policy Engine* **Shall Not** send its *Not\_Supported* Message before the *ChunkingNotSupportedTimer* expires.

### 6.6.18.2 ChunkSenderRequestTimer

The *ChunkSenderRequestTimer* is used during a *Chunked Message* transmission.

The *ChunkSenderRequestTimer* **Shall** be used by the sender's *Chunking* state machine to ensure that a *Chunk Response* is responded to within a bounded time of *tChunkSenderRequest*. Failure to receive the expected response is detected when the *ChunkSenderRequestTimer* expires.

The *ChunkSenderRequestTimer* **Shall** be started when:

- The last bit of the *GoodCRC* Message *EOP*, corresponding to the *Chunk Response Message*, is received by the *PHY* Layer. ✖



The **ChunkSenderRequestTimer** *Shall* be stopped when:

- The last bit of the **EOP** of the **GoodCRC** Message, corresponding to the *Chunk Request Message*, is transmitted by the **PHY** Layer.
- A Message other than a *Chunk Request* is received from the *Protocol Layer Rx*.

The receiver of a *Chunk Response* requiring a *Chunk Request* *Shall* respond with a *Chunk Request* within **tChunkReceiverRequest** in order to ensure that the sender's **ChunkSenderRequestTimer** does not expire.

The **tChunkReceiverRequest** time *Shall* be measured from the time the last bit of the **EOP** of the **GoodCRC** Message, corresponding to the *Chunk Response Message*, has been transmitted by the **PHY** Layer until the first bit of the response Message Preamble has been transmitted by the **PHY** Layer.

### 6.6.18.3 ChunkSenderResponseTimer

The **ChunkSenderResponseTimer** is used during a *Chunked Message* transmission.

The **ChunkSenderResponseTimer** *Shall* be used by the sender's *Chunking* state machine to ensure that a *Chunk Request* is responded to within a bounded time of **tChunkSenderResponse**. Failure to receive the expected response is detected when the **ChunkSenderResponseTimer** expires.

The **ChunkSenderResponseTimer** *Shall* be started when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the *Chunk Request Message*, is received by the **PHY** Layer.

The **ChunkSenderResponseTimer** *Shall* be stopped when:

- The last bit of the **GoodCRC** Message **EOP**, corresponding to the *Chunk Response Message*, is transmitted by the **PHY** Layer.
- A Message other than a *Chunk* is received from the *Protocol Layer*.

The receiver of a *Chunk Request* requiring a *Chunk Response* *Shall* respond with a *Chunk Response* within **tChunkReceiverResponse** in order to ensure that the sender's **ChunkSenderResponseTimer** does not expire.

The **tChunkReceiverResponse** time *Shall* be measured from the time the last bit of the **EOP** of the **GoodCRC** Message, corresponding to the *Chunk Request Message*, has been transmitted by the **PHY** Layer until the first bit of the response Message Preamble has been transmitted by the **PHY** Layer.

## 6.6.19 Programmable Power Supply Timers

### 6.6.19.1 SinkPPSPeriodicTimer

The **SinkPPSPeriodicTimer** *Shall* be used by the *Sink's Policy Engine* to ensure that communication between the *Sink* and *Source* occurs within a bounded time of **tPPSRequest** when in *SPR PPS Mode*. In the absence of any other traffic, a **Request** Message requesting an *SPR PPS APDO* is sent periodically as a keep alive mechanism.

**SinkPPSPeriodicTimer** *Shall* be re-initialized and restarted on transmission, by the **PHY** Layer, of the last bit of the **GoodCRC** Message **EOP**, corresponding to any received Message, that causes the *Sink* to enter the **PE\_SNK\_Ready** state.

The *Sink* *Shall* stop the **SinkPPSPeriodicTimer** on transmission, by the **PHY** Layer, of the last bit of the **GoodCRC** Message **EOP**, corresponding to any Message, or the last bit of any *Signaling* is received, by the **PHY** Layer, from the *Source* and by the *Sink* that causes the *Sink* to leave the **PE\_SNK\_Ready** state.

### 6.6.19.2 SourcePPSCommTimer

The **SourcePPSCommTimer** *Shall* be used by the *Source's Policy Engine* to ensure that communication between the *Sink* and *Source* occurs within a bounded time of **tPPSTimeout** when in *SPR PPS Mode*. In the absence of any other traffic, a **Request** Message requesting an *SPR PPS APDO* is received periodically as a keep alive mechanism.

**SourcePPSCmmTimer** *Shall* be re-initialized and restarted when, after receiving any *Message* that causes the *Source* to enter the **PE\_SRC\_Ready** state, the last bit of the corresponding **GoodCRC** *Message EOP* is transmitted by the *PHY* Layer.

The *Source* *Shall* stop the **SourcePPSCmmTimer** when:

- After receiving any *Message* that causes the *Source* to leave the **PE\_SRC\_Ready** state, the last bit of the corresponding **GoodCRC** *Message EOP* is sent by the *PHY* Layer, or
- The last bit of any *Signaling* is received by the *PHY* Layer from the *Sink* by the *Source* that causes the *Source* to leave the **PE\_SRC\_Ready** state.

When the **SourcePPSCmmTimer** times out the *Source* *Shall* issue **Hard Reset** *Signaling*.

## 6.6.20 tEnterUSB

The *DFP* *Shall* send the **Enter\_USB** *Message* within **tEnterUSB** of either:

- The last bit of the **GoodCRC** acknowledging the **Data\_Reset\_Complete** *Message* in response to the **Data\_Reset** *Message* or
- A PD **Connection**, specifically the last bit of the **GoodCRC** acknowledging the **Source\_Capabilities** *Message* after the initial entry into the **PE\_SRC\_Send\_Capabilities** state or
- The last bit of the **GoodCRC** acknowledging the **Accept** *Message* in response to the **DR\_Swap** *Message*

Failure by the *DFP* to meet this timeout parameter can result in the ports not transitioning into **[USB4]** operation. Any *AMS* initiated by the *UFP* prior to receiving the **Enter\_USB** *Message* will delay reception of the **Enter\_USB** *Message* and **[USB4]** operation, therefore a USB4<sup>®</sup>-capable *UFP* *Should Not* initiate any *AMS* until the *DFP* has been given time to send the **Enter\_USB** *Message*.

## 6.6.21 EPR Timers



### 6.6.21.1 SinkEPREnterTimer Timer

The **SinkEPREnterTimer** is used to ensure the *EPR Mode* entry process completes within **tEnterEPR**. The *Sink* *Shall* start the timer when it sees the last bit of the **GoodCRC** *Message* in response to the **EPR\_Mode** *Message* with the Action field set to 1 (**Enter**). The *Sink* *Shall* stop the timer when the last bit of the corresponding **GoodCRC** *Message EOP*, corresponding to the received **EPR\_Mode** *Message* with the Action field set to 3 (**Enter Succeeded**), has been transmitted by the *PHY* Layer. If the timer expires the *Sink* *Shall* send a **Soft\_Reset** *Message*.

### 6.6.21.2 SinkEPRKeepAlive Timer

The **SinkEPRKeepAliveTimer** *Shall* be used by the *Sink's* Policy Engine to ensure that communication between the *Sink* and *Source* occurs within a bounded time of **tSinkEPRKeepAlive**. The *Sink* *Shall* initialize and run this timer upon entry into the **PE\_SNK\_Ready** State when in *EPR Mode* and *Shall* stop it upon exit from the **PE\_SNK\_Ready** when in *EPR Mode*.

While operating in *EPR Mode*, the *Sink* *Shall* stop the **SinkEPRKeepAliveTimer** timer whenever:

- The last bit of the **GoodCRC** *Message EOP*, in response any *Message* from the *Source*, is transmitted by the *PHY* Layer.
- The *PHY* Layer receives the last bit of the **GoodCRC** *Message EOP* in response to any *Message* sent to the *Source*.

If the timer expires the *Sink* *Shall* send an **EPR\_KeepAlive** *Message*.

### 6.6.21.3 SourceEPRKeepAlive Timer

The **SourceEPRKeepAliveTimer** *Shall* be used by the *Source's* Policy Engine to ensure that communication between the *Sink* and *Source* occurs within a bounded time of **tSourceEPRKeepAlive**. The *Source* *Shall* initialize

and run this timer upon entry into the **PE\_SRC\_Ready** State when in **EPR Mode** and **Shall** disable it upon exit from the **PE\_SRC\_Ready** State when **EPR Mode**.

While operating in **EPR Mode**, the Source **Shall** stop the **SourceEPRKeepAliveTimer** timer whenever:

- The last bit of the **GoodCRC** Message **EOP**, in response any *Message* from the *Sink*, is transmitted by the **PHY Layer**.
- The **PHY Layer** receives the last bit of the **GoodCRC** Message **EOP** in response to any *Message* sent to the Source.

If the timer expires the Source **Shall** send **Hard Reset** Signaling.

#### 6.6.21.4 tEPRSourceCableDiscovery

After *Port Partners* are *Attached* or after a *Hard Reset* or after a *Power Role Swap* or after a *Fast Role Swap* an **EPR Source** **Shall** discover the *Cable Plug* within **tEPRSourceCableDiscovery** of entering the **First Explicit Contract**.

The **EPR Source** **Shall** send the **Discover Identity REQ** Command, to the *Cable Plug*, within **tEPRSourceCableDiscovery** of receiving the **GoodCRC** Message acknowledging the **PS\_RDY** Message as part of the *Explicit Contract Negotiation*.

**Note:** If the **EPR Source** is not the *VCONN Source*, **tEPRSourceCableDiscovery**, will also include the time needed for the *VCONN Swap*.

## 6.6.22 Time Values and Timers

[Table 6.68, "Time Values"](#) summarizes the values for the timers listed in this section. For each Timer Value, a given implementation **shall** pick a fixed value within the range specified. [Table 6.69, "Timers"](#) lists the timers.

**Table 6.68 Time Values**

| Parameter                            | Value (min) | Value (Nom) | Value (max) | Units | Reference                         |
|--------------------------------------|-------------|-------------|-------------|-------|-----------------------------------|
| <i>tACTempUpdate</i>                 |             |             | 500         | ms    | <a href="#">Section 6.5.2.2.1</a> |
| <i>tBISTContMode</i>                 | 30          | 45          | 60          | ms    | <a href="#">Section 6.6.7.2</a>   |
| <i>tBISTCarrierMode</i>              |             |             | 300         | ms    | <a href="#">Section 6.6.7.1</a>   |
| <i>tBISTSharedTestMode</i>           |             |             | 1           | s     | <a href="#">Section 6.6.7.3</a>   |
| <i>tCableMessage</i>                 | 750         |             |             | µs    | <a href="#">Section 6.6.14</a>    |
| <i>tCapabilitiesMismatchResponse</i> |             |             | 2           | s     | <a href="#">Section 6.4.2.3</a>   |
| <i>tChunkingNotSupported</i>         | 40          | 45          | 50          | ms    | <a href="#">Section 6.6.18.1</a>  |
| <i>tChunkReceiverRequest</i>         |             |             | 15          | ms    | <a href="#">Section 6.6.18.2</a>  |
| <i>tChunkReceiverResponse</i>        |             |             | 15          | ms    | <a href="#">Section 6.6.18.3</a>  |
| <i>tChunkSenderRequest</i>           | 24          | 27          | 30          | ms    | <a href="#">Section 6.6.18.2</a>  |
| <i>tChunkSenderResponse</i>          | 24          | 27          | 30          | ms    | <a href="#">Section 6.6.18.3</a>  |
| <i>tDataReset</i>                    | 200         | 225         | 250         | ms    | <a href="#">Section 6.6.10.2</a>  |
| <i>tDataResetFail</i>                | 300         |             | 400         | ms    | <a href="#">Section 6.6.10.3</a>  |
| <i>tDataResetFailUFP</i>             | 450         |             | 550         | ms    | <a href="#">Section 6.6.10.4</a>  |
| <i>tDiscoverIdentity</i>             | 40          |             | 50          | ms    | <a href="#">Section 6.6.14</a>    |
| <i>tDRSwapHardReset</i>              |             |             | 15          | ms    | <a href="#">Section 6.6.11.3</a>  |
| <i>tDRSwapWait</i>                   | 100         |             |             | ms    | <a href="#">Section 6.6.4.3</a>   |
| <i>tEnterUSB</i>                     |             |             | 500         | ms    | <a href="#">Section 6.6.20</a>    |
| <i>tEnterUSBWait</i>                 | 100         |             |             | ms    | <a href="#">Section 6.6.4.7</a>   |
| <i>tEnterEPR</i>                     | 450         | 500         | 550         | ms    | <a href="#">Section 6.6.21.1</a>  |
| <i>tEPRSourceCableDiscovery</i>      |             |             | 2           | s     | <a href="#">Section 6.6.21.4</a>  |
| <i>tFirstSourceCap</i>               |             |             | 250         | ms    | <a href="#">Section 6.6.3.3</a>   |
| <i>tFRSwap5V</i>                     |             |             | 15          | ms    | <a href="#">Section 6.6.17.1</a>  |
| <i>tFRSwapComplete</i>               |             |             | 15          | ms    | <a href="#">Section 6.6.17.2</a>  |
| <i>tFRSwapInit</i>                   |             |             | 15          | ms    | <a href="#">Section 6.6.17.3</a>  |
| <i>tHardReset</i>                    |             |             | 5           | ms    | <a href="#">Section 6.3.13</a>    |
| <i>tHardResetComplete</i>            | 4           | 4.5         | 5           | ms    | <a href="#">Section 6.6.9</a>     |
| <i>tSourceEPRKeepAlive</i>           | 0.750       | 0.875       | 1.000       | s     | <a href="#">Section 6.6.21.3</a>  |
| <i>tSinkEPRKeepAlive</i>             | 0.250       | 0.375       | 0.500       | s     | <a href="#">Section 6.6.21.2</a>  |
| <i>tNoResponse</i>                   | 4.5         | 5.0         | 5.5         | s     | <a href="#">Section 6.6.6</a>     |
| <i>tPPSRequest</i>                   |             |             | 10          | s     | <a href="#">Section 6.6.19.1</a>  |
| <i>tPPSTimeout</i>                   | 12.0        | 13.5        | 15.0        | s     | <a href="#">Section 6.6.19.2</a>  |
| <i>tProtErrHardReset</i>             |             |             | 15          | ms    | <a href="#">Section 6.6.11.4</a>  |
| <i>tProtErrSoftReset</i>             |             |             | 15          | ms    | <a href="#">Section 6.6.9.2</a>   |
| <i>tPRSwapWait</i>                   | 100         |             |             | ms    | <a href="#">Section 6.6.4.2</a>   |
| <i>tPSHardReset</i>                  | 25          | 30          | 35          | ms    | <a href="#">Section 6.6.11.2</a>  |

Table 6.68 Time Values (Continued)

| Parameter                           |                 | Value (min) | Value (Nom) | Value (max) | Units                                                                                    | Reference                                                                                                         |
|-------------------------------------|-----------------|-------------|-------------|-------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b><i>tPSSourceOff</i></b>          | <i>SPR Mode</i> | 750         | 835         | 920         | ms                                                                                       | <a href="#">Section 6.6.5.2</a>                                                                                   |
|                                     | <i>EPR Mode</i> | 1120        | 1260        | 1400        |                                                                                          |                                                                                                                   |
| <b><i>tPSSourceOn</i></b>           | <i>SPR Mode</i> | 390         | 435         | 480         | ms                                                                                       | <a href="#">Section 6.6.5.3</a>                                                                                   |
| <b><i>tPSTransition</i></b>         | <i>SPR Mode</i> | 450         | 500         | 550         | ms                                                                                       | <a href="#">Section 6.6.5.1</a>                                                                                   |
|                                     | <i>EPR Mode</i> | 830         | 925         | 1020        |                                                                                          |                                                                                                                   |
| <b><i>tReceive</i></b>              |                 | 0.9         | 1.0         | 1.1         | ms                                                                                       | <a href="#">Section 6.6.1</a>                                                                                     |
| <b><i>tReceiverResponse</i></b>     |                 |             |             | 15          | ms                                                                                       | <a href="#">Section 6.6.2</a>                                                                                     |
| <b><i>tRetry</i></b>                |                 |             |             | 195         | µs                                                                                       | <a href="#">Section 6.6.1</a>                                                                                     |
| <b><i>tSenderResponse</i></b>       |                 | 27          | 30          | 33          | ms                                                                                       | <a href="#">Section 6.6.2</a>                                                                                     |
| <b><i>tSinkDelay</i></b>            |                 |             |             | 5           | ms                                                                                       | <a href="#">Section 5.7</a>                                                                                       |
| <b><i>tSinkRequest</i></b>          |                 | 100         |             |             | ms                                                                                       | <a href="#">Section 6.6.4.1</a>                                                                                   |
| <b><i>tSinkTx</i></b>               |                 | 16          | 18          | 20          | ms                                                                                       | <a href="#">Section 6.6.16</a>                                                                                    |
| <b><i>tSoftReset</i></b>            |                 |             |             | 15          | ms                                                                                       | <a href="#">Section 6.8.1</a>                                                                                     |
| <b><i>tSrcHoldsBus</i></b>          |                 |             |             | 50          | ms                                                                                       | <a href="#">Section 8.3.3.2</a>                                                                                   |
| <b><i>tSwapSinkReady</i></b>        |                 |             |             | 15          | ms                                                                                       | <a href="#">Section 6.6.8.1</a>                                                                                   |
| <b><i>tSwapSourceStart</i></b>      |                 | 20          |             |             | ms                                                                                       | <a href="#">Section 6.6.8.1</a>                                                                                   |
| <b><i>tTransmit</i></b>             |                 |             |             | 195         | µs                                                                                       | <a href="#">Section 6.6.1</a>  |
| <b><i>tTypeC_SendSourceCap</i></b>  |                 | 100         | 150         | 200         | ms                                                                                       | <a href="#">Section 6.6.3.1</a>                                                                                   |
| <b><i>tTypeC_SinkWaitCap</i></b>    |                 | 310         | 465         | 620         | ms                                                                                       | <a href="#">Section 6.6.3.2</a>                                                                                   |
| <b><i>tVCONNSourceDischarge</i></b> |                 | 160         | 200         | 240         | ms                                                                                       | <a href="#">Section 6.6.10.1</a>                                                                                  |
| <b><i>tVCONNSourceOff</i></b>       |                 |             |             | 25          | ms                                                                                       | <a href="#">Section 6.6.13</a>                                                                                    |
| <b><i>tVCONNSourceOn</i></b>        |                 |             |             | 50          | ms                                                                                       | <a href="#">Section 6.3.11</a>                                                                                    |
| <b><i>tVCONNSourceTimeout</i></b>   |                 | 100         | 150         | 200         | ms  | <a href="#">Section 6.6.13</a>                                                                                    |
| <b><i>tVCONN_SwapWait</i></b>       |                 | 100         |             |             | ms                                                                                       | <a href="#">Section 6.6.4.4</a>                                                                                   |
| <b><i>tVCONN_SwapDelayDFP</i></b>   |                 | 100         |             |             | ms                                                                                       | <a href="#">Section 6.6.4.5</a>                                                                                   |
| <b><i>tVCONN_SwapDelayUFP</i></b>   |                 | 500         |             |             | ms                                                                                       | <a href="#">Section 6.6.4.6</a>                                                                                   |
| <b><i>tVDMBusy</i></b>              |                 | 50          |             |             | ms                                                                                       | <a href="#">Section 6.6.12.4</a>                                                                                  |
| <b><i>tVDMEnterMode</i></b>         |                 |             |             | 25          | ms                                                                                       | <a href="#">Section 6.6.12.2</a>                                                                                  |
| <b><i>tVDMExitMode</i></b>          |                 |             |             | 25          | ms                                                                                       | <a href="#">Section 6.6.12.3</a>                                                                                  |
| <b><i>tVDMReceiverResponse</i></b>  |                 |             |             | 15          | ms                                                                                       | <a href="#">Section 6.6.12.1</a>                                                                                  |
| <b><i>tVDMSenderResponse</i></b>    |                 | 24          | 27          | 30          | ms                                                                                       | <a href="#">Section 6.6.12.1</a>                                                                                  |
| <b><i>tVDMWaitModeEntry</i></b>     |                 | 40          | 45          | 50          | ms                                                                                       | <a href="#">Section 6.6.12.2</a>                                                                                  |
| <b><i>tVDMWaitModeExit</i></b>      |                 | 40          | 45          | 50          | ms                                                                                       | <a href="#">Section 6.6.12.3</a>                                                                                  |

Table 6.69 Timers

| Timer                            | Parameter                    | Used By        | Reference                        |
|----------------------------------|------------------------------|----------------|----------------------------------|
| <i>BISTContModeTimer</i>         | <i>tBISTContMode</i> ✖       | Policy Engine  | <a href="#">Section 6.6.7.2</a>  |
| <i>ChunkingNotSupportedTimer</i> | <i>tChunkingNotSupported</i> | Policy Engine  | <a href="#">Section 6.6.18.1</a> |
| <i>ChunkSenderRequestTimer</i>   | <i>tChunkSenderRequest</i>   | Protocol Layer | <a href="#">Section 6.6.18.2</a> |
| <i>ChunkSenderResponseTimer</i>  | <i>tChunkSenderResponse</i>  | Protocol Layer | <a href="#">Section 6.6.18.3</a> |
| <i>CRCReceiveTimer</i>           | <i>tReceive</i>              | Protocol Layer | <a href="#">Section 6.6.1</a>    |
| <i>DataResetFailTimer</i>        | <i>tDataResetFail</i>        | Policy Engine  | <a href="#">Section 6.6.10.3</a> |
| <i>DataResetFailUFPTimer</i>     | <i>tDataResetFailUFP</i>     | Policy Engine  | <a href="#">Section 6.6.10.4</a> |
| <i>DiscoverIdentityTimer</i>     | <i>tDiscoverIdentity</i>     | Policy Engine  | <a href="#">Section 6.6.15</a>   |
| ✖ <i>HardResetCompleteTimer</i>  | <i>tHardResetComplete</i>    | Protocol Layer | <a href="#">Section 6.6.9</a>    |
| <i>NoResponseTimer</i>           | <i>tNoResponse</i>           | Policy Engine  | <a href="#">Section 6.6.6</a>    |
| <i>PSHardResetTimer</i>          | <i>tPSHardReset</i>          | Policy Engine  | <a href="#">Section 6.6.11.2</a> |
| <i>PSSourceOffTimer</i>          | <i>tPSSourceOff</i>          | Policy Engine  | <a href="#">Section 6.6.5.2</a>  |
| <i>PSSourceOnTimer</i>           | <i>tPSSourceOn</i>           | Policy Engine  | <a href="#">Section 6.6.5.3</a>  |
| <i>PSTransitionTimer</i>         | <i>tPSTransition</i>         | Policy Engine  | <a href="#">Section 6.6.5.1</a>  |
| <i>SenderResponseTimer</i>       | <i>tSenderResponse</i>       | Policy Engine  | <a href="#">Section 6.6.2</a>    |
| <i>SinkEPREnterTimer</i>         | <i>tEnterEPR</i>             | Policy Engine  | <a href="#">Section 6.6.21.1</a> |
| <i>SinkEPRKeepAliveTimer</i>     | <i>tSinkEPRKeepAlive</i>     | Policy Engine  | <a href="#">Section 6.6.21.2</a> |
| <i>SinkPPSPeriodicTimer</i>      | <i>tPPSRequest</i>           | Policy Engine  | <a href="#">Section 6.6.19.1</a> |
| <i>SinkRequestTimer</i>          | <i>tSinkRequest</i>          | Policy Engine  | <a href="#">Section 6.6.4</a>    |
| <i>SinkWaitCapTimer</i>          | <i>tTypeCSinkWaitCap</i>     | Policy Engine  | <a href="#">Section 6.6.3.2</a>  |
| <i>SourceCapabilityTimer</i>     | <i>tTypeCSendSourceCap</i>   | Policy Engine  | <a href="#">Section 6.6.3.1</a>  |
| <i>SourceEPRKeepAliveTimer</i>   | <i>tSourceEPRKeepAlive</i>   | Policy Engine  | <a href="#">Section 6.6.21.3</a> |
| <i>SourcePPSCommTimer</i>        | <i>tPPSTimeout</i>           | Policy Engine  | <a href="#">Section 6.6.19.2</a> |
| <i>SinkTxTimer</i>               | <i>tSinkTx</i>               | Protocol Layer | <a href="#">Section 6.6.16</a>   |
| <i>SwapSourceStartTimer</i>      | <i>tSwapSourceStart</i>      | Policy Engine  | <a href="#">Section 6.6.8.1</a>  |
| <i>VCONNDischargeTimer</i>       | <i>tVCONNSourceDischarge</i> | Policy Engine  | <a href="#">Section 6.6.10.1</a> |
| <i>VCONNOnTimer</i>              | <i>tVCONNSourceTimeout</i>   | Policy Engine  | <a href="#">Section 6.6.13.1</a> |
| <i>VDMModeEntryTimer</i>         | <i>tVDMWaitModeEntry</i>     | Policy Engine  | <a href="#">Section 6.6.12.2</a> |
| <i>VDMModeExitTimer</i>          | <i>tVDMWaitModeExit</i>      | Policy Engine  | <a href="#">Section 6.6.12.3</a> |
| <i>VDMResponseTimer</i>          | <i>tVDMSenderResponse</i>    | Policy Engine  | <a href="#">Section 6.6.12.1</a> |



## 6.7 Counters

### 6.7.1 MessageID Counter

The **MessageIDCounter** is a rolling counter, ranging from 0 to **nMessageIDCount**, used to detect duplicate Messages. This value is used for the **MessageID** field in the *Message Header* of each transmitted *Message*.

Each *Port* **Shall** maintain a copy of the last **MessageID** value received from its *Port Partner*. Devices that support multiple ports, such as *Hubs*, **Shall** maintain copies of the last **MessageID** on a per *Port* basis. A *Port* which communicates using *SOP\* Packets* **Shall** maintain copies of the last **MessageID** for each type of *SOP\** it uses.

The transmitter **Shall** use the **MessageID** in a **GoodCRC** *Message* to verify that a particular *Message* was received correctly. The receiver **Shall** use the **MessageID** to detect duplicate Messages.

#### 6.7.1.1 Transmitter Usage

The Transmitter **Shall** use the **MessageID** as follows:

- Upon receiving either **Hard Reset** Signaling, or a **Soft\_Reset** *Message*, the transmitter **Shall** set its **MessageIDCounter** to zero and re-initialize its retry mechanism.
- If a **GoodCRC** *Message* with a **MessageID** matching the **MessageIDCounter** is not received before the **CRCReceiveTimer** expires, it **Shall** retry the same **Packet** up to **nRetryCount** times using the same **MessageID**.
- If a **GoodCRC** *Message* is received with a **MessageID** matching the current **MessageIDCounter** before the **CRCReceiveTimer** expires, the transmitter **Shall** re-initialize its retry mechanism and increment its **MessageIDCounter**.
- If the *Message* is aborted by the *Policy Engine*, the transmitter **Shall** delete the *Message* from its transmit buffer, re-initialize its retry mechanism and increment its **MessageIDCounter**.

#### 6.7.1.2 Receiver Usage

The Receiver **Shall** use the **MessageID** as follows:

- When the first good **Packet** is received after a reset, the receiver **Shall** store a copy of the received **MessageID** value.
- For subsequent Messages, if **MessageID** value in a received *Message* is the same as the stored value, the receiver **Shall** return a **GoodCRC** *Message* with that **MessageID** value and drop the *Message* (this is a retry of an already received *Message*).

**Note:** This **Shall Not** apply to the **Soft\_Reset** *Message* which always has a **MessageID** value of zero.

- If **MessageID** value in the received *Message* is different than the stored value, the receiver **Shall** return a **GoodCRC** *Message* with the new **MessageID** value, store a copy of the new **MessageID** value and process the *Message*.

### 6.7.2 Retry Counter

The **RetryCounter** is used by a *Port* whenever there is a *Message* transmission failure (timeout of **CRCReceiveTimer**). If the **nRetryCount** retry fails, then the link **Shall** be reset using the *Soft Reset* mechanism.

The following rules apply to retries when there is a *Message* transmission failure (see also [Section 6.12.2.2, "Protocol Layer Message Transmission"](#)):

- *Cable Plugs* **Shall Not** retry Messages.
- *Extended Messages* of **Data Size** > **MaxExtendedMsgLegacyLen** that are not *Chunked* (**Chunked** flag set to zero) **Shall Not** be retried.



- Extended Messages of **Data Size**  $\leq$  **MaxExtendedMsgLegacyLen** (**Chunked** flag set to zero or one) **Shall** be retried.
- Extended Messages of **Data Size**  $>$  **MaxExtendedMsgLegacyLen** that are **Chunked** (**Chunked** flag set to one) individual Chunks **Shall** be retried.

When **Messages** are not retried, then the **RetryCounter** is not used. Higher layer protocols are expected to accommodate **Message** delivery failure or failure to receive a **GoodCRC** Message.

### 6.7.3 Hard Reset Counter

The **HardResetCounter** is used to retry the **Hard Reset** whenever there is no response from the remote device (see [Section 6.6.6, "NoResponseTimer"](#)). Once the **Hard Reset** has been retried **nHardResetCount** times then it **Shall** be assumed that the remote device is non-responsive.

### 6.7.4 Capabilities Counter

The **CapsCounter** is used to count the number of **Source\_Capabilities** Messages which have been sent by a **Source** at power up or after a **Hard Reset**. Implementation of the **CapsCounter** is **Optional** but **May** be used by any **Source** which wishes to preserve power by not sending **Source\_Capabilities** Messages after a period of time.

When the **CapsCounter** is implemented and the **Source** detects that a **Sink** is **Attached** then after **nCapsCount** **Source\_Capabilities** Messages have been sent the **Source** **Shall** decide that the **Sink** is non-responsive, stop sending **Source\_Capabilities** Messages and disable PD.

A **Sink** **Shall** use the **SinkWaitCapTimer** to trigger the resending of **Source\_Capabilities** Messages by a USB Power Delivery capable **Source** which has previously stopped sending **Source\_Capabilities** Messages. Any **Sink** which is **Attached** and does not detect a **Source\_Capabilities** Message, **Shall** issue **Hard Reset** Signaling when the **SinkWaitCapTimer** times out in order to reset the **Source**. Resetting the **Source** **Shall** also reset the **CapsCounter** and restart the sending of **Source\_Capabilities** Messages.

### 6.7.5 Discover Identity Counter

When sending **Discover Identity** Messages to a **Cable Plug** a **Port** **Shall** maintain a count of Messages sent (**DiscoverIdentityCounter**). No more than **nDiscoverIdentityCount** **Discover Identity** Messages **Shall** be sent by the **Port** without receiving a **GoodCRC** Message response. A **VCONN Swap** **Shall** reset the **DiscoverIdentityCounter**.

### 6.7.6 VDMBusyCounter

When sending **Responder BUSY** responses to a Structured **Vendor\_Defined** Message a **UFP** or **Cable Plug** **Shall** maintain a count of Messages sent (**VDMBusyCounter**). No more than **nBusyCount** **Responder BUSY** responses **Shall** be sent. The **VDMBusyCounter** **Shall** be reset on sending a **non-BUSY** response. Products wishing to meet [\[USB Type-C 2.4\]](#) requirements for **Alternate Mode** entry **Should** use an **nBusyCount** of 1.

## 6.7.7 Counter Values and Counters

[Table 6.70, "Counter Parameters"](#) lists the counters used in this section and [Table 6.71, "Counters"](#) shows the corresponding parameters.

**Table 6.70 Counter Parameters**

| Parameter                                                                                                | Value | Reference                     |
|----------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| <i>nBusyCount</i>                                                                                        | 5     | <a href="#">Section 6.7.6</a> |
| <i>nCapsCount</i>                                                                                        | 50    | <a href="#">Section 6.7.4</a> |
| <i>nDiscoverIdentityCount</i>                                                                            | 20    | <a href="#">Section 6.7.5</a> |
|  <i>nHardResetCount</i> | 2     | <a href="#">Section 6.7.3</a> |
| <i>nMessageIDCount</i>                                                                                   | 7     | <a href="#">Section 6.7.1</a> |
| <i>nRetryCount</i>                                                                                       | 2     | <a href="#">Section 6.7.2</a> |

**Table 6.71 Counters**

| Counter                                                                                                   | Max                           | Reference                                                                                                         |
|-----------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <i>CapsCounter</i>                                                                                        | <i>nCapsCount</i>             | <a href="#">Section 6.7.4</a>  |
| <i>DiscoverIdentityCounter</i>                                                                            | <i>nDiscoverIdentityCount</i> | <a href="#">Section 6.7.5</a>                                                                                     |
|  <i>HardResetCounter</i> | <i>nHardResetCount</i>        | <a href="#">Section 6.7.3</a>                                                                                     |
| <i>MessageIDCounter</i>                                                                                   | <i>nMessageIDCount</i>        | <a href="#">Section 6.7.1</a>                                                                                     |
| <i>RetryCounter</i>                                                                                       | <i>nRetryCount</i>            | <a href="#">Section 6.7.2</a>                                                                                     |
| <i>VDMBusyCounter</i>                                                                                     | <i>nBusyCount</i>             | <a href="#">Section 6.7.6</a>                                                                                     |

## 6.8 Reset

Resets are a necessary response to protocol or other error conditions. USB Power Delivery defines four different types of reset:

- *Soft Reset*, which resets protocol.
- *Data Reset* which resets the *USB Communications*.
- *Hard Reset* which resets both the power supplies and protocol
- *Cable Reset* which resets the cable.

### 6.8.1 Soft Reset and Protocol Error

A **Soft\_Reset** Message is used to cause a *Soft Reset* of protocol communication when this has broken down in some way. It **Shall Not** have any impact on power supply operation but is used to correct a *Protocol Error* occurring during an *Atomic Message Sequence* (AMS). The *Soft Reset* **May** be triggered by either *Port Partner* in response to the *Protocol Error*.

*Protocol Errors* are any unexpected *Message* during an AMS. If the first *Message* in an AMS has been passed to the *Protocol Layer* by the *Policy Engine* but has not yet been sent (i.e., a **GoodCRC** Message acknowledging the *Message* has not been received) when the *Protocol Error* occurs, the *Policy Engine* **Shall Not** issue a *Soft Reset* but **Shall** return to the **PE\_SNK\_Ready** or **PE\_SRC\_Ready** state and then process the incoming *Message*. If the incoming *Message* is an *Unexpected Message* received in the **PE\_SNK\_Ready** or **PE\_SRC\_Ready** state, the *Policy Engine* **Shall** issue a *Soft Reset*. If the *Protocol Error* occurs during an AMS this **Shall** lead to a *Soft Reset* in order to re-synchronize the *Policy Engine* state machines (see [Section 8.3.3.4, "SOP Soft Reset and Protocol Error State Diagrams"](#)) except when the **voltage** is transition when a *Protocol Error* **Shall** lead to a *Hard Reset* (see [Section 6.6.11.4, "tProtErrHardReset"](#) and [Section 8.3.3.2, "Policy Engine Source Port State Diagram"](#)). Details of AMS's can be found in [Section 8.3.2.1.3, "Atomic Message Sequences"](#).

An *Unrecognized Message* or *Unsupported Message* received in the **PE\_SNK\_Ready** or **PE\_SRC\_Ready** states, **Shall Not** cause a **Soft\_Reset** Message to be generated but instead a **Not\_Supported** Message **Shall** be generated.

A **Soft\_Reset** Message **Shall** be sent regardless of the  $R_p$  value either **SinkTxOK** or **SinkTxNG** if it is the correct response in that state.

**Note:** This means that a **Soft\_Reset** Message can be sent during an AMS regardless of the  $R_p$  value either **SinkTxOK** or **SinkTxNG** when responding to a *Protocol Error*.

Table 6.72, "Response to an incoming Message (except VDM)" and Table 6.73, "Response to an incoming VDM" summarize the responses that **Shall** be made to an incoming Message including VDMs.

**Table 6.72 Response to an incoming Message (except VDM)**

| Recipient's Power Role | Recipient's state                                  | Incoming Message |                                       |                                          |                                                                                                                                                            |
|------------------------|----------------------------------------------------|------------------|---------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                    | Recognized       |                                       |                                          | Unrecognized                                                                                                                                               |
|                        |                                                    | Supported        |                                       | Unsupported                              |                                                                                                                                                            |
|                        |                                                    | Expected         | Unexpected                            |                                          |                                                                                                                                                            |
| Source                 | <b>PE_SRC_Ready</b>                                | Process Message  | <b>Soft_Reset Message<sup>2</sup></b> | <b>Not_Supported Message<sup>3</sup></b> | <b>Not_Supported Message<sup>3</sup></b> (except for VDM)<br>See <a href="#">Section 6.4.4.1</a> for UVDM.<br>See <a href="#">Section 6.4.4.1</a> for SVDM |
|                        | During AMS (power not transitioning <sup>1</sup> ) | Process Message  | <b>Soft_Reset Message<sup>2</sup></b> |                                          |                                                                                                                                                            |
|                        | During AMS (power transitioning <sup>1</sup> )     | Process Message  | <b>Hard_Reset Signaling</b>           |                                          |                                                                                                                                                            |
| Sink                   | <b>PE_SNK_Ready</b>                                | Process Message  | <b>Soft_Reset Message<sup>2</sup></b> | <b>Not_Supported Message<sup>3</sup></b> | <b>Not_Supported Message<sup>3</sup></b> (except for VDM)<br>See <a href="#">Section 6.4.4.1</a> for UVDM.<br>See <a href="#">Section 6.4.4.1</a> for SVDM |
|                        | During AMS (not power transitioned)                | Process Message  | <b>Soft_Reset Message<sup>2</sup></b> |                                          |                                                                                                                                                            |
|                        | During AMS (power transitioned)                    | Process Message  | <b>Hard_Reset Signaling</b>           |                                          |                                                                                                                                                            |

1)

“Power transitioning” means the *Policy Engine* is in **PE\_SRC\_Transition\_Supply** State or **PE\_SNK\_Transition\_Sink** State or **PE\_FRS\_SNK\_SRC\_Start\_AMS** State.

2)

The **Soft\_Reset** Message **Shall** be sent using the SOP\* of the incoming Message.

3)

The **Not\_Supported** Message **Shall** be sent using the SOP\* of the incoming Message.

**Table 6.73 Response to an incoming VDM**

| Recipient's Role         | Unstructured VDM  |                              |                              | Structured VDM                     |                              |                    |
|--------------------------|-------------------|------------------------------|------------------------------|------------------------------------|------------------------------|--------------------|
|                          | Supported         | Unsupported                  | Unrecognized                 | Supported                          | Unsupported                  | Unrecognized       |
| <b>DFP</b> or <b>UFP</b> | Defined by vendor | <b>Not_Supported Message</b> | <b>Not_Supported Message</b> | See <a href="#">Section 6.13.5</a> | <b>Not_Supported Message</b> | <b>NAK Command</b> |
| <b>Cable Plug</b>        | Defined by vendor | <b>Message Ignored</b>       | <b>Message Ignored</b>       | See <a href="#">Section 6.13.5</a> | <b>Message Ignored</b>       | <b>NAK Command</b> |

A failure to see a **GoodCRC** Message in response to any Message within **tReceive** (after **nRetryCount** retries), when a Port Pair is **Connected**, is indicative of a communications failure resulting in a Soft Reset (see [Section 6.6.9.1, "tSoftReset"](#)).

A Soft Reset **Shall** impact the USB Power Delivery layers in the following ways:

- **PHY Layer**: Reset not required since the **PHY Layer** resets on each **Packet** transmission/reception.
- **Protocol Layer**: Reset **MessageIDCounter**, **RetryCounter** and state machines.

- **Policy Engine:** Reset state dependent behavior by performing an *Explicit Contract Negotiation*.
- Power supply: **Shall Not** change.

**Note:** When in *SPR Mode* the Source sends a *Source\_Capabilities* Message and when in *EPR Mode* the Source sends an *EPR\_Source\_Capabilities* Message.

A *Soft Reset* is performed using an *AMS* (see [Table 8.8, "AMS: Soft Reset"](#)). Message numbers **Shall** be set to zero prior to sending the *Soft\_Reset/Accept* Message since the issue might be with the counters. The sender of a *Soft\_Reset* Message **Shall** reset its *MessageIDCounter* and *RetryCounter*, the receiver of the Message **Shall** reset its *MessageIDCounter* and *RetryCounter* before sending the *Accept* Message response. Any failure in the *Soft Reset* process will trigger a *Hard Reset* when *SOP Packets* are being used or *Cable Reset*, sent by the *DFP* only, for any other *SOP\* Packets*; for example a *GoodCRC* Message is not received during the *Soft Reset* process (see [Section 6.8.3, "Hard Reset"](#) and [Section 6.8.4, "Cable Reset"](#)).

## 6.8.2 Data Reset

A *Data\_Reset* Message is used by a *Port* to reset its USB data connection and to exit all *Alternate Modes* both with its *Port Partner* and in the *Cable Plug(s)*.

- The *Data Reset* process **May** be initiated by either *Port Partner* sending a *Data\_Reset* Message.

A *Data Reset* impacts USB Power Delivery in the following ways:

- **Shall Not** change the *Port Power Roles* (*Source/Sink*) or *Port Data Roles* (*DFP/UFP*).
- **Shall Not** change the existing *Explicit Contract*.
- **Shall** cause all *Active Modes* to be exited.
- **Shall** reset the cable by Power cycling *VCONN*.
- The *DFP* **Shall** become the *VCONN Source*.
- If the *Data Reset* process fails, then the *Port* **Shall** enter the *ErrorRecovery* State as defined in [\[USB Type-C 2.4\]](#).

See [Section 6.3.14, "Data\\_Reset Message"](#) for details of *Data Reset* operation.

## 6.8.3 Hard Reset

*Hard Resets* are signaled by an ordered set as defined in [Section 5.6.4, "Hard Reset"](#). Both the sender and recipient **Shall** cause their power supplies to return to their default states (see [Section 7.3.3.1, "Source Initiated Hard Reset"](#) and [Section 7.3.3.2, "Sink Initiated Hard Reset"](#) for details of voltage transitions). In addition, their respective *Protocol Layers* **Shall** be reset as for the *Soft Reset*. This allows the *Attached* devices to be in a state where they can re-establish USB PD communication. *Hard Reset* is retried up to *nHardResetCount* times (see also [Section 6.6.6, "NoResponseTimer"](#) and [Section 6.7.3, "Hard Reset Counter"](#)).

**Note:** Even though *VBUS* drops to *vSafe0V* during a *Hard Reset* a *Sink* will not see this as a disconnect since this is expected behavior.

A *Hard Reset* **Shall Not** cause any change to either the  $R_p/R_d$  resistor being asserted.

If there has been a *Data Role Swap* the *Hard Reset* **Shall** cause the *Port Data Role* to be changed back to *DFP* for a *Port* with the  $R_p$  resistor asserted and *UFP* for a *Port* with the  $R_d$  resistor asserted.

When *VCONN* is supported (see [\[USB Type-C 2.4\]](#)) the *Hard Reset* **Shall** cause the *Port* with the  $R_p$  resistor asserted to supply *VCONN* and the *Port* with the  $R_d$  resistor asserted to turn off *VCONN*.

In effect the *Hard Reset* will revert the *Ports* to their default state based on their *CC* line resistors. Removing and reapplying *VCONN* from the *Cable Plugs* also ensures that they re-establish their configuration as either *SOP'* or *SOP''* based on the location of *VCONN* (see [\[USB Type-C 2.4\]](#)).

If the *Hard Reset* is insufficient to clear the error condition, then the *Port* **Shall** use USB *Type-C ErrorRecovery* as defined in [\[USB Type-C 2.4\]](#).

A *Sink* **Shall** be able to send *Hard Reset Signaling* regardless of the value of  $R_p$  (see [Section 5.7, "Collision Avoidance"](#)).

### 6.8.3.1 Cable Plugs and Hard Reset

*Cable Plugs* **Shall Not** generate *Hard Reset Signaling* but **Shall** monitor for *Hard Reset Signaling* between the *Port Partners* and **Shall** reset when this is detected (see [Section 8.3.3.25.2.2, "Cable Plug Hard Reset State Diagram"](#)). The *Cable Plugs* **Shall** perform the equivalent of a power cycle returning to their initial power up state. This allows the *Port Partners* to be in a state where they can re-establish USB PD communication.

### 6.8.3.2 Modal Operation and Hard Reset

A *Hard Reset* **Shall** cause *EPR Mode* and all *Active Modes* to be exited by both *Port Partners* and any *Cable Plugs* (see [Section 6.4.4.3.4, "Enter Mode Command"](#)).

### 6.8.4 Cable Reset

*Cable Resets* are signaled by an ordered set as defined in [Section 5.6.5, "Cable Reset"](#). Both the sender and recipient of *Cable Reset Signaling* **Shall** reset their respective *Protocol Layers*. The *Cable Plugs* **Shall** perform the equivalent of a power cycle returning to their initial power up state. This allows the *Port Partners* to be in a state where they can re-establish USB PD communication.

The *DFP* must be supplying *VCONN* prior to a *Cable Reset*. If *VCONN* has been turned off the *DFP* **Shall** turn on *VCONN* prior to generating *Cable Reset Signaling*. If there has been a *VCONN Swap* and the *UFP* is currently supplying *VCONN*, the *DFP* **Shall** perform a *VCONN Swap* such that it is supplying *VCONN* prior to generating *Cable Reset Signaling*.

Only a *DFP* **Shall** generate *Cable Reset Signaling*. A *DFP* **Shall** only generate *Cable Reset Signaling* within an *Explicit Contract*.

A *Cable Reset* **Shall** cause all *Active Modes* in the *Cable Plugs* to be exited (see [Section 6.4.4.3.4, "Enter Mode Command"](#)).

## 6.9 Accept, Reject and Wait

The recipient of a *Request*, *EPR\_Request*, *PR\_Swap*, *DR\_Swap*, *VCONN\_Swap*, or *Enter\_USB Message* **shall** respond by sending one of the following responses:

- An *Accept Message* in response to a **Valid** request which can be serviced immediately (see [Section 6.3.3, "Accept Message"](#)).
- A *Wait Message* in response to a **Valid** request which cannot be serviced immediately but could be serviced at a later time (see [Section 6.3.12, "Wait Message"](#)).
- A *Reject Message* in response to an **Invalid** request or a request which is outside of the device's design Capabilities (see [Section 6.3.4, "Reject Message"](#)).



## 6.10 Collision Avoidance

To avoid *Message* collisions due to asynchronous *Messaging* sent from the *Sink*, the *Source* sets  $R_p$  to *SinkTxOK* to indicate to the *Sink* that it is *OK* to initiate an *AMS*. When the *Source* wishes to initiate an *AMS*, it sets  $R_p$  to *SinkTxNG*. When the *Sink* detects that  $R_p$  is set to *SinkTxOK* it *May* initiate an *AMS*. When the *Sink* detects that  $R_p$  is set to *SinkTxNG* it *Shall Not* initiate an *AMS* and *Shall* only send *Messages* that are part of an *AMS* the *Source* has initiated.

**Note:** This restriction applies to *SOP\*AMS's* i.e., for both *Port to Port* and *Port to Cable Plug* communications.

If a transition into the *PE\_SRC\_Ready* state will result in an immediate transition out of the *PE\_SRC\_Ready* state within *tSrcHoldsBus* e.g. it is due to a *Protocol Error* that has not resulted in a *Soft Reset*, then the notifications of the end of *AMS* and first *Message* in an *AMS* *May Not* be sent to avoid changing the  $R_p$  value unnecessarily.

**Note:** A *Sink* can still send *Hard Reset Signaling* at any time.

## 6.11 Message Discarding

On receiving a received *Message* on *SOP*, the *Protocol Layer* **Shall Discard** any pending *SOP\** *Messages*. A received *Message* on *SOP'/SOP''* **Shall Not** cause any pending *SOP\** *Messages* to be **Discarded**.

It is assumed that *Messages* using *SOP'/SOP''* constitute a simple request/response *AMS*, with the *Cable Plug* providing the response so there is no reason for a pending *SOP\** *Message* to be **Discarded**. There can only be one *AMS* between the *Port Partners*, and these also take priority over *Cable Plug* communications so a *Message* received on *SOP* will always cause a *Message* pending on *SOP\** to be **Discarded**.

[Table 6.74, "Message Discarding"](#) for details of the *Messages* that **Shall/ Shall Not** be **Discarded**.

**Table 6.74 Message Discarding**

| Message pending transmission | Message received  | Message to be Discarded |
|------------------------------|-------------------|-------------------------|
| <i>SOP</i>                   | <i>SOP</i>        | Outgoing <i>Message</i> |
| <i>SOP</i>                   | <i>SOP'/SOP''</i> | Incoming <i>Message</i> |
| <i>SOP'</i>                  | <i>SOP</i>        | Outgoing <i>Message</i> |
| <i>SOP'</i>                  | <i>SOP'</i>       | Incoming <i>Message</i> |
| <i>SOP'</i>                  | <i>SOP''</i>      | Incoming <i>Message</i> |
| <i>SOP''</i>                 | <i>SOP</i>        | Outgoing <i>Message</i> |
| <i>SOP''</i>                 | <i>SOP'</i>       | Incoming <i>Message</i> |
| <i>SOP''</i>                 | <i>SOP''</i>      | Incoming <i>Message</i> |

## 6.12 State behavior

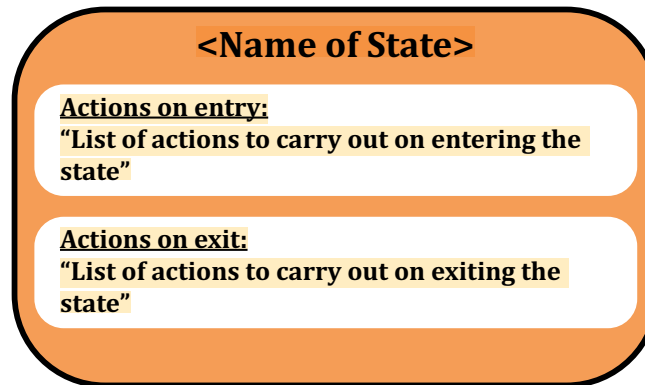
### 6.12.1 Introduction to state diagrams used in Chapter 6

The state diagrams defined in [Section 6.12, "State behavior"](#) are **Normative** and **Shall** define the operation of the Power Delivery *Protocol Layer*.

**Note:** These state diagrams are not intended to replace a well written and robust design.

[Figure 6.57, "Outline of States"](#) shows an outline of the states defined in the following sections. At the top there is the name of the state. This is followed by "Actions on entry" a list of actions carried out on entering the state and in some states "Actions on exit" a list of actions carried out on exiting the state.

Figure 6.57 Outline of States

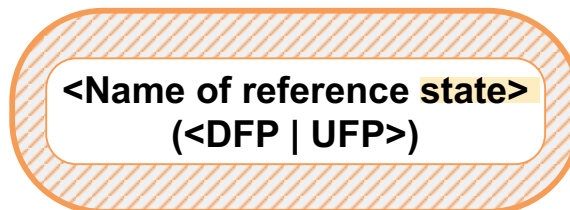


Transitions from one state to another are indicated by arrows with the conditions listed on the arrow. Where there are multiple conditions, these are connected using either a logical OR "|" or a logical AND "&." The inverse of a condition is shown with a "NOT" in front of the condition.

In some cases, there are transitions which can occur from any state to a particular state. These are indicated by an arrow which is unconnected to a state at one end, but with the other end (the point) connected to the final state.

In some state diagrams it is necessary to enter or exit from states in other diagrams. [Figure 6.57, "Outline of States"](#) indicates how such references are made. The reference is indicated with a hatched box. The box contains the name of the referenced state.

Figure 6.58 References to states



Timers are included in many of the states. Timers are initialized (set to their starting condition) and run (timer is counting) in the state it is referenced. As soon as the state is exited then the timer is no longer active. Timeouts of the timers are listed as conditions on state transitions.

Conditions listed on state transitions will come from one of three sources:

- Messages received from the *PHY Layer*.
- Events triggered within the *Protocol Layer* e.g., timer timeouts
- Message and related indications passed up to the *Policy Engine* from the *Protocol Layer* (Message sent; Message received etc.)

## 6.12.2 State Operation

The following section details *Protocol Layer* State Operation when sending and receiving *SOP\* Packets*.

For each *SOP' Communication* being sent and received there **shall** be separate *Protocol Layer* Transmission and *Protocol Layer* Reception and *Hard Reset* State Machine instances, with their own counter and timer instances. When *Chunking* is supported there **shall** be separate *Chunked Tx*, *Chunked Rx*, and *Chunked Message Router* State Machine instances.

*Soft Reset* **shall** only apply to the State Machine instances it is targeted at based on the type of *SOP\* Packet* used to send the *Soft\_Reset* Message. The *Hard Reset* State Machine (including *Cable Reset*) **shall** apply simultaneously to all *Protocol Layer* State Machine instances active in the *DFP*, *UFP* and *Cable Plug* (if present).

### 6.12.2.1 Protocol Layer Chunking

#### 6.12.2.1.1 Architecture of Device Including Chunking Layer

The *Chunking* component resides in the *Protocol Layer* between the *Policy Engine* and *Protocol Tx/Rx*. [Figure 6.59, "Chunking architecture Showing Message and Control Flow"](#) illustrates the relationship between components.

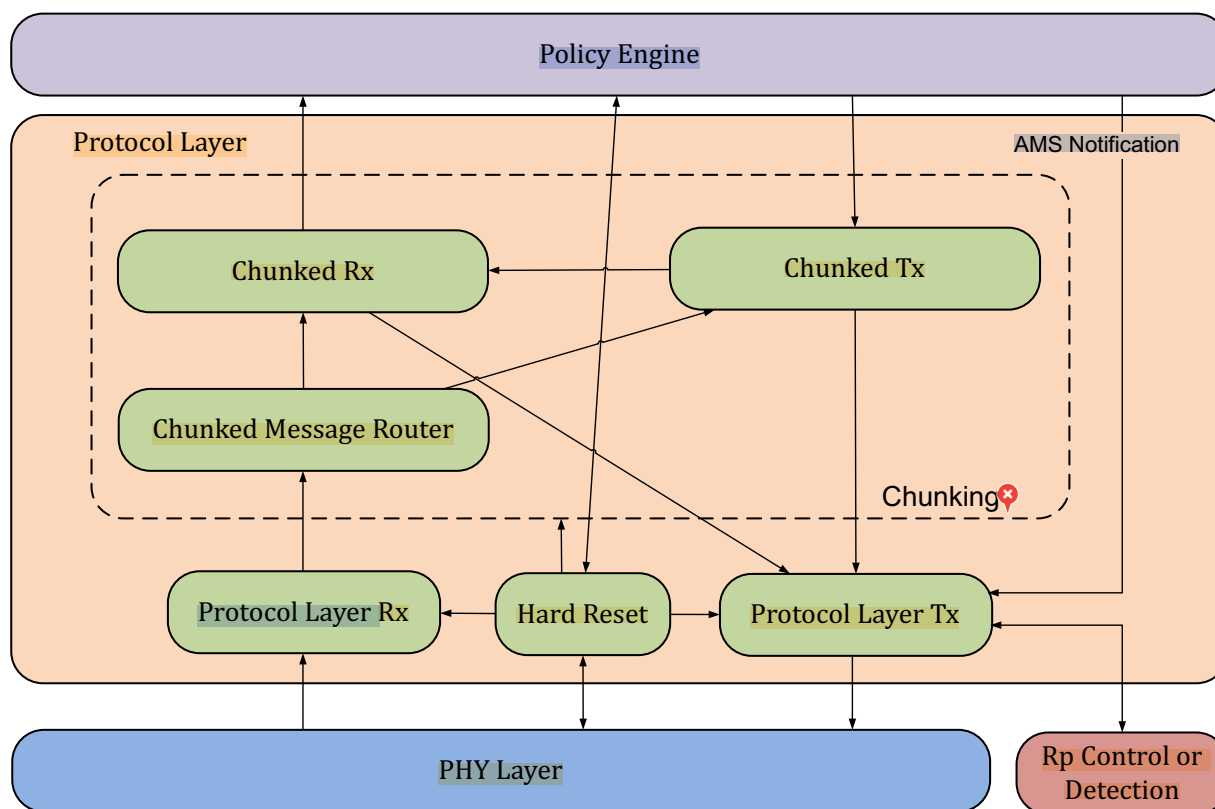
The *Chunking Layer* comprises three related state machines:

- *Chunked Rx*.
- *Chunked Tx*.
- *Chunked Message Router*.

**Note:** The consequence of this architecture is that the *Policy Engine* deals entirely in *Unchunked Messages*. It will not receive (and might not respond to) a *Message* until all the related chunks have been collated.

If a *PD device* or *Cable Plug* has no requirement to handle any *Message* requiring more than one *Chunk* of any *Extended Message*, it **may** omit the *Chunking Layer*. In this case it **shall** implement the *ChunkingNotSupportedTimer* to ensure compatible operation with partners which support *Chunking* (see [Section 6.6.18.1, "ChunkingNotSupportedTimer"](#) and [Section 8.3.3.6, "Not Supported Message State Diagrams"](#)).

Figure 6.59 Chunking architecture Showing Message and Control Flow



#### 6.12.2.1.1.1 Optional Abort Mechanism

Long *Chunked Messages* bring with them the potential problem that they could prevent urgent *Messages* from being transmitted in a timely manner. An **Optional** Abort mechanism is provided to remedy this problem.

The Abort Flag referred to in the diagrams below **May** be set and examined by the *Policy Engine*. The specific means are left to the implementer.

#### 6.12.2.1.1.2 Aborting Sending a Long-Chunked Message

A long-*Chunked Message* being sent **May** be aborted by setting the **Optional** Abort Flag. The *Message* **Shall** be considered aborted when the Abort Flag is again cleared by the *Chunked Tx* state machine.

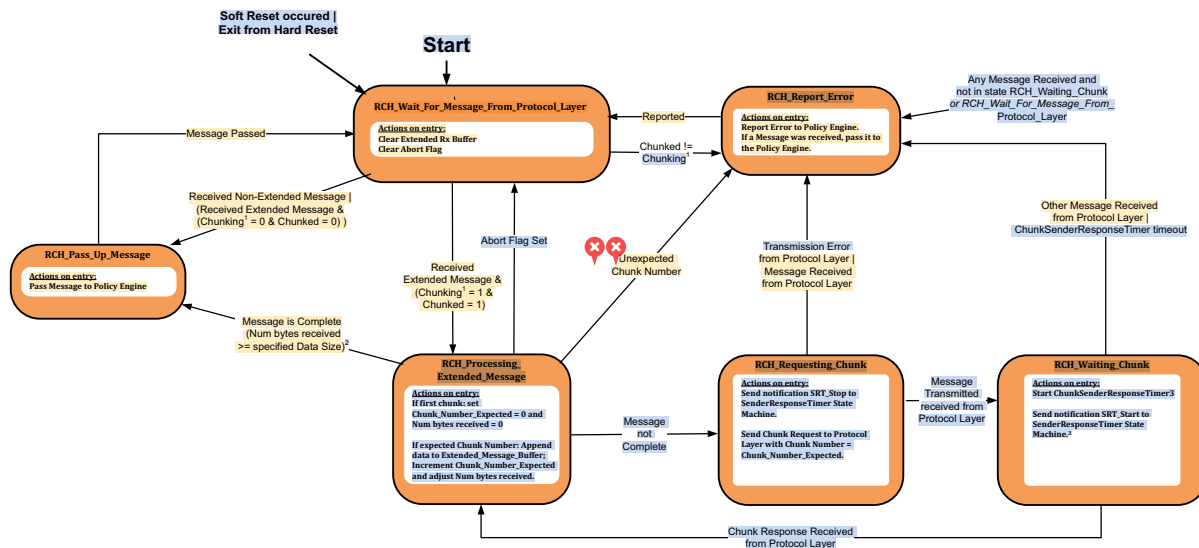
#### 6.12.2.1.1.3 Aborting Receiving a Long-Chunked Message

If the **Optional** Abort mechanism has been implemented, any *Message* sent while a *Chunked Message* receive is in progress will result in an error report being received by the *Policy Engine*, to indicate that the *Message* request has been **Discarded**. If the *Message* was urgent the *Policy Engine* might set the Abort Flag, which will result in the incoming *Chunked Message* being aborted. The Abort Flag being cleared by the *Chunked Rx* state machine indicates that the urgent *Message* can now be sent.

#### 6.12.2.1.2 Chunked Rx State Diagram

Figure 6.60, "*Chunked Rx State Diagram*" shows the state behavior for the *Chunked Rx* State Machine. This recognizes whether *Chunked* received *Messages* are involved and deals with requesting chunks when they are. It also performs validity checks on all *Messages* related to *Chunking*.

### Figure 6.60 Chunked Rx State Diagram



- <sup>1)</sup> Chunking is an internal state that is set to 1 if the "Unchunked Extended Messages Supported" bit in either Source Capabilities or Request is 0. It defaults to 0 and is set after the first exchange of Source Capabilities and Request. It is also set to 1 for *SOP*\* or *SOP*\* communication.
- <sup>2)</sup> Additional bytes received over specified *Data Size* will be because of padding in the last chunk.
- <sup>3)</sup> This state is responsible for starting two timers of similar length. The implementor **Should** mitigate against more than one of these timers resulting in recovery action.

#### 6.12.2.1.2.1 RCH\_Wait\_For\_Message\_From\_Protocol\_Layer State

The *Chunked* Rx State Machine **shall** enter the *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer* state:

- At startup.
- As a result of a *Soft Reset* occurring.
- On exit from a *Hard Reset*.

On entry to the ***RCH\_Wait\_For\_Message\_From\_Protocol\_Layer*** state the *Chunked* Rx state machine clears the Extended Rx Buffer and clears the ***Optional*** Abort Flag.

In the *RGH.Wait\_For\_Message\_From\_Protocol\_Layer* state the *Chunked* Rx state machine waits until the *Chunked* Message Router passes up a received *Message*.

The *Chunked Rx State Machine* **Shall** transition to the *RCH\_Pass\_Up\_Message* state when:

- A *non-Extended Message* is passed up from the *Chunked Message Router*.
- An *Extended Message* is passed up from the *Chunked Message Router*, and the *Policy Engine* has determined that we are not doing *Chunking*, and the *Message* has its *Chunked* bit set to 0b.

The *Chunked* Rx State Machine **shall** transition to the *RCH\_Processing\_Extended\_Message* state when:

- An *Extended Message* is passed up from the *Chunked Message Router*, and the *Policy Engine* has determined that we are doing *Chunking*, and the *Message* has its *Chunked* bit set to 1b.

#### 6.12.2.1.2.2 RCH\_Pass\_Up\_Message State

On entry to the **RCH\_Pass\_Up\_Message** state the *Chunked Rx* state machine **shall** pass the received **Message** to the *Policy Engine*.

The *Chunked Rx* State Machine **shall** transition to the *RCH Wait For Message From Protocol Layer* state when:

- The *Message* has been passed.

#### 6.12.2.1.2.3

#### RCH\_Processing\_Extended\_Message State

On entry to the *RCH\_Processing\_Extended\_Message* state the *Chunked* Rx state machine **Shall**:

- If this is the first chunk:
  - Set *Chunk\_Number\_Expected* = 0.
  - Set Num bytes received = 0.
- If chunk contains the expected *Chunk Number*:
  - Append its data to the *Extended\_Message\_Buffer*.
  - Increment *Chunk\_Number\_Expected*.
  - Adjust Num bytes received.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Pass\_Up\_Message* state when:

- The *Message* is complete (i.e., Num bytes received  $\geq$  specified *Data Size*).

**Note:** The inequality allows for padding bytes in the last chunk, which are not actually part of the *Extended Message*.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Requesting\_Chunk* state when:

- The *Message* is not yet complete.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Report\_Error* state when:

- An unexpected *Chunk Number* is received.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer* state when:

- The *Optional* Abort Flag is set.

#### 6.12.2.1.2.4

#### RCH\_Requesting\_Chunk State

On entry to the *RCH\_Requesting\_Chunk* state the *Chunked* Rx state machine **Shall**:

- Send notification SRT\_Stop to *SenderResponseTimer* state machine (see [Section 8.3.3.1.1, "SenderResponseTimer State Diagram"](#)).
- Send *Chunk Request to Protocol Layer* with *Chunk Number* = *Chunk\_Number\_Expected*.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Waiting\_Chunk* state when:

- *Message Transmitted* is received from the *Protocol Layer*.

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Report\_Error* state when:

- Transmission Error is received from the *Protocol Layer*, or
- A *Message* is received from the *Protocol Layer*.

#### 6.12.2.1.2.5

#### RCH\_Waiting\_Chunk State

On entry to the *RCH\_Waiting\_Chunk* state the *Chunked* Rx state machine **Shall**:

- Start the *ChunkSenderResponseTimer*.
- Send notification SRT\_Start to *SenderResponseTimer* state machine (see [Section 8.3.3.1.1, "SenderResponseTimer State Diagram"](#)).

The *Chunked* Rx State Machine **Shall** transition to the *RCH\_Processing\_Extended\_Message* state when:

- A *Chunk* is received from the *Protocol Layer*.





The *Chunked Rx State Machine* **Shall** transition to the *RCH\_Report\_Error* state when:

- A *Message*, other than a *Chunk*, is received from the *Protocol Layer*, or
- The *ChunkSenderResponseTimer* expires.

#### 6.12.2.1.2.6 RCH\_Report\_Error State

The *Chunked Rx State Machine* **Shall** enter the *RCH\_Report\_Error* state:

- When any *Message* is received and the *Chunked Rx State Machine* is not in one of the states *RCH\_Waiting\_Chunk* or *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer*.

On entry to the *RCH\_Report\_Error* state the *Chunked Rx state machine* **Shall**:

- Report the error to the *Policy Engine*.
- If the state was entered because a *Message* was received, this *Message* **Shall** be passed to the *Policy Engine*.

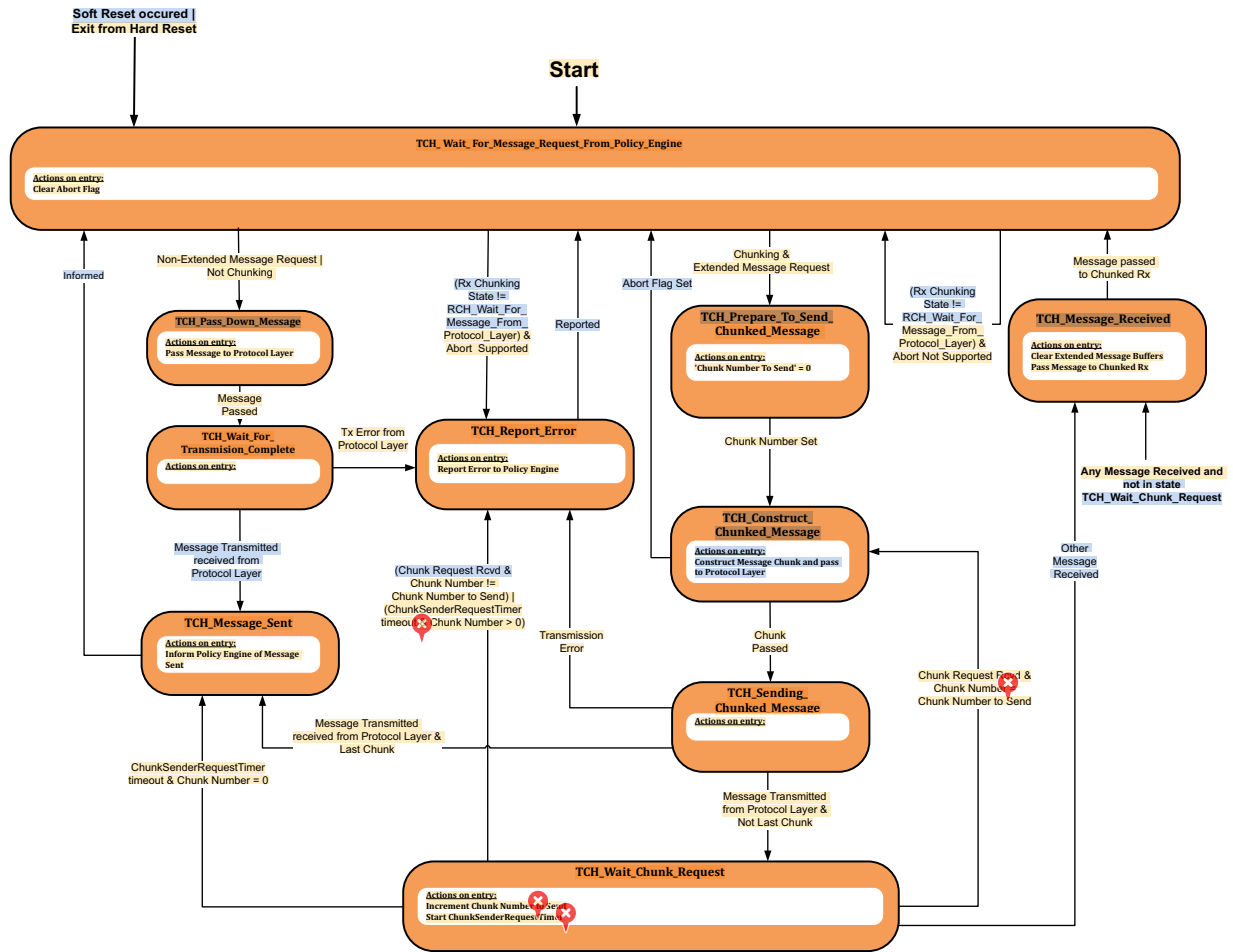
The *Chunked Rx State Machine* **Shall** transition to the *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer* state when:

- The error has been reported.
- Any *Message* received was passed to the *Policy Engine*.

### 6.12.2.1.3 Chunked Tx State Diagram

Figure 6.61, "Chunked Tx State Diagram" shows the state behavior for the *Chunked Tx State Machine*. This recognizes whether *Chunked* transmitted *Messages* are involved and deals with sending chunks and waiting for chunk requests when they are. It also performs validity checks on all related *Messages* related to *Chunking*.

Figure 6.61 Chunked Tx State Diagram



#### 6.12.2.1.3.1 TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine State

The *Chunked Tx State Machine* **shall** enter the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state:

- At startup.
- As a result of a *Soft Reset* occurring.
- On exit from a *Hard Reset*.

On entry to the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state the *Chunked Tx* state machine clears the **Optional** Abort Flag.

In the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state the *Chunked Tx* State Machine waits until the *Policy Engine* sends it a *Message Request*.

The *Chunked Tx* State Machine **shall** transition to the *TCH\_Pass\_Down\_Message* state when:

- A non-Extended *Message Request* is received from the *Policy Engine*, or
- A *Message Request* is received from the *Policy Engine* and the link is not *Chunking*.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Prepave\_To\_Send\_Chunked\_Message* state when:

- An *Extended Message* Request is received from the *Policy Engine*, and the link is *Chunking*.

The *Chunked* Tx State Machine **Shall Discard** the *Message* Request and remain in the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state when:

- The *Chunked* Rx state is any other than *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer*, and the **Optional** Abort Flag has not been implemented.

The *Chunked* Tx State Machine **Shall Discard** the *Message* Request and enter the *TCH\_Report\_Error* state when:

- The *Chunked* Rx state is any other than *RCH\_Wait\_For\_Message\_From\_Protocol\_Layer* and the **Optional** Abort Flag has been implemented.

#### 6.12.2.1.3.2 TCH\_Pass\_Down\_Message State

On entry to the *TCH\_Pass\_Down\_Message* state the *Chunked* Tx State Machine **Shall** pass the *Message* to the *Protocol Layer*.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Wait\_For\_Transmission\_Complete* state when:

- The *Message* has been passed to the *Protocol Layer*.

#### 6.12.2.1.3.3 TCH\_Wait\_For\_Transmission\_Complete State

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Message\_Sent* state when:

- *Message* Transmitted has been received from the *Protocol Layer*.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Report\_Error* state when:

- Transmission Error has been received from the *Protocol Layer*.

#### 6.12.2.1.3.4 TCH\_Message\_Sent State

On entry to the *TCH\_Message\_Sent* state the *Chunked* Tx State Machine **Shall**:

- Inform the *Policy Engine* that the *Message* has been sent.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state when:

- The *Policy Engine* has been informed.

#### 6.12.2.1.3.5 TCH\_Prepave\_To\_Send\_Chunked\_Message State

On entry to the *TCH\_Prepave\_To\_Send\_Chunked\_Message* state the *Chunked* Tx State Machine **Shall**:

- Set 'Chunk Number To *Send*' to zero.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Construct\_Chunked\_Message* state when:

- 'Chunk Number To *Send*' has been set to zero.

#### 6.12.2.1.3.6 TCH\_Construct\_Chunked\_Message State

On entry to the *TCH\_Construct\_Chunked\_Message* state the *Chunked* Tx State Machine **Shall**:

- Construct a *Message Chunk* and pass it to the *Protocol Layer*.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Sending\_Chunked\_Message* state when:

- The *Message Chunk* has been passed to the *Protocol Layer*.

The *Chunked* Tx State Machine **Shall** transition to the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state when:

- The **Optional** Abort Flag is set.

#### 6.12.2.1.3.7 TCH\_Sending\_Chunked\_Message State

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Wait\_Chunk\_Request** state when:

- *Message* Transmitted is received from *Protocol Layer* and this was not the last chunk.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Message\_Sent** state when:

- *Message* Transmitted is received from *Protocol Layer* and this was the last chunk.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Report\_Error** state when:

- Transmission Error has been received from the *Protocol Layer*.

#### 6.12.2.1.3.8 TCH\_Wait\_Chunk\_Request State

On entry to the **TCH\_Wait\_Chunk\_Request** state the *Chunked* Tx State Machine **Shall**:

- Increment *Chunk Number* to Send.
- Start **ChunkSenderRequestTimer**.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Report\_Error** state when:

- A *Chunk* Request has been received and the *Chunk Number* does not equal *Chunk Number* to Send or
- **ChunkSenderRequestTimer** has expired and *Chunk Number* is greater than zero.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Message\_Sent** state when:

- **ChunkSenderRequestTimer** has expired and *Chunk Number* equals zero.

**Note:** This is the mechanism which allows the remote *Port Partner* or *Cable Plug* to omit the *Chunking Layer*. The *Policy Engine* will receive a *Message* Sent signal if the remote *Port Partner* or *Cable Plug* is present (**GoodCRC** *Message* received) but does not send a *Chunk* Request. After this the remote *Port Partner* will send a **Not\_Supported** *Message*, or the *Cable Plug* will **Ignore** the *Chunked Message*.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Message\_Received** state when:

- Any other *Message* than *Chunk* Request is received.

#### 6.12.2.1.3.9 TCH\_Message\_Received State

The *Chunked* Tx State Machine **Shall** enter the **TCH\_Message\_Received** state:

- When any *Message* is received, and the *Chunked* Tx State Machine is not in the **TCH\_Wait\_Chunk\_Request** state.

On entry to the **TCH\_Message\_Received** state the *Chunked* Tx State Machine **Shall**:

- Clear the *Extended Message* Buffers.
- Pass the received *Message* to *Chunked Rx Engine*.

The *Chunked* Tx State Machine **Shall** transition to the **TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine** state when:

- The received *Message* has been passed to the *Chunked Rx Engine*.

#### 6.12.2.1.3.10 TCH\_Report\_Error State

On entry to the **TCH\_Report\_Error** state the *Chunked* Tx State Machine **Shall**:

- Report the error to the *Policy Engine*.

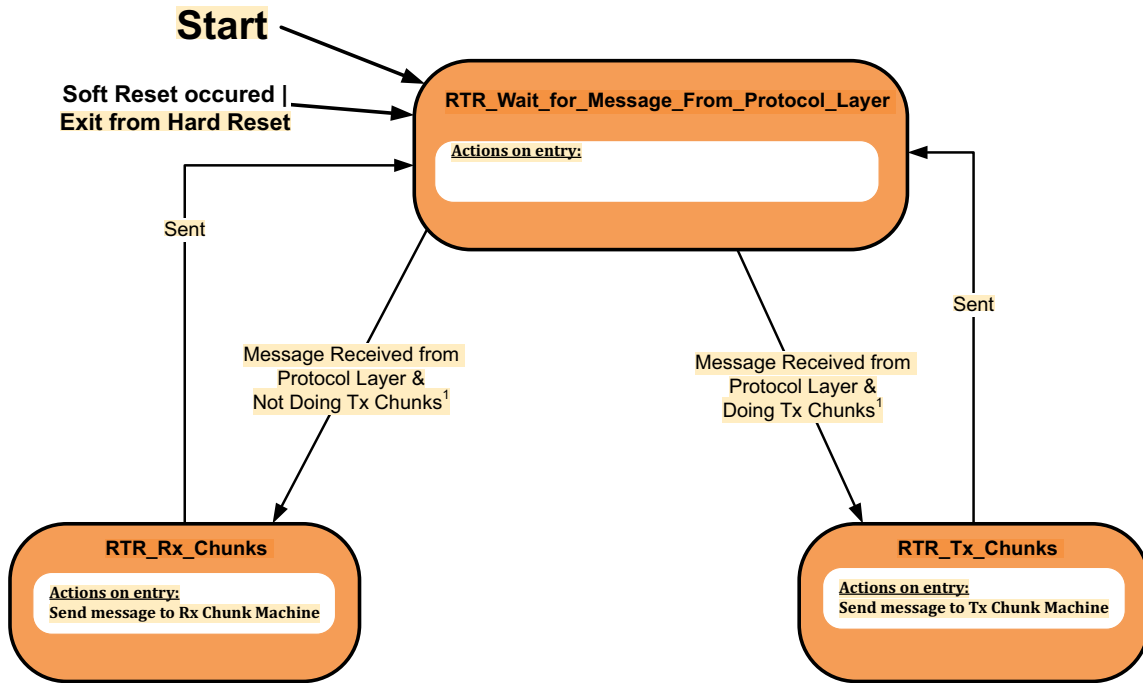
The *Chunked Tx State Machine* **Shall** transition to the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state when:

- The error has been reported.

#### 6.12.2.1.4 Chunked Message Router State Diagram

Figure 6.62, "*Chunked Message Router State Diagram*" shows the state behavior for the *Chunked Message Router*. This determines to which state machine an incoming *Message* is routed to (*Chunked Rx*, *Chunked Tx* or direct to *Policy Engine*).

Figure 6.62 Chunked Message Router State Diagram



- 1) Doing Tx Chunks means that *Chunked Tx State Machine* is not in the *TCH\_Wait\_For\_Message\_Request\_From\_Policy\_Engine* state.
- 2) Messages are taken to include notification about transmission success or otherwise of Messages.

##### 6.12.2.1.4.1 RTR\_Wait\_for\_Message\_From\_Protocol\_Layer State

In the *RTR\_Wait\_for\_Message\_From\_Protocol\_Layer* state the *Chunked Message Router* waits until the *Protocol Layer* sends it a received *Message*.

The *Chunked Message Router* **Shall** transition to the *RTR\_Rx\_Chunks* state when:

- A *Message* is received from the *Protocol Layer*, and the combined *Chunking* is not doing *Tx Chunks*.

The *Chunked Message Router* **Shall** transition to the *RTR\_Tx\_Chunks* state when:

- A *Message* is received from the *Protocol Layer*, and the combined *Chunking* is doing *Tx Chunks*.

##### 6.12.2.1.4.2 RTR\_Rx\_Chunks State

On entry to the *RTR\_Rx\_Chunks* state the *Chunked Message Router* **Shall**:

- Send the *Message* to the *Chunked Rx State Machine*.
- Transition to the *RTR\_Wait\_for\_Message\_From\_Protocol\_Layer* state.



On entry to the *RTR\_Tx\_Chunks* state the *Chunked* Message Router **Shall**:

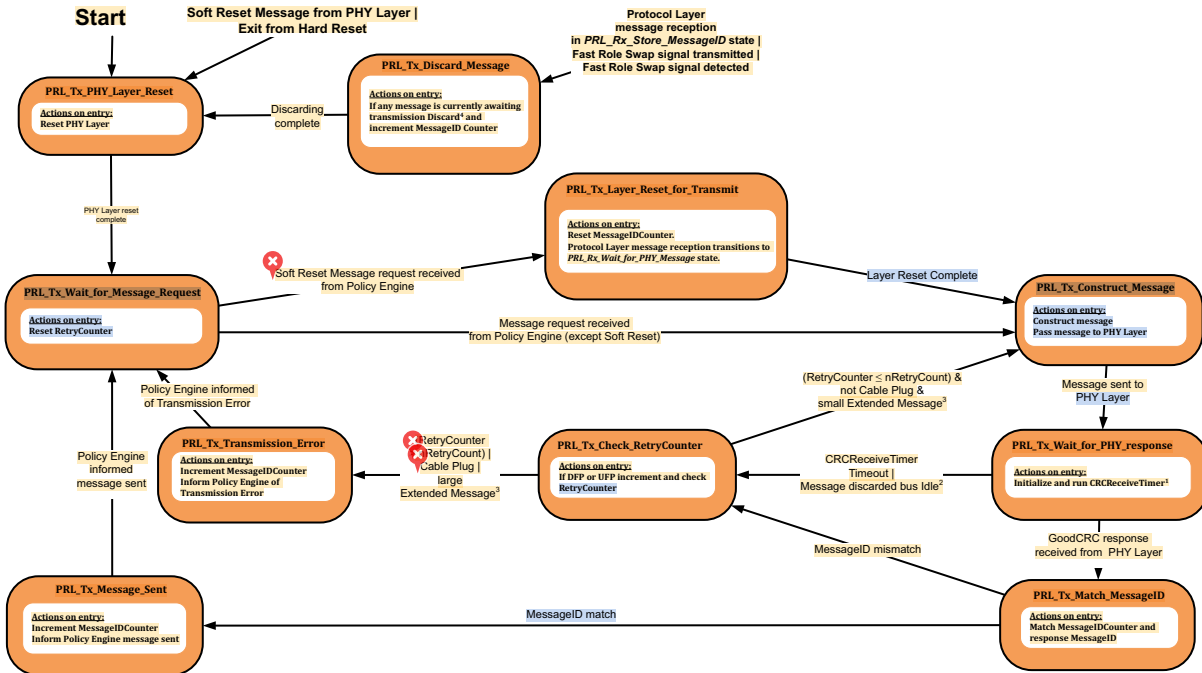
- Send the *Message* to the *Chunked* Tx State Machine.
- Transition to the *RTR\_Wait\_for\_Message\_From\_Protocol\_Layer* state.

## 6.12.2.2 Protocol Layer Message Transmission

### 6.12.2.2.1 Common Protocol Layer Message Transmission State Diagram

Figure 6.63, "Common Protocol Layer Message Transmission State Diagram" shows the state behavior, common between the *Source* and the *Sink*, for the *Protocol Layer* when transmitting a *Message*.

Figure 6.63 Common Protocol Layer Message Transmission State Diagram



- <sup>1)</sup> The **CRCReceiveTimer** is only started after the PHY has sent the message. If the message is not sent due to a busy channel, then the **CRCReceiveTimer** will not be started (see Section 6.6.1 "CRCReceiveTimer").
- <sup>2)</sup> This indication is sent by the PHY Layer when a message has been **Discarded** due to CC being busy, and after CC becomes idle again (see Section 5.7 "Collision Avoidance"). The **CRCReceiveTimer** is not running in this case since no message has been sent.
- <sup>3)</sup> A "small" Extended Message is either an Extended Message with **Data Size ≤ MaxExtendedMsgLegacyLen** bytes or an Extended Message with **Data Size > MaxExtendedMsgLegacyLen** bytes that has been Chunked. A "large" Extended Message is an Extended Message with **Data Size > MaxExtendedMsgLegacyLen** bytes that has not been Chunked.
- <sup>4)</sup> See Section 6.11 "Message Discarding" for details of when Messages are **Discarded**.

#### 6.12.2.2.1.1 PRL\_Tx\_PHY\_Layer\_Reset State

The *Protocol Layer* **Shall** enter the **PRL\_Tx\_PHY\_Layer\_Reset** state:

- At startup.
- As a result of a *Soft Reset* request being received by the *PHY Layer*.
- On exit from a *Hard Reset*.

On entry to the **PRL\_Tx\_PHY\_Layer\_Reset** state the *Protocol Layer* **Shall** reset the *PHY Layer* (clear any outstanding *Messages* and enable communications).

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Wait\_for\_Message\_Request** state when:

- When the *PHY Layer* reset is complete.

#### 6.12.2.2.1.2 PRL\_Tx\_Wait\_for\_Message\_Request State

In the **PRL\_Tx\_Wait\_for\_Message\_Request** state the *Protocol Layer* waits until the *Policy Engine* directs it to send a *Message*.

- On entry to the **PRL\_Tx\_Wait\_for\_Message\_Request** state the *Protocol Layer* **Shall** reset the **RetryCounter**.



The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Construct\_Message* state when:

- A *Message* request is received from the *Policy Engine* which is not a *Soft\_Reset* *Message*.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Layer\_Reset\_for\_Transmit* state when:

- A *Message* request is received from the *Policy Engine* which is a *Soft\_Reset* *Message*.

#### 6.12.2.2.1.3 PRL\_Tx\_Layer\_Reset\_for\_Transmit State

On entry to the *PRL\_Tx\_Layer\_Reset\_for\_Transmit* state the *Protocol Layer* **Shall** reset the *MessageIDCounter*. The *Protocol Layer* **Shall** transition *Protocol Layer Message* reception to the *PRL\_Rx\_Wait\_for\_PHY\_Message* state (see [Section 6.12.2.3.1, "PRL\\_Rx\\_Wait\\_for\\_PHY\\_Message state"](#)) in order to reset the stored *MessageID*.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Construct\_Message* state when:

- The layer reset actions in this state have been completed.

#### 6.12.2.2.1.4 PRL\_Tx\_Construct\_Message State

On entry to the *PRL\_Tx\_Construct\_Message* state the *Protocol Layer* **Shall** construct the *Message* requested by the *Policy Engine*, or resend a previously constructed *Message*, and then pass this *Message* to the *PHY Layer*.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Wait\_for\_PHY\_Response* state when:

- The *Message* has been sent to the *PHY Layer*.

#### 6.12.2.2.1.5 PRL\_Tx\_Wait\_for\_PHY\_Response State

On entry to the *PRL\_Tx\_Wait\_for\_PHY\_Response* state, once the *Message* has been sent, the *Protocol Layer* **Shall** initialize and run the *CRCReceiveTimer* (see [Section 6.6.1, "CRCReceiveTimer"](#)).

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Match\_MessageID* state when:

- A *GoodCRC* *Message* response is received from the *PHY Layer*.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Check\_RetryCounter* state when:

- The *CRCReceiveTimer* times out.
- Or the *PHY Layer* indicates that a *Message* has been **Discarded** due to the channel being busy but the channel is now *Idle* (see [Section 5.7, "Collision Avoidance"](#)).

#### 6.12.2.2.1.6 PRL\_Tx\_Match\_MessageID State

On entry to the *PRL\_Tx\_Match\_MessageID* state the *Protocol Layer* **Shall** compare the *MessageIDCounter* and the *MessageID* of the received *GoodCRC* *Message*.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Message\_Sent* state when:

- The *MessageIDCounter* and the *MessageID* of the received *GoodCRC* *Message* match.

The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Check\_RetryCounter* state when:

- The *MessageIDCounter* and the *MessageID* of the received *GoodCRC* *Message* do not match.

#### 6.12.2.2.1.7 PRL\_Tx\_Message\_Sent State

On entry to the *PRL\_Tx\_Message\_Sent* state the *Protocol Layer* **Shall** increment the *MessageIDCounter* and inform the *Policy Engine* that the *Message* has been sent.



The *Protocol Layer* **Shall** transition to the *PRL\_Tx\_Wait\_for\_Message\_Request* state when:

- The *Policy Engine* has been informed that the *Message* has been sent.

#### 6.12.2.2.1.8

#### PRL\_Tx\_Check\_RetryCounter State

On entry to the **PRL\_Tx\_Check\_RetryCounter** state the *Protocol Layer* in a *DFP* or *UFP* **Shall** increment the value of the **RetryCounter** and then check it in order to determine whether it is necessary to retry sending the *Message*.

**Note:** *Cable Plugs* do not retry *Messages* and so do not use the **RetryCounter**.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Construct\_Message** state in order to retry *Message* sending when:

- **RetryCounter**  $\leq$  **nRetryCount** and
- This is not a *Cable Plug* and
- This is an *Extended Message* with **Data Size**  $\leq$  **MaxExtendedMsgLegacyLen** or
- This is an *Extended Message* that has been *Chunked*.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Transmission\_Error** state when:

- **RetryCounter**  $>$  **nRetryCount** or
- This is a *Cable Plug*, which does not retry.
- This is an *Extended Message* with **Data Size**  $>$  **MaxExtendedMsgLegacyLen** that has not been *Chunked*.

#### 6.12.2.2.1.9

#### PRL\_Tx\_Transmission\_Error State

On entry to the **PRL\_Tx\_Transmission\_Error** state the *Protocol Layer* **Shall** increment the **MessageIDCounter** and inform the *Policy Engine* of the transmission error.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Wait\_for\_Message\_Request** state when:

- The *Policy Engine* has been informed of the transmission error.

#### 6.12.2.2.1.10

#### PRL\_Tx\_Discard\_Message State

*Protocol Layer Message* transmission **Shall** enter the **PRL\_Tx\_Discard\_Message** state whenever:

- *Protocol Layer Message* reception receives an incoming *Message* or
- The *Fast Role Swap Request* is being transmitted (see [Section 5.8.5.6, "Fast Role Swap Transmission"](#))
- The *Fast Role Swap Request* is detected (see [Section 5.8.6.3, "Fast Role Swap Detection"](#)).

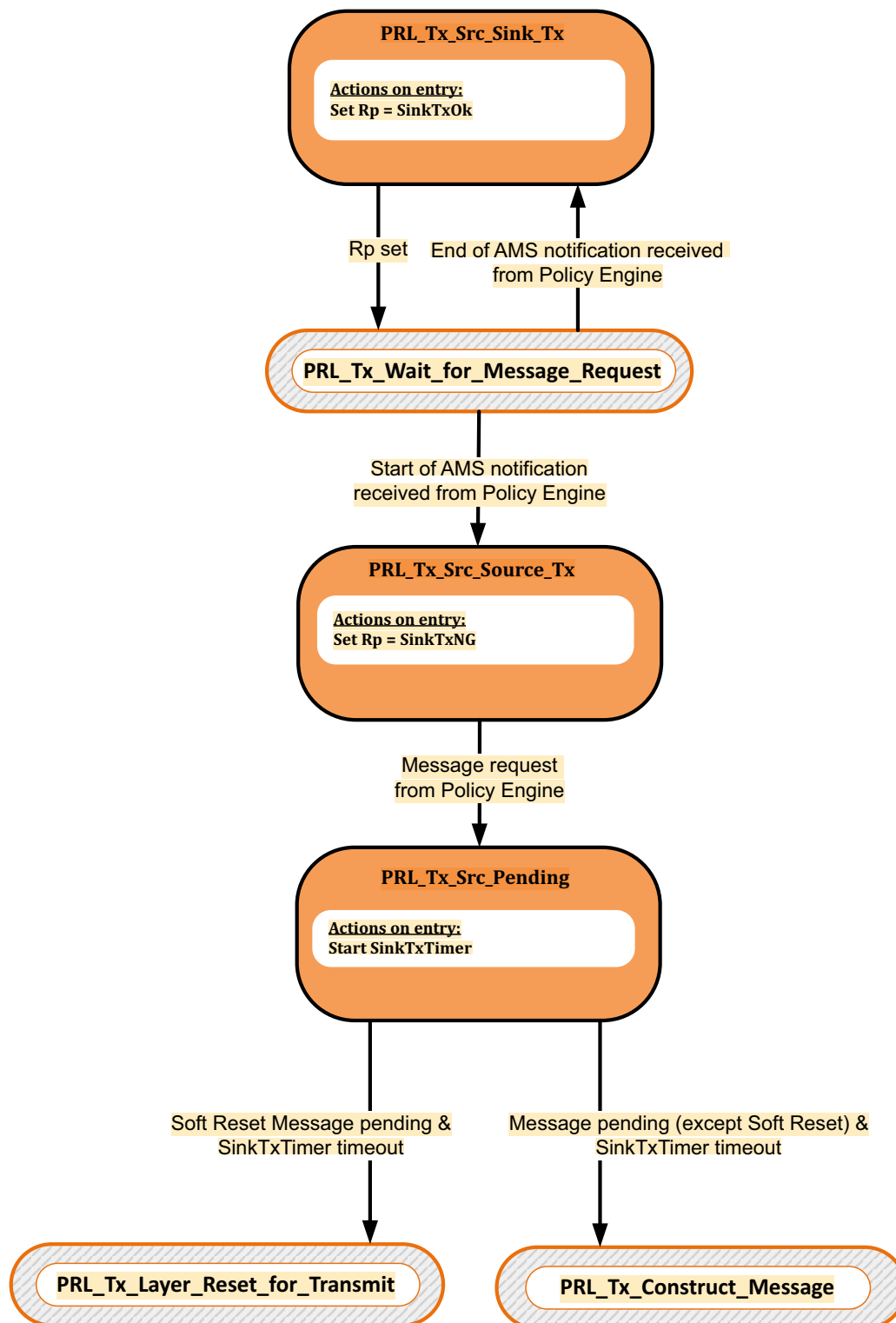
On entry to the **PRL\_Tx\_Discard\_Message** state, if there is a *Message* queued awaiting transmission, the *Protocol Layer* **Shall Discard** the *Message* according to the rules in [Section 6.11, "Message Discarding"](#) and increment the **MessageIDCounter**.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_PHY\_Layer\_Reset** state when:

- Discarding is complete i.e., the *Message* queue is empty.

Figure 6.64, "Source Protocol Layer Message Transmission State Diagram" shows the state behavior for the *Protocol Layer* in a *Source* when transmitting a *Message*.

Figure 6.64 Source Protocol Layer Message Transmission State Diagram



#### 6.12.2.2.2.1

#### PRL\_Tx\_Src\_Sink\_Tx State

In the **PRL\_Tx\_Src\_Sink\_Tx** state the *Source* sets  $R_p$  to **SinkTxOK** allowing the *Sink* to start an *Atomic Message Sequence* (AMS).

The *Protocol Layer* in a *Source* **Shall** transition from the **PRL\_Tx\_Wait\_for\_Message\_Request** state to the **PRL\_Tx\_Src\_Sink\_Tx** state when:

- A notification is received from the *Policy Engine* that the end of an AMS has been reached.

On entry to the **PRL\_Tx\_Src\_Sink\_Tx** state the *Protocol Layer* **Shall** request the *PHY Layer* to  $R_p$  to **SinkTxOK**.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Wait\_for\_Message\_Request** state when:

- $R_p$  has been set.

#### 6.12.2.2.2.2

#### PRL\_Tx\_Src\_Source\_Tx State

In the **PRL\_Tx\_Src\_Source\_Tx** state the *Source* sets  $R_p$  to **SinkTxNG** allowing the *Source* to start an *Atomic Message Sequence* (AMS).

The *Protocol Layer* in a *Source* **Shall** transition from the **PRL\_Tx\_Wait\_for\_Message\_Request** state to the **PRL\_Tx\_Src\_Source\_Tx** state when:

- A notification is received from the *Policy Engine* that an AMS will be starting.

On entry to the **PRL\_Tx\_Src\_Source\_Tx** state the *Protocol Layer* **Shall** set  $R_p$  to **SinkTxNG**.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Src\_Pending** state when:

- A *Message* request is received from the *Policy Engine*.

#### 6.12.2.2.2.3

#### PRL\_Tx\_Src\_Pending State

In the **PRL\_Tx\_Src\_Pending** state the *Protocol Layer* has a *Message* buffered ready for transmission.

On entry to the **PRL\_Tx\_Src\_Pending** state the **SinkTxTimer** **Shall** be initialized and run.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Construct\_Message** state when:

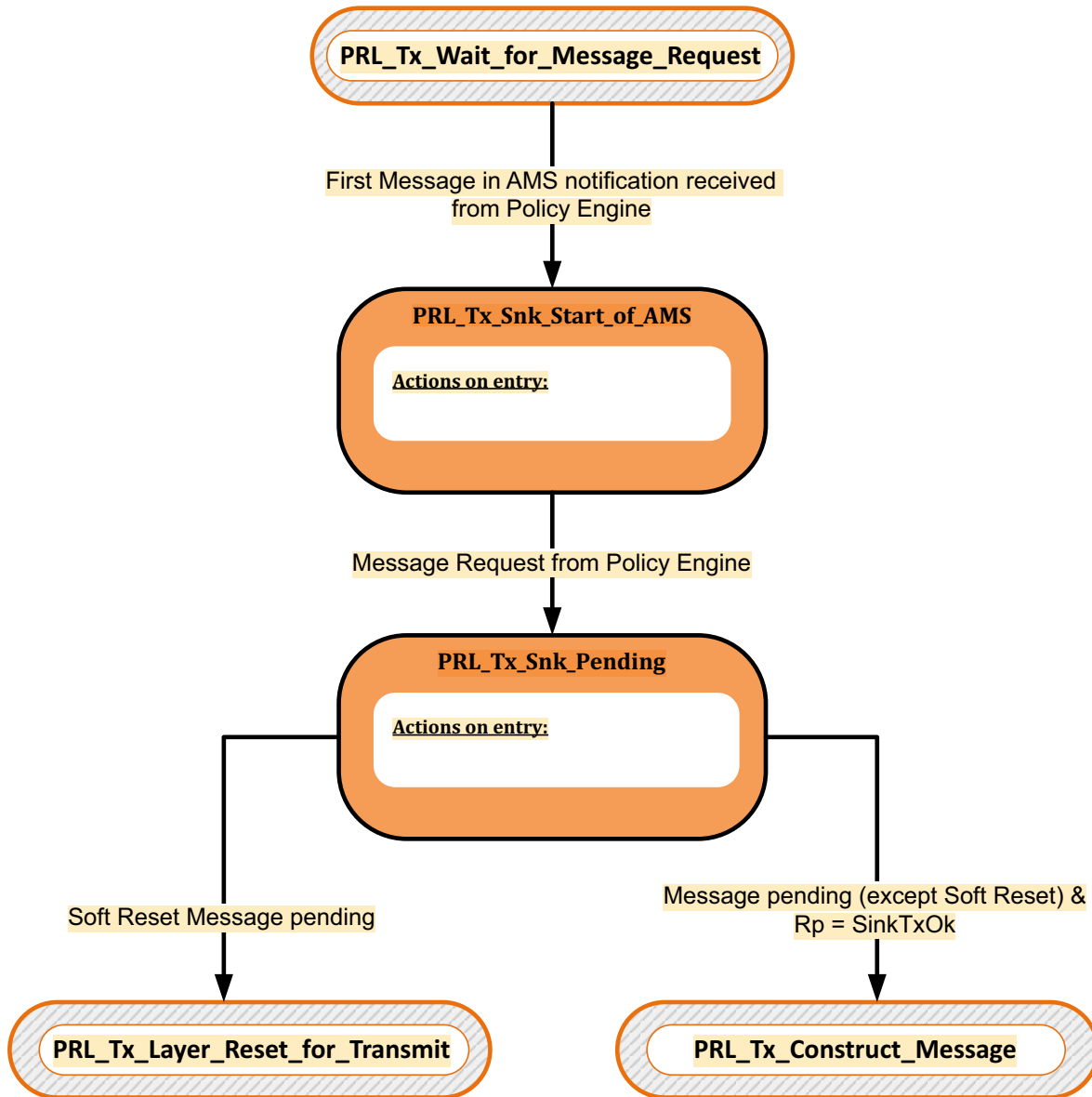
- The pending *Message* request from the *Policy Engine* is not a **Soft\_Reset** *Message* and
- The **SinkTxTimer** times out.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Layer\_Reset\_for\_Transmit** state when:

- The pending *Message* request from the *Policy Engine* is a **Soft\_Reset** *Message* and
- The **SinkTxTimer** times out.

Figure 6.65, "Sink Protocol Layer Message Transmission State Diagram" shows the state behavior for the *Protocol Layer* in a *Sink* when transmitting a *Message*.

Figure 6.65 Sink Protocol Layer Message Transmission State Diagram



#### 6.12.2.2.3.1

#### PRL\_Tx\_Snk\_Start\_of\_AMS State

In the **PRL\_Tx\_Snk\_Start\_of\_AMS** state the *Protocol Layer* waits for the first *Message* in a *Sink* initiated *AMS*.

The *Protocol Layer* in a *Sink* **shall** transition from the **PRL\_Tx\_Wait\_for\_Message\_Request** state to the **PRL\_Tx\_Snk\_Start\_of\_AMS** state when:

- A notification is received from the *Policy Engine* that the next *Message* the *Sink* will send is the start of an *AMS*.

The *Protocol Layer* **shall** transition to the **PRL\_Tx\_Snk\_Pending** state when:

- A *Message* request is received from the *Policy Engine*.

In the **PRL\_Tx\_Snk\_Pending** state the *Protocol Layer* has the first *Message* in a *Sink* initiated *AMS* ready to send and is waiting for  $R_p$  to transition to **SinkTxOK** before sending the *Message*.

The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Construct\_Message** state when:

- A *Message* is Pending that is not a **Soft\_Reset** *Message* and
- $R_p$  is set to **SinkTxOK**.

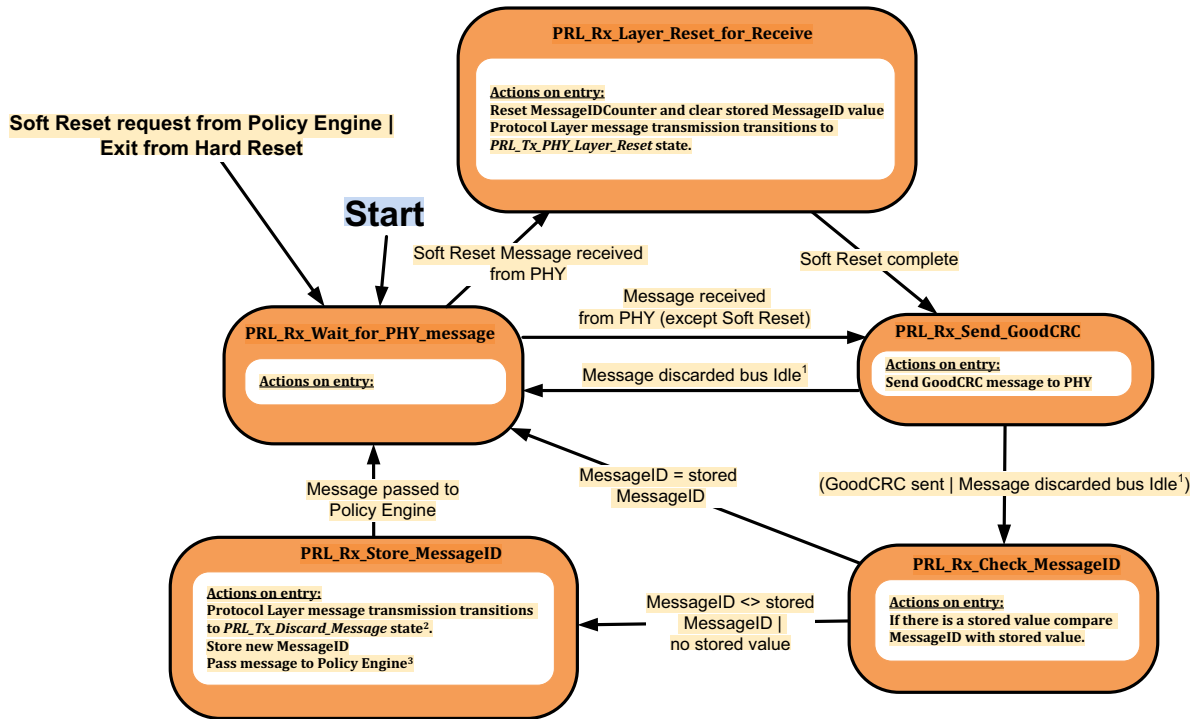
The *Protocol Layer* **Shall** transition to the **PRL\_Tx\_Layer\_Reset\_for\_Transmit** state when:

- A **Soft\_Reset** *Message* is pending.

### 6.12.2.3 Protocol Layer Message Reception

Figure 6.66, "Protocol layer Message reception" shows the state behavior for the *Protocol Layer* when receiving a Message.

Figure 6.66 Protocol layer Message reception



- 1) This indication is sent by the PHY when a message has been **Discarded** due to CC being busy, and after CC becomes idle again (see Section 5.7 "Collision Avoidance"). Two alternate allowable transitions are shown.
- 2) In the case of a Ping message being received, in order to maintain robust communications in the presence of collisions, the outgoing message **Should Not be Discarded**.
- 3) See Section 6.11 "Message Discarding" for details of when Messages are discarded.

#### 6.12.2.3.1 PRL\_Rx\_Wait\_for\_PHY\_Message state

The *Protocol Layer* **Shall** enter the *PRL\_Rx\_Wait\_for\_PHY\_Message* state:

- At startup.
- As a result of a *Soft Reset* request from the *Policy Engine*.
- On exit from a *Hard Reset*.

In the *PRL\_Rx\_Wait\_for\_PHY\_Message* state the *Protocol Layer* waits until the *PHY Layer* passes up a received Message.

The *Protocol Layer* **Shall** transition to the *PRL\_Rx\_Send\_GoodCRC* state when:

- A Message is passed up from the *PHY Layer*.

✖ The *Protocol Layer* **Shall** transition to the *PRL\_Rx\_Layer\_Reset\_for\_Receive* state when:

- A *Soft\_Reset* Message is received from the *PHY Layer*.



### 6.12.2.3.2

### PRL\_Rx\_Layer\_Reset\_for\_Receive state

On entry to the **PRL\_Rx\_Layer\_Reset\_for\_Receive** state the *Protocol Layer* **Shall** reset the **MessageIDCounter** and clear the stored **MessageID**. The *Protocol Layer* **Shall** transition *Protocol Layer Message* transmission to the **PRL\_Tx\_Wait\_for\_Message\_Request** state (see [Section 6.12.2.2.1.2, "PRL Tx Wait for Message Request State"](#)).

The *Protocol Layer* **Shall** transition to the **PRL\_Rx\_Send\_GoodCRC** State when:

- The *Soft Reset* actions in this state have been completed.

### 6.12.2.3.3

### PRL\_Rx\_Send\_GoodCRC state

On entry to the **PRL\_Rx\_Send\_GoodCRC** state the *Protocol Layer* **Shall** construct a **GoodCRC** Message and request the *PHY Layer* to transmit it.

The *Protocol Layer* **Shall** transition to the **PRL\_Rx\_Check\_MessageID** state when:

- The **GoodCRC** Message has been passed to the *PHY Layer*.

When the *PHY Layer* indicates that a Message has been **Discarded** due to CC being busy but CC is now **Idle** (see [Section 5.7, "Collision Avoidance"](#)), the *Protocol Layer* **Shall** either:

- Transition to the **PRL\_Rx\_Check\_MessageID** state or
- Transition to the **PRL\_Rx\_Wait\_for\_PHY\_Message** state.

### 6.12.2.3.4

### PRL\_Rx\_Check\_MessageID state

On entry to the **PRL\_Rx\_Check\_MessageID** state the *Protocol Layer* **Shall** compare the **MessageID** of the received Message with its stored value if a value has previously been stored.

The *Protocol Layer* **Shall** transition to the **PRL\_Rx\_Wait\_for\_PHY\_Message** state when:

- The **MessageID** of the received Message equals the stored **MessageID** value since this is a Message retry which **Shall** be **Discarded**.

The *Protocol Layer* **Shall** transition to the **PRL\_Rx\_Store\_MessageID** state when:

- The **MessageID** of the received Message does not equal the stored **MessageID** value since this is a new Message or
- This is the first received Message and no **MessageID** value is currently stored.

### 6.12.2.3.5

### PRL\_Rx\_Store\_MessageID state

On entry to the **PRL\_Rx\_Store\_MessageID** state the *Protocol Layer* **Shall** transition *Protocol Layer Message* transmission to the **PRL\_Tx\_Discard\_Message** state, replace the stored value of **MessageID** with the value of **MessageID** in the received Message and pass the Message up to the *Policy Engine*.

The *Protocol Layer* **Shall** transition to the **PRL\_Rx\_Wait\_for\_PHY\_Message** state when:

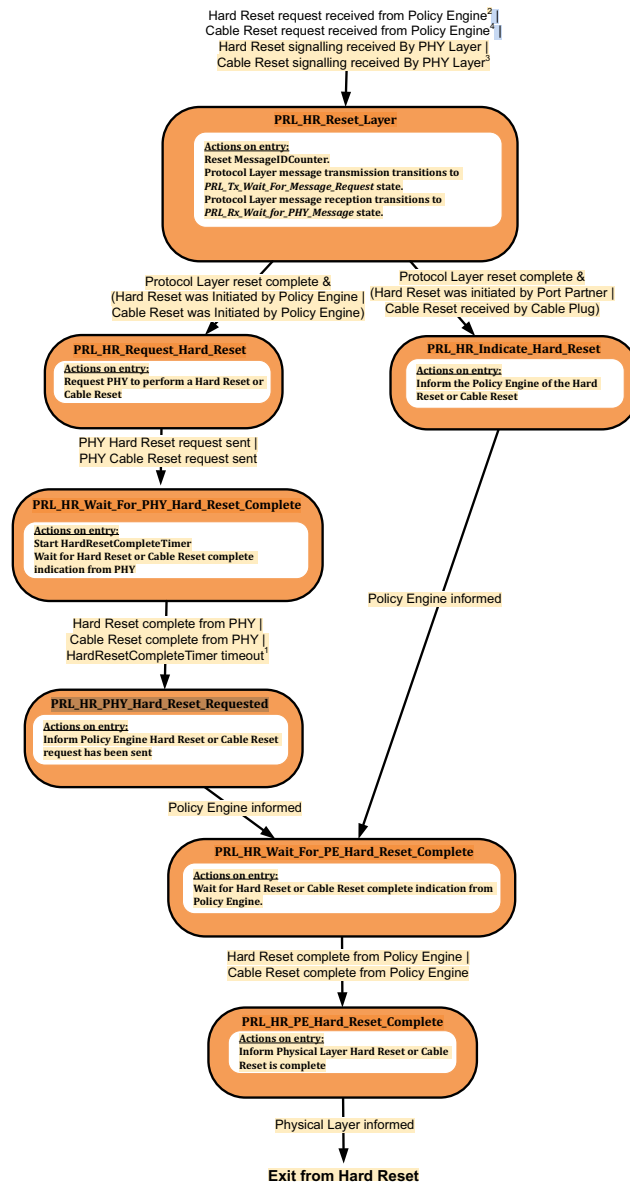
- The Message has been passed up to the *Policy Engine*.

## 6.12.2.4

## Hard Reset operation

[Figure 6.57, "Outline of States"](#) shows the state behavior for the *Protocol Layer* when receiving a **Hard Reset** or **Cable Reset** request from the *Policy Engine* or **Hard Reset** Signaling or **Cable Reset** Signaling from the *PHY Layer* (see also [Section 6.8.3, "Hard Reset"](#) and [Section 6.8.4, "Cable Reset"](#)).

**Figure 6.67 Hard/Cable Reset**



- <sup>1</sup> If the *HardResetCompleteTimer* timeout occurs this means that the PHY is still waiting to send the Hard Reset due to a non-idle channel. This condition will be cleared once the PE Hard Reset is completed.  
<sup>2</sup> Cable Plugs do not generate *Hard Reset* signaling but are required to monitor for *Hard Reset* signaling between the Port Partners and respond by resetting.  
<sup>3</sup> Cable Reset signaling is only recognized by a Cable Plug.  
<sup>4</sup> Cable Reset signaling cannot be generated by Cable Plugs.

#### 6.12.2.4.1 PRL\_HR\_Reset\_Layer state

The **PRL\_HR\_Reset\_Layer** State defines the mode of operation of both the *Protocol Layer* transmission and reception state machines during a *Hard Reset* or *Cable Reset*. During *Hard Reset* no USB Power Delivery Protocol Messages are sent or received; only *Hard Reset Signaling* is present after which the communication channel is assumed to have been disabled by the *PHY Layer* until completion of the *Hard Reset*. During *Cable Reset* no USB Power Delivery Protocol Messages are sent to or received by the *Cable Plug* but other USB Power Delivery communication *May* continue.

The *Protocol Layer* **Shall** enter the **PRL\_HR\_Reset\_Layer** state from any other state when:

- A *Hard Reset* Request is received from the *Policy Engine* or
- *Hard Reset Signaling* is received from the *PHY Layer* or

- A **Cable Reset** Request is received from the *Policy Engine* or
- **Cable Reset** Signaling is received from the *PHY Layer*.

On entry to the **PRL\_HR\_Reset\_Layer** state the *Protocol Layer* **Shall** reset the **MessageIDCounter**. It **Shall** also reset the states of the *Protocol Layer* transmission and reception state machines to their starting points. The *Protocol Layer* transmission state machine **Shall** transition to the **PRL\_Tx\_Wait\_for\_Message\_Request** state. The *Protocol Layer* reception state machine **Shall** transition to the **PRL\_Rx\_Wait\_for\_PHY\_Message** state.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_Request\_Hard\_Reset** state when:

- The *Protocol Layer*'s reset is complete and
- The **Hard Reset** request has originated from the *Policy Engine* or
- The **Cable Reset** request has originated from the *Policy Engine*.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_Indicate\_Hard\_Reset** state when:

- The *Protocol Layer*'s reset is complete and
- The **Hard Reset** request has been passed up from the *PHY Layer* or
- A **Cable Reset** request has been passed up from the *PHY Layer* (*Cable Plug* only).

#### 6.12.2.4.2 PRL\_HR\_Indicate\_Hard\_Reset state

On entry to the **PRL\_HR\_Indicate\_Hard\_Reset** state the *Protocol Layer* **Shall** indicate to the *Policy Engine* that either **Hard Reset** Signaling or **Cable Reset** Signaling has been received.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_Wait\_for\_PE\_Hard\_Reset\_Complete** state when:

- The indication to the *Policy Engine* has been sent.

#### 6.12.2.4.3 PRL\_HR\_Request\_Hard\_Reset state

On entry to the **PRL\_HR\_Request\_Hard\_Reset** state the *Protocol Layer* **Shall** request the *PHY Layer* to send either **Hard Reset** Signaling or **Cable Reset** Signaling.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_Wait\_for\_PHY\_Hard\_Reset\_Complete** state when:

- The *PHY Layer* **Hard Reset** Signaling request has been sent or
- The *PHY Layer* **Cable Reset** Signaling request has been sent.

#### 6.12.2.4.4 PRL\_HR\_Wait\_for\_PHY\_Hard\_Reset\_Complete state

In the **PRL\_HR\_Wait\_for\_PHY\_Hard\_Reset\_Complete** state the *Protocol Layer* **Shall** start the **HardResetCompleteTimer** and wait for the *PHY Layer* to indicate that the **Hard Reset** or **Cable Reset** has been completed.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_PHY\_Hard\_Reset\_Requested** state when:

- A **Hard Reset** complete indication is received from the *PHY Layer* or
- A **Cable Reset** complete indication is received from the *PHY Layer* or
- The **HardResetCompleteTimer** times out.

#### 6.12.2.4.5 PRL\_HR\_PHY\_Hard\_Reset\_Requested state

On entry to the **PRL\_HR\_PHY\_Hard\_Reset\_Requested** state the *Protocol Layer* **Shall** inform the *Policy Engine* that the *PHY Layer* has been requested to perform a **Hard Reset** or **Cable Reset**.

The *Protocol Layer* **Shall** transition to the **PRL\_HR\_Wait\_for\_PE\_Hard\_Reset\_Complete** state when:

- The Indication to the *Policy Engine* has been sent.



#### 6.12.2.4.6 PRL\_HR\_Wait\_for\_PE\_Hard\_Reset\_Complete state

In the *PRL\_HR\_Wait\_for\_PE\_Hard\_Reset\_Complete* state the *Protocol Layer* **Shall** wait for the *Policy Engine* to indicate that the *Hard Reset* or *Cable Reset* has been completed.

The *Protocol Layer* **Shall** transition to the *PRL\_HR\_PE\_Hard\_Reset\_Complete* state when:

- A *Hard Reset* complete indication is received from the *Policy Engine* or
- A *Cable Reset* complete indication is received from the *Policy Engine*.

#### 6.12.2.4.7 PRL\_HR\_PE\_Hard\_Reset\_Complete

On entry to the *PRL\_HR\_PE\_Hard\_Reset\_Complete* state the *Protocol Layer* **Shall** inform the *PHY* Layer that the *Hard Reset* or *Cable Reset* is complete.

The *Protocol Layer* **Shall** exit from the *Hard Reset* and return to normal operation when:

- The *PHY* Layer has been informed that the *Hard Reset* is complete so that it will re-enable the communications channel. If *Hard Reset Signaling* is still pending due to a *non-Idle* channel this **Shall** be cleared and not sent or
- The *PHY* Layer has been informed that the *Cable Reset* is complete.

## 6.12.3 List of Protocol Layer States

Table 6.75, "Protocol Layer States" lists the states used by the various state machines.

Table 6.75 Protocol Layer States

| State Name                                        | Section                               |
|---------------------------------------------------|---------------------------------------|
| <b>Protocol Layer Message Transmission</b>        |                                       |
| <b>Common Protocol Layer Message Transmission</b> |                                       |
| <i>PRL_Tx_PHY_Layer_Reset</i>                     | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>PRL_Tx_Wait_for_Message_Request</i>            | <a href="#">Section 6.12.2.2.1.2</a>  |
| <i>PRL_Tx_Layer_Reset_for_Transmit</i>            | <a href="#">Section 6.12.2.2.1.3</a>  |
| <i>PRL_Tx_Construct_Message</i>                   | <a href="#">Section 6.12.2.2.1.4</a>  |
| <i>PRL_Tx_Wait_for_PHY_Response</i>               | <a href="#">Section 6.12.2.2.1.5</a>  |
| <i>PRL_Tx_Match_MessageID</i>                     | <a href="#">Section 6.12.2.2.1.6</a>  |
| <i>PRL_Tx_Message_Sent</i>                        | <a href="#">Section 6.12.2.2.1.7</a>  |
| <i>PRL_Tx_Check_RetryCounter</i>                  | <a href="#">Section 6.12.2.2.1.8</a>  |
| <i>PRL_Tx_Transmission_Error</i>                  | <a href="#">Section 6.12.2.2.1.9</a>  |
| <i>PRL_Tx_Discard_Message</i>                     | <a href="#">Section 6.12.2.2.1.10</a> |
| <b>Source Protocol Layer Message Transmission</b> |                                       |
| <i>PRL_Tx_Src_Sink_Tx</i>                         | <a href="#">Section 6.12.2.2.2.1</a>  |
| <i>PRL_Tx_Src_Source_Tx</i>                       | <a href="#">Section 6.12.2.2.2.2</a>  |
| <i>PRL_Tx_Src_Pending</i>                         | <a href="#">Section 6.12.2.2.2.3</a>  |
| <b>Sink Protocol Layer Message Transmission</b>   |                                       |
| <i>PRL_Tx_Snk_Start_of_AMS</i>                    | <a href="#">Section 6.12.2.2.3.1</a>  |
| <i>PRL_Tx_Snk_Pending</i>                         | <a href="#">Section 6.12.2.2.3.2</a>  |
| <b>Protocol Layer Message Reception</b>           |                                       |
| <i>PRL_Rx_Wait_for_PHY_Message</i>                | <a href="#">Section 6.12.2.3.1</a>    |
| <i>PRL_Rx_Layer_Reset_for_Receive</i>             | <a href="#">Section 6.12.2.3.2</a>    |
| <i>PRL_Rx_Send_GoodCRC</i>                        | <a href="#">Section 6.12.2.3.3</a>    |
| <i>PRL_Rx_Check_MessageID</i>                     | <a href="#">Section 6.12.2.3.4</a>    |
| <i>PRL_Rx_Store_MessageID</i>                     | <a href="#">Section 6.12.2.3.5</a>    |
| <b>Hard Reset Operation</b>                       |                                       |
| <i>PRL_HR_Reset_Layer</i>                         | <a href="#">Section 6.12.2.4.1</a>    |
| <i>PRL_HR_Indicate_Hard_Reset</i>                 | <a href="#">Section 6.12.2.4.2</a>    |
| <i>PRL_HR_Request_Hard_Reset</i>                  | <a href="#">Section 6.12.2.4.3</a>    |
| <i>PRL_HR_Wait_for_PHY_Hard_Reset_Complete</i>    | <a href="#">Section 6.12.2.4.4</a>    |
| <i>PRL_HR_PHY_Hard_Reset_Requested</i>            | <a href="#">Section 6.12.2.4.5</a>    |
| <i>PRL_HR_Wait_for_PE_Hard_Reset_Complete</i>     | <a href="#">Section 6.12.2.4.6</a>    |
| <i>PRL_HR_PE_Hard_Reset_Complete</i>              | <a href="#">Section 6.12.2.4.7</a>    |

**Table 6.75 Protocol Layer States (Continued)**

| State Name                                             | Section                               |
|--------------------------------------------------------|---------------------------------------|
| <b>Chunking</b>                                        |                                       |
| <b>Chunked Rx</b>                                      |                                       |
| <i>RCH_Wait_For_Message_From_Protocol_Layer</i>        | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>RCH_Pass_Up_Message</i>                             | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>RCH_Processing_Extended_Message</i>                 | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>RCH_Requesting_Chunk</i>                            | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>RCH_Waiting_Chunk</i>                               | <a href="#">Section 6.12.2.2.1.1</a>  |
| <i>RCH_Report_Error</i>                                | <a href="#">Section 6.12.2.2.1.1</a>  |
| <b>Chunked Tx</b>                                      |                                       |
| <i>TCH_Wait_For_Message_Request_From_Policy_Engine</i> | <a href="#">Section 6.12.2.1.3.1</a>  |
| <i>TCH_Pass_Down_Message</i>                           | <a href="#">Section 6.12.2.1.3.2</a>  |
| <i>TCH_Wait_For_Transmission_Complete</i>              | <a href="#">Section 6.12.2.1.3.3</a>  |
| <i>TCH_Message_Sent</i>                                | <a href="#">Section 6.12.2.1.3.4</a>  |
| <i>TCH_Prepate_To_Send_Chunked_Message</i>             | <a href="#">Section 6.12.2.1.3.5</a>  |
| <i>TCH_Construct_Chunked_Message</i>                   | <a href="#">Section 6.12.2.1.3.6</a>  |
| <i>TCH_Sending_Chunked_Message</i>                     | <a href="#">Section 6.12.2.1.3.7</a>  |
| <i>TCH_Wait_Chunk_Request</i>                          | <a href="#">Section 6.12.2.1.3.8</a>  |
| <i>TCH_Message_Received</i>                            | <a href="#">Section 6.12.2.1.3.9</a>  |
| <i>TCH_Report_Error</i>                                | <a href="#">Section 6.12.2.1.3.10</a> |
| <b>Chunked Message Router</b>                          |                                       |
| <i>RTR_Wait_for_Message_From_Protocol_Layer</i>        | <a href="#">Section 6.12.2.1.4.1</a>  |
| <i>RTR_Rx_Chunks</i>                                   | <a href="#">Section 6.12.2.1.4.2</a>  |
| <i>RTR_Tx_Chunks</i>                                   | <a href="#">Section 6.12.2.1.4.3</a>  |

## 6.13 Message Applicability

The following tables outline the *Messages* supported by a given *Port*, depending on its capability.

When a *Message* is supported the feature and the *AMS* implied by the *Message* **Shall** also be supported. The abbreviations in [Table 6.76, "Message Applicability Abbreviations"](#) are used in this section to denote the level of support required.

**Table 6.76 Message Applicability Abbreviations**

| Abbreviation | Meaning                      | Description                                                                                                                                  |
|--------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <i>N</i>     | <b>Normative</b>             | <b>Shall</b> be supported by this <i>Port/Cable Plug</i> .                                                                                   |
| <i>CN</i>    | <b>Conditional Normative</b> | <b>Shall</b> supported by a given <i>Port/Cable Plug</i> based on features.                                                                  |
| <i>R</i>     | Recommended                  | <b>Should</b> be supported by this <i>Port/Cable Plug</i> .                                                                                  |
| <i>O</i>     | <b>Optional</b>              | <b>May</b> be supported by this <i>Port/Cable Plug</i> .                                                                                     |
| <i>NS</i>    | Not Supported                | <b>Shall</b> result in a <b>Not Supported</b> <i>Message</i> response by this <i>Port/Cable Plug</i> when received. ❌                        |
| <i>I</i>     | <b>Ignore</b>                | <b>Shall</b> be <b>Ignored</b> by this <i>Port/Cable Plug</i> when received.                                                                 |
| <i>NK</i>    | <b>NAK</b>                   | This <i>Port/Cable Plug</i> <b>Shall</b> return Responder <b>NAK</b> to this <i>Command</i> when received. ❌                                 |
| <i>NA</i>    | Not allowed                  | <b>Shall Not</b> be transmitted by this <i>Port/Cable Plug</i> .                                                                             |
| <i>DR</i>    | Don't Recognize              | There <b>Shall</b> be no response at all (i.e., not even a <b>GoodCRC</b> <i>Message</i> ) from this <i>Port/Cable Plug</i> when received. ❌ |

For the case of **Conditional Normative** a note has been added to indicate the condition. "*CN*/" notation is used to indicate the level of support when the condition is not present.

"*R*/" and "*O*/" notation is used to indicate the response when the Recommended or **Optional** *Message* is not supported.

**Note:** Where *NS*/*R*/*NK* is indicated for Received *Messages* this **Shall** apply to the **PE\_CBL\_Ready**, **PE\_SNK\_Ready** or **PE\_SRC\_Ready** states only since unexpected *Messages* received during an *AMS* are *Protocol Errors* (see [Section 6.8.1, "Soft Reset and Protocol Error"](#)).

This section covers *Control Message* and *Data Message* support for *Sources*, *Sink* and *Cable Plugs*. It also covers *VDM Command* support for *DFPs*, *UFPs* and *Cable Plugs*.



## 6.13.1 Applicability of Control Messages

*Table 6.77, "Applicability of Control Messages"* details *Control Messages* that **Shall/Should/Shall Not** be transmitted and received by a *Source, Sink, Cable Plug* or *VPD*. Requirements for *Dual-Role Power Ports* and *Dual-Role Data Ports* **Shall** override any requirements for *Source-only* or *Sink-Only* Ports.

**Table 6.77 Applicability of Control Messages**

| Message Type                   | Source              | Sink                | Dual-Role Power | Dual-Role Data | Cable Plug           | VPD <sup>9</sup> |
|--------------------------------|---------------------|---------------------|-----------------|----------------|----------------------|------------------|
| Transmitted Message            |                     |                     |                 |                |                      |                  |
| <i>Accept</i>                  | N                   | N                   |                 |                | N                    | N                |
| <i>Data_Reset</i>              | CN <sup>10</sup> /R | CN <sup>10</sup> /R |                 |                | NA                   | NA               |
| <i>DR_Swap</i>                 | O                   | O                   |                 | N              | NA                   | NA               |
| <i>FR_Swap</i>                 | NA                  | NA                  | R               |                | NA                   | NA               |
| <i>Get_Country_Codes</i>       | CN <sup>7</sup> /NA | CN <sup>7</sup> /NA |                 |                | NA                   | NA               |
| <i>Get_PPS_Status</i>          | NA                  | CN <sup>6</sup>     |                 |                | NA                   | NA               |
| <i>Get_Sink_Cap</i>            | R                   | NA                  | N               |                | NA                   | NA               |
| <i>Get_Sink_Cap_Extended</i>   | R                   | NA                  | R               |                | NA                   | NA               |
| <i>Get_Source_Cap</i>          | NA                  | R                   | N               |                | NA                   | NA               |
| <i>Get_Source_Cap_Extended</i> | NA                  | R                   | R               |                | NA                   | NA               |
| <i>Get_Source_Info</i>         | NA                  | R                   | R               |                | NA                   | NA               |
| <i>Get_Revision</i>            | R                   | R                   |                 |                | NA                   | NA               |
| <i>Get_Status</i>              | R                   | R                   |                 |                | NA                   | NA               |
| <i>GoodCRC</i>                 | N                   | N                   |                 |                | N                    | N                |
| <i>GotoMin (Deprecated)</i>    | NA                  | NA                  |                 |                | NA                   | NA               |
| <i>Not_Supported</i>           | N                   | N                   |                 |                | NA                   | NA               |
| <i>Ping (Deprecated)</i>       | NA                  | NA                  |                 |                | NA                   | NA               |
| <i>PR_Swap</i>                 | NA                  | NA                  | N               |                | NA                   | NA               |
| <i>PS_RDY</i>                  | N                   | CN <sup>1</sup> /NA | N               |                | NA                   | NA               |
| <i>Reject</i>                  | N                   | O                   | O               | O              | CN <sup>10</sup> /NA | NA               |
| <i>Soft_Reset</i>              | N                   | N                   |                 |                | NA                   | NA               |
| <i>VCONN_Swap</i>              | R                   | R                   |                 |                | NA                   | NA               |
| <i>Wait</i>                    | O                   | NA                  | O               | O              | NA                   | NA               |

- 1) **Shall** be supported by any *Port* that can supply *VCONN*.
- 2) **Shall** be supported products that support the *Source\_Capabilities\_Extended* Message.
- 3) **Shall** be supported by *Sources* that support the *Alert* Message.
- 4) **Shall** be supported when the *Fast Role Swap Request* is supported.
- 5) **Shall** be supported when *VCONN Swap* is supported.
- 6) **Shall** be supported when *SPR PPS Mode* is supported.
- 7) **Shall** be supported when required by a country authority.
- 8) **Shall** be supported by *Active Cables*.
- 9) *VPD* includes *CT-VPDs* when not *Connected* to a *Charger*. PD communication with a *CT-VPD* **Shall** only take place when not *Connected* to a *Charger*.
- 10) **Shall** be supported by products that support *[USB4]*.
- 11) **Shall** be supported by all *Sources* except single *Port SPR Chargers* with *Invariant PDOs*.

Table 6.77 Applicability of Control Messages (Continued)

| Message Type                   | Source              | Sink                | Dual-Role Power     | Dual-Role Data | Cable Plug         | VPD <sup>9</sup>                                                                     |
|--------------------------------|---------------------|---------------------|---------------------|----------------|--------------------|--------------------------------------------------------------------------------------|
| Received Message               |                     |                     |                     |                |                    |                                                                                      |
| <i>Accept</i>                  | N                   | N                   | N                   | N              | I                  | I                                                                                    |
| <i>Data_Reset</i>              | CN <sup>10</sup> /R | CN <sup>10</sup> /R |                     |                | I                  | I                                                                                    |
| <i>DR_Swap</i>                 | O/NS                | O/NS                |                     | N              | I                  | I                                                                                    |
| <i>FR_Swap</i>                 | NS                  | NS                  | CN <sup>4</sup> /NS |                | I                  | I                                                                                    |
| <i>Get_Country_Codes</i>       | CN <sup>7</sup> /NS | CN <sup>7</sup> /NS |                     |                | I                  | I                                                                                    |
| <i>Get_PPS_Status</i>          | CN <sup>6</sup> /NS | NS                  |                     |                | I                  | I                                                                                    |
| <i>Get_Sink_Cap</i>            | NS                  | N                   | N                   |                | I                  | I                                                                                    |
| <i>Get_Sink_Cap_Extended</i>   | NS                  | N                   | N                   |                | I                  | I                                                                                    |
| <i>Get_Source_Cap</i>          | N                   | NS                  | N                   |                | I                  | I                                                                                    |
| <i>Get_Source_Cap_Extended</i> | CN <sup>2</sup> /NS | NS                  | CN <sup>2</sup> /NS |                | I                  | I                                                                                    |
| <i>Get_Source_Info</i>         | CN <sup>11</sup>    | NS                  | N                   |                | I                  | I                                                                                    |
| <i>Get_Revision</i>            | N                   | N                   |                     |                | O/I                | O/I                                                                                  |
| <i>Get_Status</i>              | CN <sup>3</sup> /NS | CN <sup>3</sup> /NS | CN <sup>3</sup> /NS |                | CN <sup>8</sup> /I | I                                                                                    |
| <i>GoodCRC</i>                 | N                   | N                   |                     |                | N                  | N                                                                                    |
| <i>GotoMin (Deprecated)</i>    | NS                  | NS                  |                     |                | I                  | I                                                                                    |
| <i>Not_Supported</i>           | N                   | N                   |                     |                | CN <sup>8</sup> /I | I                                                                                    |
| <i>Ping (Deprecated)</i>       | NS                  | NS/I                |                     |                | I                  |  |
| <i>PR_Swap</i>                 | NS                  | NS                  | N                   |                | I                  | I                                                                                    |
| <i>PS_RDY</i>                  | CN <sup>1</sup> /NS | N                   | N                   |                | I                  | I                                                                                    |
| <i>Reject</i>                  | CN <sup>5</sup> /NS | N                   | N                   | N              | I                  | I                                                                                    |
| <i>Soft_Reset</i>              | N                   | N                   |                     |                | N                  | N                                                                                    |
| <i>VCONN_Swap</i>              | CN <sup>1</sup> /NS | CN <sup>1</sup> /NS |                     |                | I                  | I                                                                                    |
| <i>Wait</i>                    | CN <sup>5</sup> /NS | N                   | N                   | N              | I                  | I                                                                                    |

- 1) **Shall** be supported by any *Port* that can supply *VCONN*.
- 2) **Shall** be supported products that support the *Source\_Capabilities\_Extended* Message.
- 3) **Shall** be supported by *Sources* that support the *Alert* Message.
- 4) **Shall** be supported when the *Fast Role Swap Request* is supported.
- 5) **Shall** be supported when *VCONN Swap* is supported.
- 6) **Shall** be supported when *SPR PPS Mode* is supported.
- 7) **Shall** be supported when required by a country authority.
- 8) **Shall** be supported by *Active Cables*.
- 9) *VPD* includes *CT-VPDs* when not *Connected* to a *Charger*. PD communication with a *CT-VPD* **Shall** only take place when not *Connected* to a *Charger*.
- 10) **Shall** be supported by products that support *[USB4]*.
- 11) **Shall** be supported by all *Sources* except single *Port SPR Chargers* with *Invariant PDOs*.

## 6.13.2 Applicability of Data Messages

Table 6.78, "Applicability of Data Messages" details Data Messages (except for VDM Commands) that **Shall/Should/ Shall Not** be transmitted and received by a Source, Sink, Cable Plug or VPD. Requirements for Dual-Role Power Ports **Shall** override any requirements for Source-only or Sink-Only Ports.

Table 6.78 Applicability of Data Messages

| Message Type        | Source              | Sink                | Dual-Role Power | Cable Plug SOP'       | Cable Plug SOP''   | VPD <sup>6</sup> |
|---------------------|---------------------|---------------------|-----------------|-----------------------|--------------------|------------------|
| Transmitted Message |                     |                     |                 |                       |                    |                  |
| Source_Capabilities | N                   | NA                  | N               | NA                    | NA                 | NA               |
| Request             | NA                  | N                   |                 | NA                    | NA                 | NA               |
| Get_Country_Info    | CN <sup>5</sup> /O  | CN <sup>5</sup> /O  |                 | NA                    | NA                 | NA               |
| BIST                | N <sup>1</sup>      | N <sup>1</sup>      |                 | NA                    | NA                 | NA               |
| Sink_Capabilities   | NA                  | N                   | N               | NA                    | NA                 | NA               |
| Battery_Status      | CN <sup>2</sup>     | CN <sup>2</sup>     |                 | NA                    | NA                 | NA               |
| Alert               | CN <sup>11</sup> /R | CN <sup>11</sup> /R |                 | NA                    | NA                 | NA               |
| Enter_USB           | CN <sup>7</sup> /O  | CN <sup>7</sup> /O  |                 | NA                    | NA                 | NA               |
| EPR_Request         | NA                  | CN <sup>9</sup>     |                 | NA                    | NA                 | NA               |
| EPR_Mode            | CN <sup>9</sup>     | CN <sup>9</sup>     |                 | NA                    | NA                 | NA               |
| Source_Info         | CN <sup>10</sup>    | NA                  | N               | NA                    | NA                 | NA               |
| Revisor             | N                   | N                   |                 | CN <sup>12</sup> /O/I | NA                 | O                |
| Received Message    |                     |                     |                 |                       |                    |                  |
| Source_Capabilities | NS                  | N                   | N               | I                     | I                  | I                |
| Request             | N                   | NS                  |                 | I                     | I                  | I                |
| Get_Country_Info    | CN <sup>5</sup> /NS | CN <sup>5</sup> /NS |                 | I                     | I                  | I                |
| BIST                | N <sup>1</sup>      | N <sup>1</sup>      |                 | N <sup>1</sup>        | N <sup>1</sup>     | N <sup>1</sup>   |
| Sink_Capabilities   | CN <sup>4</sup>     | NS                  | CN <sup>4</sup> | I                     | I                  | I                |
| Battery_Status      | CN <sup>3</sup> /NS | CN <sup>3</sup> /NS |                 | I                     | I                  | I                |
| Alert               | R/NS                | R/NS                |                 | I                     | I                  | I                |
| Enter_USB           | CN <sup>7</sup> /O  | CN <sup>7</sup> /O  |                 | CN <sup>8</sup> /I    | CN <sup>8</sup> /I | I                |

- 1) For details of which BIST Modes and BIST Messages **Shall** be supported see [Section 5.9](#) and [Section 6.4.3](#).
- 2) **Shall** be supported by products that contain batteries.
- 3) **Shall** be supported by products that support the [Get\\_Battery\\_Status](#) Message.
- 4) **Shall** be supported by products that support the [Get\\_Sink\\_Cap](#) Message.
- 5) **Shall** be supported when required by a country authority.
- 6) VPD includes CT-VPDs when not Connected to a Charger. PD communication with a CT-VPD **Shall** only take place when not Connected to a Charger.
- 7) **Shall** be supported by products that support [\[USB4\]](#).
- 8) **Shall** be supported by Active Cables that support [\[USB4\]](#).
- 9) **Shall** be supported by products that support Source operation in *EPR Mode*.
- 10) **Shall** be supported by all Source Ports except singlePort SPR Chargers with Invariant PDOs.
- 11) **Shall** be supported when SPR PPS Mode is supported.
- 12) **Shall** be supported by Active Cables.

Table 6.78 Applicability of Data Messages (Continued)

| Message Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Source          | Sink            | Dual-Role Power | Cable Plug SOP' | Cable Plug SOP'' | VPD <sup>6</sup> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|------------------|------------------|
| <i>EPR_Request</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CN <sup>9</sup> | NA              |                 | I               | I                | I                |
| <i>EPR_Mode</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CN <sup>9</sup> | CN <sup>9</sup> |                 | I               | I                | I                |
| <i>Source_Info</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NA              | N               | N               | I               | I                | I                |
| <i>Revision</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | N               | N               |                 | I               | I                | I                |
| <p>1) For details of which <i>BIST Modes</i> and <i>BIST</i> Messages <b>Shall</b> be supported see <a href="#">Section 5.9</a> and <a href="#">Section 6.4.3</a>.</p> <p>2) <b>Shall</b> be supported by products that contain batteries.</p> <p>3) <b>Shall</b> be supported by products that support the <i>Get_Battery_Status</i> Message.</p> <p>4) <b>Shall</b> be supported by products that support the <i>Get_Sink_Cap</i> Message.</p> <p>5) <b>Shall</b> be supported when required by a country authority.</p> <p>6) VPD includes CT-VPDs when not <i>Connected</i> to a <i>Charger</i>. PD communication with a CT-VPD <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i>.</p> <p>7) <b>Shall</b> be supported by products that support <a href="#">[USB4]</a>.</p> <p>8) <b>Shall</b> be supported by <i>Active Cables</i> that support <a href="#">[USB4]</a>.</p> <p>9) <b>Shall</b> be supported by products that support <i>Source</i> operation in <i>EPR Mode</i>.</p> <p>10) <b>Shall</b> be supported by all <i>Source Ports</i> except <i>singlePort SPR Chargers</i> with <i>Invariant PDOs</i>.</p> <p>11) <b>Shall</b> be supported when <i>SPR PPS Mode</i> is supported.</p> <p>12) <b>Shall</b> be supported by <i>Active Cables</i>.</p> |                 |                 |                 |                 |                  |                  |

## 6.13.3 Applicability of Extended Messages

Table 6.79, "[Applicability of Extended Messages](#)" details *Extended Messages* (except for *VDEM Commands*) that **Shall/Should/ Shall Not** be transmitted and received by a *Source*, *Sink*, *Cable Plug* or *VPD*. Requirements for *Dual-Role Power Ports* **Shall** override any requirements for *Source-only* or *Sink-Only* Ports.

**Table 6.79 Applicability of Extended Messages**

| Message Type                        | Source                                                                                                                                                                                         | Sink                 | Dual-Role Power      | Cable Plug SOP'     | Cable Plug SOP'' | VPD <sup>13</sup> |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|---------------------|------------------|-------------------|
| Transmitted Message                 |                                                                                                                                                                                                |                      |                      |                     |                  |                   |
| <i>Battery_Capabilities</i>         | CN <sup>1</sup> /NA                                                                                                                                                                            | CN <sup>1</sup> /NA  |                      | NA                  | NA               | NA                |
| <i>Country_Codes</i>                | CN <sup>10</sup> /NA                                                                                                                                                                           | CN <sup>10</sup> /NA |                      | NA                  | NA               | NA                |
| <i>Country_Info</i>                 | CN <sup>10</sup> /NA                                                                                                                                                                           | CN <sup>10</sup> /NA |                      | NA                  | NA               | NA                |
| <i>EPR_Source_Capabilities</i>      | CN <sup>14</sup> /NA                                                                                                                                                                           | NA                   | CN <sup>14</sup> /NA | NA                  | NA               | NA                |
| <i>EPR_Sink_Capabilities</i>        | NA                                                                                                                                                                                             | CN <sup>14</sup> /NA | CN <sup>14</sup> /NA | NA                  | NA               | NA                |
| <i>Extended_Control</i>             | See <a href="#">Section 6.13.4</a> for details                                                                                                                                                 |                      |                      |                     |                  |                   |
| <i>Firmware_Update_Request</i>      | CN <sup>7</sup> /NA                                                                                                                                                                            | CN <sup>7</sup> /NA  |                      | NA                  | NA               | NA                |
| <i>Firmware_Update_Response</i>     | CN <sup>7</sup> /NA                                                                                                                                                                            | CN <sup>7</sup> /NA  |                      | CN <sup>7</sup> /NA | O                | NA                |
| <i>Get_Battery_Cap</i>              | R                                                                                                                                                                                              | R                    |                      | NA                  | NA               | NA                |
| <i>Get_Battery_Status</i>           | R                                                                                                                                                                                              | R                    |                      | NA                  | NA               | NA                |
| <i>Get_Manufacturer_Info</i>        | R                                                                                                                                                                                              | R                    |                      | NA                  | NA               | NA                |
| <i>Manufacturer_Info</i>            | R                                                                                                                                                                                              | R                    |                      | R                   | NA               | NA                |
| <i>PPS_Status</i>                   | CN <sup>8</sup> /NA                                                                                                                                                                            | NA                   |                      | NA                  | NA               | NA                |
| <i>Security_Request</i>             | CN <sup>6</sup> /NA                                                                                                                                                                            | CN <sup>6</sup> /NA  |                      | NA                  | NA               | NA                |
| <i>Security_Response</i>            | CN <sup>6</sup> /NA                                                                                                                                                                            | CN <sup>6</sup> /NA  |                      | CN <sup>6</sup> /NA | NA               | NA                |
| <i>Sink_Capabilities_Extended</i>   | NA                                                                                                                                                                                             | N                    | N                    | NA                  | NA               | NA                |
| <i>Source_Capabilities_Extended</i> | R                                                                                                                                                                                              | NA                   | R                    | NA                  | NA               | NA                |
| 1)                                  | <b>Shall</b> be supported by products that contain batteries.                                                                                                                                  |                      |                      |                     |                  |                   |
| 2)                                  | <b>Shall</b> be supported by products that can transmit the <i>Get_Source_Cap_Extended</i> Message.                                                                                            |                      |                      |                     |                  |                   |
| 3)                                  | <b>Shall</b> be supported by products that can transmit the <i>Get_Status</i> Message.                                                                                                         |                      |                      |                     |                  |                   |
| 4)                                  | <b>Shall</b> be supported by products that can transmit the <i>Get_Battery_Cap</i> Message.                                                                                                    |                      |                      |                     |                  |                   |
| 5)                                  | <b>Shall</b> be supported by products that can transmit the <i>Get_Manufacturer_Info</i> Message.                                                                                              |                      |                      |                     |                  |                   |
| 6)                                  | <b>Shall</b> be supported by products that support USB security communication as defined in <a href="#">[USBTypeCAuthentication 1.0]</a> .                                                     |                      |                      |                     |                  |                   |
| 7)                                  | <b>Shall</b> be supported by products that support USB firmware update communication as defined in <a href="#">[USBPDFirmwareUpdate 1.0]</a> .                                                 |                      |                      |                     |                  |                   |
| 8)                                  | <b>Shall</b> be supported when PPS is supported.                                                                                                                                               |                      |                      |                     |                  |                   |
| 9)                                  | <b>Shall</b> be supported by products that can transmit the <i>Get_PPS_Status</i> Message.                                                                                                     |                      |                      |                     |                  |                   |
| 10)                                 | <b>Shall</b> be supported when required by a country authority.                                                                                                                                |                      |                      |                     |                  |                   |
| 11)                                 | <b>Shall</b> be supported by products that can transmit the <i>Get_Sink_Cap_Extended</i> Message.                                                                                              |                      |                      |                     |                  |                   |
| 12)                                 | <b>Shall</b> be supported by <i>Active Cables</i> .                                                                                                                                            |                      |                      |                     |                  |                   |
| 13)                                 | VPD includes <i>CT-VPDs</i> when not <i>Connected</i> to a <i>Charger</i> . PD communication with a <i>CT-VPD</i> <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i> . |                      |                      |                     |                  |                   |
| 14)                                 | <b>Shall</b> be supported by products that support operation in <i>EPR Mode</i> .                                                                                                              |                      |                      |                     |                  |                   |
| 15)                                 | <b>Shall</b> be supported by <i>Sources</i> that support the <i>Alert</i> Message.                                                                                                             |                      |                      |                     |                  |                   |

Table 6.79 Applicability of Extended Messages (Continued)

| Message Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Source                                         | Sink                 | Dual-Role Power      | Cable Plug SOP'      | Cable Plug SOP''     | VPD <sup>13</sup>                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------------|----------------------|----------------------|----------------------|-------------------------------------------------------------------------------------|
| <i>Status</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CN <sup>15</sup> /R                            | CN <sup>15</sup> /R  | CN <sup>15</sup> /R  | CN <sup>12</sup> /NA | CN <sup>12</sup> /NA | NA                                                                                  |
| <i>Vendor_Defined_Extended</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | O                                              | O                    |                      | O                    | O                    | O                                                                                   |
| Received Message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                |                      |                      |                      |                      |                                                                                     |
|  <i>Battery_Capabilities</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CN <sup>4</sup> /NS                            | CN <sup>4</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Country_Codes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CN <sup>10</sup> /NS                           | CN <sup>10</sup> /NS |                      | I                    | I                    | I                                                                                   |
| <i>Country_Info</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CN <sup>10</sup> /NS                           | CN <sup>10</sup> /NS |                      | I                    | I                    | I                                                                                   |
| <i>EPR_Source_Capabilities</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NS                                             | CN <sup>14</sup> /NS | CN <sup>14</sup> /NS | I                    | I                    | I                                                                                   |
| <i>EPR_Sink_Capabilities</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CN <sup>14</sup> /NS                           | NS                   | CN <sup>14</sup> /NS | I                    | I                    | I                                                                                   |
| <i>Extended_Control</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | See <a href="#">Section 6.13.4</a> for details |                      |                      |                      |                      |                                                                                     |
| <i>Firmware_Update_Request</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CN <sup>7</sup> /NS                            | CN <sup>7</sup> /NS  |                      | CN <sup>7</sup> /I   | O                    | I                                                                                   |
| <i>Firmware_Update_Response</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CN <sup>7</sup> /NS                            | CN <sup>7</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Get_Battery_Cap</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CN <sup>1</sup> /NS                            | CN <sup>1</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Get_Battery_Status</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CN <sup>1</sup> /NS                            | CN <sup>1</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Get_Manufacturer_Info</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/NS                                           | R/NS                 |                      | R/I                  | I                    |  |
| <i>Manufacturer_Info</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CN <sup>5</sup> /NS                            | CN <sup>5</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>PPS_Status</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NS                                             | CN <sup>9</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Security_Request</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CN <sup>6</sup> /NS                            | CN <sup>6</sup> /NS  |                      | CN <sup>6</sup> /I   | I                    | I                                                                                   |
| <i>Security_Response</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CN <sup>6</sup> /NS                            | CN <sup>6</sup> /NS  |                      | I                    | I                    | I                                                                                   |
| <i>Sink_Capabilities_Extended</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CN <sup>11</sup> /NS                           | NS                   | CN <sup>11</sup> /NS | I                    | I                    | I                                                                                   |
| <i>Source_Capabilities_Extended</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NS                                             | CN <sup>2</sup> /NS  | CN <sup>2</sup> /NS  | I                    | I                    | I                                                                                   |
| <ol style="list-style-type: none"> <li>1) <b>Shall</b> be supported by products that contain batteries.</li> <li>2) <b>Shall</b> be supported by products that can transmit the <i>Get_Source_Cap_Extended</i> Message.</li> <li>3) <b>Shall</b> be supported by products that can transmit the <i>Get_Status</i> Message.</li> <li>4) <b>Shall</b> be supported by products that can transmit the <i>Get_Battery_Cap</i> Message.</li> <li>5) <b>Shall</b> be supported by products that can transmit the <i>Get_Manufacturer_Info</i> Message.</li> <li>6) <b>Shall</b> be supported by products that support USB security communication as defined in <a href="#">[USBTypeCAuthentication 1.0]</a>.</li> <li>7) <b>Shall</b> be supported by products that support USB firmware update communication as defined in <a href="#">[USBPDFirmwareUpdate 1.0]</a>.</li> <li>8) <b>Shall</b> be supported when PPS is supported.</li> <li>9) <b>Shall</b> be supported by products that can transmit the <i>Get_PPS_Status</i> Message.</li> <li>10) <b>Shall</b> be supported when required by a country authority.</li> <li>11) <b>Shall</b> be supported by products that can transmit the <i>Get_Sink_Cap_Extended</i> Message.</li> <li>12) <b>Shall</b> be supported by <i>Active Cables</i>.</li> <li>13) VPD includes <i>CT</i>-VPDs when not <i>Connected</i> to a <i>Charger</i>. PD communication with a <i>CT</i>-VPD <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i>.</li> <li>14) <b>Shall</b> be supported by products that support operation in <i>EPR Mode</i>.</li> <li>15) <b>Shall</b> be supported by <i>Sources</i> that support the <i>Alert</i> Message.</li> </ol> |                                                |                      |                      |                      |                      |                                                                                     |

Table 6.79 Applicability of Extended Messages (Continued)

| Message Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Source               | Sink                | Dual-Role Power | Cable Plug SOP' | Cable Plug SOP'' | VPD <sup>13</sup> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------|-----------------|-----------------|------------------|-------------------|
| <i>Status</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CN <sup>33</sup> /NS | CN <sup>3</sup> /NS |                 | I               | I                | I                 |
| <i>Vendor_Defined_Extended</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | O/NS                 | O/NS                |                 | O/I             | O/I              | O/I               |
| 1) <b>Shall</b> be supported by products that contain batteries.<br>2) <b>Shall</b> be supported by products that can transmit the <i>Get_Source_Cap_Extended</i> Message.<br>3) <b>Shall</b> be supported by products that can transmit the <i>Get_Status</i> Message.<br>4) <b>Shall</b> be supported by products that can transmit the <i>Get_Battery_Cap</i> Message.<br>5) <b>Shall</b> be supported by products that can transmit the <i>Get_Manufacturer_Info</i> Message.<br>6) <b>Shall</b> be supported by products that support USB security communication as defined in <i>[USBTypeCAuthentication 1.0]</i> .<br>7) <b>Shall</b> be supported by products that support USB firmware update communication as defined in <i>[USBPD FirmwareUpdate 1.0]</i> .<br>8) <b>Shall</b> be supported when PPS is supported.<br>9) <b>Shall</b> be supported by products that can transmit the <i>Get_PPS_Status</i> Message.<br>10) <b>Shall</b> be supported when required by a country authority.<br>11) <b>Shall</b> be supported by products that can transmit the <i>Get_Sink_Cap_Extended</i> Message.<br>12) <b>Shall</b> be supported by <i>Active Cables</i> .<br>13) VPD includes <i>CT-VPDs</i> when not <i>Connected</i> to a <i>Charger</i> . PD communication with a <i>CT-VPD</i> <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i> .<br>14) <b>Shall</b> be supported by products that support operation in <i>EPR Mode</i> .<br>15) <b>Shall</b> be supported by <i>Sources</i> that support the <i>Alert</i> Message. |                      |                     |                 |                 |                  |                   |



## 6.13.4 Applicability of Extended Control Messages

[Table 6.80, "Applicability of Extended Control Messages"](#) details *Extended Control Messages* that **Shall/Should/****Shall Not** be transmitted and received by a *Source*, *Sink*, *Cable Plug* or *VPD*. Requirements for *Dual-Role Power Ports* and *Dual-Role Data Ports* **Shall** override any requirements for *Source-only* or *Sink-Only* Ports.

**Table 6.80 Applicability of Extended Control Messages**

| Message Type                                                                                                                                                                                                                                                                        | Source          | Sink            | Dual-Role Power | Dual-Role Data | Cable Plug | VPD <sup>2</sup> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|----------------|------------|------------------|
| Transmitted Message                                                                                                                                                                                                                                                                 |                 |                 |                 |                |            |                  |
| <i>EPR_Get_Source_Cap</i>                                                                                                                                                                                                                                                           | NA              | CN <sup>1</sup> | CN <sup>1</sup> |                | NA         | NA               |
| <i>EPR_Get_Sink_Cap</i>                                                                                                                                                                                                                                                             | CN <sup>1</sup> | NA              | CN <sup>1</sup> |                | NA         | NA               |
| <i>EPR_KeepAlive</i>                                                                                                                                                                                                                                                                | NA              | CN <sup>1</sup> |                 |                | NA         | NA               |
| <i>EPR_KeepAlive_Ack</i>                                                                                                                                                                           | CN <sup>1</sup> | NA              |                 |                | NA         | NA               |
| Received Message                                                                                                                                                                                                                                                                    |                 |                 |                 |                |            |                  |
| <i>EPR_Get_Source_Cap</i>                                                                                                                                                                                                                                                           | CN <sup>1</sup> | NS              | CN <sup>1</sup> |                | I          | I                |
| <i>EPR_Get_Sink_Cap</i>                                                                                                                                                                                                                                                             | NS              | CN <sup>1</sup> | CN <sup>1</sup> |                | I          | I                |
| <i>EPR_KeepAlive</i>                                                                                                                                                                                                                                                                | CN <sup>1</sup> | NS              |                 |                | I          | I                |
| <i>EPR_KeepAlive_Ack</i>                                                                                                                                                                                                                                                            | NS              | CN <sup>1</sup> |                 |                | I          | I                |
| 1) <b>Shall</b> be supported by products that support <i>EPR Mode</i> .<br>2) <i>VPD</i> includes <i>CT-VPDs</i> when not <i>Connected</i> to a <i>Charger</i> . PD communication with a <i>CT-VPD</i> <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i> . |                 |                 |                 |                |            |                  |

## 6.13.5 Applicability of Structured VDM Commands

Table 6.81, "Applicability of Structured VDM Commands" details Structured VDM Commands that **Shall/Should/ Shall Not** be transmitted and received by a DFP, UFP, Cable Plug or VPD. If Structured VDMs are not supported, the DFP or UFP receiving a VDM Command **Shall** send a **Not\_Supported** Message in response.

Table 6.81 Applicability of Structured VDM Commands

| Command Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DFP                                      | UFP                                      | Cable Plug SOP'     | Cable Plug SOP'' | VPD <sup>4</sup> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------|---------------------|------------------|------------------|
| Transmitted Command Request                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                          |                                          |                     |                  |                  |
| <i>Discover Identity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CN <sup>1,6</sup> /R                     | R <sup>2</sup>                           | NA                  | NA               | NA               |
| <i>Discover SVIDs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CN <sup>1</sup> /O                       | O                                        | NA                  | NA               | NA               |
| <i>Discover Modes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CN <sup>1</sup> /O                       | O                                        | NA                  | NA               | NA               |
| <i>Enter Mode</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CN <sup>1</sup> /NA                      | NA                                       | NA                  | NA               | NA               |
| <i>Exit Mode</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CN <sup>1</sup> /NA                      | NA                                       | NA                  | NA               | NA               |
| <i>Attention</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | O                                        | O                                        | NA                  | NA               | NA               |
| Received Command Request/Transmitted Command Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                          |                                          |                     |                  |                  |
| <i>Discover Identity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CN <sup>5,6</sup> /R/<br>NK <sup>3</sup> | CN <sup>1,6</sup> /R/<br>NK <sup>3</sup> | N                   | I                | N                |
| <i>Discover SVIDs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | O/NK <sup>3</sup>                        | CN <sup>1</sup> /NK <sup>3</sup>         | CN <sup>1</sup> /NK | I                | NK               |
| <i>Discover Modes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | O/NK <sup>3</sup>                        | CN <sup>1</sup> /NK <sup>3</sup>         | CN <sup>1</sup> /NK | I                | NK               |
| <i>Enter Mode</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NK <sup>3</sup>                          | CN <sup>1</sup> /NK <sup>3</sup>         | CN <sup>1</sup> /NK | O                | NK               |
| <i>Exit Mode</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | NK <sup>3</sup>                          | CN <sup>1</sup> /NK <sup>3</sup>         | CN <sup>1</sup> /NK | O                | NK               |
| <i>Attention</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | O/I <sup>3</sup>                         | O/I <sup>3</sup>                         | I                   | I                | I                |
| <ol style="list-style-type: none"> <li>1) <b>Shall</b> be supported when <i>Modal Operation</i> is supported.</li> <li>2) <b>May</b> be transmitted by a UFP/Source during discovery (see <a href="#">Section 6.4.4.3.1, "Discover Identity"</a> and <a href="#">Section 8.3.3.25.3, "Source Startup Structured VDM Discover Identity of a Cable Plug State Diagram"</a>).</li> <li>3) If <i>Structured VDMs</i> are not supported, the DFP or UFP receiving a VDM Command <b>Shall</b> send a <b>Not_Supported</b> Message in response.</li> <li>4) VPD includes CT-VPDs when not <i>Connected</i> to a <i>Charger</i>. PD communication with a <b>CT-VPD</b> <b>Shall</b> only take place when not <i>Connected</i> to a <i>Charger</i>.</li> <li>5) <b>Shall</b> be supported by products with more than one DFP.</li> <li>6) <b>Shall</b> be supported by products that support <b>[USB4]</b>.</li> </ol> |                                          |                                          |                     |                  |                  |

## 6.13.6 Applicability of Reset Signaling

Table 6.82, "Applicability of Reset Signaling" details the Reset that **Shall/Should/ Shall Not** be transmitted and received by a DFP/UPF or Cable Plug.

Table 6.82 Applicability of Reset Signaling

| Reset Type                                                                                                                                                                                                                                                            | DFP             | UPF | Cable Plug SOP' | Cable Plug SOP'' | VPD <sup>2</sup> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----------------|------------------|------------------|
| Transmitted Message/Signaling                                                                                                                                                                                                                                         |                 |     |                 |                  |                  |
| <i>Soft_Reset</i>                                                                                                                                                                                                                                                     | N               | N   | NA              | NA               | NA               |
| <i>Hard_Reset</i>                                                                                                                                                                                                                                                     | N               | N   | NA              | NA               | NA               |
| <i>Cable_Reset</i>                                                                                                                                                                                                                                                    | CN <sup>1</sup> | NA  | NA              | NA               | NA               |
| Received Message/Signaling                                                                                                                                                                                                                                            |                 |     |                 |                  |                  |
|  <i>Soft_Reset</i>                                                                                                                                                                   | N               | N   | N               | N                | N                |
| <i>Hard_Reset</i>                                                                                                                                                                                                                                                     | N               | N   | N               | N                | N                |
| <i>Cable_Reset</i>                                                                                                                                                                                                                                                    | DR              | DR  | N               | N                | N                |
| 1) <b>Shall</b> be supported when transmission of SOP' Packets are supported, and the Port can supply VCONN.<br>2) VPD includes CT-VPDs when not Connected to a Charger. PD communication with a CT-VPD <b>Shall</b> only take place when not Connected to a Charger. |                 |     |                 |                  |                  |

## 6.13.7 Applicability of Fast Role Swap Request

Table 6.83, "Applicability of Fast Role Swap Request" details the Fast Role Swap Request that **Shall/Should/ Shall Not** be transmitted and received by a Source or Sink.

Table 6.83 Applicability of Fast Role Swap Request

| Command Type                                                                                              | Source | Sink | Dual-Role Power |
|-----------------------------------------------------------------------------------------------------------|--------|------|-----------------|
| Transmitted Message/Signaling                                                                             |        |      |                 |
| <i>Fast Role Swap</i>                                                                                     | NA     | NA   | R               |
| Received Message/Signaling                                                                                |        |      |                 |
|  <i>Fast Role Swap</i> | NA     | NA   | R               |

# 6.14 Value Parameters

Table 6.84, "Value Parameters" contains value parameters used in this section.

Table 6.84 Value Parameters

| Parameter                      | Description                                                                              | Value | Unit | Reference                       |
|--------------------------------|------------------------------------------------------------------------------------------|-------|------|---------------------------------|
| <i>MaxExtendedMsgLen</i>       | Maximum length of an <i>Extended Message</i> as expressed in the <i>Data Size</i> field. | 260   | Byte | <a href="#">Section 6.2.1.2</a> |
| <i>MaxExtendedMsgChunkLen</i>  | Maximum length of an <i>Extended Message Chunk</i> .                                     | 26    | Byte | <a href="#">Section 6.2.1.2</a> |
| <i>MaxExtendedMsgLegacyLen</i> | Maximum length of an <i>Extended Message</i> that can be sent without <i>Chunking</i> .  | 26    | Byte | <a href="#">Section 6.2.1.2</a> |